QGIS User Guide

*출시 **2.0.0***

QGIS Project

2013년 06월 20일

CHAPTER 1

Preamble

이 문서는 Quantum GIS 소프트웨어의 사용자가이드입니다. 이 문서에서 기술된소프트웨어와 하드웨어는 대부분 등록상표로서, 이에 따른 법적제한을 받습니다. Quantum GIS는 GNU General Public License에 따릅니다. Quantum GIS 에 관한 좀 더 자세한 내용은 홈페이지 http://www.qgis.org를 참고하시기 바랍니다.

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이 문서의 링크 (**Links in this Document**)

This document contains internal and external links. Clicking on an internal link moves within the document, while clicking on an external link opens an internet address. In PDF form, internal and external links are shown in blue and are handled by the system browser. In HTML form, the browser displays and handles both identically.

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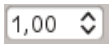
CHAPTER 2

Conventions

This section describes a collection of uniform styles throughout the manual. The conventions used in this manual are as follows:

2.1 GUI Conventions

The GUI convention styles are intended to mimic the appearance of the GUI. In general, the objective is to use the non-hover appearance, so a user can visually scan the GUI to find something that looks like the instruction in the manual.

- Menu Options: *Layer* → *Add a Raster Layer* or *Settings* → *Toolbars* → *Digitizing*
- Tool:  Add a Raster Layer
- Button : **[Save as Default]**
- Dialog Box Title: *Layer Properties*
- Tab: *General*
- Checkbox:  *Render*
- Radio Button:  *Postgis SRID*  *EPSG ID*
- Select a Number: 

- Select a String: 
- Browse for a File: 

- Select a Color: 

- Slider: 
- Input Text: 

A shadow indicates a clickable GUI component.

2.2 Text or Keyboard Conventions

The manual also includes styles related to text, keyboard commands and coding to indicate different entities, such as classes, or methods. They don't correspond to any actual appearance.

- Hyperlinks: http://qgis.org
- Keystroke Combinations: press `Ctrl+B`, meaning press and hold the Ctrl key and then press the B key.

- Name of a File: `lakes.shp`
- Name of a Class: **NewLayer**
- Method: *classFactory*
- Server: *myhost.de*
- User Text: `qgis --help`

Lines of code are indicated by a fixed-width font

```
PROJCS["NAD_1927_Albers",
  GEOGCS["GCS_North_American_1927",
```

2.3 Platform-specific instructions

GUI sequences and small amounts of text can be formatted inline: Click *File* → *QGIS* → *Quit to close QGIS*.

This indicates that on Linux, Unix and Windows platforms, click the File menu option first, then Quit from the dropdown menu, while on Macintosh OSX platforms, click the QGIS menu option first, then Quit from the dropdown menu. Larger amounts of text may be formatted as a list:

- do this;
- do that;
- do something else.

or as paragraphs.

Do this and this and this. Then do this and this and this and this and this and this and this and this and this.

Do that. Then do that and that and that and that and that and that and that and that and that and that and that and that and that and that and that and that and that and that and that.

Screenshots that appear throughout the user guide have been created on different platforms; the platform is indicated by the platform-specific icon at the end of the figure caption.

CHAPTER 3

Foreword

Welcome to the wonderful world of Geographical Information Systems (GIS)!

Quantum GIS (QGIS) is an Open Source Geographic Information System. The project was born in May of 2002 and was established as a project on SourceForge in June of the same year. We've worked hard to make GIS software (which is traditionally expensive proprietary software) a viable prospect for anyone with basic access to a Personal Computer. QGIS currently runs on most Unix platforms, Windows, and OS X. QGIS is developed using the Qt toolkit (http://qt.digia.com) and C++. This means that QGIS feels snappy to use and has a pleasing, easy-to-use graphical user interface (GUI).

QGIS aims to be an easy-to-use GIS, providing common functions and features. The initial goal was to provide a GIS data viewer. QGIS has reached the point in its evolution where it is being used by many for their daily GIS data viewing needs. QGIS supports a number of raster and vector data formats, with new format support easily added using the plugin architecture.

QGIS is released under the GNU General Public License (GPL). Developing QGIS under this license means that you can inspect and modify the source code, and guarantees that you, our happy user, will always have access to a GIS program that is free of cost and can be freely modified. You should have received a full copy of the license with your copy of QGIS, and you also can find it in Appendix *GNU General Public License*.

참고: Up-to-date Documentation

The latest version of this document can always be found in the documentation area of the QGIS website at http://documentation.qgis.org

3.1 Introduction To GIS

A Geographical Information System (GIS) (Mitchell 2005 *Literature and Web References*) is a collection of software that allows you to create, visualize, query and analyze geospatial data. Geospatial data refers to information about the geographic location of an entity. This often involves the use of a geographic coordinate, like a latitude or longitude value. Spatial data is another commonly used term, as are: geographic data, GIS data, map data, location data, coordinate data and spatial geometry data.

Applications using geospatial data perform a variety of functions. Map production is the most easily understood function of geospatial applications. Mapping programs take geospatial data and render it in a form that is viewable, usually on a computer screen or printed page. Applications can present static maps (a simple image) or dynamic maps that are customised by the person viewing the map through a desktop program or a web page.

Many people mistakenly assume that geospatial applications just produce maps, but geospatial data analysis is another primary function of geospatial applications. Some typical types of analysis include computing:

1. distances between geographic locations
2. the amount of area (e.g., square meters) within a certain geographic region
3. what geographic features overlap other features

4. the amount of overlap between features
5. the number of locations within a certain distance of another
6. and so on...

These may seem simplistic, but can be applied in all sorts of ways across many disciplines. The results of analysis may be shown on a map, but are often tabulated into a report to support management decisions.

The recent phenomena of location-based services promises to introduce all sorts of other features, but many will be based on a combination of maps and analysis. For example, you have a cell phone that tracks your geographic location. If you have the right software, your phone can tell you what kind of restaurants are within walking distance. While this is a novel application of geospatial technology, it is essentially doing geospatial data analysis and listing the results for you.

3.1.1 Why is all this so new?

Well, it's not. There are many new hardware devices that are enabling mobile geospatial services. Many open source geospatial applications are also available, but the existence of geospatially focused hardware and software is nothing new. Global positioning system (GPS) receivers are becoming commonplace, but have been used in various industries for more than a decade. Likewise, desktop mapping and analysis tools have also been a major commercial market, primarily focused on industries such as natural resource management.

What is new is how the latest hardware and software is being applied and who is applying it. Traditional users of mapping and analysis tools were highly trained GIS Analysts or digital mapping technicians trained to use CAD-like tools. Now, the processing capabilities of home PCs and open source software (OSS) packages have enabled an army of hobbyists, professionals, web developers, etc. to interact with geospatial data. The learning curve has come down. The costs have come down. The amount of geospatial technology saturation has increased.

How is geospatial data stored? In a nutshell, there are two types of geospatial data in widespread use today. This is in addition to traditional tabular data that is also widely used by geospatial applications.

3.1.2 Raster Data

One type of geospatial data is called raster data or simply "a raster". The most easily recognised form of raster data is digital satellite imagery or air photos. Elevation shading or digital elevation models are also typically represented as raster data. Any type of map feature can be represented as raster data, but there are limitations.

A raster is a regular grid made up of cells, or in the case of imagery, pixels. They have a fixed number of rows and columns. Each cell has a numeric value and has a certain geographic size (e.g. 30x30 meters in size).

Multiple overlapping rasters are used to represent images using more than one colour value (i.e. one raster for each set of red, green and blue values is combined to create a colour image). Satellite imagery also represents data in multiple "bands". Each band is essentially a separate, spatially overlapping raster, where each band holds values of certain wavelengths of light. As you can imagine, a large raster takes up more file space.

A raster with smaller cells can provide more detail, but takes up more file space. The trick is finding the right balance between cell size for storage purposes and cell size for analytical or mapping purposes.

3.1.3 Vector Data

Vector data is also used in geospatial applications. If you stayed awake during trigonometry and coordinate geometry classes, you will already be familiar with some of the qualities of vector data. In its simplest sense, vectors are a way of describing a location by using a set of coordinates. Each coordinate refers to a geographic location using a system of x and y values.

This can be thought of in reference to a Cartesian plane - you know, the diagrams from school that showed an x and y-axis. You might have used them to chart declining retirement savings or increasing compound mortgage interest, but the concepts are essential to geospatial data analysis and mapping.

There are various ways of representing these geographic coordinates depending on your purpose. This is a whole area of study for another day - map projections.

Vector data takes on three forms, each progressively more complex and building on the former.

1. Points - A single coordinate (x y) represents a discrete geographic location
2. Lines - Multiple coordinates (x_1 y_1, x_2 y_2, x_3 y_3, ... x_n y_n) strung together in a certain order, like drawing a line from Point (x_1 y_1) to Point (x_2 y_2) and so on. These parts between each point are considered line segments. They have a length and the line can be said to have a direction based on the order of the points. Technically, a line is a single pair of coordinates connected together, whereas a line string is multiple lines connected together.
3. Polygons - When lines are strung together by more than two points, with the last point being at the same location as the first, we call this a polygon. A triangle, circle, rectangle, etc. are all polygons. The key feature of polygons is that there is a fixed area within them.

CHAPTER 4

Features

QGIS offers many common GIS functionalities provided by core features and plugins. As a short summary they are presented in six categories to gain a first insight.

4.1 View data

You can view and overlay vector and raster data in different formats and projections without conversion to an internal or common format. Supported formats include:

- Spatially-enabled tables and views using PostGIS, SpatiaLite and MSSQL Spatial, vector formats supported by the installed OGR library, including ESRI shapefiles, MapInfo, SDTS, GML and many more, see section *벡터 데이터 다루기*.
- Raster and imagery formats supported by the installed GDAL (Geospatial Data Abstraction Library) library, such as GeoTiff, Erdas Img., ArcInfo Ascii Grid, JPEG, PNG and many more, see section *Working with Raster Data*.
- GRASS raster and vector data from GRASS databases (location/mapset), see section *GRASS GIS Integration*.
- Online spatial data served as OGC-compliant Web Map Service (WMS) or Web Feature Service (WFS), see section *Working with OGC Data*.
- OpenStreetMap data, see section *OpenStreetMap*.

4.2 Explore data and compose maps

You can compose maps and interactively explore spatial data with a friendly GUI. The many helpful tools available in the GUI include:

- QGIS browser
- On the fly projection
- Map composer
- Overview panel
- Spatial bookmarks
- Identify/select features
- Edit/view/search attributes
- Feature labeling
- Change vector and raster symbology
- Add a graticule layer - now via fTools plugin and as decoration

- Decorate your map with a north arrow scale bar and copyright label
- Save and restore projects

4.3 Create, edit, manage and export data

You can create, edit, manage and export vector maps in several formats. Raster data have to be imported into GRASS to be able to edit and export them into other formats. QGIS offers the following:

- Digitizing tools for OGR supported formats and GRASS vector layer
- Create and edit shapefiles and GRASS vector layers
- Geocode images with the Georeferencer plugin
- GPS tools to import and export GPX format, and convert other GPS formats to GPX or down/upload directly to a GPS unit (on Linux, usb: has been addedto list of GPS devices)
- Visualize and edit OpenStreetMap data
- Create PostGIS layers from shapefiles with the SPIT plugin
- Improved handling of PostGIS tables
- Manage vector attribute tables with the new attribute table (see section *속성테이블 작업*) or Table Manager plugin
- Save screenshots as georeferenced images

4.4 Analyse data

You can perform spatial data analysis on PostgreSQL/PostGIS and other OGR supported formats using the fTools Python plugin. QGIS currently offers vector analysis, sampling, geoprocessing, geometry and database management tools. You can also use the integrated GRASS tools, which include the complete GRASS functionality of more than 400 modules (See Section *GRASS GIS Integration*). Or you work with SEXTANTE, which provides powerful a geospatial analysis framework to call native and third party algorithms from QGIS, such as GDAL, SAGA, GRASS, fTools and more (see section *SEXTANTE*).

4.5 Publish maps on the Internet

QGIS can be used to export data to a mapfile and to publish them on the Internet using a webserver with UMN MapServer installed. QGIS can also be used as a WMS, WMS-C or WFS and WFS-T client, and as WMS or WFS server (see section *Working with OGC Data*).

4.6 Extend QGIS functionality through plugins

QGIS can be adapted to your special needs with the extensible plugin architecture. QGIS provides libraries that can be used to create plugins. You can even create new applications with C++ or Python!

4.6.1 Core Plugins

1. Add Delimited Text Layer (Loads and displays delimited text files containing x,y coordinates)
2. Coordinate Capture (Capture mouse coordinates in different CRS)
3. DB Manager (Exchange, edit and view layers and tables; execute SQL queries)

4. Diagram Overlay (Placing diagrams on vector layer)
5. Dxf2Shp Converter (Convert DXF to Shape)
6. GPS Tools (Loading and importing GPS data)
7. GRASS (GRASS GIS integration)
8. GDALTools (Integrate GDAL Tools into QGIS)
9. Georeferencer GDAL (Adding projection information to raster using GDAL)
10. Heatmap tool (Generating raster heatmaps from point data)
11. Interpolation plugin (interpolate based on vertices of a vector layer)
12. Mapserver Export (Export QGIS project file to a MapServer map file)
13. Offline Editing (Allow offline editing and synchronizing with database)
14. OpenStreetMap plugin (Viewer and editor for openstreetmap data)
15. Oracle Spatial GeoRaster support
16. Plugin Installer (Download and install QGIS python plugins)
17. Raster terrain analysis (Raster based terrain analysis)
18. Road graph plugin (Shortest Path network analysis)
19. SPIT (Import Shapefile to PostgreSQL/PostGIS)
20. SQL Anywhere Plugin (Store vector layers within a SQL Anywhere database)
21. Zonal statictics plugin (Calculate count, sum, mean of raster for each polygon of a vector layer)
22. Spatial Query plugin (Makes spatial queries on vector layers)
23. eVIS (Event Visualization Tool)
24. fTools (Tools for vector data analysis and management)

4.6.2 External Python Plugins

QGIS offers a growing number of external python plugins that are provided by the community. These plugins reside in the official plugins repository, and can be easily installed using the Python Plugin Installer (See Section *Loading an external QGIS Plugin*).

4.7 What's new in the version 1.8.0

Please note that this is a release in our 'cutting edge' release series. As such it contains new features and extends the programmatic interface over QGIS 1.0.x and QGIS 1.7.0. We recommend that you use this version over previous releases.

This release includes hundreds of bug fixes and many new features and enhancements that will be described in this manual.

QGIS Browser

A stand alone app and a new panel in QGIS. The browser lets you easily navigate your file system and connection based (PostGIS, WFS etc.) datasets, preview them and drag and drop items into the canvas.

DB Manager

The DB manager is now officially part of QGIS core. You can drag layers from the QGIS Browser into DB Manager and it will import your layer into your spatial database. Drag and drop tables between spatial databases and they will get imported. You can use the DB Manager to execute SQL queries against your spatial database

and then view the spatial output for queries by adding the results to QGIS as a query layer. You can also create, edit, delete, and empty tables, and move them to another schema.

Terrain Analysis Plugin

A new core plugin was added for doing terrain analysis (slope, aspect, hillshade, relief and ruggedness index).

New symbol layer types

- Line Pattern Fill
- Point Pattern Fill
- Ellipse renderer (render ellipse and also rectangles, triangles, crosses)

New plugin repository

Note that the old repository is now no longer supported by default; plugin authors are kindly requested to move their plugins to the new repository. Get the QGIS Plugins list at http://plugins.qgis.org/plugins/.

More new features

- Support for nesting projects within other projects to embed content from other project files
- Group Selected: Option to group layers to a group
- Message log: Lets you keep an eye on the messages QGIS generates during loading and operation
- GUI Customization: Allows setting up simplified QGIS interface by hiding various components of main window and widgets in dialogs
- Action Tool is now accessible from the map tools toolbar and allows you to click on a vector feature and execute an action
- New scale selector: select from a list of predefined scales
- Pan To Selected tool: Pans the map to selected feature(s); does not change the zoom level
- Copy and paste styles between layers
- Updated CRS selector dialog
- Define Legend-independent drawing order
- MSSQL Spatial Support - you can now connect to your Microsoft SQL Server spatial databases using QGIS
- Print Composers allows to have multiple lines on legend items using a specified character
- Expression based labeling
- Heatmap Plugin - a new core plugin has been added for generating raster heatmaps from point data
- The GPS live tracking user interface was overhauled and many fixes and improvements were added to it
- The menu was re-organised a little - we now have separate menus for Vector, Raster, Web and many plugins were updated to place their menus in the new Vector, Raster and Web top level menus
- Offset Curves - a new digitising tool for creating offset curves was added
- New tools in the Vector menu to Densify geometries and Build spatial index
- Export/add geometry column tool can export info using layer CRS, project CRS or ellipsoidal measurements
- Model/view based tree for rules in rule-based renderer
- Improvements in Spatial Bookmarks
- New Plugin metadata in metadata.txt
- Refactored postgres data provider: support for arbitrary key (including non-numeric and multi column), support for requesting a certain geometry type and/or srid in QgsDataSourceURI
- Added gdal_fillnodata to GDALTools plugin
- Support for PostGIS TopoGeometry datatype

- Python bindings for vector field symbol layer and general updates to the Python bindings
- Added a Benchmark program
- Added Row cache for attribute table
- UUID generation widget for attribute table
- Added support of editable views in SpatiaLite databases
- added expression based widget in field calculator
- Creation of event layers in analysis lib using linear referencing
- Load/save layer styles in the new symbology renderer from/to SLD document
- QGIS Server can act as WFS Server
- WFS Client support is now a core feature in QGIS
- Option to skip WKT geometry when copying from attribute table
- Support loading of zipped and gzipped layers
- The QGIS test suite now passes all tests on major platforms and nightly tests
- You can set tile size for WMS layers

CHAPTER 5

Getting Started

This chapter gives a quick overview of installing QGIS, some sample data from the QGIS web page and running a first and simple session visualizing raster and vector layers.

5.1 Installation

Installation of QGIS is very simple. Standard installer packages are available for MS Windows and Mac OS X. For many flavors of GNU/Linux binary packages (rpm and deb) or software repositories to add to your installation manager are provided. Get the latest information on binary packages at the QGIS website at http://download.qgis.org.

5.1.1 Installation from source

If you need to build QGIS from source, please refer to the installation instructions. They are distributed with the QGIS source code in a file called 'INSTALL'. You can also find it online at https://github.com/qgis/Quantum-GIS/blob/master/INSTALL

5.1.2 Installation on external media

QGIS allows to define a `--configpath` option that overrides the default path (e.g. ~/.qgis under Linux) for user configuration and forces QSettings to use this directory, too. This allows users to e.g. carry a QGIS installation on a flash drive together with all plugins and settings.

5.2 Sample Data

The user guide contains examples based on the QGIS sample dataset.

The Windows installer has an option to download the QGIS sample dataset. If checked, the data will be downloaded to your `My Documents` folder and placed in a folder called `GIS Database`. You may use Windows Explorer to move this folder to any convenient location. If you did not select the checkbox to install the sample dataset during the initial QGIS installation, you can either

- use GIS data that you already have;
- download the sample data from the qgis website at http://download.qgis.org; or
- uninstall QGIS and reinstall with the data download option checked, only if the above solutions are unsuccessful.

For GNU/Linux and Mac OSX there are not yet dataset installation packages available as rpm, deb or dmg. To use the sample dataset download the file `qgis_sample_data` as ZIP or TAR archive from http://download.osgeo.org/qgis/data/ and unzip or untar the archive on your system. The Alaska dataset includes

all GIS data that are used as examples and screenshots in the user guide, and also includes a small GRASS database. The projection for the qgis sample dataset is Alaska Albers Equal Area with unit feet. The EPSG code is 2964.

```
PROJCS["Albers Equal Area",
    GEOGCS["NAD27",
        DATUM["North_American_Datum_1927",
            SPHEROID["Clarke 1866",6378206.4,294.978698213898,
                AUTHORITY["EPSG","7008"]],
            TOWGS84[-3,142,183,0,0,0,0],
            AUTHORITY["EPSG","6267"]],
        PRIMEM["Greenwich",0,
            AUTHORITY["EPSG","8901"]],
        UNIT["degree",0.0174532925199433,
            AUTHORITY["EPSG","9108"]],
        AUTHORITY["EPSG","4267"]],
    PROJECTION["Albers_Conic_Equal_Area"],
    PARAMETER["standard_parallel_1",55],
    PARAMETER["standard_parallel_2",65],
    PARAMETER["latitude_of_center",50],
    PARAMETER["longitude_of_center",-154],
    PARAMETER["false_easting",0],
    PARAMETER["false_northing",0],
    UNIT["us_survey_feet",0.3048006096012192]]
```

If you intend to use QGIS as graphical frontend for GRASS, you can find a selection of sample locations (e.g. Spearfish or South Dakota) at the official GRASS GIS website http://grass.osgeo.org/download/data.php.

5.3 Sample Session

Now that you have QGIS installed and a sample dataset available, we would like to demonstrate a short and simple QGIS sample session. We will visualize a raster and a vector layer. We will use the landcover raster layer `qgis_sample_data/raster/landcover.img` and the lakes vector layer `qgis_sample_data/gml/lakes.gml`.

5.3.1 Start QGIS

- Start QGIS by typing: "QGIS" at a command prompt, or if using precompiled binary, using the Applications menu.
- Start QGIS using the Start menu or desktop shortcut, or double click on a QGIS project file.
- Double click the icon in your Applications folder.

5.3.2 Load raster and vector layers from the sample dataset

1. Click on the Load Raster icon.
2. Browse to the folder `qgis_sample_data/raster/`, select the ERDAS Img file `landcover.img` and click **[Open]**.
3. If the file is not listed, check if the Filetype combobox at the bottom of the dialog is set on the right type, in this case "Erdas Imagine Images (`*.img`, `*.IMG`)".
4. Now click on the Load Vector icon.
5. *File* should be selected as Source Type in the new *Add Vector Layer* dialog. Now click **[Browse]** to select the vector layer.

6. Browse to the folder `qgis_sample_data/gml/`, select "GML" from the filetype combobox, then select the GML file `lakes.gml` and click **[Open]**, then in Add Vector dialog click **[OK]**.
7. Zoom in a bit to your favorite area with some lakes.
8. Double click the `lakes` layer in the map legend to open the *Properties* dialog.
9. Click on the *Style* tab and select a blue as fill color.
10. Click on the *Labels* tab and check the *Display lables* checkbox to enable labeling. Choose NAMES field as field containing label.
11. To improve readability of labels, you can add a white buffer around them, by clicking "Buffer" in the list on the left, checking *Buffer labels?* and choosing 3 as buffer size.
12. Click **[Apply]**, check if the result looks good and finally click **[OK]**.

You can see how easy it is to visualize raster and vector layers in QGIS. Let's move on to the sections that follow to learn more about the available functionality, features and settings and how to use them.

5.4 Starting and Stopping QGIS

In Section *Sample Session* you already learned how to start QGIS. We will repeat this here and you will see that QGIS also provides further command line options.

- Assuming that QGIS is installed in the PATH, you can start QGIS by typing: `qgis` at a command prompt or by double clicking on the QGIS application link (or shortcut) on the desktop or in the application menu.
- Start QGIS using the Start menu or desktop shortcut, or double click on a QGIS project file.
- Double click the icon in your Applications folder. If you need to start QGIS in a shell, run /path-to-installation-executable/Contents/MacOS/Qgis.

To stop QGIS, click the menu options *File* *QGIS* → *Quit*, or use the shortcut `Ctrl+Q`.

5.5 Command Line Options

QGIS supports a number of options when started from the command line. To get a list of the options, enter `qgis --help` on the command line. The usage statement for QGIS is:

```
qgis --help
Quantum GIS - 1.8.0-Lisboa 'Lisboa' (exported)
Quantum GIS (QGIS) is a viewer for spatial data sets, including
raster and vector data.
Usage: qgis [options] [FILES]
  options:
        [--snapshot filename]           emit snapshot of loaded datasets to given file
        [--width width]                 width of snapshot to emit
        [--height height]               height of snapshot to emit
        [--lang language]               use language for interface text
        [--project projectfile]         load the given QGIS project
        [--extent xmin,ymin,xmax,ymax]  set initial map extent
        [--nologo]                      hide splash screen
        [--noplugins]                   don't restore plugins on startup
        [--nocustomization]             don't apply GUI customization
        [--optionspath path]            use the given QSettings path
        [--configpath path]             use the given path for all user configuration
        [--help]                        this text

  FILES:
```

```
Files specified on the command line can include rasters,
vectors, and QGIS project files (.qgs):
1. Rasters - Supported formats include GeoTiff, DEM
   and others supported by GDAL
2. Vectors - Supported formats include ESRI Shapefiles
   and others supported by OGR and PostgreSQL layers using
   the PostGIS extension
```

참고: Example Using command line arguments

You can start QGIS by specifying one or more data files on the command line. For example, assuming you are in the qgis_sample_data directory, you could start QGIS with a vector layer and a raster file set to load on startup using the following command: `qgis ./raster/landcover.img ./gml/lakes.gml`

Command line option `--snapshot`

This option allows you to create a snapshot in PNG format from the current view. This comes in handy when you have a lot of projects and want to generate snapshots from your data.

Currently it generates a PNG-file with 800x600 pixels. This can be adapted using the `--width` and `--height` command line arguments. A filename can be added after `--snapshot`.

Command line option `--lang`

Based on your locale QGIS, selects the correct localization. If you would like to change your language, you can specify a language code. For example: `--lang=it` starts QGIS in italian localization. A list of currently supported languages with language code and status is provided at http://hub.qgis.org/wiki/quantum-gis/GUI_Translation_Progress

Command line option `--project`

Starting QGIS with an existing project file is also possible. Just add the command line option `--project` followed by your project name and QGIS will open with all layers loaded described in the given file.

Command line option `--extent`

To start with a specific map extent use this option. You need to add the bounding box of your extent in the following order separated by a comma:

```
--extent xmin,ymin,xmax,ymax
```

Command line option `--nologo`

This command line argument hides the splash screen when you start QGIS.

Command line option `--noplugins`

If you have trouble at startup with plugins, you can avoid loading them at startup. They will still be available in Plugins Manager after-wards.

Command line option `--nocustomization`

Using this command line argument existing GUI customization will not be applied at startup.

Command line option `--optionspath`

You can have multiple configurations and decide which one to use when starting QGIS using this option. See *Options* to check where does the operating system save the settings files. Presently there is no way to specify in which file where to write the settings, therefore you can create a copy of the original settings file and rename it.

Command line option `--configpath`

This option is similar to the one above, but furthermore overrides the default path (~/.qgis) for user configuration and forces QSettings to use this directory, too. This allows users to e.g. carry QGIS installation on a flash drive together with all plugins and settings.

5.6 Projects

The state of your QGIS session is considered a Project. QGIS works on one project at a time. Settings are either considered as being per-project, or as a default for new projects (see Section *Options*). QGIS can save the state of your workspace into a project file using the menu options *File* → *Save Project* or *File* → *Save Project As*.

Load saved projects into a QGIS session using *File* → *Open Project* or *File* → *Open Recent Project*.

If you wish to clear your session and start fresh, choose *File* → *New Project*. Either of these menu options will prompt you to save the existing project if changes have been made since it was opened or last saved.

The kinds of information saved in a project file include:

- Layers added
- Layer properties, including symbolization
- Projection for the map view
- Last viewed extent

The project file is saved in XML format, so it is possible to edit the file outside QGIS if you know what you are doing. The file format was updated several times compared to earlier QGIS versions. Project files from older QGIS versions may not work properly anymore. To be made aware of this, in the *General* tab under *Settings* → *Options* you can select:

Prompt to save project changes when required

Warn when opening a project file saved with an older version of QGIS

5.7 Output

There are several ways to generate output from your QGIS session. We have discussed one already in Section *Projects* saving as a project file. Here is a sampling of other ways to produce output files:

- Menu option *File* → Save as Image opens a file dialog where you select the name, path and type of image (PNG or JPG format). A world file with extension PNGW or JPGW saved in the same folder georeferences the image.
- Menu option *File* → *New Print Composer* opens a dialog where you can layout and print the current map canvas (see Section *인쇄 조판기*).

CHAPTER 6

QGIS GUI

When QGIS starts, you are presented with the GUI as shown below (the numbers 1 through 5 in yellow ovals refer to the six major areas of the interface as discussed below):

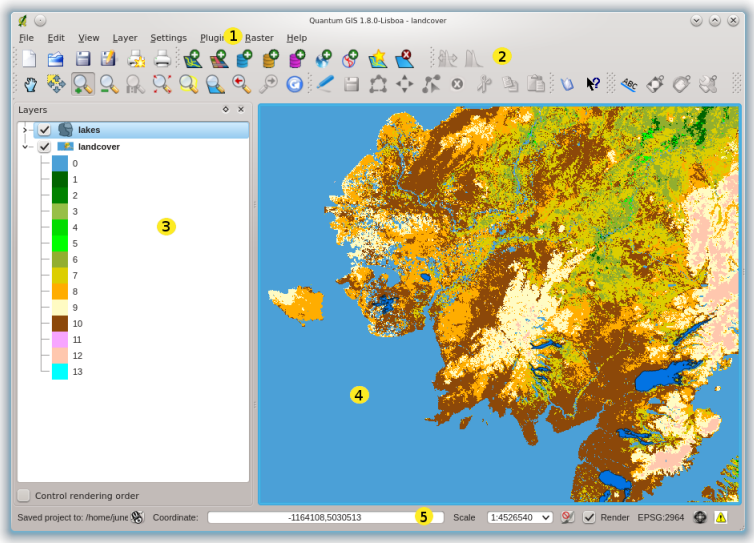


Figure 6.1: QGIS GUI with Alaska sample data

주석: Your window decorations (title bar, etc.) may appear different depending on your operating system and window manager.

The QGIS GUI is divided into five areas:

1. Menu Bar
2. Tool Bar
3. Map Legend
4. Map View
5. Status Bar

These five components of the QGIS interface are described in more detail in the following sections. Two more sections present keyboard shortcuts and context help.

6.1 Menu Bar

The menu bar provides access to various QGIS features using a standard hierarchical menu. The top-level menus and a summary of some of the menu options are listed below, together with the icons of the corresponding tools as they appear on the toolbar, as well as keyboard shortcuts. Keyboard shortcuts can also be configured manually (shortcuts presented in this section are the defaults), using the [**Configure Shortcuts**] tool under *Settings*.

Although most menu options have a corresponding tool and vice-versa, the menus are not organized quite like the toolbars. The toolbar containing the tool is listed after each menu option as a checkbox entry. Some menu options only appear, if the corresponding plugin is loaded. For more information about tools and toolbars, see Section *Toolbar*.

6.1.1 File

Menu Option	Shortcut	Reference	Toolbar
New Project	`Ctrl+N`	see *Projects*	*File*
Open Project	`Ctrl+O`	see *Projects*	*File*
Open Recent Projects →		see *Projects*	
Save Project	`Ctrl+S`	see *Projects*	*File*
Save Project As	`Ctrl+Shift+S`	see *Projects*	*File*
Save as Image		see *Output*	
New Print Composer	`Ctrl+P`	see *인쇄 조판기*	*File*
Composer manager		see *인쇄 조판기*	*File*
Print Composers →		see *인쇄 조판기*	
Exit	`Ctrl+Q`		

6.1.2 Edit

Menu Option	Shortcut	Reference	Toolbar
Undo	Ctrl+Z	see *고급 디지타이징 기능*	*Advanced Digitizing*
Redo	Ctrl+Shift+Z	see *고급 디지타이징 기능*	*Advanced Digitizing*
Cut Features	Ctrl+X	see *1.1.1. 기존 레이어의 디지타이징*	*Digitizing*
Copy Features	Ctrl+C	see *1.1.1. 기존 레이어의 디지타이징*	*Digitizing*
Paste Features	Ctrl+V	see *1.1.1. 기존 레이어의 디지타이징*	*Digitizing*
Add Feature	Ctrl+.	see *1.1.1. 기존 레이어의 디지타이징*	*Digitizing*
Move Feature(s)		see *1.1.1. 기존 레이어의 디지타이징*	*Digitizing*
Delete Selected		see *1.1.1. 기존 레이어의 디지타이징*	*Digitizing*
Simplify Feature		see *고급 디지타이징 기능*	*Advanced Digitizing*
Add Ring		see *고급 디지타이징 기능*	*Advanced Digitizing*
Add Part		see *고급 디지타이징 기능*	*Advanced Digitizing*
Delete Ring		see *고급 디지타이징 기능*	*Advanced Digitizing*
Delete Part		see *고급 디지타이징 기능*	*Advanced Digitizing*
Reshape Features		see *고급 디지타이징 기능*	*Advanced Digitizing*
Offset Curves		see *고급 디지타이징 기능*	*Advanced Digitizing*
Split Features		see *고급 디지타이징 기능*	*Advanced Digitizing*
Merge selected Features		see *고급 디지타이징 기능*	*Advanced Digitizing*
Merge attr. of selected Features		see *고급 디지타이징 기능*	*Advanced Digitizing*
Node Tool		see *1.1.1. 기존 레이어의 디지타이징*	*Digitizing*
Rotate Point Symbols		see *고급 디지타이징 기능*	*Advanced Digitizing*

After activating Toggle editing mode for a layer, you will find the `Add Feature` icon in the *Edit* menu depend-

ing on the layer type (point, line or polygon).

6.1.3 Edit (extra)

Menu Option	Shortcut	Reference	Toolbar
Add Feature		see *1.1.1. 기존 레이어의 디지타이징*	*Digitizing*
Add Feature		see *1.1.1. 기존 레이어의 디지타이징*	*Digitizing*
Add Feature		see *1.1.1. 기존 레이어의 디지타이징*	*Digitizing*

6.1.4 View

Menu Option	Shortcut	Reference	Toolbar
Pan Map			*Map Navigation*
Pan Map to Selection			*Map Navigation*
Zoom In	`Ctrl++`		*Map Navigation*
Zoom Out	`Ctrl+-`		*Map Navigation*
Select →		see *Select and deselect features*	*Attributes*
Identify Features	`Ctrl+Shift+I`		*Attributes*
Measure →		see *Measuring*	*Attributes*
Zoom Full	`Ctrl+Shift+F`		*Map Navigation*
Zoom To Layer			*Map Navigation*
Zoom To Selection	`Ctrl+J`		*Map Navigation*
Zoom Last			*Map Navigation*
Zoom Next			*Map Navigation*
Zoom Actual Size			*Map Navigation*
Decorations →		see *Decorations*	
Map Tips			*Attributes*
New Bookmark	`Ctrl+B`	see *Spatial Bookmarks*	*Attributes*
Show Bookmarks	`Ctrl+Shift+B`	see *Spatial Bookmarks*	*Attributes*
Refresh	`Ctrl+R`		*Map Navigation*
Tile scale slider		see *Tilesets*	*Tile scale*

6.1.5 Layer

Menu Option	Shortcut	Reference	Toolbar
New →		see *새로운 벡터 레이어 생성*	*Manage Layers*
Embed Layers and Groups ...		see *Nesting Projects*	
Add Vector Layer	`Ctrl+Shift+V`	see *벡터 데이터 다루기*	*Manage Layers*
Add Raster Layer	`Ctrl+Shift+R`	see *Loading raster data in QGIS*	*Manage Layers*
Add PostGIS Layer	`Ctrl+Shift+D`	see *PostGIS Layers*	*Manage Layers*

Table 6.1 – 이전 페이지에서 계속

Menu Option	Shortcut	Reference	Toolbar
Add SpatiaLite Layer	Ctrl+Shift+L	see *SpatiaLite Layers*	*Manage Layers*
Add MSSQL Spatial Layer	Ctrl+Shift+M	see *MSSQL Spatial Layers*	*Manage Layers*
Add WMS Layer	Ctrl+Shift+W	see *WMS Client*	*Manage Layers*
Add Delimited Text Layer		see *Delimited Text Plugin*	*Manage Layers*
Create new GPX layer		see *GPS Plugin*	*Manage Layers*
Add Oracle GeoRaster layer		see *Oracle GeoRaster Plugin*	*Manage Layers*
Add SQL Anywhere Layer		see *SQL Anywhere Plugin*	*Manage Layers*
Add WFS Layer			*Manage Layers*
Copy style		see 스타일 탭	
Paste style		see 스타일 탭	
Open Attribute Table			*Attributes*
Save edits			*Digitizing*
Toggle editing			*Digitizing*
Save as...			
Save selection as vector file...		See 속성테이블 작업	
Remove Layer	Ctrl+D		
Set CRS of Layer(s)	Ctrl+Shift+C		
Set project CRS from Layer			
Properties			
Query...			
Labeling			
Add to Overview	Ctrl+Shift+O		*Manage Layers*
Add All To Overview			
Remove All From Overview			
Show All Layers	Ctrl+Shift+U		*Manage Layers*
Hide All Layers	Ctrl+Shift+H		*Manage Layers*

6.1.6 Settings

Menu Option	Shortcut	Reference	Toolbar
Panels →		see *Panels and Toolbars*	
Toolbars →		see *Panels and Toolbars*	
Toggle Full Screen Mode	Ctrl-F		
Project Properties ...	Ctrl+Shift+P	see *Projects*	
Custom CRS ...		see *Custom Coordinate Reference System*	
Style Manager...		see *Style Manager*	
Configure shortcuts ...			
Customization ...		see *Customization*	
Options ...		see *Options*	
Snapping Options ...			

6.1.7 Plugins

Menu Option	Shortcut	Reference	Toolbar
 *Fetch Python Plugins*		see *QGIS Plugins*	
 *Manage Plugins*		see *Managing Plugins*	
Python Console			
GRASS →		see *GRASS GIS Integration*	*GRASS*

6.1.8 Vector

Menu Option	Shortcut	Reference	Toolbar
Analysis Tools →		see *fTools Plugin*	
Coordinate Capture →		see *Coordinate Capture Plugin*	
Data Management Tools →		see *fTools Plugin*	
Dxf2Shp →		see *Dxf2Shp Converter Plugin*	*Vector*
Geometry Tools →		see *fTools Plugin*	
Geoprocessing Tools →		see *fTools Plugin*	
GPS →		see *GPS Plugin*	*Vector*
Research Tools →		see *fTools Plugin*	
Road Graph →		see *Road Graph Plugin*	
Spatial Query →		see *Spatial Query Plugin*	*Vector*

6.1.9 Raster

Menu Option	Shortcut	Reference	Toolbar
Raster calculator		see *Raster Calculator*	
Georeferencer →		see *Georeferencer Plugin*	*Raster*
Heatmap →		see *Heatmap Plugin*	*Raster*
Interpolation →		see *Interpolation Plugin*	*Raster*
 *Terrain Analysis*		see *Raster Terrain Analysis Plugin*	
Zonal Statistics →		see *Zonal Statistics Plugin*	*Raster*
Projections →		see *GDAL Tools Plugin*	
Conversion →		see *GDAL Tools Plugin*	
Extraction →		see *GDAL Tools Plugin*	
Analysis →		see *GDAL Tools Plugin*	
Miscellaneous →		see *GDAL Tools Plugin*	
GdalTools settings		see *GDAL Tools Plugin*	

6.1.10 Database

Menu Option	Shortcut	Reference	Toolbar
DB manager →		see *DB Manager Plugin*	*Database*
eVis →		see *eVis Plugin*	*Database*
Offline Editing →		see *Offline Editing Plugin*	*Database*
Spit →		see *SPIT Plugin*	*Database*

6.1.11 Web

Menu Option	Shortcut	Reference	Toolbar
MapServer Export ... →		see *MapServer Export Plugin*	*Web*
OpenStreetMap →		see *OpenStreetMap*	*OpenStreetMap*

6.1.12 Help

Menu Option	Shortcut	Reference	Toolbar
Help Contents	F1		*Help*
What's This?	Shift+F1		*Help*
API Documentation			
QGIS Home Page	Ctrl+H		
Check QGIS Version			
About			
QGIS Sponsors			

Please not that for Linux the Menu Bar items listed above are the default ones in KDE window manager. In GNOME, Settings menu is missing and its items are to be found here:

Project Properties	*File*
Options	*Edit*
Configure Shortcuts	*Edit*
Style Manager	*Edit*
Custom CRS	*Edit*
Panels →	*View*
Toolbars →	*View*
Toggle Full Screen Mode	*View*
Tile scale slider	*View*
Live GPS tracking	*View*

6.2 Toolbar

The toolbar provides access to most of the same functions as the menus, plus additional tools for interacting with the map. Each toolbar item has popup help available. Hold your mouse over the item and a short description of the tool's purpose will be displayed.

Every menubar can be moved around according to your needs. Additionally every menubar can be switched off using your right mouse button context menu holding the mouse over the toolbars (read also *Panels and Toolbars*).

참고: Restoring toolbars

If you have accidentally hidden all your toolbars, you can get them back by choosing menu option *Settings* → *Toolbars* →. If a toolbar disappears under Windows, which seems to be a problem in QGIS from time to time, you have to remove `\HKEY_CURRENT_USER\Software\QuantumGIS\qgis\UI\state` in the registry. When you restart QGIS, the key is written again with the default state, and all toolbars are visible again.

6.3 Map Legend

The map legend area lists all the layers in the project. The checkbox in each legend entry can be used to show or hide the layer.

A layer can be selected and dragged up or down in the legend to change the z-ordering. Z-ordering means that layers listed nearer the top of the legend are drawn over layers listed lower down in the legend.

Layers in the legend window can be organised into groups. There are two ways to do so:

1. Right click in the legend window and choose *Add Group*. Type in a name for the group and press `Enter`. Now click on an existing layer and drag it onto the group.
2. Select some layers, right click in the legend window and choose *Group Selected*. The selected layers will automatically be placed in a new group.

To bring a layer out of a group you can drag it out, or right click on it and choose *Make to toplevel item*. Groups can be nested inside other groups.

The checkbox for a group will show or hide all the layers in the group with one click.

The content of the right mouse button context menu depends on whether the selected legend item is a raster or a vector layer. For GRASS vector layers  Toggle editing is not available. See section *Digitizing and editing a GRASS vector layer* for information on editing GRASS vector layers.

Right mouse button menu for raster layers

- *Zoom to layer extent*
- *Zoom to best scale (100 %)*
- *Show in overview*
- *Remove*
- *Set Layer CRS*
- *Set Project CRS from Layer*
- *Properties*
- *Rename*
- *Copy Style*
- *Add New Group*
- *Expand all*
- *Collapse all*
- *Update Drawing Order*

Additionally, according to layer position and selection

- *Make to toplevel item*
- *Group Selected*

Right mouse button menu for vector layers

- *Zoom to layer extent*
- *Show in overview*
- *Remove*
- *Set Layer CRS*
- *Set Project CRS from Layer*
- *Open attribute table*
- *Toggle editing* (not available for GRASS layers)
- *Save as*
- *Save selection as*
- *Query*
- *Show Feature Count*
- *Properties*

- *Rename*
- *Copy Style*
- *Add New Group*
- *Expand all*
- *Collapse all*
- *Update Drawing Order*

Additionally, according to layer position and selection

- *Make to toplevel item*
- *Group Selected*

Right mouse button menu for layer groups

- *Zoom to group*
- *Remove*
- *Set group CRS*
- *Rename*
- *Add New Group*
- *Expand all*
- *Collapse all*
- *Update Drawing Order*

It is possible to select more than one layer or group at the same time by holding down the `Ctrl` key while selecting the layers with the left mouse button. You can then move all selected layers to a new group at the same time.

You are also able to delete more than one Layer or Group at once by selecting several Layers with the `Ctrl` key and pressing `Ctrl+D` afterwards. This way all selected Layers or groups will be removed from the layerlist.

6.3.1 Working with the Legend independent layer order

Since QGIS 1.8 there is a widget that allows to define a legend independent drawing order. You can activate it in the menu *Settings* → *Panels*. Determine the drawing order of the layers in the map view here. Doing so makes it possible to order your layers in order of importance, for example, but to still display them in the correct order (see figure_layer_order). Checking the  *control rendering order* box underneath the list of layers will cause a revert to default behavior.

6.4 Map View

This is the "business end" of QGIS - maps are displayed in this area! The map displayed in this window will depend on the vector and raster layers you have chosen to load (see sections that follow for more information on how to load layers). The map view can be panned (shifting the focus of the map display to another region) and zoomed in and out. Various other operations can be performed on the map as described in the toolbar description above. The map view and the legend are tightly bound to each other - the maps in view reflect changes you make in the legend area.

참고: Zooming the Map with the Mouse Wheel

You can use the mouse wheel to zoom in and out on the map. Place the mouse cursor inside the map area and roll the wheel forward (away from you) to zoom in and backwards (towards you) to zoom out. The mouse cursor position is the center where the zoom occurs. You can customize the behavior of the mouse wheel zoom using the *Map tools* tab under the *Settings* → *Options* menu.

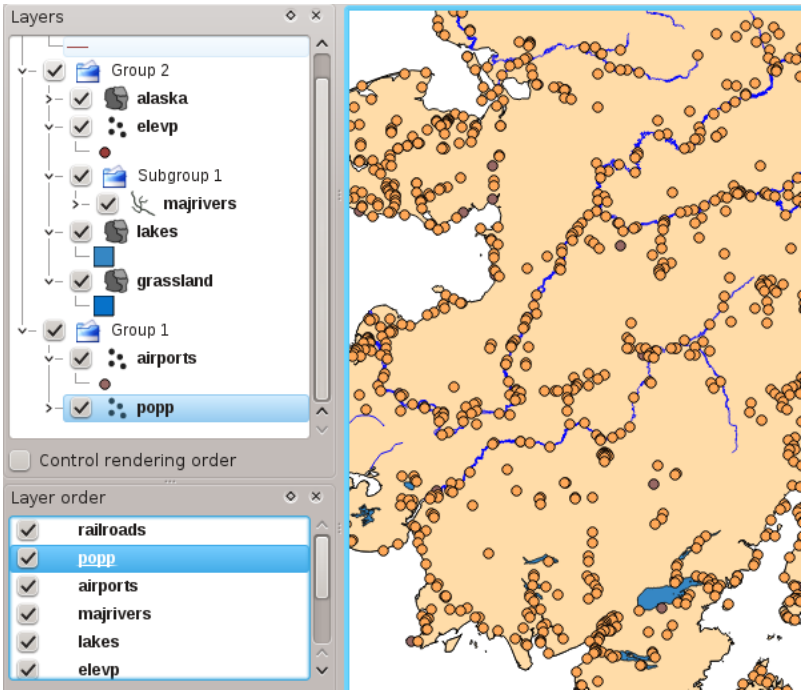


Figure 6.2: Define a legend independent layer order

참고: Panning the Map with the Arrow Keys and Space Bar

You can use the arrow keys to pan in the map. Place the mouse cursor inside the map area and click on the right arrow key to pan East, left arrow key to pan West, up arrow key to pan North and down arrow key to pan South. You can also pan the map using the space bar: just move the mouse while holding down space bar.

6.5 Status Bar

The status bar shows you your current position in map coordinates (e.g. meters or decimal degrees) as the mouse pointer is moved across the map view. To the left of the coordinate display in the status bar is a small button that will toggle between showing coordinate position or the view extents of the map view as you pan and zoom in and out.

Next to the coordinate display you find the scale display. It shows the scale of the map view. If you zoom in or out QGIS shows you the current scale. Since QGIS 1.8 there is a scale selector which allows you to choose between predefined scales from 1:500 until 1:1000000.

A progress bar in the status bar shows progress of rendering as each layer is drawn to the map view. In some cases, such as the gathering of statistics in raster layers, the progress bar will be used to show the status of lengthy operations.

If a new plugin or a plugin update is available, you will see a message at the far right of the status bar. On the right side of the status bar is a small checkbox which can be used to temporarily prevent layers being rendered to the map view (see Section *Rendering* below). The icon immediately stops the current map rendering process.

To the right of the render functions you find the EPSG code of the current project CRS and a projector icon. Clicking on this opens the projection properties for the current project.

참고: Calculating the correct Scale of your Map Canvas

When you start QGIS, degrees is the default unit, and it tells QGIS that any coordinate in your layer is in degrees.

To get correct scale values, you can either change this to meter manually in the *General* tab under *Settings* → *Project Properties* or you can select a project Coordinate Reference System (CRS) clicking on the  CRS status icon in the lower right-hand corner of the statusbar. In the last case, the units are set to what the project projection specifies, e.g. '+units=m'.

CHAPTER 7

General Tools

7.1 Keyboard shortcuts

QGIS provides default keyboard shortcuts for many features. You find them in Section *Menu Bar*. Additionally the menu option *Settings* → *Configure Shortcuts* allows to change the default keyboard shortcuts and to add new keyboard shortcuts to QGIS features.

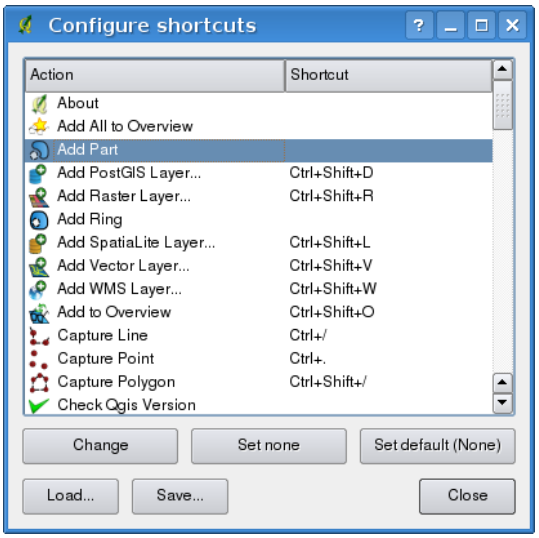


Figure 7.1: Define shortcut options (KDE)

Configuration is very simple. Just select a feature from the list and click on **[Change]**, **[Set none]** or **[Set default]**. Once you have found your configuration, you can save it as XML file and load it to another QGIS installation.

7.2 Context help

When you need help on a specific topic, you can access context help via the *Help* button available in most dialogs - please note that third-party plugins can point to dedicated web pages.

7.3 Rendering

By default, QGIS renders all visible layers whenever the map canvas must be refreshed. The events that trigger a refresh of the map canvas include:

- Adding a layer

- Panning or zooming
- Resizing the QGIS window
- Changing the visibility of a layer or layers

QGIS allows you to control the rendering process in a number of ways.

7.3.1 Scale Dependent Rendering

Scale dependent rendering allows you to specify the minimum and maximum scales at which a layer will be visible. To set scale dependency rendering, open the *Properties* dialog by double-clicking on the layer in the legend. On the *General* tab, set the minimum and maximum scale values and then click on the *Use scale dependent rendering* checkbox.

You can determine the scale values by first zooming to the level you want to use and noting the scale value in the QGIS status bar.

7.3.2 Controlling Map Rendering

Map rendering can be controlled in the following ways:

Suspending Rendering

To suspend rendering, click the *Render* checkbox in the lower right corner of the statusbar. When the *Render* checkbox is not checked, QGIS does not redraw the canvas in response to any of the events described in Section *Rendering*. Examples of when you might want to suspend rendering include:

- Add many layers and symbolize them prior to drawing
- Add one or more large layers and set scale dependency before drawing
- Add one or more large layers and zoom to a specific view before drawing
- Any combination of the above

Checking the *Render* checkbox enables rendering and causes an immediate refresh of the map canvas.

Setting Layer Add Option

You can set an option to always load new layers without drawing them. This means the layer will be added to the map, but its visibility checkbox in the legend will be unchecked by default. To set this option, choose menu option *Settings* → *Options* → and click on the *Rendering* tab. Uncheck the *By default new layers added to the map should be displayed* checkbox. Any layer added to the map will be off (invisible) by default.

Stopping Rendering

To stop the map drawing, press the `ESC` key. This will halt the refresh of the map canvas and leave the map partially drawn. It may take a bit of time between pressing `ESC` and the time the map drawing is halted.

주석: It is currently not possible to stop rendering - this was disabled in qt4 port because of User Interface (UI) problems and crashes.

Updating the Map Display During Rendering

You can set an option to update the map display as features are drawn. By default, QGIS does not display any features for a layer until the entire layer has been rendered. To update the display as features are read from the datastore, choose menu option *Settings* → *Options* click on the *Rendering* tab. Set the feature count to an appropriate value to update the display during rendering. Setting a value of 0 disables update during drawing (this is the default). Setting a value too low will result in poor performance as the map canvas is continually updated during the reading of the features. A suggested value to start with is 500.

Influence Rendering Quality

To influence the rendering quality of the map you have 2 options. Choose menu option *Settings* → *Options* click on the *Rendering* tab and select or deselect following checkboxes.

- *Make lines appear less jagged at the expense of some drawing performance*
- *Fix problems with incorrectly filled polygons*

7.4 Measuring

Measuring works within projected coordinate systems (e.g., UTM) and unprojected data. If the loaded map is defined with a geographic coordinate system (latitude/longitude), the results from line or area measurements will be incorrect. To fix this you need to set an appropriate map coordinate system (See Section *Working with Projections*). All measuring modules also use the snapping settings from the digitizing module. This is useful, if you want to measure along lines or areas in vector layers.

To select a measure tool click on and select the tool you want to use.

7.4.1 Measure length, areas and angles

QGIS is able to measure real distances between given points according to a defined ellipsoid. To configure this, choose menu option *Settings* → *Options*, click on the *Map tools* tab and choose the appropriate ellipsoid. There you can also define a rubberband color and your preferred measurement units (meters or feet) and angle units (degrees, radians and gon). The tools then allows you to click points on the map. Each segment-length as well as the total shows up in the measure-window. To stop measuring click your right mouse button.

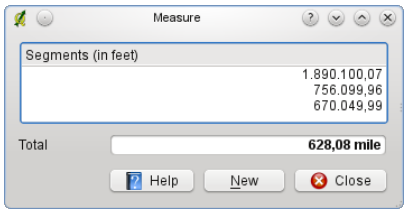


Figure 7.2: Measure Distance (KDE)

Areas can also be measured. In the measure window the accumulated area size appears. In addition, the measuring tool will snap to the currently selected layer, provided that layer has its snapping tolerance set. (See Section *검색 반경 및 스냅 공차 설정*). So if you want to measure exactly along a line feature, or around a polygon feature, first set its snapping tolerance, then select the layer. Now, when using the measuring tools, each mouse click (within the tolerance setting) will snap to that layer.

You can also measure angles, selecting Measure Angle tool. The cursor becomes cross-shaped. Click to draw the first segment of the angle you wish to measure, then move the the cursor to draw the desired angle. The measure is displayed in a popup dialog.

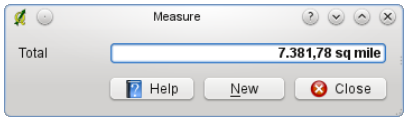


Figure 7.3: Measure Area (KDE)

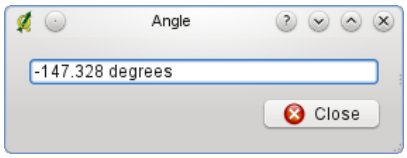


Figure 7.4: Measure Angle (KDE)

7.4.2 Select and deselect features

The QGIS toolbar provides several tools to select features in the map canvas. To select one or several features just click on and select your tool:

- Select single feature
- Select features by rectangle
- Select features by polygon
- Select features by freehand
- Select features by radius

To deselect all selected features click on Deselect features from all layers.

7.5 Decorations

The Decorations of QGIS includes the Copyright Label, the North Arrow and the Scale Bar. They are used to 'decorate' the map by adding cartographic elements.

주석: Before QGIS 1.8 the same functionality was provided by the Decoration Plugin.

7.5.1 Copyright Label

Copyright label adds a Copyright label using the text you prefer to the map.

1. Select from menu *View* → *Decorations* → *Copyright Label*. The dialog starts (see figure_decorations_1).
2. Enter the text you want to place on the map. You can use HTML as shown in the example
3. Choose the placement of the label from the *Placement* 'Bottom Right' drop-down box
4. Make sure the *Enable Copyright Label* checkbox is checked
5. Click **[OK]**

In the example above (default) QGIS places a copyright symbol followed by the date in the lower right hand corner of the map canvas.

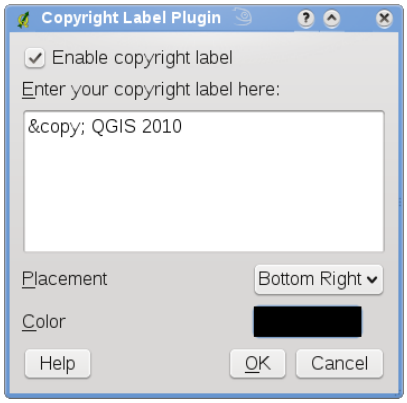


Figure 7.5: The copyright Dialog

7.5.2 North Arrow

North Arrow places a simple north arrow on the map canvas. At present there is only one style available. You can adjust the angle of the arrow or let QGIS set the direction automatically. If you choose to let QGIS determine the direction, it makes its best guess as to how the arrow should be oriented. For placement of the arrow you have four options, corresponding to the four corners of the map canvas.

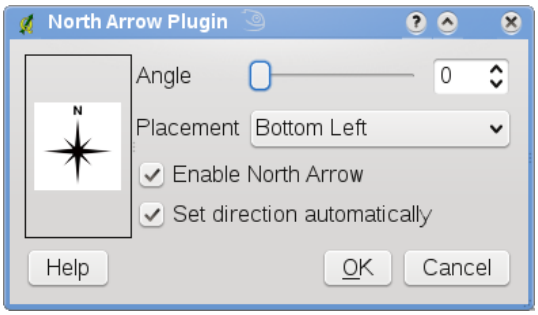


Figure 7.6: The North Arrow Dialog

7.5.3 Scale Bar

Scale Bar adds a simple scale bar to the map canvas. You control the style and placement, as well as the labeling of the bar.

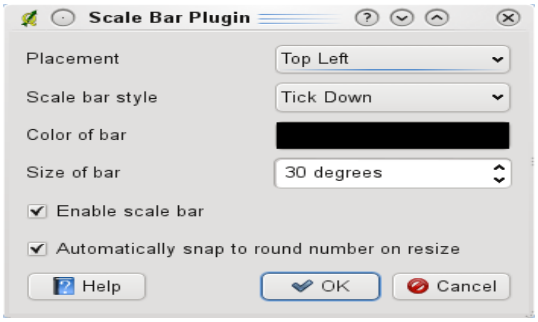


Figure 7.7: The Scale Bar Dialog

QGIS only supports displaying the scale in the same units as your map frame. So if the units of your layers are in

meters, you can't create a scale bar in feet. Likewise if you are using decimal degrees, you can't create a scale bar to display distance in meters.

To add a scale bar:

1. Select from menu *View* → *Decorations* → *Scale Bar* The dialog starts (see figure_decorations_3)
2. Choose the placement from the *Placement* 'Bottom Left' drop-down list
3. Choose the style from the *Scale bar style* 'Tick Down' list
4. Select the color for the bar *Color of bar* 'black' or use the default black color
5. Set the size of the bar and its label *Size of bar* '30 degrees'
6. Make sure the *Enable scale bar* checkbox is checked
7. Optionally choose to automatically snap to a round number when the canvas is resized *Automatically snap to round number on resize*
8. Click **[OK]**

참고: Settings of Decorations

When you save a .qgs project, any changes you have made to NorthArrow, ScaleBar and Copyright will be saved in the project and restored the next time you load the project.

7.6 Annotation Tools

The *Text Annotation* tools in the attribute toolbar provides the possibility to place formatted text in a balloon on the QGIS map canvas. Use the *Text Annotation* tool and click into the map canvas.

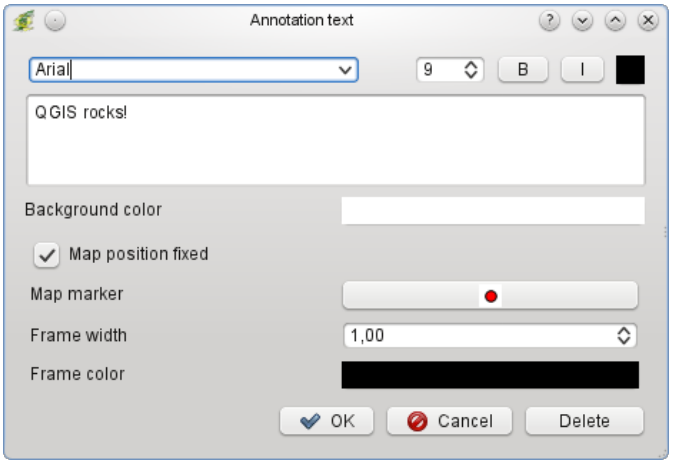


Figure 7.8: Annotation text dialog

Double click on the item opens a dialog with various options. There is the text editor to enter the formatted text and other item settings. E.g. there is the choice of having the item placed on a map position (displayed by a marker symbol) or to have the item on a screen position (not related to the map). The item can be moved by map position (drag the map marker) or by moving only the balloon. The icons are part of GIS theme, and are used by default in the other themes too.

The *Move Annotation* tool allows to move the annotation on the map canvas.

7.6.1 Form annotations

Additionally you can also create your own annotation forms. The *Form Annotation* tool is useful to display attributes of a vector layer in a customized qt designer form (see figure_custom_annotation). It is similar to the designer forms for the *Identify features* tool, but displayed in an annotation item. Also see QGIS blog http://blog.qgis.org/node/143 for more information.

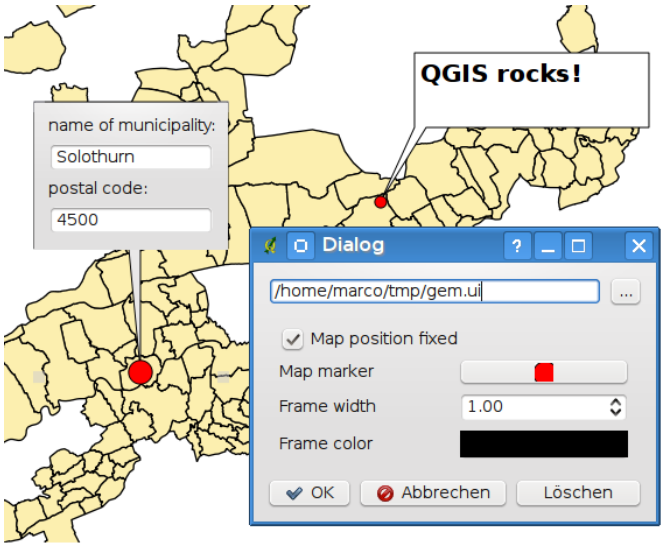


Figure 7.9: Customized qt designer annotation form

주석: If you press `Ctrl+T` while an *Annotation* tool is active (move annotation, text annotation, form annotation), the visibility states of the items are inverted.

7.7 Spatial Bookmarks

Spatial Bookmarks allow you to "bookmark" a geographic location and return to it later.

7.7.1 Creating a Bookmark

To create a bookmark:

1. Zoom or pan to the area of interest.
2. Select the menu option *View* → *New Bookmark* or press `Ctrl-B`.
3. Enter a descriptive name for the bookmark (up to 255 characters).
4. Press `Enter` to add the bookmark or **[Delete]** to remove the bookmark.

Note that you can have multiple bookmarks with the same name.

7.7.2 Working with Bookmarks

To use or manage bookmarks, select the menu option *View* → *Show Bookmarks*. The *Geospatial Bookmarks* dialog allows you to zoom to or delete a bookmark. You can not edit the bookmark name or coordinates.

7.7.3 Zooming to a Bookmark

From the *Geospatial Bookmarks* dialog, select the desired bookmark by clicking on it, then click **[Zoom To]**. You can also zoom to a bookmark by double-clicking on it.

7.7.4 Deleting a Bookmark

To delete a bookmark from the *Geospatial Bookmarks* dialog, click on it then click **[Delete]**. Confirm your choice by clicking **[Yes]** or cancel the delete by clicking **[No]**.

7.8 Nesting Projects

If you want to embed content from other project files into your project you can choose *Layer* → *Embed Layers and Groups*.

7.8.1 Embedding layers

The following dialog allows you to embed layers from other projects:

1. Press  to look for another project from the Alaska dataset.
2. Select the project file grassland. You can see the content of the project (see figure_embed_dialog).
3. Press `Ctrl` and klick on the layers grassland and regions. The layers are embedded in the map legend and the map view now.

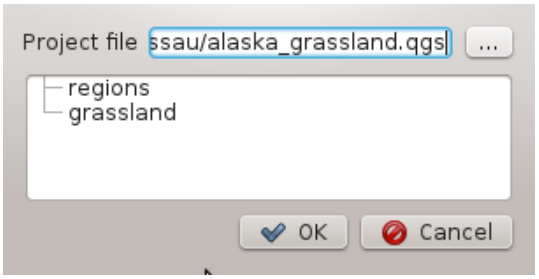


Figure 7.10: Select layers and groups to embed

While the embedded layers are editable you can't change it's properties like Style and Labeling.

Removing embedded layers

Right-click on the embedded layer and choose *Remove* .

CHAPTER 8

QGIS Configuration

QGIS is highly configurable through the *Settings* menu. Choose between Panels, Toolbars, Project properties, Options and Customization.

8.1 Panels and Toolbars

In the *Panels*→ menu you can switch on and off QGIS widgets. The *Toolbars*→ menu provides the possibility to switch on and off icon groups in the QGIS toolbar (see figure_panels_toolbars).

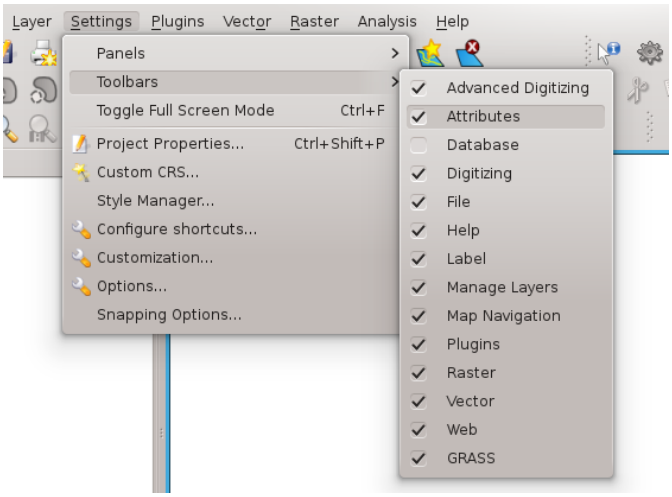


Figure 8.1: The Panels and Toolbars menu

참고: Activating the QGIS Overview

In QGIS you can use an overview panel that provides a full extent view of layers added to it. It can be selected under the menu *Settings* → *Panels*. Within the view is a rectangle showing the current map extent. This allows you to quickly determine which area of the map you are currently viewing. Note that labels are not rendered to the map overview even if the layers in the map overview have been set up for labeling. If you click and drag the red rectangle in the overview that shows your current extent, the main map view will update accordingly.

참고: Show Log Messages

Since QGIS 1.8 it's possible to track the QGIS messages. You can activate *Log Messages* in the menu *Settings* → *Panels* and follow the messages in the *General* and *Plugin* tab during loading and operation.

8.2 Project Properties

In the properties window for the project under *Settings* → *Project Properties* or *Settings* → *Project Properties* you set project specific options. These include:

- In the *General* tab the project title, selection and background color, layer units, precision, and the option to save relative paths to layers can be defined. You can define the layer units (only used when CRS transformation is disabled) and the precision of decimal places to use.
- The *CRS Coordinate Reference System* tab enables you to choose the CRS for this project, and to enable on-the-fly re-projection of raster and vector layers when displaying layers from a different CRS.
- With the third *Identifiable layers* tab you set (or disable) which layers will respond to the identify tool. (See the Map tools paragraph from the *Options* section to enable identifying of multiple layers.)
- The tab *OWS Server* allows to define information about the QGIS mapserver Service Capabilities, the Extent and the CRS Restrictions as well as the WFS Capabilities. Activating the *Add WKT geometry to feature info response* will allow to query the WMS layers.

8.3 Options

Some basic options for QGIS can be selected using the *Options* dialog. Select the menu option *Settings* → *Options*. The tabs where you can optimize your options are:

8.3.1 General Tab

- *Prompt to save project changes when required*
- *Warn when opening a project file saved with an older version of QGIS*
- Change Selection and Background color
- Change the QGIS Style
- Change the icon theme (choose between 'default', 'classic' and 'gis')
- Change icon size between 16, 24 and 32 pixel.
- Change the menu size
- Define double click action in legend (choose between 'open layer properties' and 'open ') attribute table.
- *Capitalize layer names in legend*
- *Display classification attribute names in legend*
- *Create raster icons in legend*
- *Hide splash screen at startup*
- *Show tips at startup*
- *Open identify results in a dock window (QGIS restart required)*
- *Open snapping options in a dock window (QGIS restart required)*
- *Open attribute table in a dock window*
- *Add PostGIS layers with double click and select in extended mode*
- *Add new layers to selected or current group*

- *Copy geometry in WKT representation from attribute table*
- *Attribute table behavior* (choose between 'Show all features' (default), 'Show selected features','Show features in current canvas')
- *Attribute table row cache*
- Define *Representation for NULL values*
- *Prompt for raster sublayers*  . Some rasters support sublayers - they are called subdatasets in GDAL. An example is netcdf files - if there are many netcdf variables, GDAL sees every variable as a subdataset. The option is to control how to deal with sublayers when a file with sublayers is opened. You have the following choices:
  - 'Always': always ask (if there are existing sublayers)
  - 'If needed': ask if layer has no bands, but has sublayers
  - 'Never': never prompt, will not load anything
  - 'Load all': never prompt, but load all sublayers
- *Scan for valid items in the browser dock*  . The 'Check extension' option was designed to speed up the loading of a directory, which can be time-consuming when 'Check file contents' is activated and there are many files (tens or hundreds).
- *Scan for contents of compressed files (.zip) in browser dock*  This option was also designed to speed up the loading of a directory. You have the following choices:
  - 'Basic Scan': checks that the extension is supported by one of the drivers
  - 'Full Scan': opens every file to check it is valid
  - 'Passthru': don't use this option it will be removed in the next QGIS version

8.3.2 GDAL tab

GDAL is a data exchange library for raster files. In this tab you can define which GDAL driver to be used for a raster format as in some cases more than one GDAL driver is available.

8.3.3 Plugins tab

- Add *Path(s) to search for additional C++ plugin libraries*.

8.3.4 Rendering Tab

- *By default new layers added to the map should be displayed*
- Define *Number of features to draw before updating the display*.
- *Use render caching where possible to speed up redraws*
- *Make lines appear less jagged at the expense of some drawing performance*
- *Fix problems with incorrectly filled polygons*
- *Use new generation symbology for rendering*
- Define the default Raster settings for visualisation: 'RGB band selection', *Use standard deviation* and 'Contrast Enhancement'

- Add/remove *Path(s) to search for Scalable Vector Graphics (SVG) symbols*

Additionally you can define whether to save the path for svg textures absolute or relative in the *General* tab of the *Settings → Project Properties* menu.

8.3.5 Map tools Tab

- The *Mode* setting determines which layers will be shown by the Identify tool. By switching to 'Top down' or 'Top down, stop at first' instead of 'Current layer' attributes for all identifiable layers (see the Project properties section under: *Projects* to set which layers are identifiable) will be shown with the Identify tool.
- *Open feature form, if a single feature is identified*
- Define *Search radius for identifying and displaying map tips as a percentage of the map width*
- Define *Ellipsoid for distance calculations*
- Define *Rubberband color* for measure tools
- Define *Decimal places*
- *Keep base unit*
- *Define preferred measurement units ('meters' or 'feet')*
- *Define preferred angle units ('Degrees', 'Radians' or 'Gon')*
- Define *Mouse wheel action* ('Zoom', 'Zoom and recenter', 'Zoom to mouse cursor', 'Nothing')
- Define *Zoom factor* for wheel mouse

8.3.6 Overlays Tab

- Define *Placement algorithm* for labels, symboly and diagrams (choose between 'Central point' (standard), 'Chain', 'Popmusic tabu chain', 'Popmusic tabu' and 'Popmusic chain')

8.3.7 Digitizing Tab

- Define Rubberband *Line color* and *Line width*
- Define *Default snap mode* ('To vertex', 'To segment', 'To vertex and segment')
- Define *Default snapping tolerance* in map units or pixel
- Define search radius for vertex edits in 'map units' or 'pixel'
- *Show markers only for selected features*
- Define vertex *Marker style* ('Cross' (default), 'Semi transparent circle' or 'None') and vertex *Marker size*.
- *Suppress attributes pop-up windows after each created feature*
- *Reuse last entered attribute values*
- *Validate geometries* Editing complex lines/polygons with many nodes can end up with very slow rendering. This is because the default validation procedures in QGIS can use a lot of time. To speed up rendering it is possible to select GEOS geometry validation (starting from GEOS 3.3) or to switch it off. GEOS geometry validation is much faster, but the disadvantage is that only the first geometry problem will be reported.

The next 3 options refer to the Offset Curve tool in 고급 디지타이징 기능. Through the various settings, it is possible to influence the shape of the line offset. These options are possible from GEOS 3.3 .

- *Join style for curve offset*
- *Quadrantsegments for curve offset*
- *Miter limit for curve offset*

8.3.8 CRS Tab

The CRS tab is divided in two areas. The first area allows to define the default CRS for new projects.

- Select a CRS and *Always start new projects with this CRS*.
- *Enable on the fly re-projection by default*

The second area allows to define the action, when a new layer is created, or when a layer without CRS is loaded.

- *Prompt for Coordinate Reference System (CRS)*
- *Use project Coordinate Reference System (CRS)*
- *Use default Coordinate Reference System (CRS) displayed below*

8.3.9 Locale Tab

- *Overwrite system locale* and *Locale to use instead*
- Information about active system locale

8.3.10 Network Tab

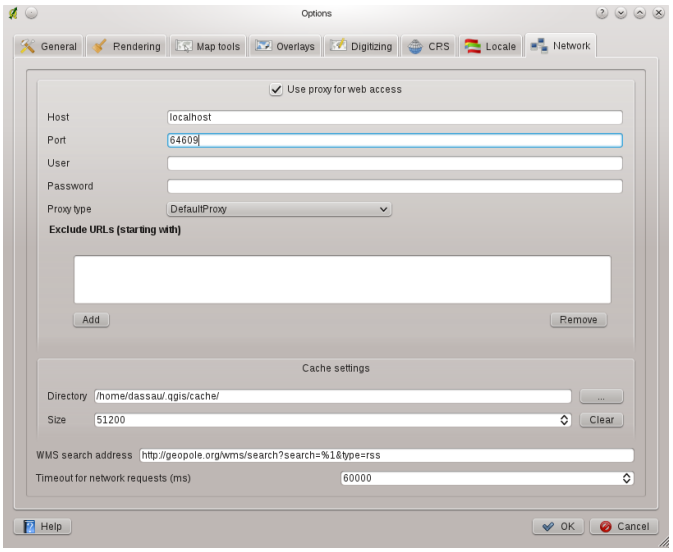


Figure 8.2: Proxy-settings in QGIS

- *Use proxy for web access* and define 'Host', 'Port', 'User', and 'Password'.
- Set the *Proxy type* according to your needs.
  - *Default Proxy*: Proxy is determined based on the application proxy set using

  - *Socks5Proxy*: Generic proxy for any kind of connection. Supports TCP, UDP, binding to a port (incoming connections) and authentication.
  - *HttpProxy*: Implemented using the "CONNECT" command, supports only outgoing TCP connections; supports authentication.
  - *HttpCachingProxy*: Implemented using normal HTTP commands, it is useful only in the context of HTTP requests
  - *FtpCachingProxy*: Implemented using an FTP proxy, it is useful only in the context of FTP requests
- Define *Cache settings* (path and size)
- Define *WMS search address*, default is `http://geopole.org/wms/search?search=\%1\&type=rss`
- Define *Timeout for network requests (ms)* - deeingebundenfault is 60000

Excluding some URLs can be added to the text box below the proxy-settings (see Figure_Network_Tab) by pressing the **[Add]** button. After that double-click into the just created URL-field and enter the URL you would like to exclude from using the proxy. Obviously the button **[Remove]** removes the selected entry.

If you need more detailed information about the different proxy-settings, please refer to the manual of the underlying QT-library-documentation at http://doc.trolltech.com/4.5/qnetworkproxy.html#ProxyType-enum.

참고: Using Proxies

Using proxies can sometimes be tricky. It is useful to 'trial and error' the above proxy types and check if they succeed in your case.

You can modify the options according to your needs. Some of the changes may require a restart of QGIS before they will be effective.

- settings are saved in a text file: `$HOME/.config/QuantumGIS/qgis.conf`
- you can find your settings in: `$HOME/Library/Preferences/org.qgis.qgis.plist`
- settings are stored in the registry under: `HKEY\CURRENT_USER\Software\QuantumGIS\qgis`

8.4 Customization

The customization tool is a new development in QGIS 1.8.. It lets you (de)activate almost every element in the QGIS user interface. This can get very useful if you have a lot of plug-ins installed that you never use and that are filling your screen.

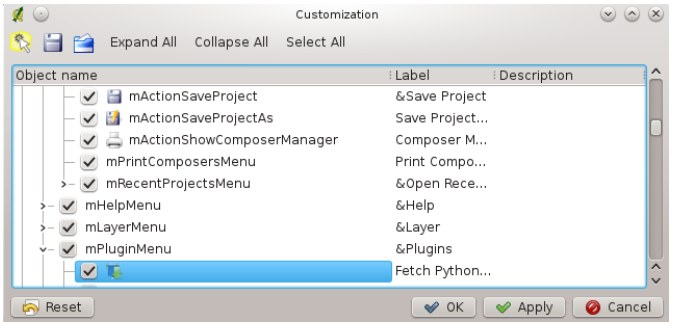


Figure 8.3: The Customization dialog

QGIS Customization is divided into five groups. In *Docks* you find the dock windows. Dock windows are applications that can be started and used as a floating, top-level window or embedded to the QGIS main window as a docked widget (see also *Panels and Toolbars*). In *Menus* you can hide entries in the Menu bar. In the

Status Bar features like the coordinate information can be daectivated. In *Toolbars* you can (de)activate the toolbar icons of QGIS and in *Widgets* you can (de)activate dialogs as well as their buttons.

With *Switch to catching widgets in main application* you can click on elements in QGIS you want to be hidden and find the corresponding entry in Customization (see figure_customization). You can also save your various different setups for different use cases as well. Before your changes are applied, you need to restart QGIS.

CHAPTER 9

Working with Projections

QGIS allows users to define a global and project-wide CRS (Coordinate Reference System) for layers without a pre-defined CRS. It also allows the user to define custom coordinate reference systems and supports on-the-fly (OTF) projection of vector and raster layers. All these features allow the user to display layers with different CRS and have them overlay properly.

9.1 Overview of Projection Support

QGIS has support for approximately 2,700 known CRS. Definitions for each of these CRS are stored in a SQLite database that is installed with QGIS. Normally you do not need to manipulate the database directly. In fact, doing so may cause projection support to fail. Custom CRS are stored in a user database. See Section *Custom Coordinate Reference System* for information on managing your custom coordinate reference systems.

The CRS available in QGIS are based on those defined by the European Petroleum Search Group (EPSG) and the Institut Geographique National de France (IGNF) and are largely abstracted from the spatial reference tables used in GDAL. EPSG identifiers are present in the database and can be used to specify a CRS in QGIS.

In order to use OTF projection, your data must contain information about its coordinate reference system or you have to define a global, layer or project-wide CRS. For PostGIS layers QGIS uses the spatial reference identifier that was specified when the layer was created. For data supported by OGR, QGIS relies on the presence of a recognized means of specifying the CRS. In the case of shapefiles, this means a file containing the Well Known Text (WKT) specification of the CRS. This projection file has the same base name as the shapefile and a prj extension. For example, a shapefile named `alaska.shp` would have a corresponding projection file named `alaska.prj`.

Whenever you select a new CRS, the used layer units will automatically be changed in the *General* tab of the *Project Properties* dialog under the *Edit* (Gnome, OSX) or *Settings* (KDE, Windows) menu.

9.2 Specifying a Projection

QGIS starts each new project using the global default projection.The global default CRS is EPSG:4326 - WGS 84 (`proj=longlat +ellps=WGS84 +datum=WGS84 +no_defs`) and comes predefined in QGIS. This default can be changed using the **[Select...]** button in the first section, used to defining the Default Coordinate Reference System to use when starting new projects, as shown in figure_projection_1. This choice will be saved for use in subsequent QGIS sessions.

When you use layers that do not have a CRS, you need to define how QGIS responds to these layers. This can be done globally or project-wide in the *CRS* tab under *Edit* → *Options* (Gnome, OSX) or *Settings* → *Options* (KDE, Windows).

The options shown in figure_projection_1 are:

- *Prompt for CRS*

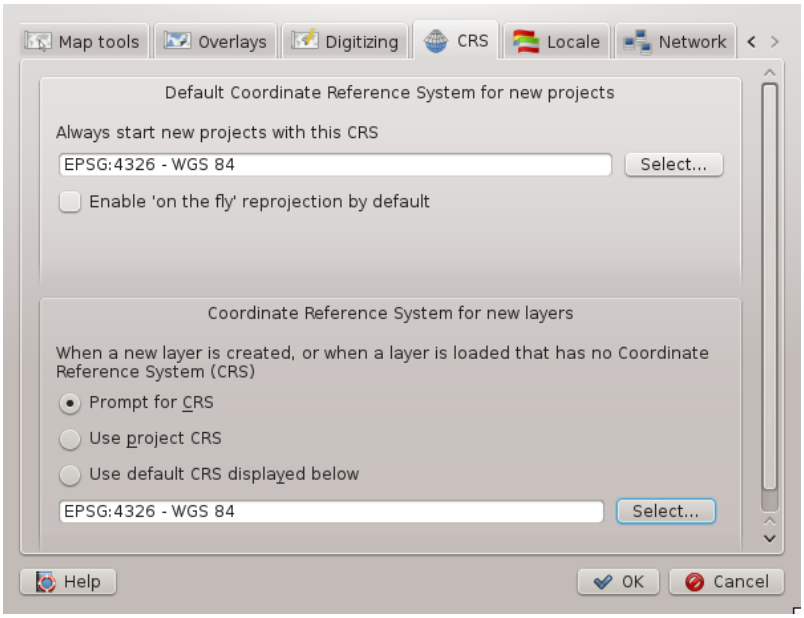


Figure 9.1: CRS tab in the QGIS Options Dialog

- *Use project CRS*
- *Use default CRS displayed below*

If you want to define the coordinate reference system for a certain layer without CRS information, you can also do that in the *General* tab of the raster (see *General Tab*) and vector (see *일반정보 탭(General Tab)*) properties dialog. If your layer already has a CRS defined, it will be displayed as shown in *벡터 레이어 속성 대화상자* .

참고: CRS in the Map Legend

Right-clicking on a layer in the Map Legend (Section *Map Legend*) provides two CRS short cuts. *Set layer CRS* takes you directly to the Coordinate Reference System Selector dialog (see figure_projection_2). *Set project CRS from Layer* redefines the project CRS using the layer's CRS

9.3 Define On The Fly (OTF) Reprojection

QGIS supports OTF reprojection for both raster and vector data. However, OTF is not activated by default. To use OTF projection, you must activate the *Enable on the fly CRS transformation* checkbox in the *CRS* tab of the *Project Properties* dialog.

There are three ways to achieve this end:

1. Select *Project Properties* from the *Edit* (Gnome, OSX) or *Settings* (KDE, Windows) menu.
2. Click on the CRS status icon in the lower right-hand corner of the statusbar.
3. Turn OTF on by default, by selecting the *CRS* tab of the *Options* dialog and selecting *Enable 'on the fly' reprojection by default*.

If you have already loaded a layer, and want to enable OTF projection, the best practice is to open the *Coordinate Reference System* tab of the *Project Properties* dialog, select a CRS, and activate the *Enable on the fly CRS transformation* checkbox. The CRS status icon will no longer be greyed-out and all layers will be OTF projected to the CRS shown next to the icon.

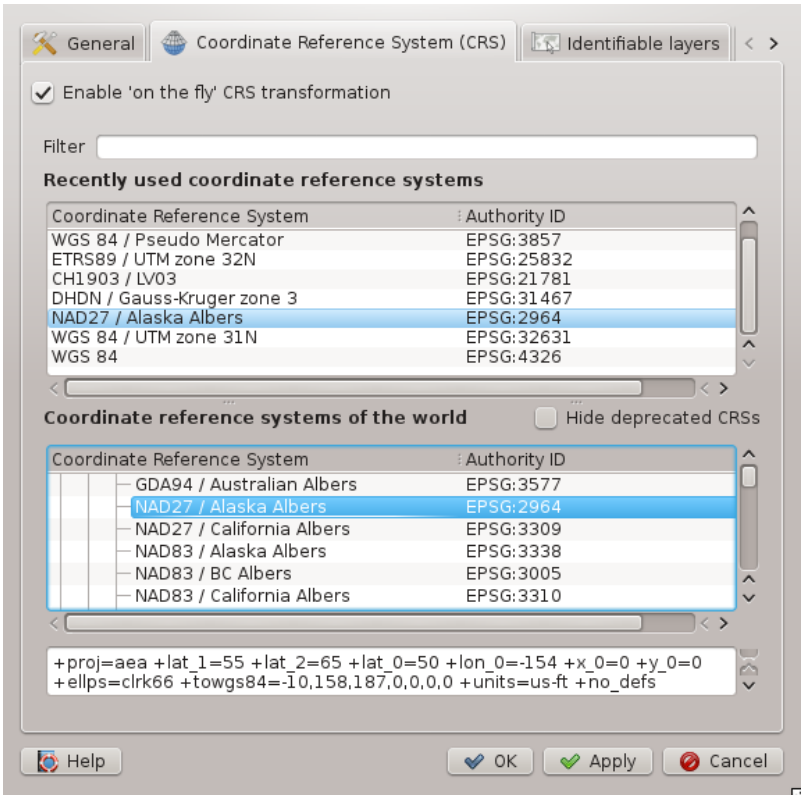


Figure 9.2: Projection Dialog

The *Coordinate Reference System* tab of the *Project Properties* dialog contains five important components as shown in Figure_projection_2 and described below.

1. **Enable on the fly CRS transformation** - this checkbox is used to enable or disable OTF projection. When off, each layer is drawn using the coordinates as read from the data source and the components described below are inactive.When on, the coordinates in each layer are projected to the coordinate reference system defined for the map canvas.
2. **Coordinate Reference System** - this is a list of all CRS supported by QGIS, including Geographic, Projected and Custom coordinate reference systems. To define a CRS, select it from the list by expanding the appropriate node and selecting the CRS. The active CRS is preselected.
3. **Proj4 text** - this is the CRS string used by the Proj4 projection engine. This text is read-only and provided for informational purposes.
4. **Filter** - if you know the EPSG code, the identifier or the name for a Coordinate Reference System, you can use the search feature to find it. Enter the EPSG code, the identifier or the name.
5. **Recently used CRS** - if you have certain CRS that you frequently use in your everyday GIS work, these will be displayed in the table at the bottom of the Projection Dialog. Click on one of these buttons to select the associated CRS.

참고: Project Properties Dialog

If you open the *Project Properties* dialog from the *Edit* (Gnome, OSX) or *Settings* (KDE, Windows) menu, you must click on the *Coordinate Reference System* tab to view the CRS settings.

Opening the dialog from the *CRS status* icon will automatically bring the *Coordinate Reference System* tab to the front.

9.4 Custom Coordinate Reference System

If QGIS does not provide the coordinate reference system you need, you can define a custom CRS. To define a CRS, select *Custom CRS* from the *Edit* (Gnome, OSX) or *Settings* (KDE, Windows) menu. Custom CRS are stored in your QGIS user database. In addition to your custom CRS, this database also contains your spatial bookmarks and other custom data.

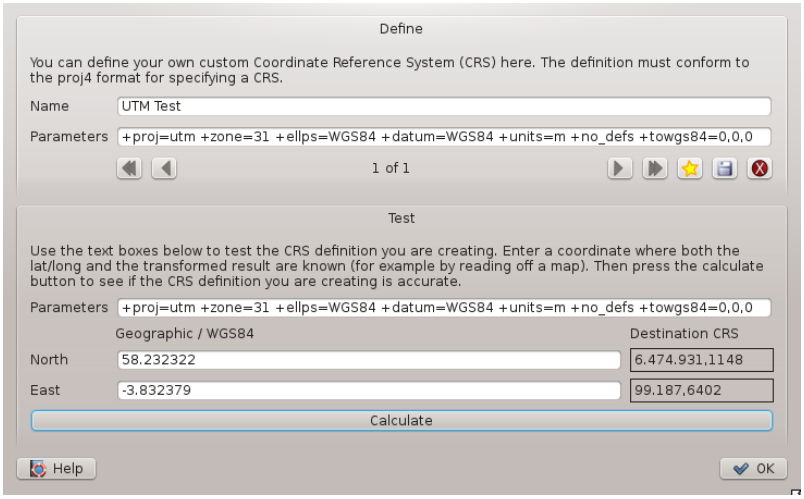


Figure 9.3: Custom CRS Dialog

Defining a custom CRS in QGIS requires a good understanding of the Proj.4 projection library. To begin, refer to the Cartographic Projection Procedures for the UNIX Environment - A User's Manual by Gerald I. Evenden, U.S. Geological Survey Open-File Report 90-284, 1990 (available at ftp://ftp.remotesensing.org/proj/OF90-284.pdf).

This manual describes the use of the `proj.4` and related command line utilities. The cartographic parameters used with `proj.4` are described in the user manual and are the same as those used by QGIS.

The *Custom Coordinate Reference System Definition* dialog requires only two parameters to define a user CRS:

1. a descriptive name and
2. the cartographic parameters in PROJ.4 format.

To create a new CRS, click the New button and enter a descriptive name and the CRS parameters. After that you can save your CRS by clicking the Save button.

Note that the *Parameters* must begin with a `+proj`=-block, to represent the new coordinate reference system.

You can test your CRS parameters to see if they give sane results by clicking on the **[Calculate]** button inside the *Test* block and pasting your CRS parameters into the *Parameters* field. Then enter known WGS 84 latitude and longitude values in *North* and *East* fields respectively. Click on **[Calculate]** and compare the results with the known values in your coordinate reference system.

CHAPTER 10

QGIS Browser

The QGIS Browser is a new panel in QGIS that lets you easily navigate in your database. You can have access to common vector files (e.g. ESRI shapefile or MapInfo files), databases (e.g.PostGIS or MSSQL Spatial) and WMS/WFS connections. You can also view your GRASS data (to get the data into QGIS see *GRASS GIS Integration*).

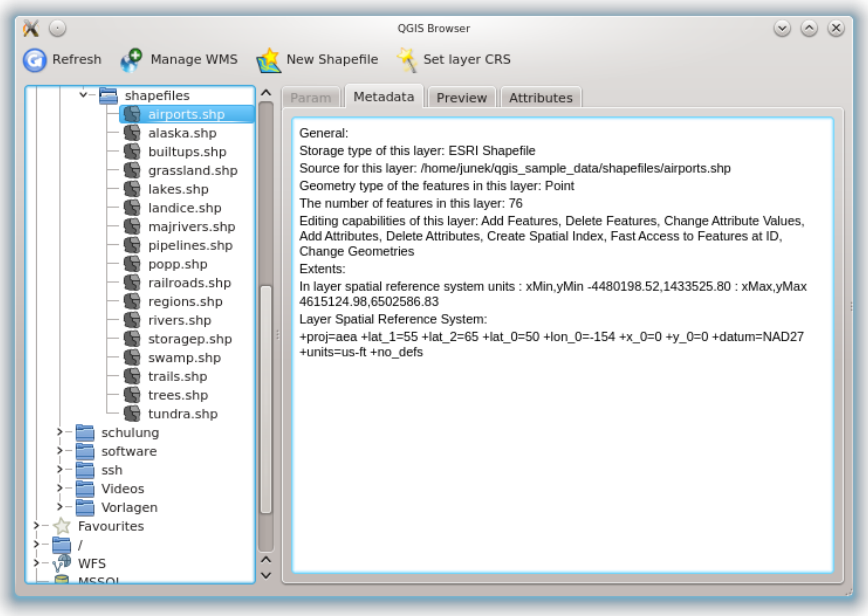


Figure 10.1: Qgis browser as a standalone application to view metadata, preview and attributes

Use QGIS Browser to preview your data. The drag and drop function makes it easy to get your data into the Map view and the Map legend.

1. Activate QGIS Browser: Right-click on the tool bar and click *Browser*.
2. Drag the panel into the legend window.
3. Klick on the *Browser* tab.
4. Browse in your database and choose the shapefile folder from qgis_sample_data.
5. Press the `Shift` key and klick on airports.shp and alaska.shp .
6. Press the left mouse button then drag and drop the files into the map canvas.
7. Right-click on a layer and choose *Set project CRS from layer*. For more information see *Working with Projections*.

8. Click on  *Zoom Full* to make the layers visible.

It's also possible to run QGIS Browser as a standalone application.

Start qgis browser

- Type in "qbrowser" at a command prompt.
- Start QGIS browser using the Start menu or desktop shortcut, or double click on a QGIS project file.
- QGIS browser is not available yet from your your Applications folder. However it can easily be made available. In Finder use *Go → Go to map...* and use it to find the folder `/Applications/QGIS.app/Contents/MacOS/bin`. Use the key combination `option – command` and drag **qbrowser.app** while holding down the left mouse button to the Applications folder. This will create a link that you can use to start QGIS browser.

In figure_browser_standalone_metadata you can see the enhanced functionality of Qgis browser. The *Param* tab provides the details of your connection based datasets like PostGIS or MSSQL Spatial. The *Metadata* tab contains general information about the file (see *메타데이터 탭(Metadata Tab)*). With the *Preview* tab you can have a look at your files without importing them into your QGIS project. It's also possible to preview the attributes of your files in the *Attributes* tab.

CHAPTER 11

벡터 데이터 다루기

11.1 Supported Data Formats

QGIS uses the OGR library to read and write vector data formats (GRASS vector and PostgreSQL support is supplied by native QGIS data provider plugins),including ESRI Shapefiles, MapInfo and Microstation file formats; PostGIS, SpatiaLite, Oracle Spatial databases and many more. The vector data can also be loaded in read mode from zip and gzip archives into QGIS. At the date of this document, 69 vector formats are supported by the OGR library (see OGR-SOFTWARE-SUITE *Literature and Web References*). The complete list is available at http://www.gdal.org/ogr/ogr_formats.html.

주석: 나열된 모든 포맷들 중에는 QGIS 에서 사용할 수 없는 것들도 있습니다. 예를 들어, 어떤 포맷은 상업용 라이브러리를 설치해야 할 수도 있고, 사용하고 계신 OS에서는 GDAL/OGR가 지원하지 못할 수도 있습니다. QGIS 를 통해 벡터를 불러올 때 테스트가 완료된 포맷들만이 파일타입 리스트에 나타납니다. 테스트가 안 된 포맷들은 파일타입을 `*.*` 로 지정하면 불러올 수 있습니다.

GRASS 벡터데이터를 사용하는 방법은 *GRASS GIS Integration* 장을 참고하시기 바랍니다.

이번 장에서는 일반적으로 널리 사용되는 포맷들의 작업방법을 설명합니다.(ESRI shape files, PostGIS layers, SpatialLite layers) QGIS의 많은 기능들은 어떤 벡터 데이터 포맷인가에 관계없이 동일한 형태로 작동됩니다. 이는 QGIS 를 설계할 때부터 고려한 사항으로, 식별(identify), 선택(select), 라벨 달기(labeling)와 속성(attribute) 기능 등이 모두 포함됩니다.

11.1.1 ESRI Shapefiles

QGIS 에서 사용되는 표준 벡터포맷은 ESRI의 Shapefile입니다. OGR의 OGR Simple Feature Library (http://www.gdal.org/ogr/)를 통하여 지원됩니다.

실제로 Shapefile은 여러 개의 파일로 구성되는데, 다음과 같은 3가지의 파일이 필요합니다:

1. `.shp` 파일은 기하학적 요소들을 포함.
2. `.dbf` 파일은 dBase 포맷으로 속성들을 포함.
3. `.shx` 파일은 인덱스 파일.

또한 Shapefile 에는 투영 정보을 담고 있는 `.prj` 파일이 추가될 수 있습니다. 투영 파일은 매우 유용하지만, 필수적인 것은 아닙니다. Shapefile 데이터세트는 추가적인 파일들을 포함할 수 있습니다. 자세한 사항은 ESRI 기술사양서를 참조하시기 바랍니다. http://www.esri.com/library/whitepapers/pdfs/shapefile.pdf.

Shapefile 불러오기

 To load a shapefile, start QGIS and click on the  Add Vector Layer toolbar button or simply type `Ctrl+Shift+V`. This will bring up a new window (see figure_vector_1).

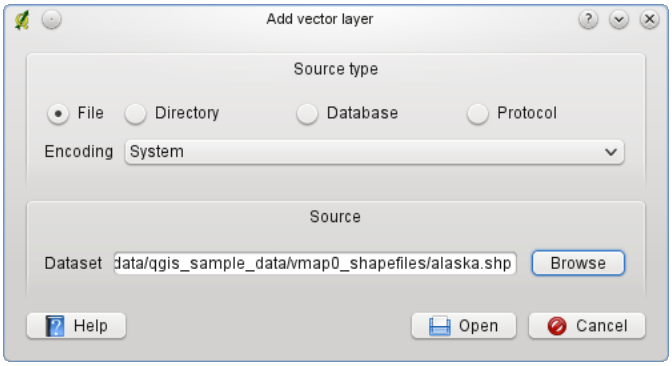


Figure 11.1: 벡터레이어 추가 대화상자

여러 가지 옵션 중 *File* [파일]을 체크하고 **[Browse]** [탐색]를 클릭합니다. 그러면 표준 파일열기 대화상자가 나타나게 되는데 (figure_vector_2 참조) 이를 통해 파일시스템을 돌아다니면서 Shape File이나 다른 데이터를 불러 올 수 있습니다. *Filter* 선택 박스를 사용하면 OGR에서 지원하는 파일 포맷을 미리 선택해 둘 수 있습니다.

원한다면 Shapefile 타입을 선택해두시면 됩니다.

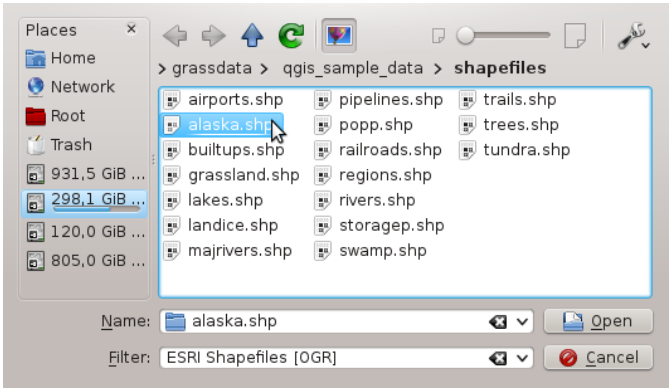


Figure 11.2: OGR이 지원되는 벡터 레이어 대화상자 열기

리스트에서 파일을 선택하고 **[Open]** [열기] 버튼을 누르면 해당 파일이 QGIS 로 불러지게 됩니다. Figure_vector_3_은 `alaska.shp` 파일을 불러온 모습입니다.

참고: 레이어 색상 (**Layer Colors**)

지도에 레이어를 추가하면 색상은 임의로 지정됩니다. 한번에 여러 개의 레이어를 불러오면 레이어별로 각각 다른 색상이 설정됩니다.

파일을 불러 온 후에는 맵 내비게이션 툴을 이용하여 파일을 확대/축소할 수 있습니다. 레이어의 스타일을 변경하고자 한다면 범례에 있는 레이어 이름을 두 번 클릭합니다. 마우스 오른쪽 버튼을 클릭한 후 팝업 메뉴에서 Properties[속성]▶ 을 선택하여 *Properties* [레이어속성] 대화상자를 여는 방법도 있습니다. 벡터 레이어 심볼 설정 방법에 대한 자세한 정보는 ***스타일 탭*** 절을 참조하시기 바랍니다.

참고: OS X에서 휴대용 드라이브에 있는 레이어와 프로젝트 불러오기 (****Load layer and project from mounted external drives on OS X**)

OS X에서는 기본 하드드라이브 이외에 연결된 휴대용 드라이브는 *File* → *Open Project* 를 선택해도 표시되지 않습니다. 우리는 현재 OS X의 열기/저장 대화상자를 수정하기 위해 노력하고 있습니다. 이 문제가 해결되기 전까지는 파일이름 박스에 '/Volumes' 하고 입력하고 `return` 를 눌러야 외부 드라이브나 네트워크상의 파일을 찾아볼 수 있습니다.

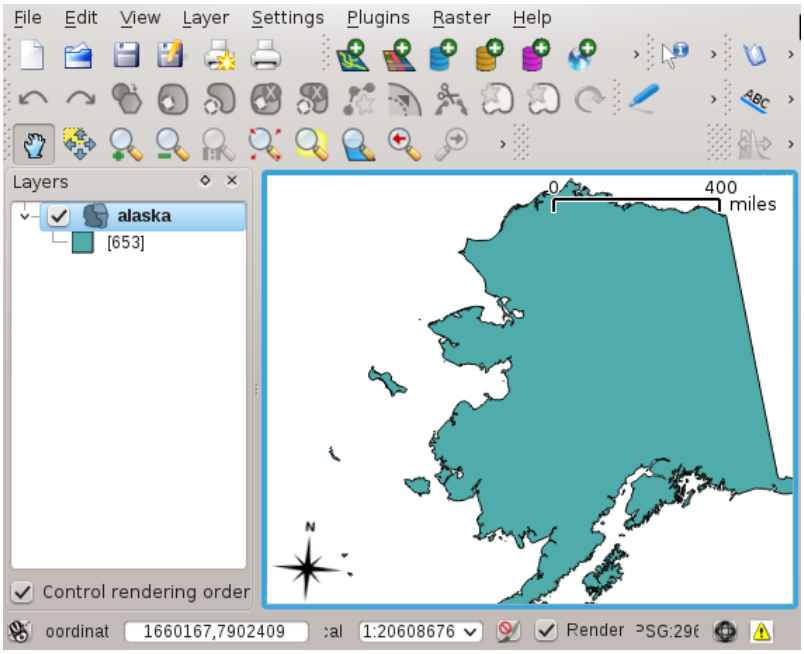


Figure 11.3: QGIS 로 불러온 Alaska Shape File

Shape 성능 향상

Shapefile이 그려지는 성능을 향상하기 위하여, 공간인덱스를 만들 수 있습니다. 공간인덱스는 확대와 이동 성능을 향상 시키게 됩니다. QGIS 에서 사용되는 공간 인덱스는 `.qix` 확장자를 가집니다.

인덱스를 만들기 위해서는 다음과 같은 과정을 사용합니다:

- Shapefile 불러오기.
- *Layer Properties* [레이어속성] 대화상자 열기(범례에서 Shapefile 이름을 두 번 클릭하거나 오른쪽을 클릭하여 팝업 메뉴에서 *Properties* 를 선택).
- *General* [일반정보] 탭에서 **[Create Spatial Index]** [공간인덱스 만들기] 버튼을 누른다.

Shape.prj 파일 불러오기 문제

.prj 파일과 함께 shape 파일을 불러들이려는데, QGIS 에서 좌표계를 읽지 못하는 경우에는, 레이어의 *Layer Properties* [레이어 속성] 대화상자 중 *General* [일반정보] 탭에 들어가서 수동으로 적절한 투영법을 정의해야 합니다. 이러한 현상은 .prj 파일이 경우에 따라 QGIS 에서 사용하는, 즉 *CRS* [좌표계] 대화상자에 목록화 되어있는 것과 같이 완벽한 투영 파라미터를 모두 포함하지 않고 배포되기 때문입니다.

이러한 이유 때문에 QGIS 에서 새로운 Shapefile을 생성하면 두 개의 다른 투영법 파일이 만들어 지게 됩니다. 즉, ESRI 소프트웨어와 호환되지만, 제한적인 파라미터만 들어있는 `.prj` 파일과, QGIS 좌표계 정의 에서 사용되는, 완벽한 파라미터를 제공하는 `.qpj` 파일이 그것입니다. QGIS가 `.qpj` 파일을 발견하면 `.prj` 파일의 존재 여부와 관계없이 `.qpj` 파일을 사용하게 됩니다.

11.1.2 MapInfo 레이어 불러오기

MapInfo 레이어를 불러오기 위해 Add Vector Layer 또는 간단하게 단축키 `Ctrl+Shift+V` 를 누릅니다. 파일 형식 필터를 *Filter* ‘Mapinfo File [OGR]’ 로 변경하여 불러오고자 하는 MapInfo layer를 선택합니다.

11.1.3 ArcInfo Binary Coverage 불러오기

 ArcInfo Binary Coverage 레이어를 불러오기 위해  Add Vector Layer 도구모음 버튼 또는 간단하게 단축키 `Ctrl+Shift+V` 로 Add Vector Layer 대화상자를 엽니다. Directory를 선택하여 *Filter* |selectstring|를 'Arc/Info Binary Coverage'로 변경합니다. 커버리지 파일이 들어있는 디렉토리에서 파일을 찾아 선택합니다.

비슷하게 UK National Transfer Format 뿐만 아니라 미국 센서스국의 초기 TIGER Format 등의 기본 벡터파일 디렉토리를 불러올 수 있습니다.

11.1.4 PostGIS Layers

PostGIS 레이어는 PostgreSQL 데이터베이스에 쌓여 있습니다. PostGIS의 이점은 공간 인덱싱, 필터링과 검색 능력들을 제공하는 것입니다. PostGIS를 사용할 경우 선택, 인식 작업 같은 벡터 기능이 QGIS 에서 OGR 레이어를 사용하는 것보다 더 정밀하게 작동합니다.

저장된 연결 만들기

 PostGIS의 데이터 소스를 처음 사용할 때에는 데이터가 저장되어 있는 PostgreSQL 데이터베이스와 연결을 만들어야 합니다.  Add PostGIS Layer 도구모음 버튼을 클릭하여 시작하고 *Layer* [레이어] 메뉴 또는 단축키 `Ctrl+Shift+D`'를 누른 다음 |mActionAddLayer| `:menuselection:'Add PostGIS Layer...` 옵션을 선택합니다. 또는 *Add Vector Layer* 대화상자를 열고 *Database*'***를 선택합니다. 그 다음*** *:guilabel:'Add PostGIS Table(s)* 대화상자가 나타날 것입니다. 연결관리자에 접근하기 위하여 **[New]** 버튼을 클릭하면 *Create a New PostGIS Connection* 대화상자가 나옵니다. The parameters required for a connection are:

- **Name**: 연결을 위한 이름. *Database*와 같은 것을 쓸 수 있습니다
- **Service**: 서비스 파라메타는 호스트이름/포트를 번갈아 사용할 수 있음.(잠정적으로 데이터베이스) 이것은 `pg_service.conf` 에서 정의할 수 있습니다
- **Host**: 데이터베이스 호스트의 이름. 이것은 텔넷연결이나 Ping으로 호스트를 여는데 사용하는 것과 같이 호스트 이름을 확인할 수 있어야 합니다 검증필요 !!.
- **Port**: PostgreSQL 데이터베이스 서버에서 대기중인 포트번호. 기본포트는 5432입니다.
- **Database**: Database 이름.
- **SSL mode**: SSL연결은 미리 설정된 서버와 가능하다. Note that massive speedups in PostGIS layer rendering can be achieved by disabling SSL in the connection editor. 옵션들은 다음과 같습니다:
  - disable : 암호화 되지 않은 SSL 연결 시도
  - Non-SSL 연결 시도, 만약 연결실패이면 SSL연결 시도
  - prefer(기본) : SSL연결시도, 만약 연결실패이면 non-SSL 연결시도;
  - require : 하나의 SSL연결만 시도.
- **Username**: 데이터베이스 로그인에 사용되는 사용자 이름.
- **Password**: 데이터베이스 연결을 위해 *Username* (사용자 이름)과 함께 사용되는 암호.

다음 체크박스를 선택적으로 활성화 할 수 있습니다:

- *Save Username* [사용자 이름 저장]
- *Save Password* [암호 저장]
- *Only look in the geometry_columns table* [geometry_columns 테이블에서만 찾기]
- *Only look in the 'public' schema* ['public' 스키마에서만 검색]

- *Also list tables with no geometry* [geometry 가 없는 테이블도 목록화]
- *Use estimated table metadata* [추정된 테이블 메타데이터 사용]

든 파라미터와 옵션들이 선택되면 **[Test Connect]** 버튼을 클릭하여 연결을 시험할 수 있습니다.

참고: 사용자 설정 및 보안 (**QGIS User Settings and Security**)

컴퓨팅 환경에 따라 QGIS 설정에 암호저장은 보안에 위험할 수 있습니다. QGIS 의 사용자 설정은 기본적으로 운영체계에 저장합니다:

- , 홈 디렉토리의 .QGIS .
- , 레지스트리에 저장.

1.1.1. PostGIS 레이어 불러오기

일단 정의된 하나 또는 그 이상의 연결로 PostgreSQL 데이터베이스로부터 레이어를 불러올 수있습니다. 물론 여기에는 PostgreSQL에 있는 Data 여야 하는 조건이 있습니다. 데이터베이스에 데이터를 입력하는 것에 대해서는 *1.1.1. PostgreSQL에 데이터 넣기* 절을 참조하십시오.

PostGIS에서 레이어를 불러오기 위해서는 아래의 단계를 밟습니다:

- If the *Add PostGIS layers* dialog is not already open, click on the Add PostGIS Layer toolbar button.
- 드롭다운 리스트에서 연결을 선택하고 **[Connect]** 를 클릭합니다.
- *Also list tables with no geometry* [geometry 가 없는 테이블도 목록화] 체크박스를 필요에 따라 선택 또는 선택 해제 합니다
- 옵션으로 레이어로부터 정의된 객체들을 불러오기 위하여 *Search Options* [옵션 검색]체크박스를 사용 하거나 **[Build Query]** 대화상자를 시작하게 하는 *Query builder* 아이콘을 사용합니다.
- 원하는 레이어를 찾고자 한다면 가능한 레이어 목록에 추가합니다.
- 레이어 클릭을 통하여 레이어를 선택합니다. `Shift` 키를 누르고 레이어를 클릭하여 다중 레이어 선택을 할 수 있습니다. 미리 레이어를 정의하기위한 PostgreSQL Query Bulider 사용정보에 대해서는 ***쿼리 빌더*** 절을 참조하십시오.
- 지도에 레이어를 추가할 때는 **[Add]** 버튼을 클릭합니다.

참고: PostGIS 레이어 (**PostGIS Layers**)

보통 PostGIS 레이어는 geometry_columns table 항목에 정의됩니다. 0.9.0 버전부터 QGIS 는 geometry_columns table에 항목에 빠져 있는 레이어들를 불러올 수 있습니다. 이것은 테이블들과 뷰들을 포함합니다. 공간 뷰의 정의는 데이터를 시각화 하는데 강력한 수단을 제공합니다. 뷰의 생성에 대한 것은 PostgreSQL 매뉴얼을 참고하십시오.

1.1.1. PostgreSQL에 대한 세부사항

이 절은 QGIS 에서 PostgreSQL 레이어에 액세스 방법에 대한 세부 정보를 포함하고 있습니다. QGIS 의 대부분은 단순히 불러올 수 있는 데이터베이스 테이블 목록을 제공하고, 요청한 것들을 불러오는 것입니다. 그러나 QGIS 로 PostgreSQL 테이블을 불러오는데 문제가 있다면, 이 내용은 QGIS 메시지를 이해하는데 도움을 주고, QGIS 에서 불러올 수 있는 PostgreSQL 테이블 또는 뷰의 정의를 변경하는 방안도 제공합니다.

QGIS 는 레이어 사용에 있어서 고유 키로 사용할 수 있는 컬럼을 포함하는 PostgreSQL 레이어를 필요로 합니다. 테이블은 일반적으로 프라이머리 키나 그것에 대해 특별히 제약하는 컬럼이 필요합니다. QGIS 에서는 그 컬럼이 int4형식(정수로 크기가 4byte)을 필요로 합니다. 그 대신에 ctid를 프라이머리 키로 사용할 수 있습니다. 테이블에 이러한 아이템이 부족하면, oid 컬럼이 대신 사용됩니다. 컬럼이 색인되어 있다면 성능은 증대 될 것입니다.(PostgreSQL에서 프라이머리 키는 자동으로 색인됨).

PostgreSQL 레이어가 뷰라고 하면 같은 요구조건이 존재하지만, 뷰는 프라이머리 키 또는 특별하게 제약받는 컬럼을 가지고 있는 않습니다. 이러한 경우에 QGIS는 적정한 테이블 컬럼을 가진 뷰에서 컬럼을 찾는 것을 시도할 것입니다. 뷰 정의 SQL를 분석하는 것이 그 시도 입니다. 그렇지만 QGIS 에서 무시하는 SQL에 대한 몇가지 경우가 있습니다. 이것들은 테이블 별칭과 SQL 함수에 의하여 만들어진 컬럼들을 포함합니다.

적정한 컬럼을 찾지 못하면, QGIS는 레이어를 불러오지 못할 것입니다. 이러한 일이 발생 했을 경우 해결책은 적절한 컬럼을 포함하도록 뷰를 변경하는 것입니다.(int4 형식과 프라이머리 키, 또는 유일조건과 함께 가능한 색인되도록).

11.1.5 1.1.1. PostgreSQL에 데이터 넣기

Data can be imported into PostgreSQL/PostGIS using several tools, such as the SPIT plugin, or the command line tools shp2pgsql or ogr2ogr.

SPIT Plugin

QGIS 는  SPIT 라는 이름의 플러그인을 제공합니다(PostGIS로 Shapefile 가져오기 도구). SPIT는 한 번에 여러 shapefiles를 로드하는 데 사용하고 스키마에 대한 지원을 포함합니다. 자세한 내용은 :ref:‘label_spit‘를 참조하십시오.

shp2pgsql

PostGIS에는 활성화된 PostGIS 데이터베이스로 shapefiles를 가져오는 데 사용할 수 있는 **shp2pgsql** 라는 유틸리티가 있습니다. 예를 들어, `gis_data` 라는 이름의 PostgreSQL 데이터베이스에 `lakes.shp` 라는 shapefile를 가져오기 위해, 다음 명령을 사용합니다:

```
shp2pgsql -s 2964 lakes.shp lakes_new | psql gis_data
```

이것은 `gis_data` 데이터베이스에 `lakes_new` 라는 새 레이어를 생성합니다. 새 레이어는 2964라는 공간적 참조 식별자(SRID)를 가지게 됩니다. 공간 참조 체계와 투영법에 대한 자세한 내용은 *Working with Projections* 장을 참조하십시오.

참고: PostGIS 데이터세트 내보내기 (**Exporting datasets from PostGIS**)

불러오기 도구인 **pgsql2shp**와 같이 PostGIS - 데이터세트인 shapefile을 내보낼 수 있는 도구인 **shp2pgsql**이 있습니다. 이것은 배포용 PostGIS에 포함되어 있습니다.

---

ogr2ogr

PostGIS에 공간데이터를 가져오기 위해서는 **shp2pgsql**과 **SPIT** 외에도 다른 도구가 있습니다: **ogr2ogr** 이것은 GDAL을 설치하면 같이 설치 됩니다.

PostGIS로 shapefile을 가져오려면 다음을 수행합니다:

```
ogr2ogr -f "PostgreSQL" PG:"dbname=postgis host=myhost.de user=postgres \
password=topsecret" alaska.shp
```

이것은 `alaska.shp` 라는 shapefile을 PostGIS 데이터베이스 *postgis*로 가져올 것입니다.(호스트 *myhost.de*, 사용자 *postgres*, 암호 *topsecret*).

OGR은 PostGIS을 지원하기 위해 PostgreSQL을 함께 구축되어야 합니다. 아래와 같이 입력하여 그것을 볼 수 있습니다

```
ogrinfo --formats | grep -i post
```

PostgreSQL의 **COPY**-명령 대신 기본 **INSERT INTO** 메소드를 사용하려는 경우에는 다음과 같은 환경-변수를 내보낼 수 있습니다(최소한 와 가능):

```
export PG_USE_COPY=YES
```

ogr2ogr은 **shp2pgsl**이 하는 것처럼 공간인덱스를 만들지 못합니다. 공간인덱스를 만들기 위하여 추가적 단계로 일반적인 SQL 명령어 **CREATE INDEX**를 사용하여 수동 작업이 필요하다. (다음 *1.1.1. **성능 향상*** 절에 설명된 대로).

1.1.1. 성능 향상

특히 네트워크를 통해 PostgreSQL 데이터베이스에서 요소들을 검색하는 것은 많은 시간이 소요될 수 있습니다. 데이터베이스의 각 레이어가 index:*PostGIS spatial index* (공간인덱스)를 가지고 있으면 postgreSQL 레이어들의 그리기 성능을 향상시킬 수 있습니다. PostGIS 는 데이터의 공간검색을 빠르게 하는 GiST (Generalized Search Tree) index (일반화된 검색트리) 인덱스를 지원합니다 (http://postgis.refractions.net).

GiST 인덱스를 만들기 위한 구문은 다음과 같습니다:

```
CREATE INDEX [indexname] ON [tablename]
  USING GIST ( [geometryfield] GIST_GEOMETRY_OPS );
```

큰 테이블에 대한 인덱스를 만들기 위해서는 시간 오래 걸리 수 있습니다. 인덱스가 만들어 지면, ‘‘VACUUM ANALYZE‘‘를 수행 합니다. 자세한 내용은 POSTGIS-PROJECT *Literature and Web References* 를 참조하십시오.

아래는 GiST 인덱스를 만드는 예입니다:

```
gsherman@madison:~/current$ psql gis_data
Welcome to psql 8.3.0, the PostgreSQL interactive terminal.

Type:  \copyright for distribution terms
       \h for help with SQL commands
       \? for help with psql commands
       \g or terminate with semicolon to execute query
       \q to quit

gis_data=# CREATE INDEX sidx_alaska_lakes ON alaska_lakes
gis_data-# USING GIST (the_geom GIST_GEOMETRY_OPS);
CREATE INDEX
gis_data=# VACUUM ANALYZE alaska_lakes;
VACUUM
gis_data=# \q
gsherman@madison:~/current$
```

11.1.6 경도 180°를 교차하는 벡터 레이어

많은 GIS 패키지는 지리 좌표계(경위도)로 crossing the 180 degrees longitude line (180 경도선)을 넘어서 벡터지도를 포함하지 않습니다(http://postgis.refractions.net/documentation/manual-1.4/ST_Shift_Longitude.html). 결국 QGIS 에서 이런 지도를 열면, 서로 가까이 보였던 것이 별개의 위치로 멀리 떨어져 있는 것을 볼 수 있습니다. Figure_vector_4_에서 지도의 멀리 왼쪽 작은 점(Chatham 섬)은 뉴질랜드 주 섬의 오른쪽과 같은 격자 안에 있어야합니다.



Figure 11.4: 경도 180도를 가로지르는 경위도 지도

A workaround is to transform the longitude values using PostGIS and the **ST_Shift_Longitude** function This function reads every point/vertex in every component of every feature in a geometry, and if the longitude coordinate is < 0° adds 360° to it. The result would be a 0° - 360° version of the data to be plotted in a 180° centric map.

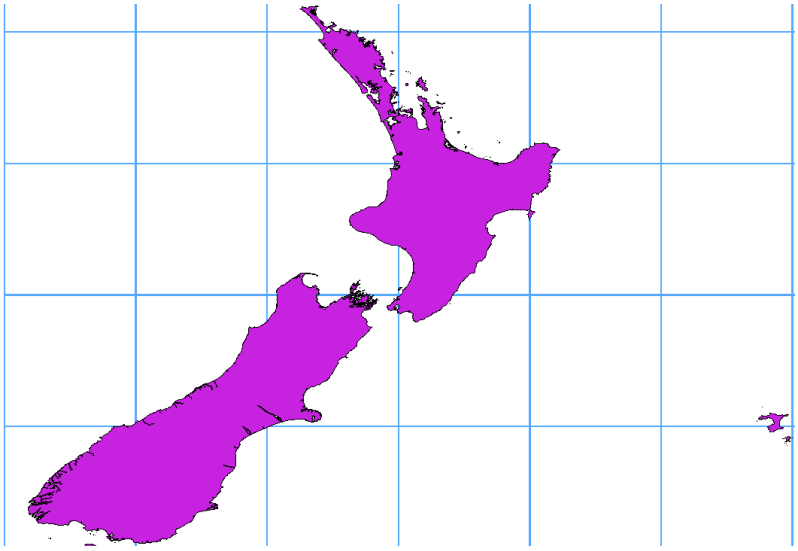


Figure 11.5: **ST_Shift_Longtitude** 기능을 적용한 180° 경도를 지도 교차

사용법

- PostGIS 관리자 플러그인이나 SPIT을 플러그인을 사용하여 PostGIS에 데이터를 넣습니다.(*1.1.1. PostgreSQL에 데이터 넣기* 절 참조)
- 아래의 명령어 문제로 PostGIS 명령어 라인 인터페이스를 사용합니다. (한 예로 "TABLE"은 PostGIS 테이블의 실제 이름입니다)

```
gis_data=# update TABLE set the_geom=ST_Shift_Longitude(the_geom);
```

- 모든 것이 정상적으로 이루어졌으면, 업데이트된 객체 개수에 대한 확인 메시지가 나타나고, 그러면 (Figure_vector_5)지도를 로드하고 그 차이를 확인할 수 있습니다

11.1.7 SpatiaLite Layers

SpatiaLite 데이터베이스로부터 처음 데이터를 불러오려면, Add SpatiaLite Layer 도구모음 버튼을 클릭하거나 *Layer* [레이어] 메뉴 또는 `Ctrl+Shift+L`을 타이핑하여 |mActionAddSpatiaLiteLayer| `:menuselection:'Add SpatiaLite Layer...` [SpatiaLite 레이어 추가]옵션을 선택하여 시작합니다.그러면 이미 |qg|에 알려진 SpatiaLite 데이터베이스에 연결할 수 있도록 허가하는 하나의 윈도우가 나타나게 되며, 드롭 다운 메뉴에서 선택하거나 새 데이터베이스에 새로운 연결을 정의할 수 있습니다. 새로운 연결을 정의하려면, **[New]** 를 선택하고 SpatiaLite 데이터베이스를 가리키도록 파일 브라우저를 사용하여, :file:'.sqlite'는 확장명을 가진 파일 찾습니다.

SpatiaLite 포맷으로 벡터 레이어를 저장하고자 한다면, 범례에서 레이어의 오른쪽을 클릭하여 수행할 수 있습니다. 그 다음에 *Save as..* [다른 이름으로 저장]를 클릭하고, 저장할 파일 이름, 'SpatiaLite' 포맷과 CRS[좌표계]를 설정하고, OGR 소스 생성옵션 필드에 ''SPATIALITE=YES''를 추가합니다. 이것은 SpatiaLite 데이터베이스를 생성하는 OGR이라는 것을 말해줍니다. http://www.gdal.org/ogr/drv_sqlite.html 를 참조하십시오.

QGIS also supports editable views in SpatiaLite.

새로운 SpatiaLite 레이어 만들기

새로운 SpatiaLite 레이어를 만들고자 하면, ***새로운** SpatiaLite **생성*** 절을 참조하십시오.

참고: SpatiaLite 데이터관리 플러그인 (**SpatiaLite data management Plugins**)

SpatiaLite 데이터 관리를 위해서 'QGIS Contributed Repository'로부터 Python plugins: QSpatiaLite, SpatiaLite Manager or DB Manager (core plugin, recommended) 또한 사용할 수 있습니다. They can be downloaded and installed with the Plugin Installer.

11.1.8 MSSQL Spatial Layers

QGIS는 또한 native MS SQL 2008을 지원합니다. The Add MSSQL Spatial Layer is part of the new toolbar button or available in the MS SQL node in the QBrowser tree, providing drag and drop import support.

11.2 벡터 속성 대화상자

The *Layer Properties* dialog for a vector layer provides information about the layer, symbology settings and labeling options. If your vector layer has been loaded from a PostgreSQL/PostGIS datastore, you can also alter the underlying SQL for the layer by invoking the *Query Builder* dialog on the *General* tab. To access the *Layer Properties* dialog, double-click on a layer in the legend or right-click on the layer and select *Properties* from the popup menu.

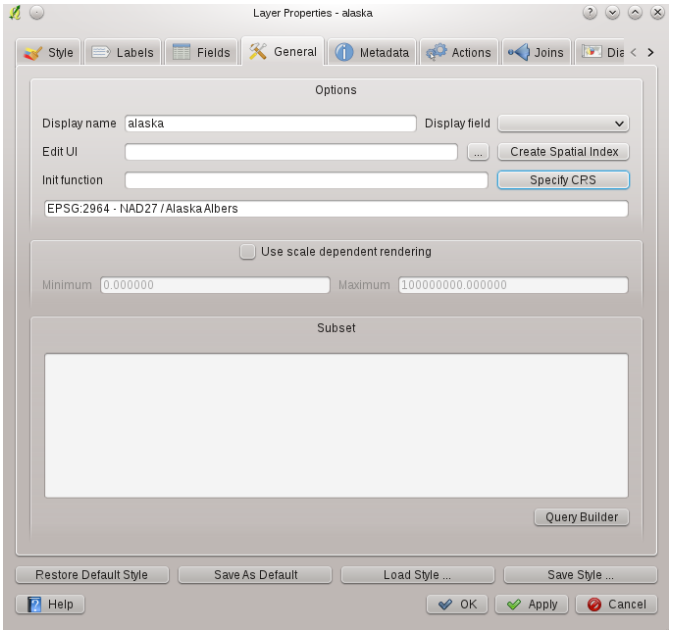


Figure 11.6: 벡터 레이어 속성 대화상자

11.2.1 스타일 탭

QGIS 1.4.0 이후부터 새로운 심볼은 기능을 개선하고 최종적으로 과거 심볼을 대체하기 위하여 개발 되었습니다. 현재 QGIS 1.7.0은 다양한 기능 개선과 새로운 요소를 제공하는 새로운 심볼을 기본으로 사용합니다.

과거 심볼에 대한 설명은 ***구 버전 심볼로지*** 절을 참조하십시오.

심볼에는 세가지 타입이 있습니다: 마커 심볼(포인트을 위함), 라인 심볼(라인을 위함), 채우기와 외곽선 심볼(폴리곤을 위함). 심볼은 하나 또는 그 이상의 심볼 레이어로 구성되어 있습니다. 이것들은 심볼의 색상 정의가 가능하고, 그러면 이러한 색상은 모든 심볼 레이어를 정의합니다. 일부 레이어는 색상이 변경되지 않도록 잠글 수 있습니다. 이것은 다중레이어 심볼의 색상을 정의했을 때 유용합니다. 마찬가지로 라인 심볼의 굵기뿐만 아니라 마커심볼의 크기와 회전 정의도 가능합니다.

가능한 심볼 레이어 형식 (**Available symbol layer types**)

- 포인트 레이어
  - **Font marker**: 폰트의 렌더링.
  - **Simple marker**: 하드코드된 마커의 렌더링.
  - **SVG marker**: SVG 그림의 렌더링.
  - **Ellipse marker**: Rendering with ellipse shapes derived from attributes.
  - **Vector field marker**: Rendering with vector fields derived from attributes.
- 라인 레이어
  - **Line decoration**: 라인 장식의 추가, 예: 라인방향을 나타내는 화살표.
  - **Marker line**: 마커심볼로 반복되어 만들어진 라인.
  - **Simple line**: 일반적인 라인의 렌더링(특정한 굵기, 컬러와 펜의 스타일).
- 폴리곤 레이어
  - **Centroid fill**: 하드코드 된 마커로 폴리곤의 중심을 채움.
  - **SVG fill**: SVG 심볼로 폴리곤의 중심을 채움.
  - **Simple fill**: 일반적인 폴리곤의 랜더링(정의된 채우기 색깔, 패턴, 외각선).
  - **Line pattern fill**: Fill a polygon with a line pattern.
  - **Point pattern fill**: Fill a polygon with a point pattern.
  - **Outline: Line decoration**: 라인 장식의 추가, 예: 라인방향을 나타내는 화살표.
  - **Outline: Marker line**: Area의 외곽선을 하드코드 된 마커로 사용.
  - **Outline: Simple line**: Area의 외곽선에 대한 굵기, 색깔, 펜 스타일 정의.

Color ramps

컬러 램프는 표시를 만드는데 사용할 색깔의 범위를 설정하는데 사용됩니다. 심볼의 색깔은 컬러 램프로 설정할 것입니다.

3가지 형식의 컬러 램프가 있습니다:

- **Gradient**: 한 색에서 다른 색으로 선형 그라디언트.
- **Random**: 정의된 색 영역에서 무작위로 만들어지는 색깔.
- **ColorBrewer**: 컬러 스키마와 컬러 클래스의 범위를 정의하여 만드는 색 범위.

컬러 램프는 *Style Manager* 의 *Color ramp* 탭에서 **[Add]** 버튼을 클릭하고 컬러 램프 형식을 선택하여 정의할 수 있습니다.(*Style Manager* 절 참조).

Styles

스타일은 다양한 심볼과 컬러램프로 나누어 집니다. 사용자가 선호하거나 자주 사용하는 심볼들을 정의할 수도 있고 이를 매번 다시 만들지 않고 사용할 수도 있습니다. 스타일 아이템(심볼과 칼라램프)은 항상 스타일에서 조회할 수 있는 이름을 가집니다. 스타일은 QGIS에서 제공하는 기본 스타일과 사용자가 추가한 스타일이 있습니다. In the lower part of the Style tab there are four buttons for managing styles: Use **[Restore Default Style]** to get back to your default settings, **[Save As Default]** to save your style as default, **[Load Style...]** to get to your own styles and **[Save Style]** to save your own styles. Layer styles can also be transferred from one layer to another layer. Activate a layer and choose *Layer* →  *Copy style* and switch to another layer. Then choose *Layer* →  *Paste style*.

Renderers

Renderer는 정확한 심볼과 함께 객체를 그리는 것을 담당합니다. Renderer에는 4가지 형식이 있습니다.(single symbol, categorized (과거 심볼에서는 unique로 불림), graduated and rule-based) Renderer는 사실 단계적 Renderer가 특별한 경우이기 때문에, 지속적인 컬러 Renderer는 없습니다. Categorized와 Graduated Renderer는 지정된 심볼과 컬러 램프로 만들어질 수 있습니다. 그것들은 적절한 심볼의 색상으로 설정될 것입니다. 새로운 심볼로 작업하기 (**Working with the New Generation Symbology**)

In the *Style* tab you can choose one of the five renderers: single symbol, categorized, graduated, rule-based and point displacement. For each data type (points, lines and polygons) vector symbol layer types are available (see vector_symbol_types). Depending on the chosen renderer, the symbology Depending on the chosen renderer, the *Style* tab provides different following sections. The new generation symbology dialog also provides a **[Style Manager]** button which gives access to the Style Manager (see Section *Style Manager*). The Style Manager allows you to edit and remove existing symbols and add new ones.

참고: Select and change multiple symbols

The New Generation Symbology allows to select multiple symbols and right click to change color, transparency, size, or width of selected entries.

Single Symbol Renderer

Single Symbol Renderer는 하나의 사용자 정의기호를 사용하여 레이어의 모든 기능을 렌더링하는데 사용됩니다. 속성은 부분적으로 레이어 타입에 따라 `Style` [스타일] 탭에서 조정되나 모든 종류에 다음과 같은 구조를 공유합니다. 탭의 왼쪽 상단 부분에 렌더링된 현재의 심볼을 미리 볼 수 있습니다. 탭의 아래 부분에 이미 목록에서 선택하여 사용할 수 있도록 준비되었고 현재의 스타일에 대한 정의된 심볼의 목록이 있습니다. 현재의 심볼은 미리보기 아래의 **[Change]** 버튼을 이용하여 수정할 수 있습니다. *Symbol Properties* [심볼속성] 대화상자를 열거나 미리보기 오른쪽 아래의 **[Change]** 버튼을 누르면 일반적인 *Color* 대화상자가 열립니다.

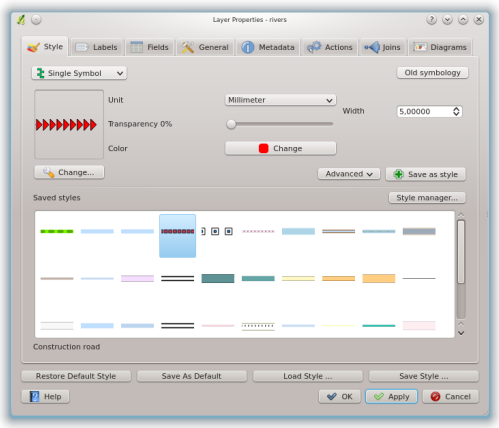


Figure 11.7: Single symbol line properties

In the *Style* tab you can apart from a general layer transparency also define to use millimeter or map units for the size scale. In the **[Advanced]** button next to the **[Save as style]** button you can use data-defined size scale and rotation. Here the *Symbol levels* → menu allows to enable and define the order in which the symbol layers are rendered (if the symbol consists of more than one layer).

필요한 부분을 변경한 심볼은 현재 스타일 심볼 리스트에 추가해서(**[Save as style]** 버튼을 사용), 후에 쉽게 사용할 수 있습니다. Furthermore you can use the **[Save Style]** button to save the symbol as a QGIS layer style file (.qml) or SLD file(.sld). Currently in version 1.8 SLDs can be exported from any type of renderer: single symbol, categorized, graduated or rule-based, but when importing an SLD, either a single symbol or rule-based renderer is created. That means that categorized or graduated styles are converted to rule-based. If you want to preserve those renderers, you have to stick to the QML format. On the other hand, it could be very handy sometimes to have this easy way of converting styles to rule-based.

Categorized Renderer

Categorized Renderer는 선택된 객체가 속성값을 색깔로 나타낸 레이어, 단일사용자 정의 심볼의 사용 등 모든 객체를 렌더링 하는데 사용합니다. *Style* [스타일] 탭에서 선택할 수 있습니다:

- 속성(Column listbox 사용)
- 심볼(Symbol dialog 사용)
- 색상(Color Ramp listbox 사용)

대화상자 아래 오른쪽 코너에 있는 **[Advanced]** 버튼은 회전과 축척 크기 정보를 포함하는 필드를 설정할 수 있게 합니다. 편의를 위해 탭의 하단부분에 현재 함께 선택된 모든 속성 값의 목록과 이를 포함된 심볼들이 렌더링 됩니다.

figure_symbology_2 은 QGIS 샘플 데이터세트의 강 레이어를 category rendering 대화상자에서 사용하는 예를 보여주고 있습니다.

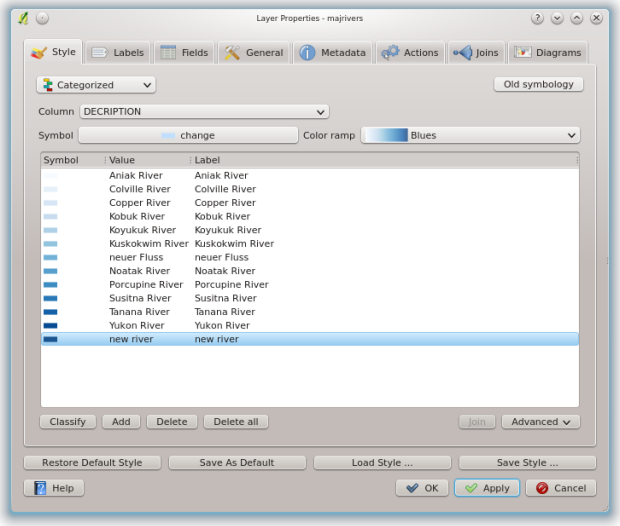


Figure 11.8: Categorized Symbolizing options

컬러램프 드롭다운 메뉴에서 *New color ramp...* 를 선택하여 사용자 컬러램프를 만들 수 있습니다. 대화상자는 ramp 형식(Gradient, Random, ColorBrewer)에 따라 다르게 실행될 것입니다. 그리고 각각의 대화상자는 컬러램프에 대해 단계별 옵션이나 다양한 스톱들의 옵션을 가지고 있습니다. figure_symbology_3_에서 사용자 컬러램프의 예를 참조하십시오.

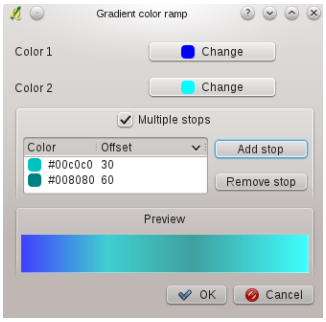


Figure 11.9: 다양한 스톱들과 사용자 그라디언트 컬러 램프의 예

Graduated Renderer

Graduated Renderer는 선택된 객체의 속성의 분류를 반영하는 색상을 가진 단일 사용자 정의 심볼을 사용한 레이어로부터 나오는 모든 객체들을 렌더링하는데 쓰입니다.

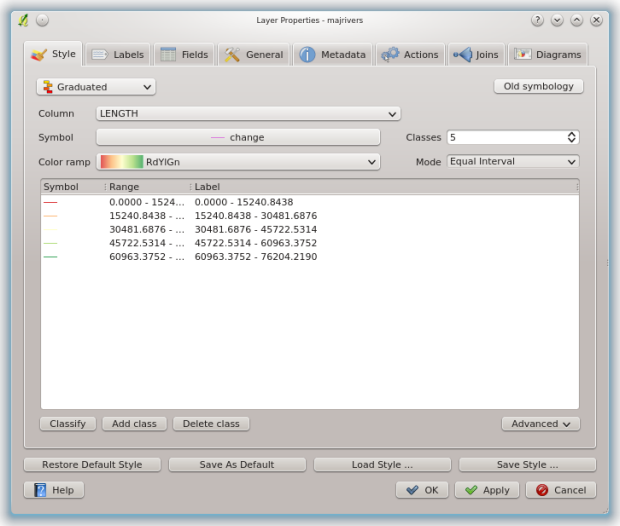


Figure 11.10: 단계별 심볼 옵션

Categorized Renderer 처럼 특정한 컬럼으로 부터 회전과 스케일 크기를 정의할 수 있습니다.

아날로그에서 the categorized 렌더링을 하려면, *Style* [스타일] 탭에서 선택하여 조정합니다:

- 속성(Column listbox 사용)
- 속성(Column list box 사용)
- 심볼(Symbol Properties button 사용)

추가적으로 클래스의 개수와 또한 어떻게 클래스 내부의 객체들을 분리할 것인가의 모드를(Mode list를 사용) 명시할 수 있습니다. 가능한 모드는 다음과 같습니다:

- Equal Interval
- Quantile
- Natural Breaks (Jenks)
- Standard Deviation
- Pretty Breaks

Style [스타일] 탭 아래 부분의 리스트박스는 범위, 라벨과 렌더링 될 심볼들의 클래스를 함께 나열합니다.

figure_symbology_4 의 예제는 QGIS의 샘플 데이터세트의 강 레이어를 단계적 렌더링 대화상자를 보여줍니다.

Rule-based rendering

Rule-based rendering은 선택된 객체의 속성 클레스를 분류하여 색상으로 표현한 Rule-based 심볼을 사용하여 레이어의 모든 객체들을 렌더링에 씁니다. 이러한 규칙은 SQL문을 기반으로 합니다. 또한 Query Builder를 이용하여 규칙을 만들 수 있습니다. 대화상자는 필터 또는 축척으로 그룹핑 할 수 있으며, 원하는 심볼 수준을 설정하거나 처음 일치하는 규칙을 사용할지를 결정할 수 있습니다.

figure_symbology_5_는 예제 QGIS 샘플 데이터세트의 하천 레이어에 대한 Rule-based rendering 대화상자를 보여줍니다.

To create a rule, activate an existing row by clicking on it or click on '+' and click on the new rule. Then press the **[Edit]** button. In the *Rule properties* dialog you can define a label for the rule. Press the  button to open the Expression builder. In the **Function List**, click on *Fields and Values* to view all attributes of the attribute table to be searched. To add an attribute to the Field calculator **Expression** field, double click its name in the *Fields and Values* list. Generally you can use the various fields, values and functions to construct the calculation expression or you can just type it into the box (see *필드 계산기*).

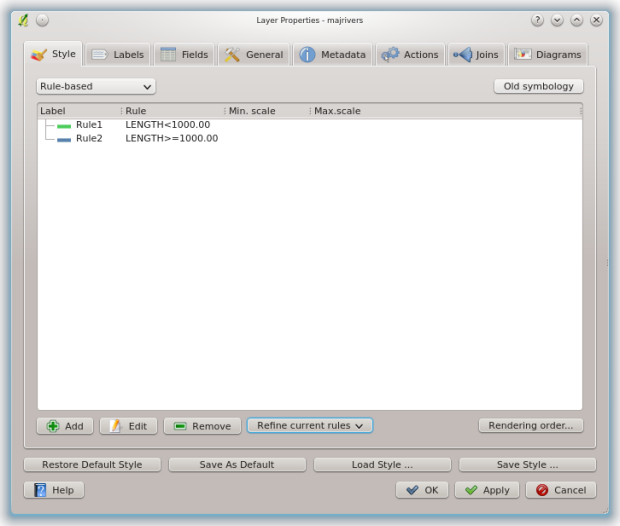


Figure 11.11: Rule-based Symbolizing options

Point displacement

포인트 디스플레이스먼트 렌더러는 QGIS 플러그인 메니저에서 디스플레이스먼트 플러그인을 로드했을 경우에만 사용할 수 있다. 이것은 포인트들이 같은 위치에 있다고 하더라도 모든 포인트 레이어의 객체들을 시각화하는 것을 제공한다. 이렇게 하면 포인트의 심볼들은 중앙심볼 주변으로 원형으로 배치됩니다.

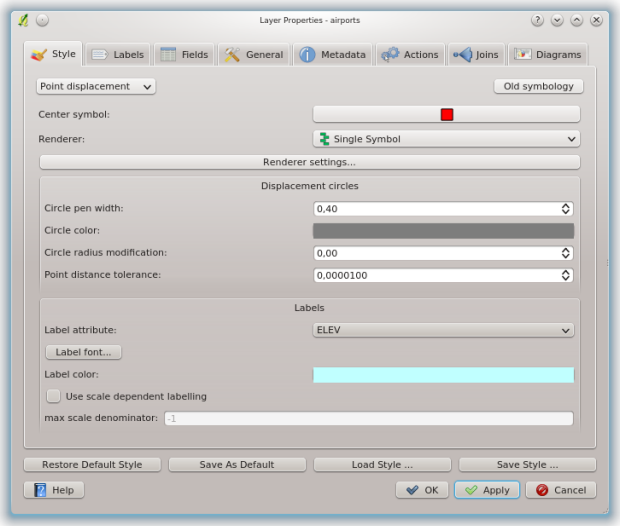


Figure 11.12: 포인트 이동 대화상자

Symbol Properties

심볼 프로퍼티 대화상자는 렌더링 된 심볼을 다른 속성으로 지정할 수 있다. 대화상자의 왼쪽 상단에서 지도 캔버스에 디스플레이 될 현재의 심볼을 미리보기 할 수 있습니다. 미리보기 아래에는 심볼 레이어의 리스트가 있습니다. *Symbol properties* 대화상자를 시작하려면, *Layer Properties* 대화상자의 *Style* [스타일] 탭에서 [**Change...**] 버튼을 클릭합니다.

컨트롤 판넬에서 레이어를 추가하거나 제거하는 것, 레이어의 위치를 바꾸는 것, 또는 레이어의 색상변경 잠그기를 조정합니다. 대화상자의 오른쪽 부분에서 심볼 레이어 리스트에서 선택된 단일 심볼 레이어에

적용되는 설정을 볼 수 있습니다. 중요한 것은 레이어 타입을 선택할 수 있는 *Symbol Layer Type*  콤보박스 입니다. 가능한 옵션들은 레이어 타입에 따라 다릅니다(Point, Line, Polygon). The symbol layer type options are described in section vector_symbol_types. You can also change the symbol layer properties in the right part of the dialog. For example if you have chosen an SVG marker for a point layer it is now possible to change its color using *Color* button.

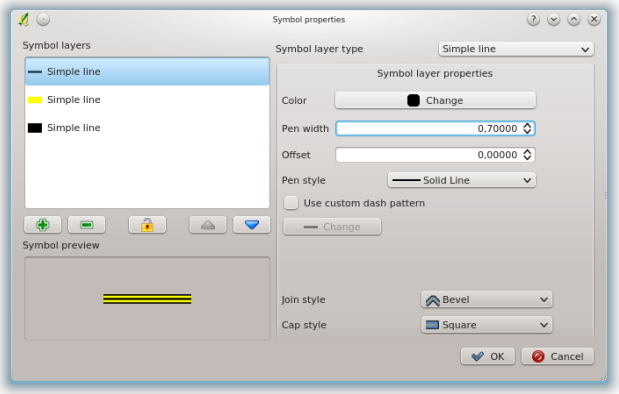


Figure 11.13: 단순한 세 개 줄로 구성된 라인

11.2.2 Style Manager

Style Manager는 작은 도우미 어플리케이션으로, 심볼과 컬러 램프에 적용 가능한 스타일에 대한 리스트를 제공합니다. 또한 아이템들을 추가 또는 삭제할 수 있습니다. Style Manager를 실행하기 위하여 메인 메뉴의 *Settings* → *Style Manager* 를 클릭합니다. Alternatively, you can access it via the *Style* tab.

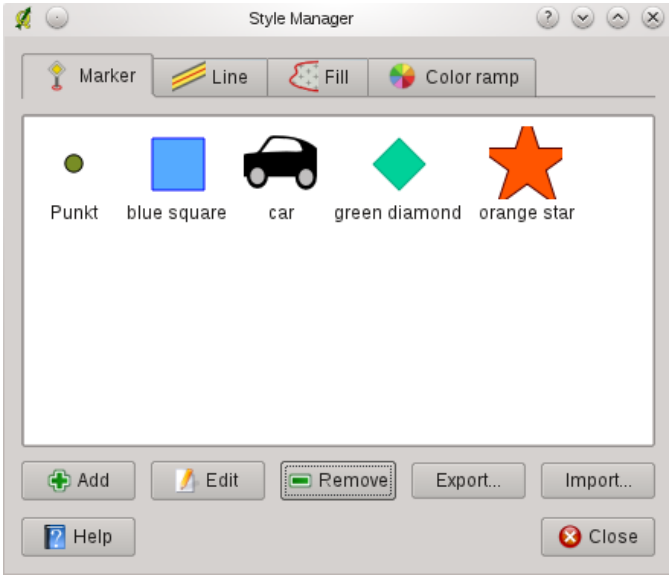


Figure 11.14: 심볼과 컬러 램프 제어를 위한 Style Manager

11.2.3 구 버전 심볼로지

주석: QGIS 1.8은 구 버전 심볼로지가 앞으로의 릴리즈 때 사라질 것이기 때문에 새로운 심볼로지로 전

환하는 것을 추천하지만, 여전히 구 버전 심볼로지를 지원합니다(vector_new_symbology 에 설명되어있듯이).

만약 구 버전 심볼로지로 돌아갈 필요가 있으면 *Layer Properties* 대화상자의 *Style* [스타일] 탭에 있는 **[Old symbology]** 버튼을 클릭합니다.

또한 구 버전 심볼을 기본설정으로 할 수도 있는데, *Settings* → *Options'의 :guilabel:'Rendering* 탭에 있는 *Use new generation symbology for rendering* [렌더링에 차세대 심볼로지를 이용] 를 비활성화 하면 됩니다.

QGIS 구 버전 심볼로지는 아래의 renderer를 제공합니다:

- **Single symbol** ? 레이어의 모든 오브젝트에 대하여 하나의 스타일을 적용함.
- **Graduated symbol** ? 레이어 내의 오브젝트가 특정 필드의 값에 의해 분류되어 다른 심볼로 표현됨.
- **Continuous color** ? 레이어 내의 오브젝트는 특정 필드내에서 수치적으로 분류된 색상으로 표현됨.
- **Unique value** ? 오브젝트는 서로 다른 심볼을 가진 각각의 값으로 지정된 필드 내에서 고유한 값을 기준으로 분류됨.

레이어의 change the symbology (심볼로지)를 변경하려면, *Layer Properties* 대화상자의 범례와 벡터를 더블클릭하기만 하면 됩니다.

스타일 옵션 (**Style Options**)

이 대화상자에서는 벡터 레이어의 스타일을 설정할 수 있습니다. 선택된 렌더링 옵션에 따라 맵객체(map feature)를 분류할 수도 있습니다.

최소한 아래의 스타일링 옵션은 거의 모든 renderer에 적용이 됩니다:

- **Fill style** ? 채움 형식. 선택할 수 있는 주어진 브러쉬 *Fill style:*  외에도  버튼을 클릭하여 사용자 자신의 텍스처 파일을 선택할 수 있습니다. 최근에는 `*.jpeg`, `*.xpm`, and `*.png` 의 파일포맷에 대하여 이 기능이 지원된다.
- **Fill color** ? 객체의 채움 색상.
- **Outline options**
  - Outline style ? 객체의 외곽선에 대한 펜 형식. 'no pen'으로 설정할 수 있습니다.
  - Outline color ? 객체의 외곽선 색상.
  - Outline width ? 객체의 폭.

사용자가 레이어에 대한 스타일을 설정하면, 이에 대한 설정값을 저장할 수 있습니다(`*.qml`). **[Save Style...]** 버튼을 사용하여 설정값을 저장하며, **[Load Style...]**은 저장된 설정값 파일을 불러오는데 사용합니다.

만약 레이어를 불러올 때마다 항상 특정한 스타일을 사용하고자 한다면 **[Save As Default]** 버튼을 사용하여 사용자의 스타일값을 기본으로 설정하면 됩니다. 또한, 변경한 스타일이 마음에 들지 않으면 **[Restore Default Style]** 버튼을 이용하여 기본값으로 복원할 수 있습니다.

벡터 투명도 (**Vector transparency**)

|qg|는 모든 벡터 레이어에 대하여 투명도를 설정할 수 있습니다. *Style* [스타일] 탭의 슬라이더 *Transparency* |slider|를 이용하여 투명도를 설정합니다.(검증필요!! 참조) 이 기능은 여러 벡터 레이어를 중첩할 때 유용하게 사용할 수 있습니다.

11.2.4 라벨 탭 (Labels Tab)

심볼로지에 대하여 QGIS 1.8.0는 현재 old/new labeling 엔진을 병렬로 제공하고 있습니다. *Labels* [라벨] 탭은 여전히 old labeling을 포함하고 있습니다. new labeling은 핵심 어플리케이션으로 구현되며 다음 버전 중 하나에 old label 탭의 기능을 대체하게 됩니다.

따라서 new labeling으로의 전환을 추천하며, 이에 대한 내용은 :ref:‘vector_new_labeling‘에 설명되어 있습니다.

The old labeling in the *Labels* tab allows you to enable labeling features and control a number of options related to fonts, placement, style, alignment and buffering. We will illustrate this by labeling the lakes shapefile of the QGIS sample dataset:

1. Shapefile(`alaska.shp`)과 GML파일 `lakes.gml` 을 |qg|에서 엽니다
2. 호수에 대한 관심지역을 확대합니다
3. `lakes` 레이어를 활성화 합니다
4. *Layer Properties* 대화상자를 엽니다
5. *Labels* [라벨] 탭을 클릭합니다
6. *Display labels* [라벨 표시]체크박스를 체크하여 라벨링이 가능하게 합니다
7. 라벨이 포함된 필드 *Field containing label* “NAMES“를 이용하여 라벨링할 필드를 선택합니다
8. 이름이 없는 호수에 대한 기본을 입력합니다. |qg|가 `NAMES` 필드에 값이 없는 호수를열 때 마주치게 될 때마다 기본 라벨을 사용합니다.
9. 레이블의 길이가 여러 줄이 될 경우 *Multiline labels? [다중행 라벨?]를 선택합니다. |qg|는 label field의 실제 라인 엔터키 수를 확인하고, 그에 맞추어 행 바꿈 넣을 것입니다. 실제 라인의 엔터키는 단일 문자* ‘n“입니다(별개의 두 문자가 아닙니다. 벡슬러시 “‘“에 문자 “n“이 따라오는 것처럼). 속성필드에 라인리턴을 넣으려면 텍스트 위젯에서 텍스트를 편집 환경설정을 합니다. (라인편집이 아님).
10. **[Apply]**를 클릭합니다.

이제 label이 생성되었습니다. 어떻게 보입니까? 아마도 호수에 대한 marker symbol과 비교하여 너무 크거나 이상한 위치에 있을 것입니다.

`Font` [글꼴] 목록을 선택하고 **[Font]** [글꼴]와 **[Color]** [색상] 버튼을 이용하여 폰트와 색상을 설정합니다. 또한 텍스트 라벨의 각도와 위치에 대해서도 변경이 가능합니다.

객체에 대한 텍스트의 상대적인 위치를 설정하려면:

1. Beneath the Basic label options change the placement by selecting one of the radio buttons in the *Placement* group. To fix our labels, choose the *Right* radio button.
2. `Font size units` (글꼴 크기)에서 *Points* (포인트 단위) 또는 *Map units* (지도 단위) 선택.
3. **[Apply]** 를 눌러서 대화상자를 닫지 않고 변경내용을 확인.

이전 보다 더 낫게 보이지만, label은 아직도 marker에 너무 가까이 있습니다. 이것을 해결하기 위해 Position 에서 옵션을 사용할 수 있습니다. X, Y방향으로 `Offset` (이격거리)을 추가 하는 방법입니다. X에 offset 5를 더하면 label이 marker에서 멀어지고 훨씬 판독하기 쉬워집니다. 물론 marker symbol이나 폰트가 크면 오프셋 값을 더 줘야 합니다.

마지막 조정과정은 label의 `Buffer` 입니다. 이는 Buffer를 둠으로써 배경에서 조금 더 두드러지게 강조하는 것을 뜻합니다. 호수 label에 버퍼를 주기 위하여:

1. *Buffer Labels* [라벨 테두리?]체크박스를 체크하여 버퍼링이 가능하게 합니다.
2. 스핀박스로 버퍼값을 선택합니다.
3. **[Color]** [색상]를 클릭하여 가장 알맞은 색상을 색상표에서 선택하고, 원한다면 버퍼에 대한 투명도를 설정할 수 있습니다.
4. 결과에 만족한다면, **[Apply]** 를 클릭합니다.

결과에 만족하지 않는다면, 설정값을 다시 조정하고 **[Apply]** 를 클릭합니다.

1 포인트(point)의 버퍼는 괜찮은 결과를 보여줍니다. 사용자가 편하다면 버퍼사이즈를 map unit으로 명기해도 좋습니다.

Label [라벨] 탭의 나머지 부분들은 레이어에 저장된 속성을 사용하여 라벨의 모양을 제어할 수 있습니다. "Data defined"은 레이어의 필드를 사용하여 label의 파라미터를 설정할 수 있습니다.

Label [라벨]은 선택한 라벨에 대한 `preview-box` (미리보기창)을 제공합니다.

11.2.5 새로운 라벨링

새로운 Labeling [라벨링]은 벡터 포인트, 라인 및 폴리곤 레이어를 위한 스마트 라벨을 제공하며 단지 몇 개의 변수가 필요합니다. 이 새로운 어플리케이션은 ***라벨 탭*** *(Labels Tab)* 절에서 언급했듯이 현재의 QGIS labeling을 대신할 것이며, 실시간으로(on-the-fly) 변환되는 레이어를 제공할 것입니다.

새로운 라벨링 사용하기 (**Using new labeling**)

1. QGIS를 실행하고, 벡터 포인트, 라인 또는 폴리곤 레이어를 불러옵니다.
2. 범례의 레이어를 활성화 시킨 후 QGIS 도구모음 메뉴의 Labeling [라벨링] 아이콘을 클릭합니다.

포인트 레이어 라벨링 (**Labeling point layers**)

첫번째 단계는 *Label this layer* [이 레이어에 라벨링]체크박스를 활성화시키고, 라벨링에 사용할 속성 컬럼을 선택하는 것입니다. Click ... if you want to define labels based on expressions. After that you can define the text style and the scale-based visibility in the *Label settings* tab (see Figure_labels_1). Choose the *Advanced* tab for the label placement and the labeling priority. You can define if every part of a multipart feature is to be labeled here. With the wrap label on character function you can define a character for a line break in the labels. The *Data defined settings* tab provides you with the attribute-based definition of Font properties, Buffer properties and Position.

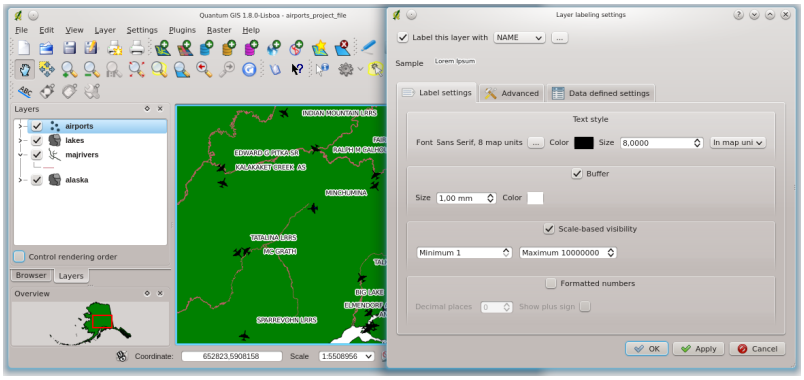


Figure 11.15: Smart labeling of vector point layers

Labeling line layers

첫번째 단계는 *Label this layer* [이 레이어에 라벨링]체크박스를 활성화시키고, 라벨링에 사용할 속성 컬럼을 선택하는 것입니다. Here you can also define labels based on expressions. After that you can define the text style and the scale-based visibility. Further labeling options are available through the *Advanced* tab. You can define the label placement and label distance, a line orientation dependend position and the labeling priority here. Furthermore you can define if every part of a multipart line is to be labeled, if lines shall be merged to avoid duplicate labels and if a direction symbol is added (see Figure_labels_2). It is also possible to supress labeling of features and wrap lables on characters. Use *Data defined settings* for attribute-based or database-connection-based settings.

Labeling polygon layers

첫번째 단계는 *Label this layer* [이 레이어에 라벨링]체크박스를 활성화시키고, 라벨링에 사용할 속성 컬럼을 선택하는 것입니다. Here you can also define labels based on expressions. In *Label settings* define the text style and the scale-based visibility (see Figure_labels_3). Use the *Advanced* tab for label placement, label distance and labeling priority. Define if every part of a multipart feature is to be labeled, suppress labeling of

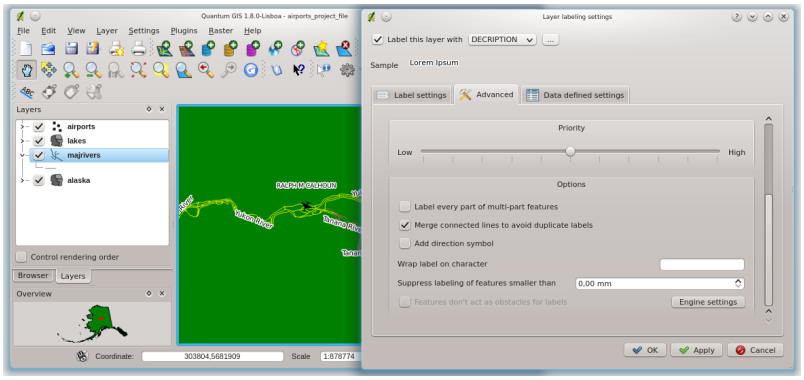


Figure 11.16: Smart labeling of vector line layers

features and wrap labels on characters here. Use *Data defined settings* for attribute-based or database-connection-based settings.

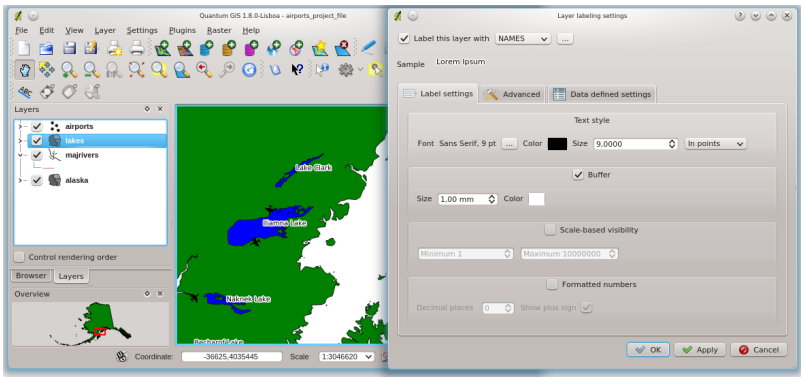


Figure 11.17: Smart labeling of vector polygon layers

설정변경 (**Change engine settings**)

또한 **[Engine settings]** 버튼을 클릭하고 검색 방법을 선택함으로써 최고의 label placement를 찾을 수 있습니다. Chain, Popmusic Tabu, Popmusic Chain, Popmusic Tabu Chain and FALP를 사용할 수 있습니다.

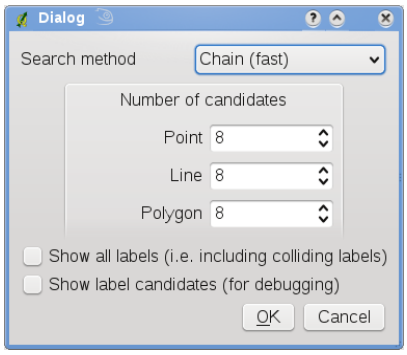


Figure 11.18: Dialog to change label engine settings

candidate은 점, 선, 및 다각형 기능에 대해 정의될 수 있으며 모든 label (colliding label 포함)및 디버깅을 위한 레벨 candidate을 표시할지에 대한 여부를 정의할 수 있습니다.

라벨링을 위한 속성 컬럼 사용의 키워드 (**Keywords to use in attribute columns for labeling**)

다음은 정의된 속성 컬럼의 레이블의 배치를 위해 사용할 수 있는 키워드입니다.

- 수평 정렬(**For horizontal alignment**): `left`, `center`, `right`

- 수직 정렬 (**For vertical alignment**): `bottom`, `base`, `half`, `top`
- **Colors can be specified in svg notation**, 예: #ff0000
- **for bold, underlined, strikeout and italic**: 0 = false 1 = true

한 컬럼에서 키워드의 조합도 가능합니다. 예: `base right` or `bottom left`.

11.2.6 Fields Tab

Within the *Fields* tab the field attributes of the selected dataset can be manipulated. The buttons New Column and Delete Column can be used, when the dataset is Editing mode.

이때에는 PostGIS 레이어의 Column만 없애거나 더할 수 있습니다. 만약 GDAL version >= 1.9이 설치되어 있다면 OGR Library는 새로운 column를 추가와 없애는 것을 지원합니다.

Edit Widget

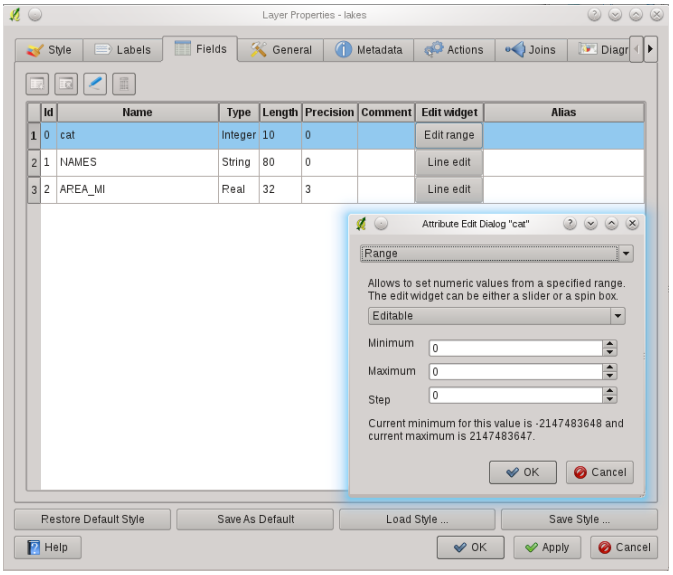


Figure 11.19: 속성 컬럼을 위한 편집 도구를 선택하는 대화상자

Fields 탭 안에서 `edit widget` 이란 Column를 발견할 수 있습니다. 이 Column은 정해진 attribute table column에 더할 수 있게 허용된 값이나 값의 범위를 정의할 때 쓰입니다. **[Edit widget]** 버튼을 클릭하게 되면, 여러 개의 widgets를 확인할 수 있는 창이 하나 뜹니다. 아래는 그 widgets들을 설명하고 있습니다:

- **Line edit**: 간단한 텍스트를 입력할 수 있는 편집 필드 (또는 숫자 속성에 대한 숫자 제한).
- **Classification**: *Style* [스타일] 탭의 범례 유형을 'unique value'로 선택했을 경우에는 분류 (Classification)에 사용되는 값을 콤보 상자로 표현됩니다.
- **Range**: 특정 범위의 숫자 값을 설정할 수 있습니다. 이 edit widget은 슬라이더이거나 spin box 형태로 될 수 있습니다.
- **Unique values**: 사용자가 이미 속성 테이블에 사용되는 값 중 하나를 선택할 수 있습니다. 만약 편집 가능이 활성화 된 경우에는 Autocompletion이 지원되는 line edit이 보여지지만 다른 경우에는 combo box가 사용됩니다.
- **File name**: 파일 선택 dialog를 더함으로써 선택을 간소화 합니다.
- **Value map**: 미리 정의된 항목들을 가진 콤보 박스. 값(Value)들은 attribute에 저장되어 있고 설명은 combo box에 나와있습니다. 수동으로 값을 정의하거나 레이어 또는 CSV파일에서 로딩할 수 있습니다.

- **Enumeration**: Columns Type내에서 사용할 수 있는 값들을 가진 combo box를 엽니다. 이것은 현재 POSTGRES 공급자에 의해서만 지원이 됩니다.
- **Immutable**: The immutable attribute column 은 읽기 전용입니다. 사용자들은 콘텐츠를 수정할 수 없습니다.
- **Hidden**: 숨은 attribute column은 눈에 보이지 않습니다. 사용자가 내용을 볼 수 없습니다.
- **Checkbox**: 체크박스 (확인란)을 표시합니다. 사용자는 체크박스가 활성화 되었거나 되지 않았을 때 어떠한 attribute가 column에 더해졌는지 확인할 수 있습니다.
- **Text edit**: 이것은 다수의 라인을 사용할 수 있도록 텍스트 편집 필드가 열립니다.
- **Calendar**: 날짜를 입력하기 위한 달력 widget를 엽니다. Column type은 꼭 텍스트여야 합니다.
- **Value Relation**: Offers values from a related table in a combobox. You can select layer, key column and value column.
- **UUID Generator**: Generates a read-only UUID (Universally Unique Identifiers) field, if empty.

11.2.7 일반정보 탭(General Tab)



The *General* tab is essentially like that of the raster dialog. There are several options available:

- Change the display name of the layer
- Set a display field to be used for the *Identify Results* dialog
- Define a certain Edit User Interface for the vector layer written with the Qt Creator IDE and tools at http://qt.digia.com/Product/Developer-Tools/
- Create a *Spatial Index* (only for OGR supported formats)
- Add an *Init function* for the layer. They will overwrite existing QGIS widget initializations, if applied
- *Update Extents* information for a layer
- View or change the projection of the specific vector layer, clicking on *Specify CRS*

Furthermore you can activate and set  *Use scale dependent rendering*, define provider specific options (e.g. encoding) and with the **[Query Builder]** button you can create a subset of the features in the layer that will be visualized (also refer to section *선택*).

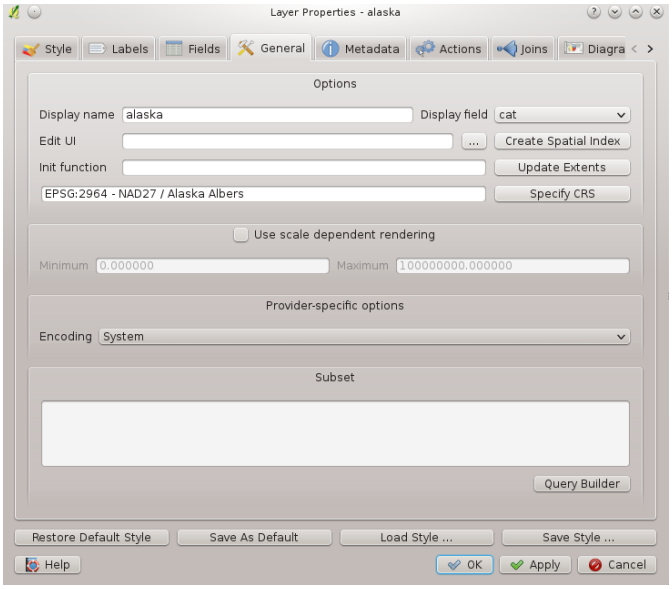


Figure 11.20: General tab in vector layers properties dialog 

11.2.8 메타데이터 탭(Metadata Tab)

The *Metadata* tab contains general information about the layer, including specifics about the type and location, number of features, feature type, and the editing capabilities. The *Extents* section, providing layer extent information, and the *Layer Spatial Reference System* section, providing information about the CRS of the layer. This is a quick way to get information about the layer.

Additionally you can add/edit a title for the layer and some abtract information. These information will be saved in the QGIS project file for following sessions and will be used for QGIS server.

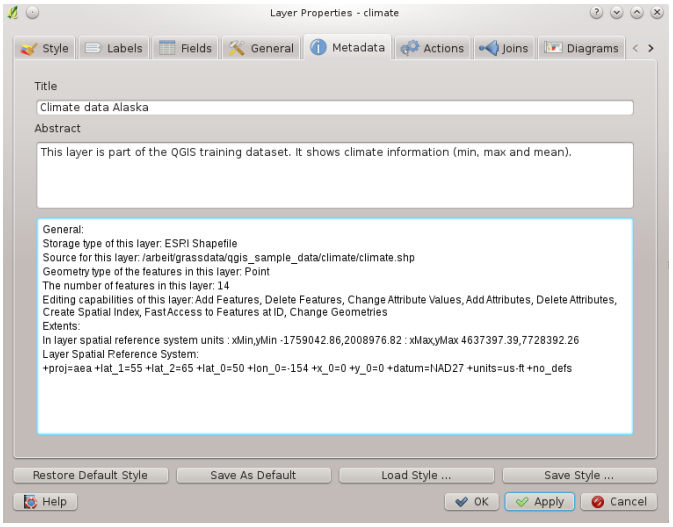


Figure 11.21: Metadata tab in vector layers properties dialog

11.2.9 액션 탭(Action Tab)

QGIS는 기능의 특성을 바탕으로 하는 action를 수행할 수 있는 기능을 제공합니다. 이것은 기능의 특성으로부터 만들어진 argument로 프로그램을 실행하거나 웹 리포팅 도구에 매개 변수를 전달하는 것과 같은 action도 실행할 수 있습니다.

Action은 사용자가 자주 외부 프로그램을 실행하거나 하나를 기반으로 웹 페이지를 보려고 할 때 굉장히 요긴하게 쓰이곤 합니다. They are devided into 6 types and can be used like this:

- Generic, Mac, Windows and Unix actions start an external process,
- Python actions execute a python expression,
- Generic and Python actions are visible everywhere,
- Mac, Windows and Unix actions are visible only on the respective platform (i.e. you can define three 'Edit' actions to open an editor and the users can only see and execute the one 'Edit' action for their platform to run the editor).

There are several examples included in the dialog. You can load them clicking on **[Add default actions]**. An example is performing a search based on an attribute value. This concept is used in the following discussion.

액션 정의하기 (**Defining Actions**)

속성 action들은 *Layer Properties* 대화상자에서 정의됩니다. define an action'을 정의하기 위해선 :guilabel:'Layer Properties 대화상자를 열고 *Actions* [액션] 탭을 클릭합니다. Select 'Generic' as type and provide a descriptive name for the action. Action자체는 Action을 불러내었을 때 실행될 응용프로그램의 이름을 포함하고 있어야 합니다. 사용자는 하나 이상의 속성값을 응용 프로그램의 하나의 인자로 추가할 수 있습니다. Action이 적용되었을 때 ''%''로 시작되고 필드의 이름이 뒤에 오는 해당 필드의 집합체는 그 필드의 값으로 대체됩니다. :index:'%%'라는 특수문자는 identify results나 속성테이블에서 선택된 필드 값에 의

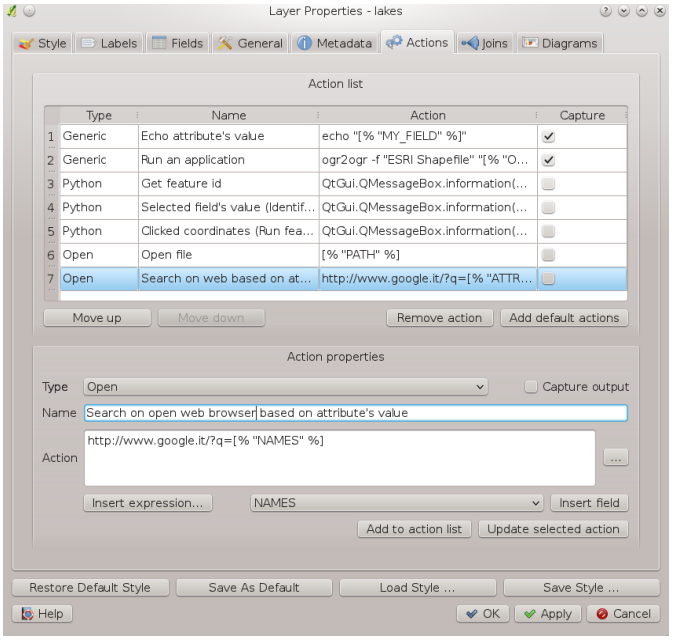


Figure 11.22: Overview action dialog with some sample actions

해 대체됩니다. (아래 using_actions below 참고). Double quote mark는 프로그램, 스크립트 또는 command에 텍스트들을 단일 인수로 그룹화하게 할 수 있습니다. 역 슬래쉬로 Double quote를 무시할 수도 있습니다.

만약 사용자가 다른 필드 name의 substrings인 필드 name을 가지고 있다면(예: `col1` and `col10`) 해당 필드 name (and % 문자)을 사각 브라켓 [ ]로 표현해야 합니다(예: `[%col10]`). 이러한 방법은 `%col10` 필드 name이 `%col1` 과 같은 필드 name 헷갈리지 않기 위해서 쓰이는 방법입니다. 브라켓은 QGIS이 필드 값을 대체할때에 제거됩니다. 만약 대체되는 값에 브라켓이 포함되길 원한다면 `[[%col10]]` 이렇게 하시면 됩니다.

Using the *Identify Features* tool you can open *Identify Results* dialog. It includes a *(Derived)* item that contains information relevant to the layer type. 이 항목의 값들은 derived된 필드name에 *(Derived)*를 씀으로써 다른 필드와 비슷한 방법으로 엑세스할 수 있습니다.예를 들자면 `X``와 ``Y` 필드 값을 가지고 있는 점 레이어와 그 값들은 `%(Derived).X``와 ``%(Derived).Y` 라는 Action으로 사용될 수 있습니다. Derived 속성들은 오로지 Identify Result 대화상자에서만 가능하며 *Attribute Table* 대화상자에서는 불가능합니다.

아래는 2개의 example actions (예)를 보여주고 있습니다:

- `konqueror http://www.google.com/search?q=%nam`
- `konqueror http://www.google.com/search?q=%%`

첫 번째 예에서는 웹브라우저 Konqueror가 호출되었고 해당 URL이 불러졌습니다. 이때 구글은 벡터 레이어에서의 `nam` 필드 값을 검색하게 됩니다. Action에 의해서 불려지는 어플리케이션이나 스크립트는 경로에 있거나 전체 경로를 제공해야 합니다. 보다 더 정확히 하기 위해 첫 번째 예를 다음과 같이 적을 수 있습니다. `/opt/kde3/bin/konqueror http://www.google.com/search?q=%nam` 이것은 actions이 불렸을 때에 konqueror가 실행된다는 것을 보장할 수 있습니다.

두 번째 예는 값에 대한 특정한 필드에 의존하지 않는 %%기호를 사용합니다. Action이 불렸을 때 %%가 식별결과 또는 속성에서 선택된 필드의 값에 의해 대체됩니다. 액션 사용하기 (**Using Actions**)

Action은 *Identify Results* 대화상자나 *Attribute Table* 대화상자에서 불러올 수 있습니다(이러한 대화상자들은 Identify Features [객체 정보표시] 또는 Open Attribute Table [속성 테이블 열기] 또는 *Run Feature Action* 를 클릭해서 열 수 있습니다).Action을 수행하기 위해선 클릭후 팝업메뉴에서 작업을 선택하시면 됩니다. Action들은 사용자가 정의한 action의 이름대로 팝업메뉴에 나열됩니다. 불러오고 싶은 Action을 클릭합니다.

만약 사용자가 ‘‘%%‘‘를 사용하는 Action을 사용한다면 응용 프로그램이나 스크립트로 전달하고자 하는 Identify Result 대화상자나 Attribute Table 대화상자의 필드값을 오른쪽 클릭하시면 됩니다.

다음은 bash와 `echo` command(이 명령어는 또는 에서만 작동할 겁니다)를 사용해 벡터 레이어에서 데이터를 가져와 파일에 삽입하는 예입니다. 여기서 쓰일 레이어는 종의 이름, Taxon_name, 위도 "lat"과 경도 "long"이 있습니다. 여기서 하고 싶은 것은 장소 (localities)에 대한 공간적 선택과 선택된 record를 위한 텍스트파일로 해당 값을 export하는 것입니다. (QGIS 지도에서 노란색으로 표현). 이것을 이루기 위한 Action이 다음과 같습니다:

```
bash -c "echo \"%taxon_name %lat %long\" >> /tmp/species_localities.txt"
```

몇몇개의 장소를 선택한 후 각각에 Action를 실행시키면 나오게 될 output 파일은 다음과 같이 나타날 것입니다:

```
Acacia mearnsii -34.0800000000 150.0800000000
Acacia mearnsii -34.9000000000 150.1200000000
Acacia mearnsii -35.2200000000 149.9300000000
Acacia mearnsii -32.2700000000 150.4100000000
```

이러한 연습방법으로써 `lakes` 레이어에 구글 검색을 수행해보겠습니다. 첫 번째로 우리는 키워드 검색을 수행하기 위해 필요한 URL을 결정해야 합니다. 이것은 쉽게 그냥 구글에 들어가 검색을 하고 주소바에서 URL하나를 집어 오면 됩니다. 이러한 간단한 노력으로 "QGIS"를 검색하였을 때 우리는 URL 포맷은 http://google.com/search?q+qgis 라는 것을 알게 됩니다. 이 정보를 가지고 우리는 다음과 같이 할 수 있습니다:

1. `lakes` 레이어가 로딩 된 것을 확인.
2. Legend에 더블클릭하거나 오른쪽 클릭 후 팝업메뉴에서 *Properties* 를 선택하여 *Layer Properties* 대화상자를 엽니다.
3. *Actions* [액션] 탭 클릭.
4. `Google Search` 같은 Action의 이름을 입력합니다.
5. Action작업을 위해 실행하는 외부 프로그램의 이름을 제공해야 합니다. 지금 이 경우에는 Firefox를 사용할 수 있습니다. 프로그램이 경로에 없다면 전체 경로를 제공해야 합니다.
6. 외부 응용 프로그램의 이름에 따라 구글 검색에 쓰일 URL를 입력합니다. 하지만 검색용어는 포함하지 않습니다. `http://google.com/search?q=`
7. *Action* 필드에 있는 Text는 이제 다음과 같이 보일 겁니다. `firefox http://google.com/search?q=`
8. `lakes` 레이어에 대한 필드 이름을 포함하는 드롭다운 상자를 클릭해야 합니다. **[Insert Field]** [필드 삽입] 버튼 왼쪽에 위치하고 있습니다.
9. *The valid attribute names for this layer*  드롭다운 박스에서 'NAMES'를 선택하고 **[Insert Field]** [필드 삽입]를 클릭합니다.
10. 그 결과 Action Text가 다음과 같이 바뀝니다:

    ```
    firefox http://google.com/search?q=%NAMES
    ```

11. 마지막으로 **[Insert action]** [액션 삽입] 를 클릭합니다.

이것으로 Action은 완료되고 사용할 준비가 되었습니다. 마지막 텍스트는 다음과 같아야 합니다:

```
firefox http://google.com/search?q=%NAMES
```

이제 Action을 사용할 준비가 되었습니다. *Layer Properties* 창을 닫고 가고싶은 지역으로 확대합니다. `lakes` 레이어가 활성화 되어있는 것을 확인하고 호수를 확인합니다. Result창에 action이 보일 겁니다:

Action을 클릭해서 Firefox가 불러지고 URL http://www.google.com/search?q=Tustumena로 이동을 할 것 입니다. 또한 Action에 대한 속성 필드를 추가할 수도 있습니다. 따라서 사용자는 Action 텍스트의 끝부분에 "+"를 추가하여 다른 필드를 선택하고 **[Insert Field]** [필드 삽입]를 클릭하면 됩니다. 이 예제에서는 검색을 할 의미를 주는 다른 필드가 없습니다.

또한 사용자는 레이어에 대해 여러 동작을 정의할 수 있으며 *Identify Results* 대화상자에 나타나게 됩니다.

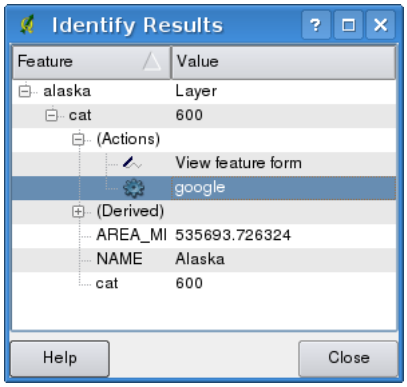


Figure 11.23: 객체 선택과 액션 선택

Action의 용도는 다양합니다. 예를 들자면 만약 사용자가 사진이나 이미지의 위치와 파일이름이 함께 있는 점 레이어를 가지고 있다면 해당 이미지를 볼수 있는 viewer를 실행하는 Action도 만들수 있습니다. 또한 지정 필드의 속성필드 또는 조합에 대한 웹기반보고서를 실행하는 Action도 사용할 수 있습니다 (검증 필요 !!!).

We can also make more complex examples, for instance on how to use **Python** actions.

Usually when we create an action to open a file with an external application we can use absolute paths, or eventually relative paths, in the second case the path is relative to the location of the external program executable file. But what about we need to use relative paths, relative to the selected layer (a file based one, like a shapefile or spatialite)? The following code will do the trick:

```
command = "firefox";
imagerelpath = "images_test/test_image.jpg";
layer = qgis.utils.iface.activeLayer();
import os.path;
layerpath = layer.source() if layer.providerType() == 'ogr' else (qgis.core.QgsDataSourceURI(layer
path = os.path.dirname(str(layerpath));
image = os.path.join(path,imagerelpath);
import subprocess;
subprocess.Popen( [command, image ] );
```

we have to just remember that the action is one of type *Python* and to change the *command* and *imagerelpath* variables to fit our needs.

But what about if the relative path need to be relative to the (saved) project file? The code of the Python action would be:

```
command="firefox";
imagerelpath="images/test_image.jpg";
projectpath=qgis.core.QgsProject.instance().fileName();
import os.path; path=os.path.dirname(str(projectpath)) if projectpath != '' else None;
image=os.path.join(path, imagerelpath);
import subprocess;
subprocess.Popen( [command, image ] );
```

Another Python actions example if the one that allows us to add new layers to the project. For instance the following examples will add to the project respectively a vector and a raster. The name of files to be added to the project and the name to be given to the layer are data driven (*filename* and *layname* are column names of the table of attributes of the vector where the action was created):

```
qgis.utils.iface.addVectorLayer('/yourpath/[% "filename" %].shp','[% "layername" %]', 'ogr')
```

To add a raster (a tif image in this example) it becomes:

```
qgis.utils.iface.addRasterLayer('/yourpath/[% "filename" %].tif','[% "layername" %]')
```

11.2.10 결합 탭(Joins Tab)

Joins [결합] 탭은 join a loaded attribute table to a loaded vector layer (로드된 속성 테이블과 로도된 벡터 레이어)를 합칠 수 있게 해줍니다. 사용자는 join layer, join field와 target field등을 key column으로써 정의를 해야합니다. QGIS는 현재 OGR이 지원하는 non spatial table formats, 구분된 텍스트, PostgreSQL Provider들의 join을 지원합니다. (figure_joins_1 참조).

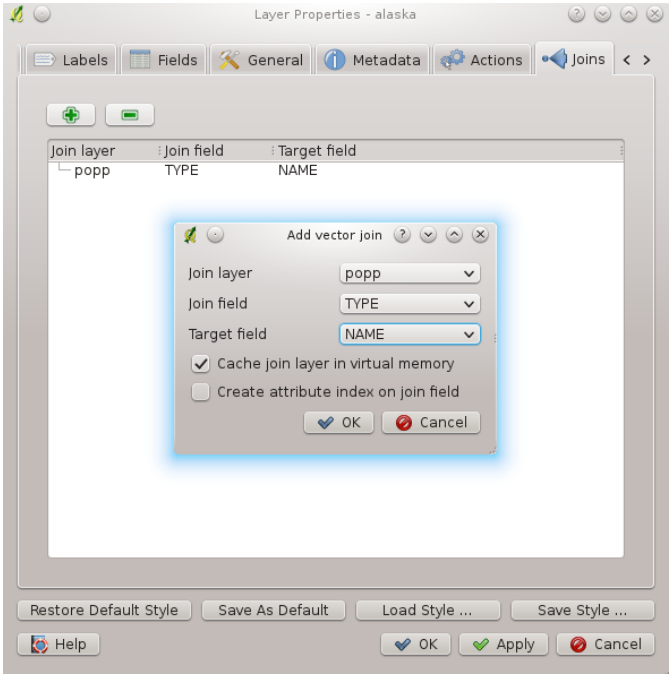


Figure 11.24: 기존 벡터 레이어에 속성 테이블 결합

또한 Add vector join[벡터 조인 추가] 대화상자는 다음을 가능하게 합니다:

- *Cache join layer in virtual memory* [조인 레이어를 가상 메모리에 캐시]
- *Create attribute index on the join field* [조인된 필드에 속성 인덱스 만들기]

11.2.11 다이어그램 탭(Diagram Tab)

Diagrams [다이어그램] 탭을 사용하면 벡터 레이어에 Graphic 오버레이를 추가할 수 있습니다.(figure_diagrams_1 참조).

The current core implementation of diagrams provides support for piecharts and text diagrams. Text values ??of different data columns are displayed one below the other with a circle or a box and dividers. Diagram size is based on a fixed size or on linear scaling according to a classification attribute. The placement of the diagrams interacts with the new labeling, so position conflicts between diagrams and labels are detected and solved. In addition to chart positions can be fixed by the users hand.

We will demonstrate an example and overlay the alaska boundary layer a text diagram showing some temperature data from a climate vector layer. Both vector layers are part of the QGIS sample dataset (see Section *Sample Data*).

1. Load Vector [벡터 레이어 추가] 아이콘을 클릭합니다. QGIS 샘플 데이터 세트를 찾고 두 벡터 레이어 :file:‘alaska.shp‘와 :file:‘climate.shp‘를 불러옵니다.
2. 지도 범례에 `climate` 레이어를 더블클릭하여 *Layer Properties* 를 불러옵니다.

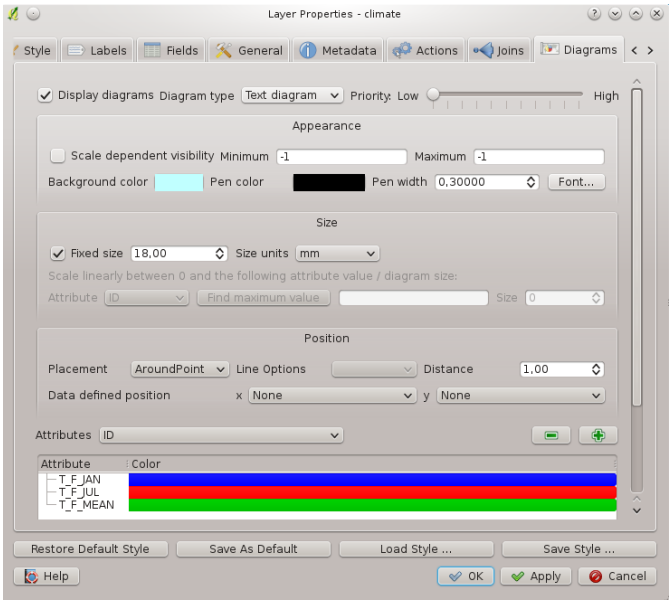


Figure 11.25: 벡터 속성 대화상자의 다이어그램 탭

3. Click on the *Diagrams* tab, activate *Display diagrams* and from *Diagram type*  combobox select 'Text diagram'
4. As Background color we choose a light blue and set a fixed size to 18 mm.
5. Placement could be set to AroundPoint.
6. 우리는 다이어그램에 3개의 컬럼 `T_F_JAN`, "T_F_JUL"와 "T_F_MEAN"을 표현하고 싶습니다. 먼저 "T_F_JAN"를 선택한 후 초록색 **[+]** 버튼을 클릭합니다. "T_F_JUL"와 "T_F_MEAN"을 같은 방법으로 입력합니다.
7. [qg] 메인창에서 **[Apply]** 버튼을 클릭합니다.
8. 이제는 차트의 크기를 바꾸거나 색상 값에 대한 색상을 바꿀 수 있습니다. Figure_diagrams_2 를 참고하십시오.
9. 끝으로 **[Ok]** 버튼을 누릅니다.

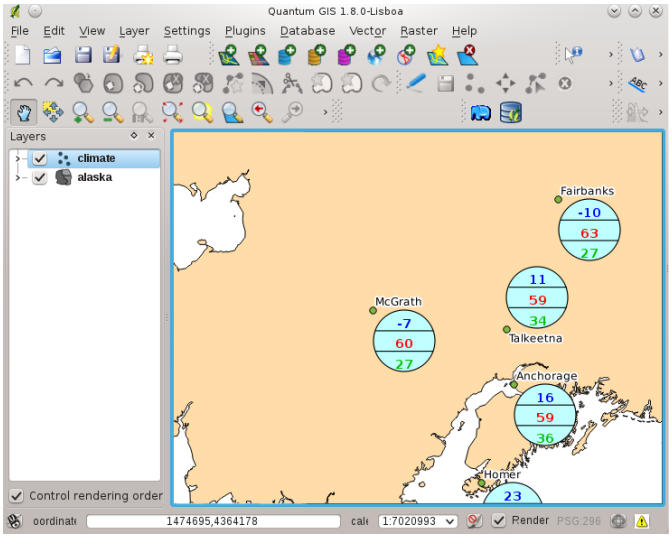


Figure 11.26: 지도 위에 오버랩 된 기온 데이터 다이어그램

Additionally in the *Settings* → *Options* dialog, there is a `Overlay` tab where it is possible to select the placement

algorithm of the diagrams. The central point method is a generic one, the others use algorithms of the PAL library. They also consider diagram objects and labels in different layers. Also see section *Diagram Overlay Plugin* for additional diagram features.

11.3 편집

QGIS 는 editing , PostGIS 등 공간 벡터 레이어들을 편집할 수 있는 다양한 기능들을 제공합니다.

주석: GRASS 레이어들은 편집하는 절차가 조금 다릅니다. 상세한 내용은 *Digitizing and editing a GRASS vector layer* 을 참조하십시오.

참고: Concurrent Edits

QGIS 버전은 같은 객체를 당신과 동시에 다른 사람이 편집하고 있다면 편집 내역을 tracking하지 않습니다. 단지 마지막에 저장한 사람만의 편집내역을 유지할 뿐입니다.

11.3.1 검색 반경 및 스냅 공차 설정

벡터 레이어 지오메트리를 최적의 상태로 편집하기 위해 버텍스 편집을 하기 전에 검색반경값과 스냅 공차값을 설정해야 합니다.

스냅 공차

스냅공차는 새로운 버텍스를 추가하거나 기존 버텍스를 이동시키고자 할 때 사용되는 가장 가까운 버텍스나 연결하고자 하는 세그먼트 검색 시 사용되는 거리값입니다. 만약 스냅공차 범위 바깥에서 버텍스를 스냅하기위해 마우스 버튼을 클릭한다면 QGIS 는 세그먼트나 버텍스를 선택되지 않을 것입니다. 스냅공차 설정값은 공차값과 관련이 있는 툴의 모든 기능에 영향을 미칩니다 '검수필요'.

1. 일반적으로 와이드 스냅 공차는 *Settings* → *Options*. 맥: go to *QIS* → *Preferences...*, 리눅스: *Edit* → *Options* 메뉴를 선택해서 그 값을 정의 할 수 있습니다. *Digitizing* 탭에서 버텍스간, 세그먼트간 또는 버텍스와 세그먼트간 중에서 기본 스냅모드를 선택할 수 있습니다. 또한 버텍스 편집에 사용되는 기본 검색 반경값과과 스냅공차 값을 정의할 수 있습니다. 공차는 픽셀 또는 지도 단위값 중에서 선택하여 세팅할 수 있습니다. 픽셀값으로 선택시의 이점은 지도 확대/축소 후에도 스냅공차값이 바뀌지 않는다는 것입니다. 샘플 프로젝트(알래스카 데이터세트)에서 스냅 단위를 feet로 설정합니다. 결과는 다를 수 있지만 1:10,000 축척에서 300ft의 설정은 합리적인 값이라 할 수 있습니다.
2. 스냅공차를 베이스로 하는 레이어는 스냅을 활성화하고 스냅모드를 조절하기 위해서 *Settings* → (또는 *File* →) *Snapping options...* 를 통해 정의할 수 있습니다.(figure_edit_1 참조).

이러한 스내핑 베이스 레이어는 디지타이징 탭에서 설정할 수 있는 글로벌 스내핑을 중단합니다. 그래서 만약 하나의 레이어만의 편집이 필요하다면 레이어 `snap to` 만을 활성화해야만 글로벌 스냅 공차를 더 작은 값으로 감소시킬 수 있습니다. 더 나아가 스내핑은 스냅 옵션창에서 체크되지 않은 레이어에게는 영향을 미치지 않을 것입니다. 그래서 스내핑하기 원하는 레이어들의 체크박스만을 마크했는지 확인하는 것이 좋습니다.

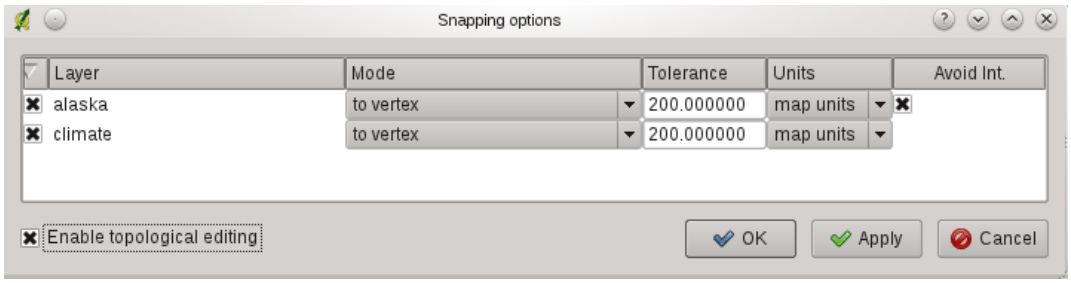


Figure 11.27: 레이어 기반 편집 스냅 옵션

검색반경

검색반경은 지도를 클릭했을 때 이동하기를 원하는 근접한 버텍스를 찾을 때 사용하는 거리값입니다. 검색반경안에 위치하지 않은 어떠한 버텍스도 QGIS 는 선택하지 않을 것입니다. 스냅공차와 검색반경은 픽셀 또는 지도 유닛값으로 설정할 수 있으며, 경우에 따라 적절한 값으로 설정을 해야 합니다. 만약 너무 큰 공차를 설정하면 QGIS 는 잘못된 버텍스를 스내핑할 수 있습니다. 또한 너무 작은 검색 반경값을 설정하면 이동시킬 어떠한 점도 선택하지 못할 것입니다.

버텍스 편집을 위한 검색반경 유닛과 프로젝트 스냅 공차는 *Settings* → *Options* 아래의 *Digitizing* 탭에서 정의할 수 있습니다.

11.3.2 확대/축소와 이동

레이어를 편집하기 전에, 편집하고자하는 관심영역으로 확대/축소 해야합니다. 이것은 모든 버텍스가 드로잉되는 것을 피할 수 있습니다.

마우스로 도구모음에 있는 pan and zoom-in / zoom-out 아이콘을 사용하거나 마우스휠과 스페이스바나 화살표키를 이용해서 기능을 사용할 수 있습니다.

마우스휠을 이용한 확대/축소와 이동

편집하는 동안 마우스휠을 눌러서 메인 창 안으로 패닝할 수 있으며, 휠을 돌려서 지도를 확대 또는 축소할 수 있습니다. 확대/축소를 하기 위해서 마우스 커서를 지도영역안에 위치시키고 휠을 앞쪽으로 조작하면 확대 되고, 뒤쪽으로 조작하면 축소가 됩니다. 마우스커서 위치가 줌영역의 중심이 됩니다. 마우스휠의 역할을 *Settings* → *Options* 메뉴의 *Map tools* 탭에서 조절할 수 있습니다.

화살표키를 이용한 이동

편집하는 동안 지도의 이동은 화살표키로 가능합니다. 마우스 커서를 지도 안에 위치시키고 오른쪽 화살표키를 누르면 동쪽으로 이동하고, 왼쪽 화살표키를 누르면 서쪽으로, 윗방향 화살표키를 누르면 북쪽, 아랫방향 화살표키를 누르면 남쪽으로 이동합니다.

임시적으로 지도말고 마우스 이동을 위해서는 스페이스바를 누르면 됩니다. `PgUp` 키와 `PgDown` 키를 누르면 확대/축소가 됩니다.

11.3.3 위상관계 편집

스냅 옵션외에도 Setting[설정] (또는 *Settings* (or *File*) 메뉴의 *Snapping options...* 대화상자에서 위상과 관련된 기능들을 정의할 수 있습니다. 여기에서 *Enable topological editing* [위상관계 편집 활성화]와 새로운 폴리곤의 인터섹션을 피할 수 있는 *Avoid Int.* [교차 금지] 옵션을 활성화 시킬 수 있습니다.

위상관계 편집 활성화

The option *Enable topological editing* is for editing and maintaining common boundaries in polygon mosaics. QGIS 'detects' a shared boundary in a polygon mosaic and you only have to move the vertex once and QGIS will take care about updating the other boundary.

새로운 폴리곤의 인터섹션 피하기

두번째 위상관계 옵션은 폴리곤 모자이크상에서 오버랩을 피하기위한 '*Avoid intersections of new polygons* 새로운 폴리곤의 인터섹션 피하기'로 불리는 *Avoid Int.* [교차 금지]입니다. 이것은 인접한 폴리곤의 빠른 디지타이징을 위한 것입니다 검정필요!!!.

11.3.4 1.1.1. 기존 레이어의 디지타이징

기본적으로 QGIS 는 읽기전용으로 레이어를 로드합니다. 이는 실수로 레이어를 편집하지 않도록 보호하려는 의도입니다. 그러나 데이터 공급자가 지원하고, 기본 데이터 원본에 쓰기 권한이 있다면(읽지 전용이 아니라면), 어떠한 레이어도 수정하도록 선택할 수 있습니다. PostgreSQL/PostGIS 데이터 소스를 사용하면 레이어 편집이 가장 다양하게 방식으로 지원됩니다.

일반적으로 편집 벡터 레이어는 ***고급 디지타이징 기능*** 절에서 설명하겠지만, 디지타이징 도구모음과 확장 디지타이징 도구모음으로 나눠집니다. *Settings* → *Toolbars* → 에 들어가면 두 개 모두 선택/취소할 수 있습니다. 기본 디지타이징 도구는 아래의 기능을 지원합니다:

아이콘	역할	아이콘	역할
	편집기능 켜기/끄기		객체 추가: 포인트 캡쳐
	객체추가: 라인 캡쳐		객체 추가: 폴리곤 캡쳐
	객체 이동		노드 도구
	선택 객체 삭제		객체 잘라내기
	객체 복사		객체 붙여넣기
	편집 저장 후 계속 편집		

Table Editing: 벡터 레이어 기본 디지타이징 도구모음

모든 편집 세션은 Toggle editing 을 선택해야만 시작됩니다. 이 메뉴는 레이어 목록창에서 마우스 오른쪽 버튼을 클릭하면 나타나는 컨텍스트 메뉴에 들어 있습니다.

다른 방법으로는 디지타이징 도구모음에 있는 Toggle Editing Toggle editing [편집모드 전환] 버튼을 클릭하면 됩니다. 레이어가 편집모드로 들어가면 모든 꼭지점들에 심볼이 그려지며, 편집 도구모음에는 레이의 유형에 따라 적당한 도구 버튼이 추가됩니다.

참고: 정기적으로 저장하십시오 (**Save Regularly**)

Remember to Save Edits regularly. This will also check that your data source can accept all the changes.

객체 추가

도구모음의 Add Feature, Add Feature, Add Feature 아이콘을 클릭하면 QGIS 가 디지타이징 모드로 들어가게 됩니다.

모든 객체는 먼저 도형을 디지타이징 한 다음 속성을 입력해야합니다. 도형을 입력하려면 지도 영역에서 마우스 왼쪽 버튼을 누르면 됩니다.

라인과 폴리곤에 대해서는 마우스 왼쪽 버튼을 계속 누르면 새로운 꼭지점이 추가됩니다. 포인트의 추가가 끝나면 지도의 영역 아무데서나 마우스 오른쪽 버튼을 클릭하면 됩니다.

도형의 입력이 끝나면 새로운 객체의 속성을 입력할 수 있는 속성창이 나타날 것입니다. Figure_edit_2 은 알래스카 rivers 레이어에 새롭게 입력된 강 객체에 대한 속성을 설정하는 예입니다. *Settings* → *Options* 메뉴의 *Digitizing* 탭에 들어가면 *Suppress attributes pop-up windows after each created feature* *Reuse last entered attribute values* 을 설정할 수 있습니다.

도구 모음의 Move Feature(s) 아이콘을 사용하면 기존 객체를 이동시킬 수 있습니다.

참고: Attribute Value Types

For editing the attribute types are validated during the entry. Because of this, it is not possible to enter a number into the text-column in the dialog *Enter Attribute Values* or vice versa. If you need to do so, you should edit the attributes in a second step within the *Attribute table* dialog.

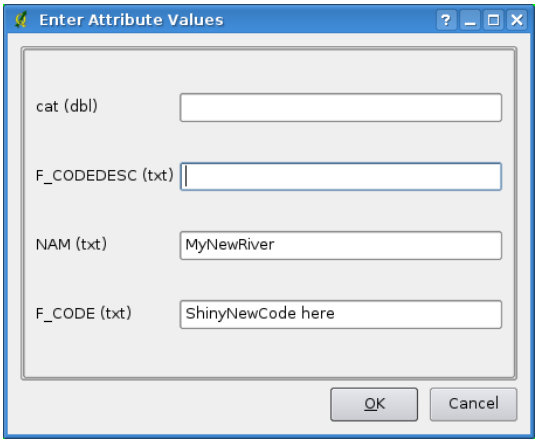


Figure 11.28: 새로운 벡터 객체를 디지타이징 후 속성값을 입력하는 창

노드 도구

Shape-file과 PostgreSQL/PostGIS 기반 레이어의 경우에는 Node Tool [노드 도구] 를 사용하여 CAD 프로그램과 유사하게 객체의 꼭지점을 편집할 수 있습니다. 한번에 여러 개의 꼭지점을 선택해서 이동시키거나 추가, 삭제할 수도 있습니다. 노드툴은 또한 실시간 투영변환이 켜진 상태에서도 사용할 수 있으며, 위상관계 편집 기능도 지원합니다. 노드도구도 다른 QGIS 도구와 마찬가지로 영속적이어서, 어떤 작업을 마친 후에도 선택이 유지됩니다. 노드도구가 아무런 객체도 찾지 못했다면 경고 메시지가 뜨게 됩니다.

중요한 것은 *Settings* → *Options* → *Digitizing* → *Search Radius:* 1,00 값을 0보다 크게 설정해야 한다는 것입니다. 그렇지 않으면 어떠한 꼭지점을 편집해야 할지 알 수 없게 됩니다.

참고: 꼭지점 마커 (**Vertex Markers**)

QGIS 현재 버전은 반투명 원, 십자모양, 아무것도 그리지 않기와 같은 3가지 종류의 마커를 지원합니다. 마커 모양을 변경시키려면 *Settings'의* |*mActionOptions*| *:menuselection:'Options* 에서 *Digitizing* 를 클릭하고 적절한 것을 선택하시면 됩니다.

기본 동작

Start by activating the Node Tool and selecting a feature by clicking on it. Red boxes will appear at each vertex of this feature.

- **Selecting vertices** (꼭지점 선택): 꼭지점을 클릭하면 그 꼭지점이 선택되고, 선분을 클릭하면 선분의 양쪽 끝점이 선택됩니다. 임의의 한점을 클릭하고 드래그하면 네모 안에 들어오는 모든 꼭지점이 선택됩니다. 선택된 꼭지점은 색상이 파란색으로 변경됩니다. `Ctrl` 키를 누른 상태로 꼭지점을 클릭하면 현재 선택에서 새롭게 선택한 꼭지점이 추가됩니다. `Ctrl` 또는 `Shift` 를 누른 상태에서 클릭을 하면 선택 상태를 바꿀 수 있습니다. (즉, 현재 선택되지 않은 꼭지점은 선택에 추가되며, 현재 선택된 꼭지점은 선택에서 제외됩니다).
- **Adding vertices** (꼭지점 추가): 꼭지점을 추가하기 위해서는 선분 근처를 더블클릭하면 하면 됩니다. 새로운 꼭지점은 커서 위치가 아니라 선분위에 나타나는데, 필요에 따라 새로운 꼭지점을 새로운 위치로 이동시키면 됩니다.
- **Deleting vertices** (꼭지점 삭제): 삭제할 꼭지점들을 선택하고 `Delete` 키를 누르면 꼭지점이 삭제됩니다. 객체를 완전히 삭제할 때는 Node Tool [노드 도구]를 사용해서는 안됩니다.|qg| 는 작업하고 있는 객체타입을 고려하여 최소 숫자의 꼭지점은 남겨둡니다. 객체를 완전히 삭제하기 위해서는 :sup:'Delete Selected [선택 제거] 툴을 사용하십시오.
- **Moving vertices** (꼭지점 이동): 이동하고자 하는 모든 꼭지점들을 선택합니다. 그다음 이동하고자 하는 위치로 선택된 꼭지점이나 엣지들을 클릭해서 드래그하면, 선택된 모든 꼭지점들이 한꺼번에

이동됩니다. 스냅이 활성화 되어 있다면 선택된 것들이 가장 가까운 꼭지점이나 라인으로 이동될 것입니다.

노드툴과의 모드 변경은 undo 대화상자에 저장됩니다. 아울러 해당 옵션을 활성화시켜두면 모든 동작은 위상관계 편집을 지원합니다. 실시간 투영변환도 지원되며, 노드툴은 툴팁을 지원합니다.

객체 잘라내기, 복사, 붙여넣기

대상 레이어들이  Toggle editing [편집모드 전환]로 설정되어 있으면, 객체를 선택한 후 동일한 QGIS 프로젝트내에 있는 레이어들 사이에서 잘라내기, 복사, 붙여넣기가 가능합니다.

아울러 객체를 텍스트 형태로 외부 애플리케이션에 붙여넣을 수 있습니다. 이때 객체들은 CSV 포맷 형태로 표현되는데, 도형은 OGC WKT(Well-Known Text)포맷으로 나타나게 됩니다.

그러나 QGIS 버전에서는 QGIS 외부의 텍스트 객체를 내부 레이어에 붙여넣을 수 없습니다. 복사와 붙여넣기 기능은 언제 유용할까요? 동시에 여러 개의 레이어를 편집하면서 레이어들 간에 객체를 복사/붙여넣기 할 때 유용할 수 있습니다. 왜 이런 기능이 필요할까요? 5,000개 이상의 호수가 포함된 레이어가 아니라, 한두 개의 호수만을 대상으로 작업해야 한다고 생각해 봅시다. 이때 새로운 레이어를 생성한 후, 복사/붙여넣기 기능을 활용하면 작업을 간단히 할 수 있게 됩니다.

예제로 새로운 레이어에 약간의 호수를 복사해 봅시다:

1. 소스(원래) 레이어를 불러옵니다
2. 타겟(목표) 레이어를 생성하거나 불러옵니다
3. 타겟 레이어를 편집 모드로 설정합니다
4. 범례 창에서 소스 레이어를 더블클릭하여 활성화시킵니다
5.  Select Single Feature [싱글 객체 선택] 툴를 사용하여 원하는 객체들을 선택합니다
6.  :sup:‘Copy Features 툴을 클릭합니다
7. 범례 창에서 타겟 레이어를 활성화합니다
8.  Paste Features 툴을 클릭합니다
9. 편집모드를 중지하고 저장합니다

소스 레이어와 타겟 레이어의 스키마가 다르면 어떻게 될까요?(필드이름이나 타입이 다를 경우) QGIS에서는 일치하는 스키마의 내용만 채우고 나머지는 무시합니다. 타겟 레이어에 대한 속성은 중요하지 않다면 필드와 데이터 타입 설계는 중요하지 않습니다. 모든 것(도형과 속성)을 모두 확실하게 옮기고 싶다면 스키마가 일치하도록 주의하셔야 합니다.

참고: 붙여넣기한 객체의 적합성 (**Congruency of Pasted Features**)

소스 레이어와 타겟 레이어가 같은 좌표계를 사용하고 있다면 붙여넣기한 객체가 원래의 객체와동일하게 됩니다. 그러나 타겟 레이어가 다른 좌표계를 사용한다면 붙여넣기한 도형이 원래 도형과 동일하다는 보장을 할 수 없습니다. 좌표계를 변환할 때 작은 반올림 오류가 포함되어 있을 수 있기 때문입니다.

선택된 객체의 삭제

폴리곤 전체을 삭제하고 싶다면 먼저  Select Single Feature [싱글 객체 선택] 툴을 사용해서 폴리곤을 선택하십시오. 복수의 객체들을 선택할 수도 있습니다. 선택이 끝났다면  Delete Selected [선택 제거] 툴을 클릭하면 객체들을 삭제할 수 있습니다.

디지타이징 도구모음의  Cut Features [객체 잘라내기]를 사용하여 객체를 삭제할 수도 있습니다. 이 도구는 현재 레이어에서 선택된 객체를 삭제함과 동시에 “공간 클립보드”에 저장합니다. 잘라낸 객체는  Paste Features [객체 붙여넣기]를 사용하여 다시 되돌릴 수 있습니다. 하거나, undo를 통해서도 한번은 불러낼

수 있습니다. 잘라내기, 복사, 붙여넣기는 현재 선택된 객체들에 대해서 작동합니다. 따라서 여러 개의 객체에 대해서도 한꺼번에 처리할 수 있습니다.

참고: 객체 삭제 지원 (**Feature Deletion Support**)

ESRI shape file을 편집할 때,|qg| 가 GDAL 1.3.2 이상의 버전과 링크가 되어 있어야 객체를 삭제할 수 있습니다. 다운로드 사이트에 있는 OS X 버전 및 윈도 버전 QGIS 는 GDAL 1.3.2 이상을 사용하여 제작되었습니다.

편집된 레이어 저장

레이어가 편집모드일 때에는 모든 변화가 QGIS 의 메모리에만 저장됩니다. 즉, 데이터 소스나 디스크에 즉각적으로 저장되지 않습니다. Save Edits'를 클릭하면 현재 레이어에 대한 편집내용이 저장되며, 계속해서 편집할 수 있습니다. |mActionToggleEditing| :sup:'Toggle editing [편집모드 전환] 버튼으로 편집모드를 변경시키거나 (또는 |qg|를 종료시킬 때) 변화된 내용을 저장할 것인지 아닌지에 대해 물어오게 됩니다.

변경 내용이 저장되지 않았을 경우 (디스크가 꽉 찼거나 속성값이 허용 범위를 벗어났을 때) QGIS 메모리 상태는 저장됩니다. 따라서 문제를 해결한 뒤 다시 저장하시면 됩니다.

참고: 데이터 무결성 (**Data Integrity**)

편집을 시작하기 전에 항상 데이터 소스를 백업해 두시는 것이 좋습니다. QGIS 개발자들은 데이터의 무결성을 이루기 위해 모든 노력을 다하고 있기는 하지만, 이러한 사항에 대해 어떠한 책임도 지지 않습니다.

11.3.5 고급 디지타이징 기능

아이콘	역할	아이콘	역할
	실행 취소		다시 실행
	객체 단순화		링 추가
	파트 추가		링 제거
	파트 제거		객체 변형
	Offset Curve		객체 분할
	선택된 객체 합병		포인트 심볼 회전
	선택된 객체의 속성 합병		

Table Advanced Editing: 벡터 레이어 확장 편집 도구모음

실행 취소와 다시 실행

Undo 와 Redo 툴을 사용하면 사용자가 데이터를 편집하다 발생한 실수를 적절히 처리할 수 있습니다. 도킹 가능한 위젯이 역시 존재하며, 여기에는 실행취소/다시실행 가능한 모든 작업의 히스토리가 보입니다.(Figure_edit_3 참조) 이 위젯은 기본적으로 나타나지 않습니다. 이 위젯은 도구모음에서 마우스 오른쪽 버튼을 클릭한 후 실행취소/다시실행 체크박스를 켜야 보이게 됩니다. 그러나 위젯이 표시되지 않은 상태에서도 실행취소/다시실행 기능은 사용할 수 있습니다.

Undo[실행취소]를 실행하면, 모든 객체와 속성들의 상태는 복귀 전의 상태로 되돌아갑니다. 일반적인 벡터 편집 작업이 아닐 경우(예: 플러그인에 의한 변화) 어떤 작업이 실행되었는지에 따라 다르지만, 상태가 복귀되지 않을 수도 있습니다.

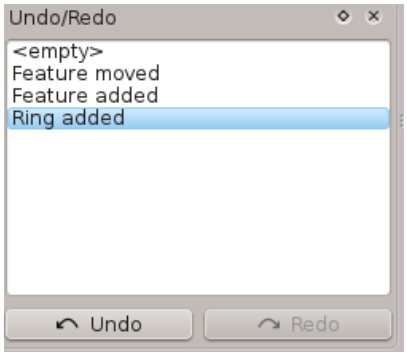


Figure 11.29: Redo와 Undo 디지타이징 절차

실행취소/다시실행 히스토리 위젯을 사용하는 방법은 간단합니다. 지금까지 실행한 작업 목록중 원하는 작업을 선택해 클릭하시면, 모든 객체들의 상태가 그 시점으로 복귀됩니다.

객체 단순화

:sup:‘Simplify Feature [객체 단순화] 툴을 사용하면 도형의 형태를 가능하면 변경하지 않은 상태에서 객체의 꼭지점 수를 줄일 수 있습니다. 객체를 선택하면 붉은색 보조선과 슬라이더가 나타날 것입니다. 슬라이더를 움직이면 붉은색 보조선이 변경되면서 객체가 어떻게 단순화 되는지 그 형태를 보여줍니다. **[OK]** 를 클릭하면 단순화된 도형이 저장됩니다. 객체를 단순화할 수 없다면(예를 들어 MultiPolygon[멀티폴리곤]) 메시지가 나타납니다.

링 추가

도구모음에 있는 using the Add Ring 아이콘을 이용하면 ring polygons (링 폴리곤)을 생성할 수 있습니다. 링폴리곤이란, 도우넛과 같이 내부를 비울 수 있는 폴리곤입니다. 기존 폴리곤 안에 링 폴리곤을 추가하면 구멍 형태로 나타나게 되고, 바깥 폴리곤과 안쪽 폴리곤의 경계 영역만 폴리곤의 영역으로 남게 됩니다.

파트 추가

add part 아이콘을 이용하면 선택된 멀티 폴리곤(여러 개의 도형으로 구성된 폴리곤)에 파트 폴리곤을 추가 할 수 있습니다. 단, 새로운 파트는 선택된 멀티 폴리곤의 바깥쪽에 입력해야 합니다.

링 제거

Delete Ring [링 삭제] 도구을 사용하면 링 폴리곤을 삭제할 수 있습니다. 이 도구는 폴리곤 레이어에서만 작동합니다. 폴리곤의 바깥쪽 폴리곤 경계에 대해서는 작동하지 않습니다. 이 도구는 일반 폴리곤과 멀티 폴리곤 객체에 대해서 사용할 수 있습니다. 적절한 링을 선택하시려면 꼭지점 편집 허용치를 조절하셔야 합니다.

파트 제거

Delete Part [파트 삭제] 도구는 멀티 폴리곤 객체로 부터 파트 폴리곤을 삭제하는데 사용합니다. 여러 개의 파트로 구성된 폴리곤의 경우, 맨 마지막 파트를 제외한 모든 파트를 삭제할 수 있습니다. 즉, 마지

막 하나의 파트는 삭제할 수 없습니다. 이 도구는 모든 멀티 파트 도형 즉, 포인트, 라인, 폴리곤에 대해 작동합니다. 파트의 꼭지점을 선택하기 전 꼭지점 편집 허용치를 적절히 선택하여야 합니다.

객체 변형

도구모음에 있는  Reshape Features [객체 변형] 아이콘을 사용하면 라인이나 폴리곤 객체의 형태를 변경할 수 있습니다. 이 도구로 입력하는 선의 첫번째 교차점에서 다음 교차점까지의 부분을 새로운 선으로 대체합니다. 폴리곤의 경우 때로는 의도하지 않은 결과를 만들 수 있으므로 조심해서 사용해야 합니다. 주로 약간의 편집에 사용하는 게 좋으며, 많은 변경이 필요한 경우에는 피하는 게 좋습니다. 또한 선을 입력할 때 여러 개의 폴리곤을 넘나드는 것은 피하셔야 합니다. 잘못된 폴리곤이 생성될 수 있기 때문입니다.

예를 들어 이 툴을 가지고 폴리곤의 경계를 편집할 수 있습니다. 먼저 폴리곤의 안쪽 영역에서 새로운 꼭지점을 생성하고자하는 위치에서 가까운 곳을 클릭합니다. 폴리곤 바깥쪽으로 새로운 꼭지점을 추가해 갑니다. 폴리곤의 안쪽영역을 오른쪽 버튼으로 클릭합니다. 원래의 폴리곤과 새로 입력한 선이 만나는 곳에 자동적으로 새로운 노드가 추가됩니다. 폴리곤 일부를 제거할 때도 사용할 수 있습니다. 먼저 폴리곤 바깥쪽을 클릭한 후, 내부에서 원하는 지점을 클릭하고 마지막으로 폴리곤 바깥쪽을 마우스 오른쪽 버튼으로 클릭하면 됩니다.

주석: reshape 툴은 폴리곤의 시작 지점을 바꿀 수 있습니다. 대부분의 애플리케이션에서 문제시 되지는 않지만, 이를 고려해 두시는 게 좋습니다.

Offset Curves

The  Offset Curve tool is a new editing tool. It creates parallel shifts of lines and polygon rings. The tool can be applied to the edited layer (the geometries are modified) or also to background layers (creates copies of the lines / rings and adds it to the the edited layer). It is thus ideally suited for the creation of distance line layers.The displacement is shown at the bottom left of the taskbar.

객체 분할

도구모음에 있는  Split Features [객체 분할] 아이콘을 이용하면 객체를 분할 할 수 있습니다. 분할하기를 원하는 객체를 가로지르는 라인만 그리면 됩니다.

선택한 객체 병합

 Merge Selected Features [선택한 객체 병합] 도구는 경계를 공유하고 속성이 동일한 객체들을 병합하는 데 사용합니다.

선택된 객체의 속성 병합

The  Merge Attributes of Selected Features tool allows to merge attributes of features with common boundaries and attributes without merging their boundaries.

포인트 심볼 회전

The  Rotate Point Symbols tool is currently only supported by the old symbology engine. It allows to change the rotation of point symbols in the map canvas, if you have defined a rotation column from the attribute table of the point layer in the *Style* tab of the *Layer Properties*. Otherwise the tool is inactive.

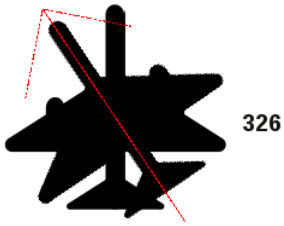


Figure 11.30: 포인트 심볼 회전

심볼을 회전시키려면 지도 캔버스 상에 있는 포인트 객체를 선택하고, 마우스 왼쪽버튼을 누른채로 회전시키면 됩니다. 회전 값과 붉은 화살표가 표시됩니다 (Figure_edit_4 참조). 왼쪽 마우스 버튼을 놓으면 속성 테이블에 회전 값이 갱신됩니다.

주석: `Ctrl` 키를 누른 상태로 회전 시키면 15도 단위로 회전됩니다.

11.3.6 새로운 벡터 레이어 생성

QGIS 에서는 새로운 Shapefile과 SpatiaLite 레이어를 생성 할 수 있습니다. GRASS의 경우에는 GRASS 플러그인을 설치해야 새로운 레이어를 생성할 수 있습니다. GRASS 벡터 레이어 생성에 대한 자세한 사항은 *Creating a new GRASS vector layer* 절을 참조하십시오.

새로운 Shapefile 레이어 생성

Shape 레이어를 생성하기 위해서는 *Layer* [레이어] 메뉴에 있는 *New* → *New Shapefile Layer...* 를 선택하시면 됩니다. Figure_edit_5 와 같이 *New Vector Layer* [새 벡터 레이어] 대화상자가 나타납니다. 여기에서 레이어의 타입(포인트, 라인, 폴리곤)과 좌표계를 선택하셔야 합니다.

QGIS는 현재 2.5D 객체와 측정 객체(measure feature) 생성은 지원하지 않습니다.

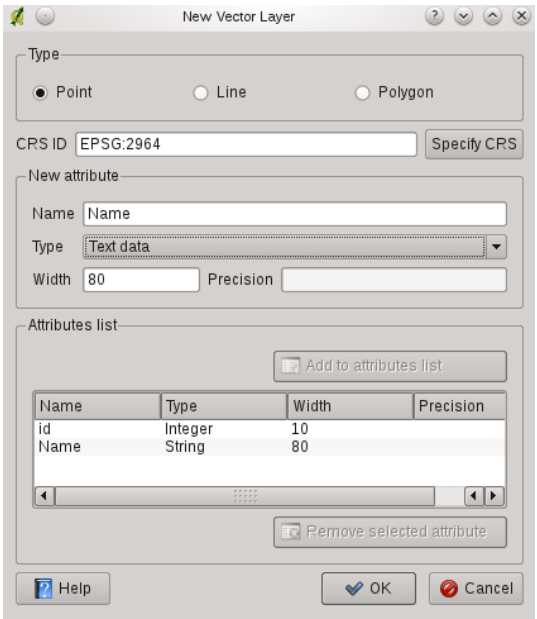


Figure 11.31: 레이어 기반 편집 스냅 옵션

새로운 shapefile 레이어를 생성하려면 적절한 속성의 이름과 타입, 길이를 지정하신 후, **[Add]** [속성 목록에 추가] 버튼을 클릭해서 속성을 추가합니다. 원하시는 만큼 반복하실 수 있습니다. 첫 번째 컬럼인 'id' 컬럼은 기본으로 생성되나 원하신다면 삭제하실 수 있습니다. 현재 *Type: real* (실수)  , *Type: integer* (정수)  , and *Type: string* (텍스트데이터)|selectstring| 만 지원됩니다. 추가하고자 하는 속성의 유형에 따라 속성 컬럼의 길이나 정밀도 지정 또한 가능합니다. 속성 지정이 끝나면 **[OK]** 버튼을 클릭하고 새롭게 뜨는 다이얼로그에서 shapefile 이름을 지정하십시오. 확장자는 .shp으로 자동지정 됩니다. 레이어가 생성되면 지도에 추가되고, *1.1.1. **기존 레이어의 디지타이징*** 절에서 설명했던 방법을 사용해서 편집할 수 있습니다.

새로운 SpatiaLite 생성

Layer [레이어] 메뉴에 있는 *New* →  *New SpatiaLite Layer...* [새 Spatialite 레이어] 를 선택하면 새로운 spatialite 레이어를 생성하고 편집할 수 있습니다. Figure_edit_6.와 같이 *New SpatiaLite Layer* [새 Spatialite 레이어] 대화상자가 나타날 것입니다.

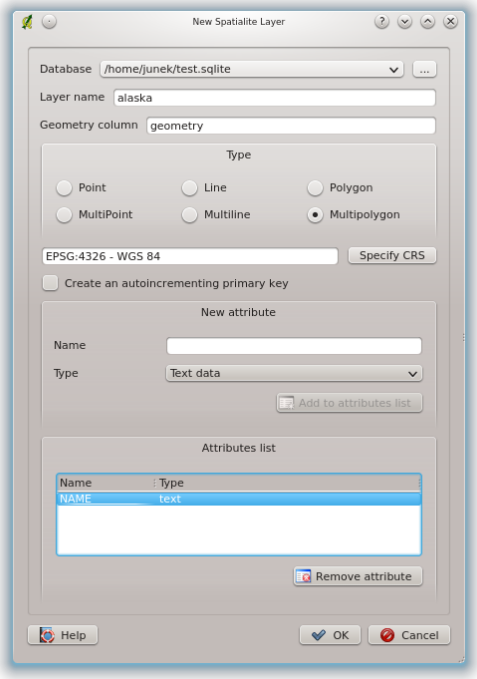


Figure 11.32: 새로운 shapefile 레이어 생성 대화상자

먼저 기존의 SpatiaLite 데이터베이스를 선택하거나 새로운 SpatiaLite 데이터베이스를 생성합니다. 데이터베이스 필드의 오른쪽에 있는 버튼을 클릭하면 됩니다. 그 다음 새로운 레이어의 이름을 추가하고 레이어 타입과 좌표계를 정의합니다. 원한다면  *Create an autoincrementing primary key* [자동 증가 기본키 만들기]를 선택할 수 있습니다.

음으로 새로운 SpatiaLite 레이어의 속성 테이블을 정의하기 위해서는, 생성하고자하는 속성컬럼 이름과 유형을 추가한 후 **[Add to attribute list]** [속성 목록에 추가] 버튼을 클릭하시면 됩니다. 속성 생성이 끝났다면 **[OK]** 를 클릭합니다. 그러면 범례에 자동으로 새로운 레이어를 추가되고, 이제 *1.1.1. **기존 레이어의 디지타이징*** 절에서 설명했던 방법을 사용해서 편집할 수 있습니다.

Further management of SpatiaLite-Layers can be done with the DB Manager see *DB Manager Plugin*.

11.3.7 속성테이블 작업

attribute table (속성테이블) 에는 선택된 레이어의 객체가 들어있습니다. 속성 테이블의 각 행은 지도의 각 객체와 대응되며, 각 컬럼은 객체별로 특별한 정보를 담고 있습니다. 테이블에서 객체들을 검색, 선택, 이동 편집할 수 있습니다.

벡터 레이어의 속성 테이블을 열기 위해서는, 먼저 지도 범례 영역에서 레이어를 클릭해서 레이어를 활성화시킵니다. 다음으로 메인 메뉴에 있는 *Layer* 를 클릭하고 *Open Attribute Table* [속성 테이블 열기]을 선택하면 됩니다. 범례에서 레이어를 마우스 오른쪽 클릭한 후, 드롭다운 메뉴에서 *Open Attribute Table* [속성 테이블 열기]를 선택하는 방법도 있습니다.

이렇게 하면 객체 속성이 표시된 새로운 창이 열립니다. (figure_attributes_1 참조) 이 창 맨 위의 타이틀에는 전체 객체의 수와 선택된 객체의 수가 표시됩니다.

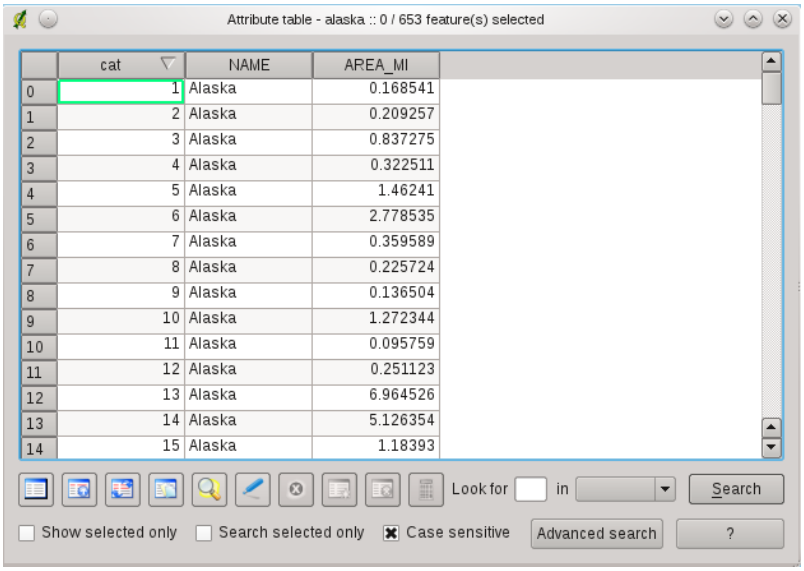


Figure 11.33: 알래스카 레이어의 속성 테이블

속성테이블에서 객체 선택

속성 테이블에서 **Each selected row** (선택된 행)에는 지도 레이어에서의 선택된 객체의 속성을 디스플레이합니다. 메인 창에서 선택된 객체가 변경되면 속성 테이블에서도 선택된 행이 변경 됩니다. 마찬가지로 속성테이블에서 행 선택을 변경하면 메인 윈도에서 선택된 객체도 변경됩니다.

행을 선택하려면 왼편에 있는 행 번호를 클릭하면 됩니다. **Multiple rows** (다중 행 선택)은 `Ctrl` 키를 누른 상태로 선택하면 가능하고, `Shift` 키를 누른상태에서 클릭하면 연속된 선택이 가능합니다. **continuous selection** (연속된 선택)은 현재 커서 위치와 이전에 클릭한 행 사이의 모든 행이 선택되는 것을 말합니다. 속성테이블에서 원하는 셀을 클릭하여 커서 위치를 바꾸더라도 행 선택은 변경되지 않습니다. 거꾸로 메인 캔버스에서 선택을 변경하더라도, 속성테이블에서 커서 위치는 바꾸지 않습니다.

컬럼 헤더를 클릭하면 테이블을 정렬할 수 있습니다. 작은 화살표가 정렬 순서를 나타냅니다.(아래방향 표시는 내림차순, 위방향 표시는 오름차순).

어떤 하나의 컬럼으로부터 원하는 값을 검색 (**simple search by attributes**) 하려면 먼저 *Look for* [찾을 값]을 사용합니다. 드롭다운 메뉴에서 검색할 필드(컬럼)을 선택하고 **[Search]** [검색] 버튼을 누릅니다. 검색이 끝나면 일치하는 행이 선택되고 검색된 행의 총 수가 속성테이블의 타이틀바와 메인 창의 상태바에 표시됩니다. 좀 더 복잡한 겁색은 확장 검색인를 사용합니다. 확장검색에 대해서는 ***쿼리 빌더*** 절에서 설명할 것입니다.

선택된 레코드만 보이기를 원할 때는 checkbox *Show selected only* [선택된 부분만 표시]를 이용하면 됩니다. 검색된 레코드만을 대상으로 새로 검색하려면 *Search selected only* [선택된 부분만 검색]를 이용

하면 됩니다. :guilabel:'Case sensitive'는 선택 작업 시 대소문자 구분여부를 설정할 때 사용합니다. 속성테이블 좌하단에 있는 다른 버튼들은 아래와 같은 기능을 제공합니다:

- ctionOpenTable| Unselect all [모두 선택 해제] `Ctrl+U` 동일기능
- Move selected to top `Ctrl+T` 동일기능
- Invert selection `Ctrl+S` 동일기능
- Copy selected rows to clipboard `Ctrl+C` 동일기능
- Zoom map to the selected rows `Ctrl+J` 동일기능
- Pan map to the selected rows `Ctrl+P` 동일기능
- Toggle editing mode [편집 모드 전환] 속성테이블의 편집: `Ctrl+E` 동일기능
- Delete selected features `Ctrl+D` 동일기능
- New Column [새 컬럼] GDAL 버전 1.6 이상의 PostGIS 레이어와 OGR 레이어: `Ctrl+W` 동일기능
- Delete Column [컬럼 삭제] GDAL 버전 1.9 이상의 PostGIS 레이어와 OGR 레이어: `Ctrl+L` 동일기능
- Open field calculator `Ctrl+I` 동일

참고: Skip WKT geometry

If you want to use attribute data in external programs (such as Excel) use the Copy selected rows to clipboard button. You can copy the information without vector geometries if you deactivate *Settings* → *Options* → General tab *Copy geometry in WKT representation from attribute table*.

선택된 객체를 새로운 레이어로 저장

선택된 객체들은 OGR에서 지원되는 벡터 포맷으로 저장할 수 있고 다른 좌표계로 변환할 수도 있습니다. 레이어의 마우스 오른쪽 메뉴를 열어서 *Save selection as* → [선택을 다른 이름으로 저장]을 클릭하고 저장할 파일 이름과 포맷, 좌표계(:ref:'label_legend 절 참조)를 지정합니다. 대화상자에서 OGR 생성 옵션도 지정할 수 있습니다.

비 공간 속성테이블 작업

QGIS는 일반 속성테이블도 불러올 수 있습니다. 현재 지원되는 포맷은 OGR에서 지원하는 테이블, 콤마로 분리한 텍스트, PostgresSQL 프로바이더에 의한 비 공간 테이블 등입니다. 이러한 테이블들은 참고용 테이블로 사용할 수도 있고, 그냥 단순히 속성테이블을 사용하여 보거나 편집할 수도 있습니다. 비공간 테이블을 불러오면 범례에 나타나게 됩니다. 비공간 테이블은 :sup:'Open Attribute Table'[속성 테이블 열기] 도구를 이용해 열 수 있고, 연 뒤에는 다른 레이어 속성 테이블과 같이 편집이 가능합니다.

예를 들어 벡터 레이어를 디지타이징하는 동안 추가되는 범위 값이나 속성값을 정의하는 비 공간 테이블의 컬럼을 사용할 수 있습니다. 보다 자세한 정보는 :ref:'vector_attributes_tab'의 편집 위젯을 참고하세요.

11.4 쿼리 빌더

The Query Builder allows you to define a subset of a table using a SQL-like WHERE clause and display the result in the main window. The query result then can be saved as a new vector layer.

11.4.1 쿼리

Open the attribute table choosing Open Attribute Table from the menu bar. Click on the **[Advanced Search]** button to open the *Query builder*. 예를 들면, `population` 필드를 가진 `towns` 레이어를 가지고 있다고 가정하고, 검색쿼리 생성기의 SQL박스에 `population > 100000` 를 입력하면 인구가 10만명이상의 마을을 선택할 수 있습니다. Figure_attributes_2 는 PostgresSQL에 저장된 PostGIS 레이어로부터 검색 쿼리를 생성한 예입니다. 필드와 값 그리고 연산자 섹션을 이용하여 SQL문을 작성할 수 있습니다.

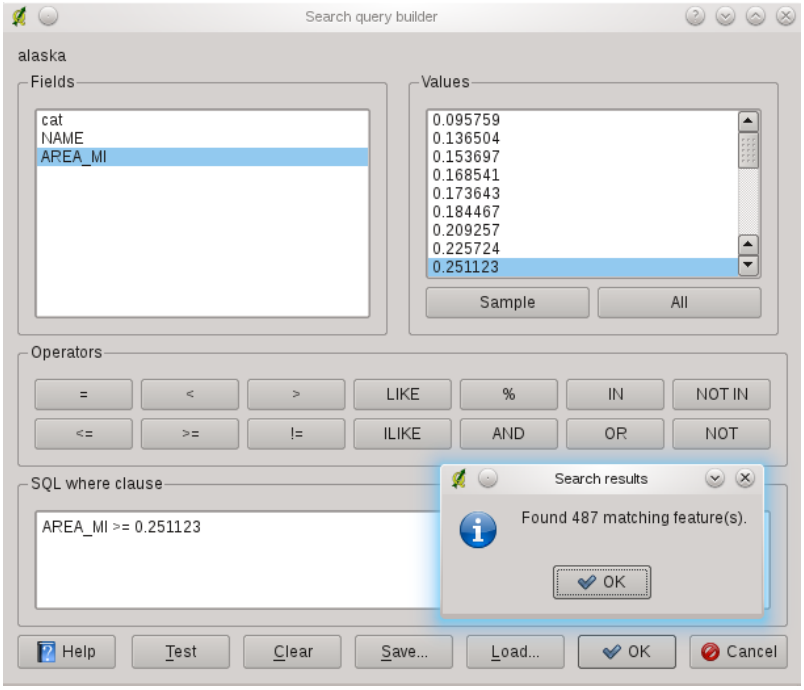


Figure 11.34: 쿼리 빌더

Fields list [필드]에는 검색된 속성 테이블의 모든 속성이 포함되어 있습니다. SQL의 필드구문에 속성을 추가하려면 필드 목록에서 추가하고자 하는 속성의 이름을 더블클릭하면 됩니다. 일반적으로 여러가지 필드와 값, 연산자를 이용하여 쿼리를 구성할 수 있으며, SQL where 절에 직접 입력할 수도 있습니다.

Values list [값]에는 속성 값이 표시됩니다. 속성값으로 가능한 모든 값들을 보려면, 필드 목록에서 속성을 선택한 뒤, **[all]** [모두] 버튼을 클릭하시면 됩니다. **[Sample]** 을 클릭하면, 속성값중 일부만 표시됩니다. 값 목록에서 원하는 속성값을 더블클릭하면 SQL where 절에 해당 값이 추가됩니다.

Operators section 은 모든 사용가능한 쿼리 연산자가 나열되어 있습니다. SQL where 절에 연산자를 추가하려면 적절한 연산자 버튼을 클릭하면 됩니다. 비교 오퍼레이터 (= , > , ...), 문자 비교 오퍼레이터 (`LIKE`), 논리 오퍼레이터 (`AND`, `OR`, ...) 등이 있습니다.

[Test] 버튼을 누르면 현재 쿼리를 만족하는 객체의 수를 메시지 박스로 보여주는데, 쿼리문 작성에 사용할 수 있습니다. **[Clear]** 버튼은 SQL 구문 박스를 깨끗하게 지웁니다. **[Save]** 와 **[Load]** 버튼은 각각 SQL 쿼리를 저장하고 로드합니다. The **[OK]** button closes the window and selects the features satisfying the query. **[Cancel]** 버튼은 현재 선택에 영향을 미치지 않고 쿼리 윈도창을 닫습니다.

참고: Changing the Layer Definition

레이어를 불러들인 후, 레이어 정의에 사용되는 SQL 쿼리를 변경하여 레이어 정의를 변경할 수 있습니다. 범례창에서 레이어를 더블클릭하여 벡터 *Layer Properties* [레이어 속성] 대화상자를 열고 *General* [일반정보] 탭의 **[Query Builder]** 버튼을 클릭하여 SQL문을 적절히 정의하면 됩니다. 좀 더 자세한 정보는 ***벡터 속성 대화상자*** 을 참조하십시오.

11.4.2 선택

QGIS 는 ***쿼리 빌더*** 에서 설명한 쿼리 빌더 인터페이스를 이용해 객체들을 선택하는 것도 가능합니다. In the above section the purpose of the query builder is to highlight all selected features meeting the filter criteria in yellow. The rest of the features do not change their appearance.

Another possibility is to make a selection by right clicking on a layer in the legend and choosing *Query ...*. If you perform a SQL query here only the selected features are displayed in the map view. Features that don't match the SQL query are not visualised in the map view. The same function is available from the *General* tab in the vector *Layer Properties* dialog. You can find a **[Query builder]** button right below. The selection described above can be used with all vector data providers.

11.4.3 선택된 개체들을 새 레이어로 저장

선택된 객체들은 OGR에서 지원되는 벡터 포맷으로 저장할 수 있으며, 이와 함께 다른 좌표계 (CRS, Coordinate Reference System)로 변환할 수 있습니다. 레이어의 마우스 오른쪽 메뉴를 열어서 *Save selection as* → 를 클릭하고 저장할 파일 이름과 포맷, 좌표계(see Section *Map Legend* 참조)를 지정합니다. 대화상자에서 OGR 생성 옵션을 선택하는 것도 가능합니다.

11.5 필드 계산기

속성 테이블에 있는 Field Calculator [필드 계산기 열기] 버튼은 기존 존재하는 속성 값이나 함수(예를 들면 도형의 길이를 계산하거나 면적을 계산하는 등)를 이용하여 계산을 수행할 수 있습니다. 계산결과는 새로운 속성 컬럼에 저장할 수도 있고 이미 존재하는 컬럼의 값을 갱신하는데 사용할 수도 있습니다.

필드계산기 아이콘을 클릭하여 필드 계산기 대화상자(figure_attributes_3 참조)를 열려면 먼저 그전에 벡터 레이어를 편집모드로 바꾸어야 합니다. 필드 계산기 대화상자가 뜨면, 계산결과를 사용하여 이미 존재하는 필드의 값을 갱신할 것인지, 선택된 객체만 갱신할 것인지, 또는 새로운 필드를 생성할 것인지 등을 먼저 선택해야합니다.

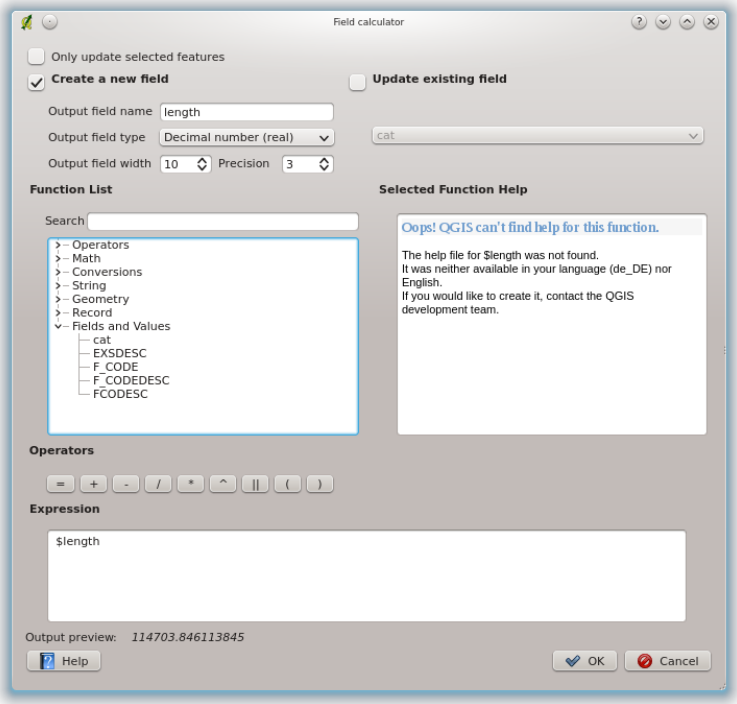


Figure 11.35: 필드계산기

새로운 필드를 추가하겠다면 필드이름, 필드타입(정수, 실수, 문자열), 필드 전체길이, 필드 정밀도를 입력해야 합니다 (figure_attributes_3 참조). 예를 들어 필드 길이를 10을 선택하고 필드 정밀도를 3으로 선택했다면 정수부분은 6자리, 소수점 아래 부분은 3자리인 실수를 사용한다는 뜻입니다.

The **Function List** contains functions as well as fields and values. View the help function in the **Selected Function Help**. In **Expression** you see the calculation expressions you create with the **Function List**. The most commonly used operators, see **Operators**.

In the **Function List**, click on *Fields and Values* to view all attributes of the attribute table to be searched. To add an attribute to the Field calculator **Expression** field, double click its name in the *Fields and Values* list. Generally you can use the various fields, values and functions to construct the calculation expression or you can just type it into the box. To display the values ??of a field, you just right click on the appropriate field. You can choose between *Load top 10 unique values* and *Load all unique values*. On the right side opens the **Field Values** list with the unique values. To add a value to the Field calculator **Expression** box, double click its name in the **Field Values** list.

The *Operators*, *Math*, *Conversions*, *String*, *Geometry* and *Record* groups provides several functions. In *Operators* you find mathematical operators. Find *Math* for mathematical functions. The *Conversions* group contains functions that convert one data type to another. The *String* group provides functions for data strings. In the *Geometry* group you find functions for geometry objects. With *Record* group functions you can add a numeration to your data set. To add a function to the Field calculator **Expression** box, click on the > and then doubleclick the function.

다음은 필드 계산기를 활용하는 간단한 예제입니다. QGIS _example_dataset 중에 포함된 ‘‘railroads‘ 레이어의 길이를 계산하기를 원한다고 가정합니다:

1. QGIS 에서 `railroads.shp` 파일을 QGIS로 로딩하고  Open Attribute Table 대화상자를 엽니다.
2.  Toggle editing mode [편집 모드 전환]를 클릭하고  Field Calculator [필드 계산기 열기] 대화상자를 엽니다.
3. 새로운 필드박스를 생성할 예정이므로  *Create a new field* [기존의 필드를 업데이트]의 체크박스를 해제합니다.
4. 결과를 저장할 필드 이름을 `length` 로 지정해서 추가하고, 필드 타입은 `real`, 필드 길이와 정밀도는 각각 10과 3으로 지정합니다.
5. 이제 *Geometry* 그룹에서 `length` 기능을 클릭하고 `$length` 를 필드 계산 입력 박스에 추가한 다음 **[Ok]** 버튼을 클릭하십시오.
6. 속성 테이블에는 새로운 `length` 컬럼을 볼 수 있습니다.

다음의 표는 사용 가능한 모든 연산자의 목록입니다.

```
column name "column name"           value of the field column name
'string'                            a string value
NULL                                null value
a IS NULL                           a has no value
a IS NOT NULL                       a has a value
a IN (value[,value])                a is below the values listed
a NOT IN (value[,value])            a is not below the values listed
a OR b                              a or *b* is true
a AND b                             a and *b* is true
NOT a                               inverted truth value of a
sqrt(a)                             square root of a
sin(a)                              sinus of a
cos(a)                              cosinus of b
tan(a)            tangens of a
asin(a)           arcussinus of a
acos(a)           arcuscosinus of a
atan(a)           arcustangens of a
to int(a)           convert string a to integer
to real(a)            convert string a to real
to string(a)          convert number a to string
```

```
lower(a)                convert string a to lower case
upper(a)              convert string a to upper case
length(a)               length of string a
atan2(y,x)                arcustangens of y/x using the signs of the two
                                    arguments to determine the quadrant of the result
replace(*a*, replacethis, withthat)   replace this with that in string a
regexp_replace(a,this,that)          replace the regular expression this with that
substr(*a*,from,len)                  len characters of string *a* starting from from
                                    (first character index is 1)
a || b                               concatenate strings a and b
$rownum                 number current row
$area                 area of polygon
$perimeter              perimeter of polygon
$length                 length of line
$id                   feature id
$x                  x coordinate of point
$y                  y coordinate of point
xat(n)                               X coordinate of the point of an n-th line (indeces
                                    start at 0; negative values refer to the line end)
yat(n)                               y coordinate of the point of an n-th line (indeces
                                    start at 0; negative values refer to the line end)
a = b                                a and b are equal
a != b                               a and b are not equal
a <> b
a >= b                                a is larger than or equal to b
a <= b                                a is less than or equal to b
a > b                                 a is larger than b
a < b                                 a is smaller than b
a ~ b                                a matches the regular expression b
a LIKE b                             a equals b
a ILIKE b                            a equals b (without regard to case-sensitive)
a |wedge| b                       a raised to the power of b
a * b                              a multiplied by b
a / b                 a divided by b
a + b                 a plus b
a - b                 a minus b
+ a                   positive sign
- a                         negative value of a
```

The field calculator **Function list** with the **Selected Function Help** , **Operators** and **Expression** menu are also available through the rule-based rendering in the Style tab of the Layer properties and the expression based labeling ... in the Labeling core application.

CHAPTER 12

Working with Raster Data

12.1 Working with Raster Data

This Section describes how to visualize and set raster layer properties. QGIS uses the GDAL library to read and write raster data formats, including Arc/Info Binary Grid, Arc/Info ASCII Grid, GeoTIFF, Erdas Imagine and many more. GRASS raster support is supplied by a native QGIS data provider plugin. The raster data can also be loaded in read mode from zip and gzip archives into QGIS.

At the date of this document, more than 100 raster formats are supported by the GDAL library (see GDAL-SOFTWARE-SUITE *Literature and Web References*). A complete list is available at http://www.gdal.org/formats_list.html.

주석: Not all of the listed formats may work in QGIS for various reasons. For example, some require external commercial libraries or the GDAL installation of your OS was not built to support the format you want to use. Only those formats that have been well tested will appear in the list of file types when loading a raster into QGIS. Other untested formats can be loaded by selecting the `[GDAL] All files (*)` filter.

Working with GRASS raster data is described in Section *GRASS GIS Integration*.

12.1.1 What is raster data?

Raster data in GIS are matrices of discrete cells that represent features on, above or below the earth's surface. Each cell in the raster grid is the same size, and cells are usually rectangular (in QGIS they will always be rectangular). Typical raster datasets include remote sensing data such as aerial photography or satellite imagery and modelled data such as an elevation matrix.

Unlike vector data, raster data typically do not have an associated database record for each cell. They are geocoded by its pixel resolution and the x/y coordinate of a corner pixel of the raster layer. This allows QGIS to position the data correctly in the map canvas.

QGIS makes use of georeference information inside the raster layer (e.g. GeoTiff) or in an appropriate world file to properly display the data.

12.1.2 Loading raster data in QGIS

Raster layers are loaded either by clicking on the 

icon or by selecting the *Layer* →  *Add Raster Layer* menu option. More than one layer can be loaded at the same time by holding down the `Control` or `Shift` key and clicking on multiple items in the dialog *Open a GDAL Supported Raster Data Source*.

Once a raster layer is loaded in the map legend you can click on the layer name with the right mouse button to select and activate layer specific features or to open a dialog to set raster properties for the layer.

Right mouse button menu for raster layers

- *Zoom to layer extent*
- *Zoom to Best Scale (100%)*
- *Show in Overview*
- *Remove*
- *Set Layer CRS*
- *Set Project CRS from Layer*
- *Properties*
- *Rename*
- *Copy Style*
- *Add New Group*
- *Expand all*
- *Collapse all*
- *Update Drawing Order*

12.2 Raster Properties Dialog

To view and set the properties for a raster layer, double click on the layer name in the map legend or right click on the layer name and choose *Properties* from the context menu:

This will open the *Raster Layer Properties* dialog, (see figure_raster_1).

There are several tabs on the dialog:

- *Style*
- *Transparency*
- *Colormap*
- *General*
- *Metadata*
- *Pyramids*
- *Histogram*

12.2.1 Style Tab

QGIS can render raster layers in two different ways:

1. Single band - one band of the image will be rendered as gray or in pseudocolor or a freak out.
2. Three band color - three bands from the image will be rendered, each band representing the red, green or blue component that will be used to create a color image.

Within both render types you can invert the color output using the  *Invert color map* checkbox.

Single Band Rendering

This selection offers you two possibilites to choose. At first you can select which band you like to use for rendering (if the dataset has more than one band).

The second option offers a selection of available colortables for rendering.

The following settings are available through the dropdownbox *Color map* 

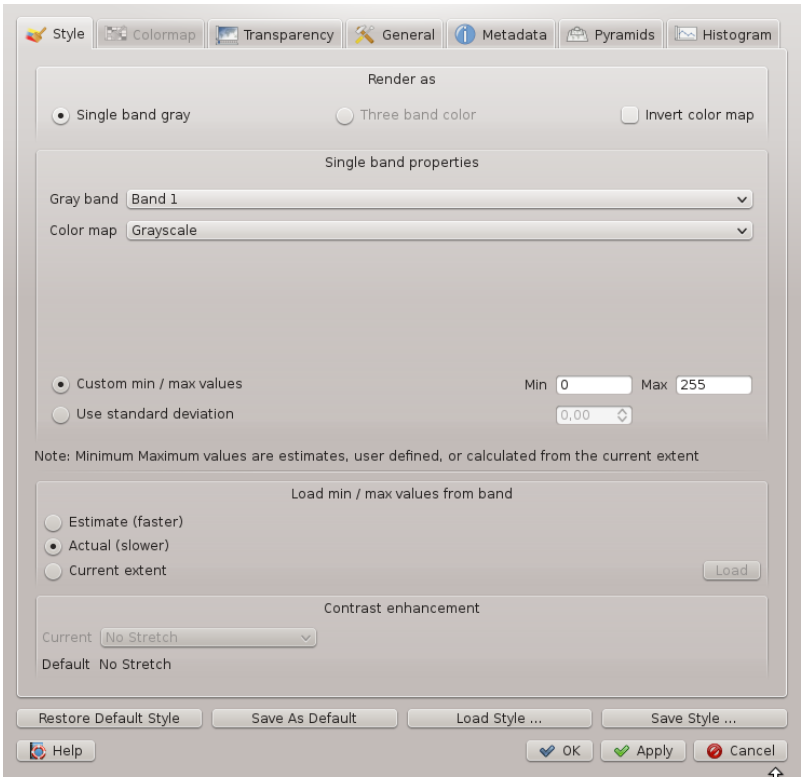


Figure 12.1: Raster Layers Properties Dialog

- Grayscale (default)
- Pseudocolor
- Freak Out
- Colormap

When selecting the entry 'Colormap' from *Color map*  combobox, the tab *Colormap* becomes available. For more information see *Colormap*.

QGIS can restrict the data displayed to only show cells whose values are within a given number of standard deviations of the mean for the layer.

This is useful when you have one or two cells with abnormally high values in a raster grid that are having a negative impact on the rendering of the raster. This option is only available for pseudocolor and freak out images.

Three band color

This selection offers you a wide range of options to modify the appearance of your rasterlayer. For example you could switch color-bands from the standard RGB-order to something else.

Also scaling of colors are available.

참고: Viewing a Single Band of a Multiband Raster

If you want to view a single band (for example Red) of a multiband image, you might think you would set the Green and Blue bands to "Not Set". But this is not the correct way. To display the Red band, set the image type to grayscale, then select Red as the band to use for Gray.

Contrast enhancement

주석: When adding GRASS rasters the option *Contrast enhancement* will be always set to automatically to *stretch to min max* regardless if the Quantum GIS general options this is set to another value.

12.2.2 Transparency Tab

QGIS has the ability to display each raster layer at varying transparency levels. Use the transparency slider to indicate to what extent the underlying layers (if any) should be visible though the current raster layer. This is very useful, if you like to overlay more than one rasterlayer, e.g. a shaded relief map overlayed by a classified rastermap. This will make the look of the map more three dimensional.

Additionally you can enter a rastervalue, which should be treated as *NODATA*. This can be done manually or with the  Add values from display icon.

An even more flexible way to customize the transparency can be done in the *Custom transparency options* section. The transparency of every pixel can be set here.

As an example we want to set the water of our example raster file `landcover.tif` to a transparency of 20 %. The following steps are neccessary:

1. Load the rasterfile `landcover`.
2. Open the *Properties* dialog by double-clicking on the raster name in the legend or by right-clicking and choosing *Properties* from the popup menu.
3. Select the *Transparency* tab
4. Click the  Add values manually button. A new row will appear in the pixel-list.
5. Enter the raster-value (we use 0 here) and adjust the transparency to 20 %.
6. Press the **[Apply]** button and have a look at the map.

You can repeat the steps 4 and 5 to adjust more values with custom transparency.

As you can see this is quite easy to set custom transparency, but it can be quite a lot of work. Therefore you can use the button  Export to file to save your transparency list to a file. The button  Import from file loads your transparency settings and applies them to the current raster layer.

12.2.3 Colormap

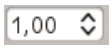
The *Colormap* tab is only available, when you have selected a singleband rendering within the *Style* tab (see *Style Tab*).

Three ways of color interpolation are available:

1. Discrete
2. Linear
3. Exact

The button **[Add Entry]** adds a color to the individual color table. Button **[Delete Entry]** deletes a color from the individual color table and the **[Sort]** button sorts the color table according to the pixel values in the value column. Double clicking on the value-column lets you insert a specific value. Double clicking on the color-column opens the dialog *Select color* where you can select a color to apply on that value. Further you can also add labels for each color but this value won't be displayed when you use the identify feature tool.

You can also click on the button  Load color map from band, which tries to load the table from the band (if it has any). And you can use the buttons  Load color map from file or  Export color map to file to load an existing color table or to save the defined color table for other sessions.

The block *Generate new color map* allows you to create newly categorized colormaps. You only need to select the *number of entries*  and press the button *Classify*. Currently only one *Classification mode*  is supported - 'Equal interval'

12.2.4 General Tab

The *General* tab displays basic information about the selected raster, including the layer source and display name in the legend (which can be modified). This tab also shows a thumbnail of the layer, its legend symbol, and the palette.

Additionally scale-dependent visibility can be set in this tab. You need to check the checkbox and set an appropriate scale where your data will be displayed in the map canvas.

Also the coordinate reference system (CRS) is printed here as a PROJ.4-string. This can be modified by hitting the **[Specify]** button.

12.2.5 Metadata Tab

The *Metadata* tab displays a wealth of information about the raster layer, including statistics about each band in the current raster layer. Statistics are gathered on a 'need to know' basis, so it may well be that a given layers statistics have not yet been collected.

This tab is mainly for information. You cannot change any values printed inside this tab.

12.2.6 Pyramids Tab

Large resolution raster layers can slow navigation in QGIS. By creating lower resolution copies of the data (pyramids), performance can be considerably improved as QGIS selects the most suitable resolution to use depending on the level of zoom.

You must have write access in the directory where the original data is stored to build pyramids.

Several resampling methods can be used to calculate the pyramids:

- Average
- Nearest Neighbour

When checking the checkbox  *Build pyramids internally if possible* QGIS tries to build pyramids internally.

Please note that building pyramids may alter the original data file and once created they cannot be removed. If you wish to preserve a 'non-pyramided' version of your raster, make a backup copy prior to building pyramids.

12.2.7 Histogram Tab

The *Histogram* tab allows you to view the distribution of the bands or colors in your raster. It is generated automaticaly when you open *Histogram* tab. All existing bands will be displayed together. You can save the histogram as an image with the  button.

12.3 Raster Calculator

The *Raster Calculator* in the *Raster* menu (see figure_raster_2) allows to perform calculations on basis of existing raster pixel values. The results are written to a new raster layer with a GDAL supported format.

The **Raster bands** list contains all loaded raster layers that can be used. To add a raster to the raster calculator expression field, double click its name in the Fields list. You can then use the operators to construct calculation expressions or you can just type it into the box.

In the **Result layer** section you have to define an output layer. You can then define the extent of the calculation area based on an input raster layer or based on X,Y coordinates and on Columns and Rows to set the resolution of the output layer. If the input layer has a different resolution, the values will be resampled with nearest neighbor algorithm.

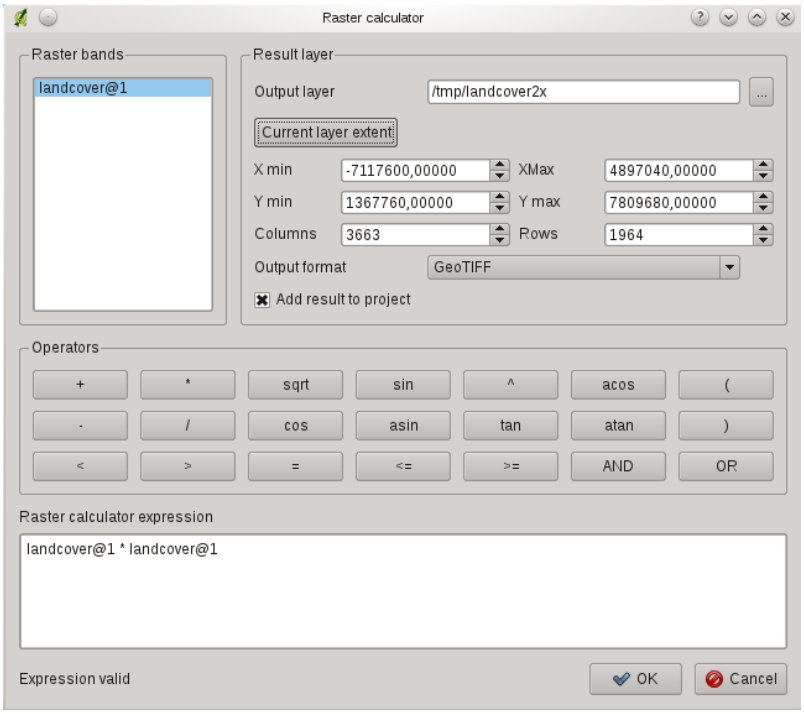


Figure 12.2: Raster Calculator

The **Operators section** contains all usable operators. To add an operator to the raster calculator expression box, click the appropriate button. Mathematical calculations (+, −, ⋆, ...) and trigonometric functions (`sin`, `cos`, `tan`, ...) are available. Stay tuned for more operators to come!

With the *Add result to project* checkbox the result layer will automatically added to the legend area and can be visualized.

12.3.1 Examples

Convert elevation values from meter to feet

Creating an elevation raster feet from a raster in meter, you need to use the conversion factor for meters to feet: 3.28. The expression is:

```
elevation@1 * 3.28
```

Using a mask

If you want to mask out parts of a raster, because you are only interested in elevations above 0 meter, you can use following expression to create a mask and apply the result to a raster in one step.

```
(elevation@1 >= 0) * elevation@1
```

For every cell greater than or equal to 0, set its value to 1, otherwise set it to 0. This creates the mask on the fly.

CHAPTER 13

Working with OGC Data

13.1 Working with OGC Data

QGIS supports WMS and WFS as data sources.

13.1.1 What is OGC Data

The Open Geospatial Consortium (OGC), is an international organization with more than 300 commercial, governmental, nonprofit and research organizations worldwide. Its members develop and implement standards for geospatial content and services, GIS data processing and exchange.

Describing a basic data model for geographic features an increasing number of specifications are developed to serve specific needs for interoperable location and geospatial technology, including GIS. Further information can be found under http://www.opengeospatial.org/.

Important OGC specifications are:

- **WMS** — Web Map Service
- **WFS** — Web Feature Service
- **WCS** — Web Coverage Service
- **CAT** — Web Catalog Service
- **SFS** — Simple Features for SQL
- **GML** — Geography Markup Language

OGC services are increasingly being used to exchange geospatial data between different GIS implementations and data stores. QGIS can now deal with three of the above specifications, being SFS (through support of the PostgreSQL / PostGIS data provider, see Section *PostGIS Layers*), WFS and WMS as a client.

13.1.2 WMS Client

Overview of WMS Support

QGIS currently can act as a WMS client that understands WMS 1.1, 1.1.1 and 1.3 servers. It has particularly been tested against publicly accessible servers such as DEMIS.

WMS servers act upon requests by the client (e.g. QGIS) for a raster map with a given extent, set of layers, symbolization style, and transparency. The WMS server then consults its local data sources, rasterizes the map, and sends it back to the client in a raster format. For QGIS this would typically be JPEG or PNG.

WMS is generically a REST (Representational State Transfer) service rather than a fully-blown Web Service. As such, you can actually take the URLs generated by QGIS and use them in a web browser to retrieve the same

images that QGIS uses internally. This can be useful for troubleshooting, as there are several brands of WMS servers in the market and they all have their own interpretation of the WMS standard.

WMS layers can be added quite simply, as long as you know the URL to access the WMS server, you have a serviceable connection to that server, and the server understands HTTP as the data transport mechanism.

Selecting WMS Servers

The first time you use the WMS feature, there are no servers defined.

Begin by clicking the  Add WMS layer button inside the toolbar, or through the *Layer* → *Add WMS Layer...* menu.

The dialog *Add Layer(s) from a Server* for adding layers from the WMS server appears. You can add some servers to play with by clicking the **[Add default servers]** button. This will add two WMS demo servers for you to use, the WMS servers of the DM Solutions Group and Lizardtech. To define a new WMS server in the tab *Layers*, select the **[New]** button. Then enter the parameters to connect to your desired WMS server, as listed in table_OGC_1:

Name	A name for this connection. This name will be used in the Server Connections drop-down box so that you can distinguish it from other WMS Servers.
URL	URL of the server providing the data. This must be a resolvable host name; the same format as you would use to open a telnet connection or ping a host.
Username	Username to access a secured WMS-server. This parameter is optional.
Password	Password for a basic authenticated WMS-server. This parameter is optional.
Ignore GetMap URI	 *Ignore GetMap URI reported in capabilities*, use given URI from URL-field above.
Ignore GetFeatureInfo URI	 *Ignore GetFeatureInfo URI reported in capabilities* , use given URI from URL-field above

Table OGC 1: WMS Connection Parameters

If you need to set up a proxy-server to be able to receive WMS-services from the internet, you can add your proxy-server in the options. Choose menu *Settings* → *Options* and click on the tab *Network & Proxy*. There you can add your proxy-settings and enable them by setting the  *Use proxy for web access*. Make sure that you select the correct proxy type from the *Proxy type*  dropdown menu.

Once the new WMS Server connection has been created, it will be preserved for future QGIS sessions.

참고: On WMS Server URLs

Be sure, when entering in the WMS server URL, that you have the base URL. For example, you shouldn't have fragments such as `request=GetCapabilities` or `version=1.0.0` in your URL.

Loading WMS Layers

Once you have successfully filled in your parameters you can use the **[Connect]** button to retrieve the capabilities of the selected server. This includes the Image encoding, Layers, Layer Styles and Projections. Since this is a network operation, the speed of the response depends on the quality of your network connection to the WMS server. While downloading data from the WMS server, the download progress is visualized in the left bottom of the WMS dialog.

Your screen should now look a bit like figure_OGR_1, which shows the response provided by the DM Solutions Group WMS server.

Image Encoding

The *Image encoding* section now lists the formats that are supported by both the client and server. Choose one depending on your image accuracy requirements.

참고: Image Encoding

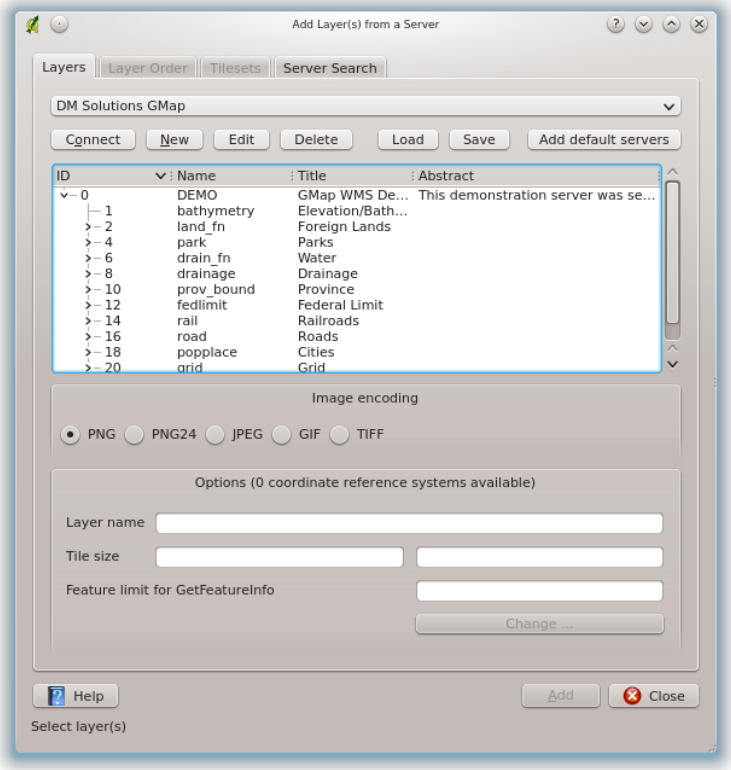


Figure 13.1: Dialog for adding a WMS server, showing its available layers

You will typically find that a WMS server offers you the choice of JPEG or PNG image encoding. JPEG is a lossy compression format, whereas PNG faithfully reproduces the raw raster data.

Use JPEG if you expect the WMS data to be photographic in nature and/or you don't mind some loss in picture quality. This trade-off typically reduces by 5 times the data transfer requirement compared to PNG.

Use PNG if you want precise representations of the original data, and you don't mind the increased data transfer requirements.

Options

The Options field provides a text field where you can add a *Layer name* for the WMS-layer. This name will be presented in the legend after loading the layer.

Below the layer name you can define *Tile size*, if you want to set tile sizes (eg. 256x256) to split up the WMS request into multiple requests.

The *Feature limit for GetFeatureInfo* defines what features from the server to query.

If you select a WMS from the list a field with the default projection, provided by the mapserver, appears. If the **[Change...]** button is active, you can click on it and change the default projection of the WMS to another CRS, provided by the WMS server.

Layer Order

The tab *Layer Order* lists the selected layers available from the current connected WMS server. You may notice that some layers are expandable, this means that the layer can be displayed in a choice of image styles.

You can select several layers at once, but only one image style per layer. When several layers are selected, they will be combined at the WMS Server and transmitted to QGIS in one go.

참고: WMS Layer Ordering

WMS layers rendered by a server are overlaid in the order listed in the Layers section, from top to bottom of the list. If you want to change the overlay order, you can use the tab *Layer Order*.

Transparency

In this version of QGIS, the *Global transparency* setting from the *Layer Properties* is hard-coded to be always on, where available.

참고: WMS Layer Transparency

The availability of WMS image transparency depends on the image encoding used: PNG and GIF support transparency, whilst JPEG leaves it unsupported.

Coordinate Reference System

A Coordinate Reference System (CRS) is the OGC terminology for a QGIS Projection.

Each WMS Layer can be presented in multiple CRSs, depending on the capability of the WMS server.

To choose a CRS, select **[Change...]** and a dialog similar to Figure Projection 3 in *Working with Projections* will appear. The main difference with the WMS version of the screen is that only those CRSs supported by the WMS Server will be shown.

Server search

Within QGIS you can search for WMS-servers. Figure_OGC_2 shows the tab *Server Search* with the *Add Layer(s) from a Server* dialog.

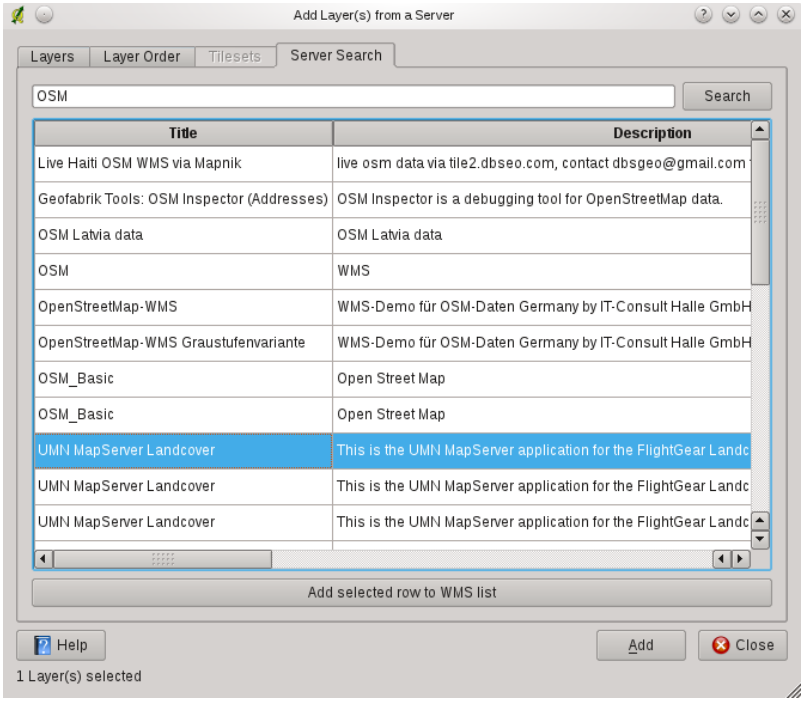


Figure 13.2: Dialog for searching WMS servers after some keywords

As you can see it is possible to enter a search-string in the text field and hit the **[Search]** button. After a short while the search result will be populated into the list below the text field. Browse the result list and inspect your search results within the table. To visualize the results, select a table entry, press the **[Add selected row to WMS-list]** button and change back to the tab *Layers*. QGIS automatically has updated your server list and the selected search result is already enabled in the list of saved WMS-servers in the *Layers* tab. You only need to request the list of

layers by clicking the **[Connect]** button. This option is quite handy when you want to search maps by specific keywords.

Basically this option is a frontend to the API of http://geopole.org.

Tilesets

When using WMS-C (Cached WMS) Services like http://labs.metacarta.com/wms-c/Basic.py you are able to browse through the tab *Tilesets* given by the server. Additional information like tile size, formats and supported CRS are listed in this table. In combination with this feature you can use the tile scale slider from the *View* → *Tile scale slider*, which gives you the available scales from the tileserver with nice slider docked in.

Using the Identify Tool

Once you have added a WMS server, and if any layer from a WMS server is queryable, you can then use the  Identify tool to select a pixel on the map canvas. A query is made to the WMS server for each selection made. The results of the query are returned in plain text. The formatting of this text is dependent on the particular WMS server used. **Format selection**

If multiple output formats are supported by the server, a combo box with supported formats is automatically added to the identify results dialog and the selected format will is stored in project for the layer. **GML format support**

The  Identify tool supports WMS server response (GetFeatureInfo) in GML (it is called Feature in QGIS GUI in this context) format. If "Feature" format is supported by the server and selected, results of the Identify tool are vector features like from regular vector layer. When a single feature is selected in the tree, it is highlighted in the map and it can be copied to clipboard and pasted to another vector layer. See example setup of UMN Mapserver below to support GetFeatureInfo GML format.

```
# in layer METADATA add which fields should be included and define geometry (example):

"gml_include_items"   "all"
"ows_geometries"      "mygeom"
"ows_mygeom_type"     "polygon"

# Then there are two possibilities/formats available, see a) and b):

# a) basic (output is generated by Mapserver and does not contain XSD)
# in WEB METADATA define formats (example):
"wms_getfeatureinfo_formatlist" "application/vnd.ogc.gml,text/html"

# b) using OGR (output is generated by OGR, it is send as multipart and contains XSD)
# in MAP define OUTPUTFORMAT (example):
OUTPUTFORMAT
    NAME "OGRGML"
    MIMETYPE "ogr/gml"
    DRIVER "OGR/GML"
    FORMATOPTION "FORM=multipart"
END

# in WEB METADATA define formats (example):
"wms_getfeatureinfo_formatlist" "OGRGML,text/html"
```

Viewing Properties

Once you have added a WMS server, you can view its properties by right-clicking on it in the legend, and selecting *Properties*. **Metadata Tab**

The tab *Metadata* displays a wealth of information about the WMS server, generally collected from the Capabilities statement returned from that server. Many definitions can be cleaned by reading the WMS standards (see OPEN-GEOSPATIAL-CONSORTIUM *Literature and Web References*), but here are a few handy definitions:

- **Server Properties**
  - **WMS Version** — The WMS version supported by the server.
  - **Image Formats** — The list of MIME-types the server can respond with when drawing the map. QGIS supports whatever formats the underlying Qt libraries were built with, which is typically at least `image/png` and `image/jpeg`.
  - **Identity Formats** — The list of MIME-types the server can respond with when you use the Identify tool. Currently QGIS supports the `text-plain` type.
- **Layer Properties**
  - **Selected** — Whether or not this layer was selected when its server was added to this project.
  - **Visible** — Whether or not this layer is selected as visible in the legend. (Not yet used in this version of QGIS.)
  - **Can Identify** — Whether or not this layer will return any results when the Identify tool is used on it.
  - **Can be Transparent** — Whether or not this layer can be rendered with transparency. This version of QGIS will always use transparency if this is `Yes` and the image encoding supports transparency
  - **Can Zoom In** — Whether or not this layer can be zoomed in by the server. This version of QGIS assumes all WMS layers have this set to `Yes`. Deficient layers may be rendered strangely.
  - **Cascade Count** — WMS servers can act as a proxy to other WMS servers to get the raster data for a layer. This entry shows how many times the request for this layer is forwarded to peer WMS servers for a result.
  - **Fixed Width, Fixed Height** — Whether or not this layer has fixed source pixel dimensions. This version of QGIS assumes all WMS layers have this set to nothing. Deficient layers may be rendered strangely.
  - **WGS 84 Bounding Box** — The bounding box of the layer, in WGS 84 coordinates. Some WMS servers do not set this correctly (e.g. UTM coordinates are used instead). If this is the case, then the initial view of this layer may be rendered with a very 'zoomed-out' appearance by QGIS. The WMS webmaster should be informed of this error, which they may know as the WMS XML elements `LatLonBoundingBox`, `EX_GeographicBoundingBox` or the CRS:84 `BoundingBox`.
  - **Available in CRS** — The projections that this layer can be rendered in by the WMS server. These are listed in the WMS-native format.
  - **Available in style** — The image styles that this layer can be rendered in by the WMS server.

WMS Client Limitations

Not all possible WMS Client functionality had been included in this version of QGIS. Some of the more notable exceptions follow.

Editing WMS Layer Settings

Once you've completed the  Add WMS layer procedure, there is no ability to change the settings. A workaround is to delete the layer completely and start again.

WMS Servers Requiring Authentication

Currently public accessible and secured WMS-services are supported. The secured WMS-servers can be accessed by public authentication. You can add the (optional) credentials when you add a WMS-server. See section *Selecting WMS Servers* for details.

참고: Accessing secured OGC-layers

If you need to access secured layers with other secured methods than basic authentication, you could use InteProxy as a transparent proxy, which does support several authentication methods. More information can be found at the InteProxy manual found on the website http://inteproxy.wald.intevation.org.

참고: QGIS WMS Mapserver

From Version 1.7.0 QGIS has its own implementation of a WMS 1.3.0 Mapserver. Read more about this at chapter *QGIS Server*.

13.1.3 WFS and WFS-T Client

In QGIS, a WFS layer behaves pretty much like any other vector layer. You can identify and select features and view the attribute table. Since QGIS 1.6 editing (WFS-T) is also supported.

In general adding a WFS layer is very similar to the procedure used with WMS. The difference is there are no default servers defined, so we have to add our own.

Loading a WFS Layer

As an example we use the DM Solutions WFS server and display a layer. The URL is: http://www2.dmsolutions.ca/cgi-bin/mswfs_gmap

1. Click on the Add WFS Layer tool on the Layers toolbar, the dialog *Add WFS Layer from a Server* appears
2. Click on **[New]**
3. Enter 'DM Solutions' as name
4. Enter the URL (see above)
5. Click **[OK]**
6. Choose 'DM Solutions' from the dropdown list *Server Connections*
7. Click **[Connect]**
8. Wait for the list of layers to be populated
9. Select the *Parks* layer in the list
10. Click **[Apply]** to add the layer to the map

Note that proxy settings you have set in your preferences are also recognized.

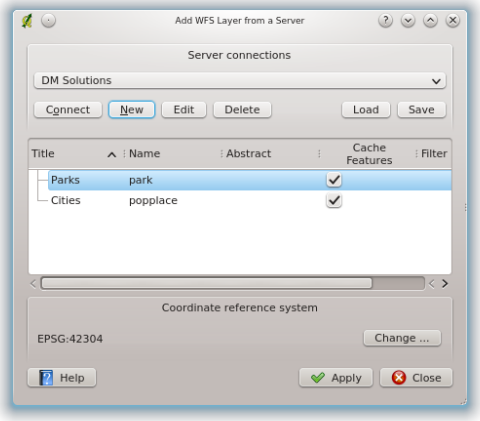


Figure 13.3: Adding a WFS layer

You'll notice the download progress is visualized in the left bottom of the QGIS main window. Once the layer is loaded, you can identify and select a province or two and view the attribute table.

Only WFS 1.0.0 is supported. At this point there have not been many tests against WFS versions implemented in other WFS-servers. If you encounter problems with any other WFS-server, please do not hesitate to contact the development team. Please refer to Section *Mailing lists* for further information about the mailinglists.

참고: Finding WFS Servers

You can find additional WFS servers by using Google or your favorite search engine. There are a number of lists with public URLs, some of them maintained and some not.

13.2 QGIS Server

QGIS Server is an open source WMS 1.3 and WFS 1.0.0 implementation which, in addition, implements advanced cartographic features for thematic mapping. The QGIS Server is a FastCGI/CGI (Common Gateway Interface) application written in C++ that works together with a webserver (e.g. Apache, Lighttpd). It is funded by the EU projects Orchestra, Sany and the city of Uster in Switzerland.

It uses QGIS as backend for the GIS logic and for map rendering. Furthermore the Qt library is used for graphics and for platform independent C++ programming. In contrast to other WMS software, the QGIS Server uses cartographic rules as a configuration language, both for the server configuration and for the user-defined cartographic rules.

Moreover, the QGIS Server project provides the 'Publish to Web' plugin, a plugin for QGIS desktop which exports the current layers and symbology as a web project for QGIS Server (containing cartographic visualization rules expressed in SLD).

As QGIS desktop and QGIS Server use the same visualization libraries, the maps that are published on the web look the same as in desktop GIS. The 'Publish to Web' plugin currently supports basic symbolization, with more complex cartographic visualization rules introduced manually. As the configuration is performed with the SLD standard and its documented extensions, there is only one standardised language to learn, which greatly simplifies the complexity of creating maps for the Web.

In one of the following manuals we will provide a sample configuration to set up a QGIS Server. But for now we recommend to read one of the following URLs to get more information:

- http://karlinapp.ethz.ch/qgis_wms/
- http://www.qgis.org/wiki/QGIS_mapserver_tutorial
- http://linfiniti.com/2010/08/qgis-mapserver-a-wms-server-for-the-masses/

13.2.1 Sample installation on Debian Squeeze

At this point we will give a short and simple sample installation howto for Debian Squeeze. Many other OS provide packages for QGIS Server, too. If you have to build it all from source, please refer to the URLs above.

Apart from qgis and qgis-mapserver you need a webserver, in our case apache2. You can install all packages with aptitude or apt-get install together with other necessary dependency packages. After installation you should test, if the webserver and qgis server works as expected. Make sure the apache server is running with `/etc/init.d/apache2 start`. Open a web browser and type URL: `http://localhost`. If apache is up, you should see the message 'It works!'.

Now we test the qgis server installation. The `qgis_mapserv.fcgi` is available at `/usr/lib/cgi-bin/qgis_mapserv.fcgi` and provides a standard wms that shows the state boundaries of Alaska. Add the WMS with the URL `http://localhost/cgi-bin/qgis_mapserv.fcgi` as described in *Selecting WMS Servers*.

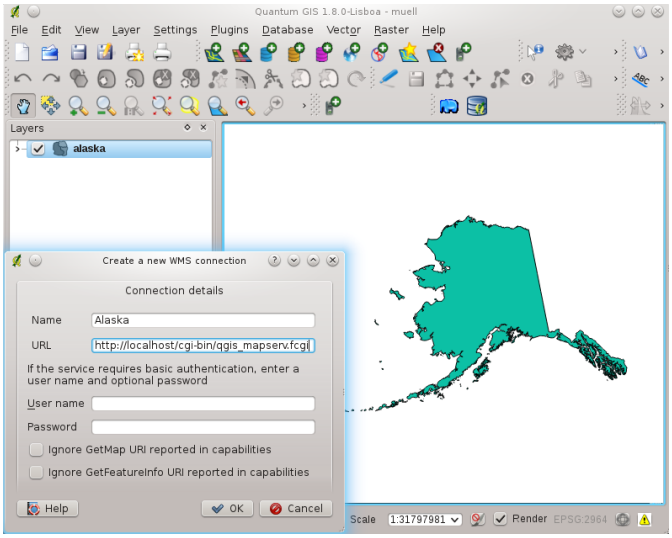


Figure 13.4: Standard WMS with USA boundaries included in the qgis server (KDE)

13.2.2 Creating a WMS/WFS from a QGIS project

To provide a new qgis wms/wfs server we have to create a qgis project file with some data. Here we use the 'alaska' shapefile from the qgis_sample_dataset. Define the colors and styles of the layers in QGIS and define the project CRS, if not already done.

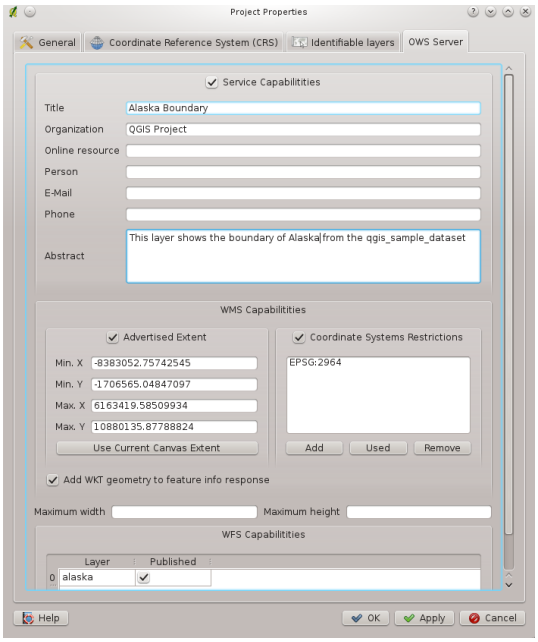


Figure 13.5: Definitions for a qgis project WMS/WFS server (KDE)

Then open the `OWS Server` tab in the menu *Settings* → *Project Properties* and define the general fields under 'Service Capabilities'. For `WMS Capabilities` define 'Coordinate System Restrictions' and 'Advertised Extend'. Additionally you can enable the checkbox ☑ *Add WKT geometry to feature into response* to make the layers queryable and add a 'Maximum width' and height that can be requested. For `WFS Capabilities` you only need to select the layers that you want to provide as WFS. Now save the session in a project file `alaska.qgs`. To provide the project as a WMS/WFS, we create a new folder `/usr/lib/cgi-bin/project` with admin privileges and add the project file `alaska.qgs` and a copy of the `qgis_mapserv.fcgi` file - that's all.

Now we test our project WMS and WFS, add the WMS and WFS as described in *Loading WMS Layers* and *WFS*

and WFS-T Client to QGIS and load the WMS. The URL is:

`http://localhost/cgi-bin/project/qgis_mapserv.fcgi`

CHAPTER 14

GPS 데이터 다루기

14.1 GPS Plugin

14.1.1 What is GPS?

GPS, the Global Positioning System, is a satellite-based system that allows anyone with a GPS receiver to find their exact position anywhere in the world. It is used as an aid in navigation, for example in airplanes, in boats and by hikers. The GPS receiver uses the signals from the satellites to calculate its latitude, longitude and (sometimes) elevation. Most receivers also have the capability to store locations (known as *waypoints*), sequences of locations that make up a planned *route* and a tracklog or *track* of the receivers movement over time. Waypoints, routes and tracks are the three basic feature types in GPS data. QGIS displays waypoints in point layers while routes and tracks are displayed in linestring layers.

14.1.2 Loading GPS data from a file

There are dozens of different file formats for storing GPS data. The format that QGIS uses is called GPX (GPS eXchange format), which is a standard interchange format that can contain any number of waypoints, routes and tracks in the same file.

To load a GPX file you first need to load the plugin. *Plugins* → *Plugin Manager...* opens the Plugin Manager Dialog. Activate the *GPS Tools* checkbox. When this plugin is loaded two buttons with a small handheld GPS device will show up in the toolbar:

- Create new GPX Layer
- GPS Tools

For working with GPS data we provide an example GPX file available in the QGIS sample dataset: `qgis_sample_data/gps/national_monuments.gpx`. See Section *Sample Data* for more information about the sample data.

1. Click on the GPS Tools icon and open the *Load GPX file* tab (see figure_GPS_1).
2. Browse to the folder `qgis_sample_data/gps/`, select the GPX file `national_monuments.gpx` and click **[Open]**.

Use the **[Browse...]** button to select the GPX file, then use the checkboxes to select the feature types you want to load from that GPX file. Each feature type will be loaded in a separate layer when you click **[OK]**. The file `national_monuments.gpx` only includes waypoints.

주석: GPS units allow to store data in different coordinate systems. When downloading a GPX file (from your GPS unit or a web site) and then loading it in Quantum GIS, be sure that the data stored in the GPX file uses WGS84 (latitude/longitude). Quantum GIS expects this and it is the official GPX specification. See http://www.topografix.com/GPX/1/1/

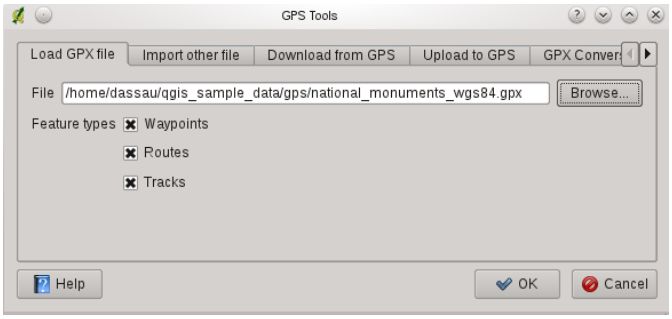


Figure 14.1: The *GPS Tools* dialog window

14.1.3 GPSBabel

Since QGIS uses GPX files you need a way to convert other GPS file formats to GPX. This can be done for many formats using the free program GPSBabel, which is available at http://www.gpsbabel.org. This program can also transfer GPS data between your computer and a GPS device. QGIS uses GPSBabel to do these things, so it is recommended that you install it. However, if you just want to load GPS data from GPX files you will not need it. Version 1.2.3 of GPSBabel is known to work with QGIS, but you should be able to use later versions without any problems.

14.1.4 Importing GPS data

To import GPS data from a file that is not a GPX file, you use the tool *Import other file* in the GPS Tools dialog. Here you select the file that you want to import (and the file type), which feature type you want to import from it, where you want to store the converted GPX file and what the name of the new layer should be. Note that not all GPS data formats will support all three feature types, so for many formats you will only be able to choose between one or two types.

14.1.5 Downloading GPS data from a device

QGIS can use GPSBabel to download data from a GPS device directly as new vector layers. For this we use the *Download from GPS* tab of the GPS Tools dialog (see Figure_GPS_2). Here, we select the type of GPS device, the port that it is connected to (or usb if your GPS supports this), the feature type that you want to download, the GPX file where the data should be stored, and the name of the new layer.

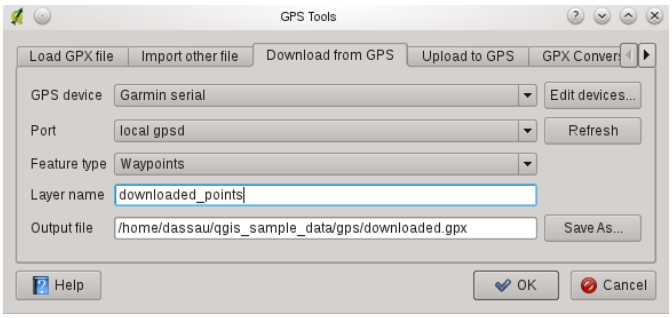


Figure 14.2: The download tool

The device type you select in the GPS device menu determines how GPSBabel tries to communicate with your GPS device. If none of the available types work with your GPS device you can create a new type (see section *Defining new device types*).

The port may be a file name or some other name that your operating system uses as a reference to the physical port in your computer that the GPS device is connected to. It may also be simply usb, for usb enabled GPS units.

- On Linux this is something like `/dev/ttyS0` or `/dev/ttyS1`
- On Windows it is `COM1` or `COM2`

When you click **[OK]** the data will be downloaded from the device and appear as a layer in QGIS.

14.1.6 Uploading GPS data to a device

You can also upload data directly from a vector layer in QGIS to a GPS device using the *Upload to GPS* tab of the GPS Tools dialog. To do this you simply select the layer that you want to upload (which must be a GPX layer), your GPS device type, and the port (or usb) that it is connected to. Just as with the download tool you can specify new device types if your device isn't in the list.

This tool is very useful in combination with the vector editing capabilities of QGIS. It allows you to load a map, create waypoints and routes, and then upload them and use them on your GPS device.

14.1.7 Defining new device types

There are lots of different types of GPS devices. The QGIS developers can't test all of them, so if you have one that does not work with any of the device types listed in the *Download from GPS* and *Upload to GPS* tools you can define your own device type for it. You do this by using the GPS device editor, which you start by clicking the **[Edit devices]** button in the download or the upload tabs.

To define a new device you simply click the **[New device]** button, enter a name, a download command and an upload command for your device, and click the **[Update device]** button. The name will be listed in the device menus in the upload and download windows, and can be any string. The download command is the command that is used to download data from the device to a GPX file. This will probably be a GPSBabel command, but you can use any other command line program that can create a GPX file. QGIS will replace the keywords `%type`, `%in`, and `%out` when it runs the command.

`%type` will be replaced by `-w` if you are downloading waypoints, `-r` if you are downloading routes and `-t` if you are downloading tracks. These are command line options that tell GPSBabel which feature type to download.

`%in` will be replaced by the port name that you choose in the download window and `%out` will be replaced by the name you choose for the GPX file that the downloaded data should be stored in. So if you create a device type with the download command `gpsbabel %type -i garmin -o gpx %in %out` (this is actually the download command for the predefined device type 'Garmin serial') and then use it to download waypoints from port `/dev/ttyS0` to the file `output.gpx`, QGIS will replace the keywords and run the command `gpsbabel -w -i garmin -o gpx /dev/ttyS0 output.gpx`.

The upload command is the command that is used to upload data to the device. The same keywords are used, but `%in` is now replaced by the name of the GPX file for the layer that is being uploaded, and `%out` is replaced by the port name.

You can learn more about GPSBabel and it's available command line options at http://www.gpsbabel.org.

Once you have created a new device type it will appear in the device lists for the download and upload tools.

14.2 라이브 GPS 추적

라이브 GPS 추적을 활성화하려면 *Settings* → *GPS information* [라이브 GPS 추적]를 선택합니다. 라이브 GPS 추적이 활성화되면 캔버스 왼쪽에 새로운 창이 나타납니다.

GPS 추적창에는 다음과 같은 4가지 화면이 있습니다:

- GPS 위치 좌표 표시. 꼭지점과 객체를 수동으로 입력할 수 있음.

- 연결되는 위성과 GPS 신호 강도.
- GPS 위성의 수와 위치 표시. 극좌표로 표시함.
- GPS 옵션 화면 (figure_gps_options 참조).

GPS 수신기(운영 체제에 의해 지원되어야함)를 연결한 후 **[Connect]** [연결]를 클릭하면 QGIS에 GPS가 연결됩니다. 한 번 더 클릭(**[Disconnect]** [연결 끊어짐]로 표시된 상태임)하면 컴퓨터와 GPS 수신기의 연결이 끊어집니다. GNU/리눅스의 경우, 대부분의 GPS 수신기가 자체적으로 지원됩니다. 따라서 먼저 GPS를 사용하려면 연결하려는 gpsd를 구성해야합니다.

경고: 캔버스에 GPS 위치를 기록하고 싶다면 먼저 새로운 벡터 레이어를 만들고 편집가능 상태로 전환해야 트랙을 기록할 수 있습니다.

14.2.1 위치와 additional attributes

GPS가 위성에서 신호를 받고 있다면 자신의 위치가 additional attributes와 함께 경위도와 높이로 나타나게 됩니다.

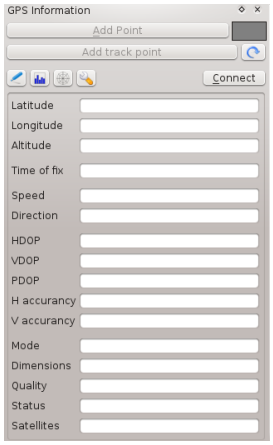


Figure 14.3: GPS 추적 위치와 additional attributes

14.2.2 GPS 신호 강도

수신하는 GPS위성의 신호 강도를 볼 수 있습니다.

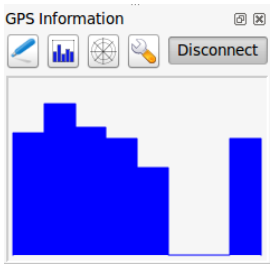


Figure 14.4: GPS 추적 신호 강도

14.2.3 GPS polar window

연결되는 위성이 어느 곳에 있는지 알고 싶다면 극좌표로 표현해야 합니다. 이 화면에서 빨간색 숫자는 신호를 받고있는 위성의 ID 번호입니다.

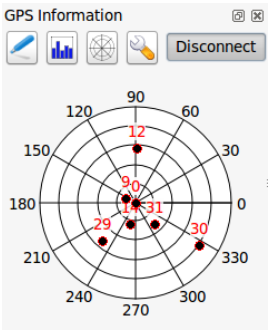


Figure 14.5: GPS tracking polar window

14.2.4 GPS options

GPS의 연결에 문제가 있을 경우 다음의 옵션에서 변경이 가능합니다:

- *Autodetect*
- *Internal*
- *Serial device*
- *gpsd* (selecting Host, Port and Device your GPS is connected to)

그 다음 **[Connect]** [연결]을 다시 클릭하면 GPS 수신기와의 연결을 시도합니다.

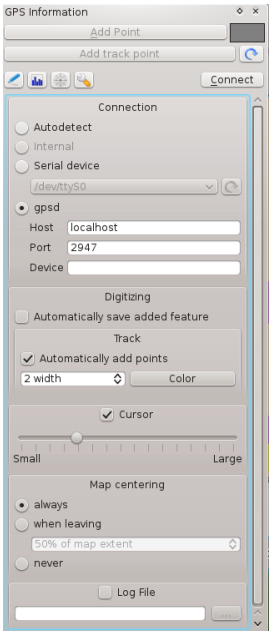


Figure 14.6: GPS 추적 옵션창

You can activate *Automatically save added features* when you are in editing mode. Or you can can activate *Automatically add points* to the map canvas with a certain width and color.

Cursor 를 활성화하면  슬라이더를 사용하여 캔버스에 있는 커서를 확대 축소 할 수 있습니다.

Activating *Map centering* allows to decide in which way the canvas will be updated. This includes 'always', 'when leaving' if your recorded coordinates start either to move out of canvas or 'never' to keep map extent.

Finally you can activate *Log file* and define a path and a file where log messages about the gps tracking a logged.

객체를 다시 수동으로 설정하고 싶으면 Position [위치 좌표]로 되돌아간 후, **[Add Point]** [객체 추가] 또는 **[Add track point]** [추적 포인트 추가] 를 한 번 더 클릭합니다.

CHAPTER 15

GRASS GIS Integration

The GRASS plugin provides access to GRASS GIS (see GRASS-PROJECT *Literature and Web References*) databases and functionalities. This includes visualization of GRASS raster and vector layers, digitizing vector layers, editing vector attributes, creating new vector layers and analysing GRASS 2D and 3D data with more than 300 GRASS modules.

In this Section we'll introduce the plugin functionalities and give some examples on managing and working with GRASS data. Following main features are provided with the toolbar menu, when you start the GRASS plugin, as described in section *Starting the GRASS plugin* :

-  Open mapset
-  New mapset
-  Close mapset
-  Add GRASS vector layer
-  Add GRASS raster layer
-  Create new GRASS vector
-  Edit GRASS vector layer
-  Open GRASS tools
-  Display current GRASS region
-  Edit current GRASS region

15.1 Starting the GRASS plugin

To use GRASS functionalities and/or visualize GRASS vector and raster layers in QGIS, you must select and load the GRASS plugin with the Plugin Manager. Therefore go to the menu *Plugins* →  *Manage Plugins*, select  *GRASS* and click **[OK]**.

You can now start loading raster and vector layers from an existing GRASS `LOCATION` (see section *Loading GRASS raster and vector layers*). Or you create a new GRASS `LOCATION` with QGIS (see section *Creating a new GRASS LOCATION*) and import some raster and vector data (see Section *Importing data into a GRASS LOCATION*) for further analysis with the GRASS Toolbox (see section *The GRASS toolbox*).

15.2 Loading GRASS raster and vector layers

With the GRASS plugin, you can load vector or raster layers using the appropriate button on the toolbar menu. As an example we use the QGIS alaska dataset (see Section *Sample Data*). It includes a small sample GRASS LOCATION with 3 vector layers and 1 raster elevation map.

1. Create a new folder grassdata, download the QGIS 'Alaska' dataset qgis_sample_data.zip from http://download.osgeo.org/qgis/data/ and unzip the file into grassdata.
2. Start QGIS.
3. If not already done in a previous QGIS session, load the GRASS plugin clicking on *Plugins* → *Manage Plugins* and activate *GRASS*. The GRASS toolbar appears in the QGIS main window.
4. In the GRASS toolbar, click the Open mapset icon to bring up the *MAPSET* wizard.
5. For Gisdbase browse and select or enter the path to the newly created folder grassdata.
6. You should now be able to select the *LOCATION* alaska and the *MAPSET* demo.
7. Click **[OK]**. Notice that some previously disabled tools in the GRASS toolbar are now enabled.
8. Click on Add GRASS raster layer, choose the map name gtopo30 and click **[OK]**. The elevation layer will be visualized.
9. Click on Add GRASS vector layer, choose the map name alaska and click **[OK]**. The Alaska boundary vector layer will be overlayed on top of the gtopo30 map. You can now adapt the layer properties as described in chapter *벡터 속성 대화상자*, e.g. change opacity, fill and outline color.
10. Also load the other two vector layers rivers and airports and adapt their properties.

As you see, it is very simple to load GRASS raster and vector layers in QGIS. See following sections for editing GRASS data and creating a new LOCATION. More sample GRASS LOCATIONs are available at the GRASS website at http://grass.osgeo.org/download/data.php.

참고: GRASS Data Loading

If you have problems loading data or QGIS terminates abnormally, check to make sure you have loaded the GRASS plugin properly as described in section *Starting the GRASS plugin*.

15.3 GRASS LOCATION and MAPSET

GRASS data are stored in a directory referred to as GISDBASE. This directory often called grassdata, must be created before you start working with the GRASS plugin in QGIS. Within this directory, the GRASS GIS data are organized by projects stored in subdirectories called LOCATION. Each LOCATION is defined by its coordinate system, map projection and geographical boundaries. Each LOCATION can have several MAPSETs (subdirectories of the LOCATION) that are used to subdivide the project into different topics, subregions, or as workspaces for individual team members (Neteler & Mitasova 2008 *Literature and Web References*). In order to analyze vector and raster layers with GRASS modules, you must import them into a GRASS LOCATION (This is not strictly true - with the GRASS modules r.external and v.external you can create read-only links to external GDAL/OGR-supported data sets without importing them. But because this is not the usual way for beginners to work with GRASS, this functionality will not be described here.).

15.3.1 Creating a new GRASS LOCATION

As an example here is how the sample GRASS LOCATION alaska, which is projected in Albers Equal Area projection with unit feet was created for the QGIS sample dataset. This sample GRASS LOCATION alaska

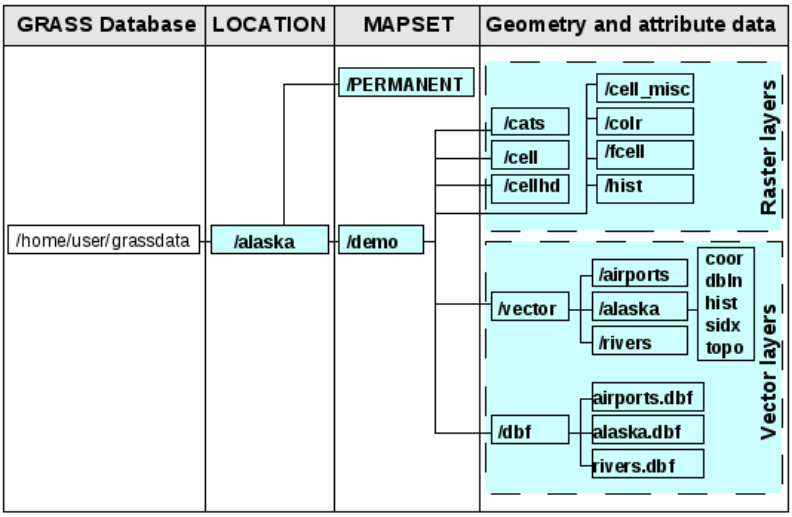


Figure 15.1: GRASS data in the alaska LOCATION

will be used for all examples and exercises in the following GRASS GIS related chapters. It is useful to download and install the dataset on your computer *Sample Data*).

1. Start QGIS and make sure the GRASS plugin is loaded.
2. Visualize the `alaska.shp` shapefile (see Section *Shapefile 불러오기*) from the QGIS alaska dataset *Sample Data*.
3. In the GRASS toolbar, click on the New mapset icon to bring up the *MAPSET* wizard.
4. Select an existing GRASS database (GISDBASE) folder `grassdata` or create one for the new `LOCATION` using a file manager on your computer. Then click **[Next]**.
5. We can use this wizard to create a new `MAPSET` within an existing `LOCATION` (see section *Adding a new MAPSET*) or to create a new `LOCATION` altogether. Select *Create new location* (see figure_grass_location_2).
6. Enter a name for the `LOCATION` - we used 'alaska' and click **[Next]**.
7. Define the projection by clicking on the radio button *Projection* to enable the projection list.
8. We are using Albers Equal Area Alaska (feet) projection. Since we happen to know that it is represented by the EPSG ID 2964, we enter it in the search box. (Note: If you want to repeat this process for another `LOCATION` and projection and haven't memorized the EPSG ID, click on the projector icon in the lower right-hand corner of the status bar (see Section *Working with Projections*)).
9. In *Filter* insert 2964 to select the projection.
10. Click **[Next]**.
11. To define the default region, we have to enter the `LOCATION` bounds in north, south, east, and west direction. Here we simply click on the button **[Set current QGIS extent]**, to apply the extend of the loaded layer `alaska.shp` as the GRASS default region extend.
12. Click **[Next]**.
13. We also need to define a `MAPSET` within our new `LOCATION`. You can name it whatever you like - we used 'demo' (when creating a new `LOCATION`). GRASS automatically creates a special `MAPSET` called `PERMANENT` designed to store the core data for the project, its default spatial extend and coordinate system definitions (Neteler & Mitasova 2008 *Literature and Web References*).
14. Check out the summary to make sure it's correct and click **[Finish]**.

15. The new LOCATION 'alaska' and two MAPSETs 'demo' and 'PERMANENT' are created. The currently opened working set is 'demo', as you defined.
16. Notice that some of the tools in the GRASS toolbar that were disabled are now enabled.

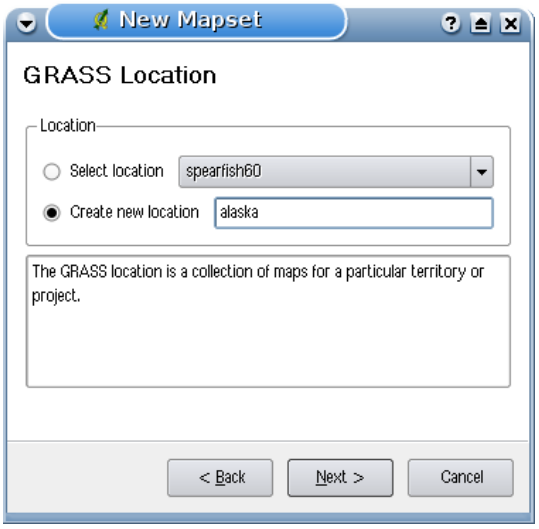


Figure 15.2: Creating a new GRASS LOCATION or a new MAPSET in QGIS

If that seemed like a lot of steps, it's really not all that bad and a very quick way to create a LOCATION. The LOCATION 'alaska' is now ready for data import (see section *Importing data into a GRASS LOCATION*). You can also use the already existing vector and raster data in the sample GRASS LOCATION 'alaska' included in the QGIS 'Alaska' dataset *Sample Data* and move on to Section *The GRASS vector data model*.

15.3.2 Adding a new MAPSET

A user has only write access to a GRASS MAPSET he created. This means that besides access to his own MAPSET, each user can read maps in other user's MAPSETs, but he can modify or remove only the maps in his own MAPSET.

All MAPSETs include a WIND file that stores the current boundary coordinate values and the currently selected raster resolution (Neteler & Mitasova 2008 *Literature and Web References*, see Section *The GRASS region tool*).

1. Start QGIS and make sure the GRASS plugin is loaded.
2. In the GRASS toolbar, click on the  New mapset icon to bring up the *MAPSET* wizard.
3. Select the GRASS database (GISDBASE) folder grassdata with the LOCATION 'alaska', where we want to add a further MAPSET, called 'test'.
4. Click **[Next]**.
5. We can use this wizard to create a new MAPSET within an existing LOCATION or to create a new LOCATION altogether. Click on the radio button  *Select location* (see figure_grass_location_2) and click **[Next]**.
6. Enter the name text for the new MAPSET. Below in the wizard you see a list of existing MAPSETs and its owners.
7. Click **[Next]**, check out the summary to make sure it's all correct and click **[Finish]**.

15.4 Importing data into a GRASS LOCATION

This Section gives an example how to import raster and vector data into the 'alaska' GRASS LOCATION provided by the QGIS 'Alaska' dataset. Therefore we use a landcover raster map landcover.img and a vector GML file

`lakes.gml` from the QGIS 'Alaska' dataset *Sample Data*.

1. Start QGIS and make sure the GRASS plugin is loaded.
2. In the GRASS toolbar, click the  Open MAPSET icon to bring up the *MAPSET* wizard.
3. Select as GRASS database the folder `grassdata` in the QGIS alaska dataset, as `LOCATION` 'alaska', as `MAPSET` 'demo' and click **[OK]**.
4. Now click the  Open GRASS tools icon. The GRASS Toolbox (see section *The GRASS toolbox*) dialog appears.
5. To import the raster map `landcover.img`, click the module `r.in.gdal` in the *Modules Tree* tab. This GRASS module allows to import GDAL supported raster files into a GRASS `LOCATION`. The module dialog for `r.in.gdal` appears.
6. Browse to the folder `raster` in the QGIS 'Alaska' dataset and select the file `landcover.img`.
7. As raster output name define `landcover_grass` and click **[Run]**. In the *Output* tab you see the currently running GRASS command `r.in.gdal -o input=/path/to/landcover.img output=landcover_grass`.
8. When it says **Succesfully finished** click **[View output]**. The `landcover_grass` raster layer is now imported into GRASS and will be visualized in the QGIS canvas.
9. To import the vector GML file `lakes.gml`, click the module `v.in.ogr` in the *Modules Tree* tab. This GRASS module allows to import OGR supported vector files into a GRASS `LOCATION`. The module dialog for `v.in.ogr` appears.
10. Browse to the folder `gml` in the QGIS 'Alaska' dataset and select the file `lakes.gml` as OGR file.
11. As vector output name define `lakes_grass` and click **[Run]**. You don't have to care about the other options in this example. In the *Output* tab you see the currently running GRASS command `v.in.ogr -o dsn=/path/to/lakes.gml output=lakes\_grass`.
12. When it says **Succesfully finished** click **[View output]**. The `lakes_grass` vector layer is now imported into GRASS and will be visualized in the QGIS canvas.

15.5 The GRASS vector data model

It is important to understand the GRASS vector data model prior to digitizing.

In general, GRASS uses a topological vector model.

This means that areas are not represented as closed polygons, but by one or more boundaries. A boundary between two adjacent areas is digitized only once, and it is shared by both areas. Boundaries must be connected and closed without gaps. An area is identified (and labeled) by the **centroid** of the area.

Besides boundaries and centroids, a vector map can also contain points and lines. All these geometry elements can be mixed in one vector and will be represented in different so called 'layers' inside one GRASS vector map. So in GRASS a layer is not a vector or raster map but a level inside a vector layer. This is important to distinguish carefully (Although it is possible to mix geometry elements, it is unusual and even in GRASS only used in special cases such as vector network analysis. Normally you should prefere to store different geometry elements in different layers.).

It is possible to store several 'layers' in one vector dataset. For example, fields, forests and lakes can be stored in one vector. Adjacent forest and lake can share the same boundary, but they have separate attribute tables. It is also possible to attach attributes to boundaries. For example, the boundary between lake and forest is a road, so it can have a different attribute table.

The 'layer' of the feature is defined by 'layer' inside GRASS. 'Layer' is the number which defines if there are more than one layer inside the dataset, e.g. if the geometry is forest or lake. For now, it can be only a number, in the future GRASS will also support names as fields in the user interface.

Attributes can be stored inside the GRASS `LOCATION` as DBase or SQLITE3 or in external database tables, for example PostgreSQL, MySQL, Oracle, etc.

Attributes in database tables are linked to geometry elements using a 'category' value.

'Category' (key, ID) is an integer attached to geometry primitives, and it is used as the link to one key column in the database table.

참고: Learning the GRASS Vector Model

The best way to learn the GRASS vector model and its capabilities is to download one of the many GRASS tutorials where the vector model is described more deeply. See http://grass.osgeo.org/gdp/manuals.php for more information, books and tutorials in several languages.

15.6 Creating a new GRASS vector layer

To create a new GRASS vector layer with the GRASS plugin click the Create new GRASS vector toolbar icon. Enter a name in the text box and you can start digitizing point, line or polygon geometries, following the procedure described in Section *Digitizing and editing a GRASS vector layer*.

In GRASS it is possible to organize all sort of geometry types (point, line and area) in one layer, because GRASS uses a topological vector model, so you don't need to select the geometry type when creating a new GRASS vector. This is different from Shapefile creation with QGIS, because Shapefiles use the Simple Feature vector model (see Section *새로운 벡터 레이어 생성*).

참고: Creating an attribute table for a new GRASS vector layer

If you want to assign attributes to your digitized geometry features, make sure to create an attribute table with columns before you start digitizing (see figure_grass_digitizing_5).

15.7 Digitizing and editing a GRASS vector layer

The digitizing tools for GRASS vector layers are accessed using the Edit GRASS vector layer icon on the toolbar. Make sure you have loaded a GRASS vector and it is the selected layer in the legend before clicking on the edit tool. Figure figure_grass_digitizing_2 shows the GRASS edit dialog that is displayed when you click on the edit tool. The tools and settings are discussed in the following sections.

참고: Digitizing polygons in GRASS

If you want to create a polygon in GRASS, you first digitize the boundary of the polygon, setting the mode to 'No category'. Then you add a centroid (label point) into the closed boundary, setting the mode to 'Next not used'. The reason is, that a topological vector model links attribute information of a polygon always to the centroid and not to the boundary.

Toolbar

In figure_grass_digitizing_1 you see the GRASS digitizing toolbar icons provided by the GRASS plugin. Table table_grass_digitizing_1 explains the available functionalities.



Figure 15.3: GRASS Digitizing Toolbar

Icon	Tool	Purpose
	New Point	Digitize new point
	New Line	Digitize new line
	New Boundary	Digitize new boundary (finish by selecting new tool)
	New Centroid	Digitize new centroid (label existing area)
	Move vertex	Move one vertex of existing line or boundary and identify new position
	Add vertex	Add a new vertex to existing line
	Delete vertex	Delete vertex from existing line (confirm selected vertex by another click)
	Move element	Move selected boundary, line, point or centroid and click on new position
	Split line	Split an existing line to 2 parts
	Delete element	Delete existing boundary, line, point or centroid (confirm selected element by another click)
	Edit attributes	Edit attributes of selected element (note that one element can represent more features, see above)
	Close	Close session and save current status (rebuilds topology afterwards)

Table GRASS Digitizing 1: GRASS Digitizing Tools

Category Tab

The *Category* tab allows you to define the way in which the category values will be assigned to a new geometry element.

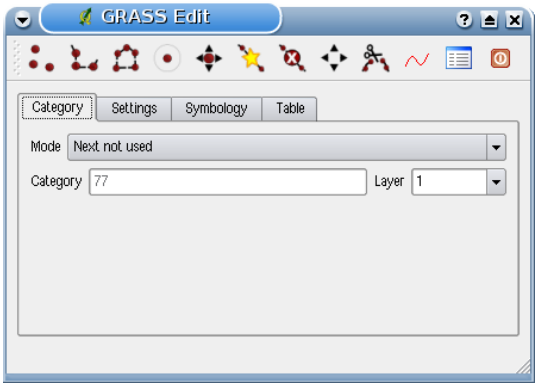


Figure 15.4: GRASS Digitizing Category Tab

- **Mode**: what category value shall be applied to new geometry elements.
  - Next not used - apply next not yet used category value to geometry element.
  - Manual entry - manually define the category value for the geometry element in the 'Category'-entry field.
  - No category - Do not apply a category value to the geometry element. This is e.g. used for area boundaries, because the category values are connected via the centroid.
- **Category** - A number (ID) is attached to each digitized geometry element. It is used to connect each geometry element with its attributes.

- **Field (layer)** - Each geometry element can be connected with several attribute tables using different GRASS geometry layers. Default layer number is 1.

참고: Creating an additional GRASS 'layer' with QGIS

If you would like to add more layers to your dataset, just add a new number in the 'Field (layer)' entry box and press return. In the Table tab you can create your new table connected to your new layer.

Settings Tab

The *Settings* tab allows you to set the snapping in screen pixels. The threshold defines at what distance new points or line ends are snapped to existing nodes. This helps to prevent gaps or dangles between boundaries. The default is set to 10 pixels.

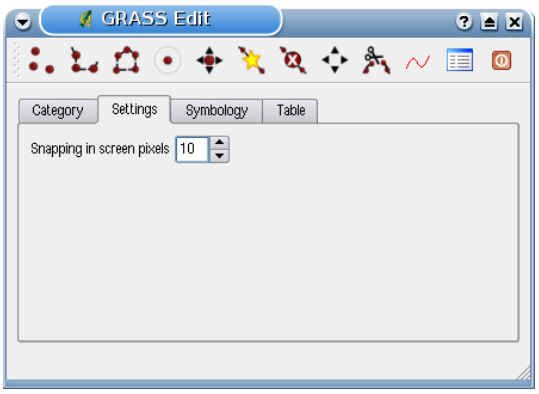


Figure 15.5: GRASS Digitizing Settings Tab

Symbology Tab

The *Symbology* tab allows you to view and set symbology and color settings for various geometry types and their topological status (e.g. closed / opened boundary).

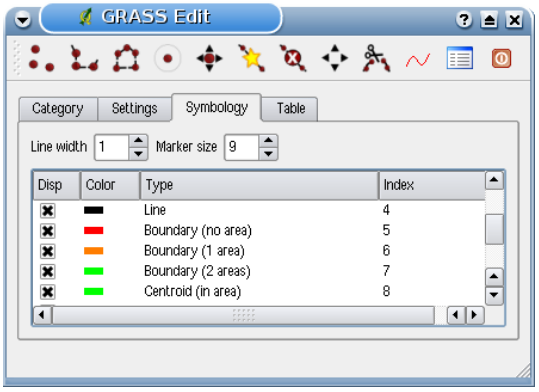


Figure 15.6: GRASS Digitizing Symbolog Tab

Table Tab

The *Table* tab provides information about the database table for a given 'layer'. Here you can add new columns to an existing attribute table, or create a new database table for a new GRASS vector layer (see Section *Creating a new GRASS vector layer*).

참고: GRASS Edit Permissions

You must be the owner of the GRASS `MAPSET` you want to edit. It is impossible to edit data layers in a `MAPSET` that is not yours, even if you have write permissions.

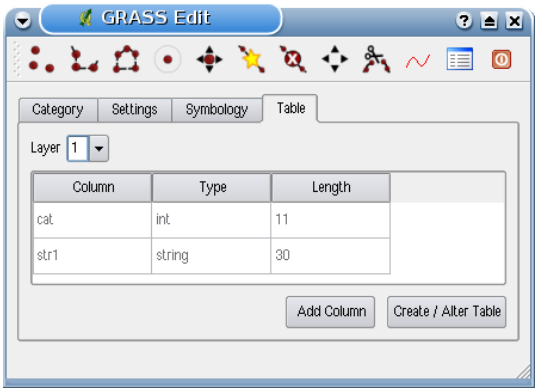


Figure 15.7: GRASS Digitizing Table Tab

15.8 The GRASS region tool

The region definition (setting a spatial working window) in GRASS is important for working with raster layers. Vector analysis is by default not limited to any defined region definitions. But all newly-created rasters will have the spatial extension and resolution of the currently defined GRASS region, regardless of their original extension and resolution. The current GRASS region is stored in the `$LOCATION/$MAPSET/WIND` file, and it defines north, south, east and west bounds, number of columns and rows, horizontal and vertical spatial resolution.

It is possible to switch on/off the visualization of the GRASS region in the QGIS canvas using the  Display current GRASS region button.

With the  Edit current GRASS region icon you can open a dialog to change the current region and the symbology of the GRASS region rectangle in the QGIS canvas. Type in the new region bounds and resolution and click **[OK]**. It also allows to select a new region interactively with your mouse on the QGIS canvas. Therefore click with the left mouse button in the QGIS canvas, open a rectangle, close it using the left mouse button again and click **[OK]**.

The GRASS module `g.region` provide a lot more parameters to define an appropriate region extend and resolution for your raster analysis. You can use these parameters with the GRASS Toolbox, described in Section *The GRASS toolbox*.

15.9 The GRASS toolbox

The  Open GRASS Tools box provides GRASS module functionalities to work with data inside a selected GRASS `LOCATION` and `MAPSET`. To use the GRASS toolbox you need to open a `LOCATION` and `MAPSET` where you have write-permission (usually granted, if you created the `MAPSET`). This is necessary, because new raster or vector layers created during analysis need to be written to the currently selected `LOCATION` and `MAPSET`.

The GRASS Shell inside the GRASS Toolbox provides access to almost all (more than 330) GRASS modules through a command line interface. To offer a more user friendly working environment, about 200 of the available GRASS modules and functionalities are also provided by graphical dialogs within the GRASS plugin Toolbox.

15.9.1 Working with GRASS modules

The GRASS Shell inside the GRASS Toolbox provides access to almost all (more than 300) GRASS modules in a command line interface. To offer a more user friendly working environment, about 200 of the available GRASS modules and functionalities are also provided by graphical dialogs.

A complete list of GRASS modules available in the graphical Toolbox in QGIS version 1.8.0 is available in the GRASS wiki (http://grass.osgeo.org/wiki/GRASS-QGIS_relevant_module_list).

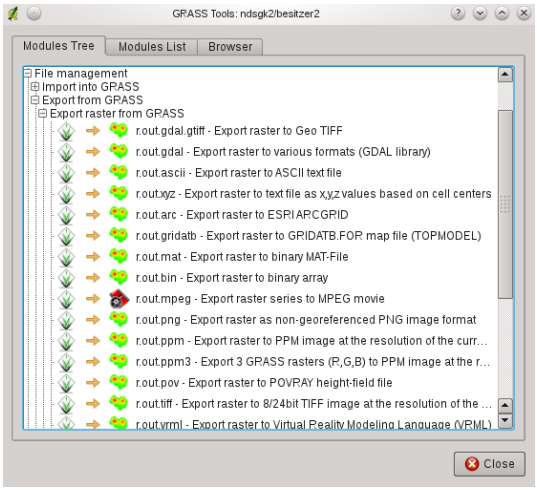


Figure 15.8: GRASS Toolbox and Module Tree

It is also possible to customize the GRASS Toolbox content. This procedure is described in Section *Customizing the GRASS Toolbox*.

As shown in figure_grass_toolbox_1 , you can look for the appropriate GRASS module using the thematically grouped *Modules Tree* or the searchable *Modules List* tab.

Clicking on a graphical module icon a new tab will be added to the toolbox dialog providing three new sub-tabs *Options*, *Output* and *Manual*.

Options

The *Options* tab provides a simplified module dialog where you can usually select a raster or vector layer visualized in the QGIS canvas and enter further module specific parameters to run the module.

The provided module parameters are often not complete to keep the dialog clear. If you want to use further module parameters and flags, you need to start the GRASS Shell and run the module in the command line.

A new feature in QGIS 1.8.0 is the support for a *show advanced options* button below the simplified module dialog in the *Options* tab. At the moment it is only added to the module v.in.ascii as an example use, but will probably be part of more / all modules in the GRASS toolbox in future versions of QGIS. This allows to use the complete GRASS module options without the need to switch to the GRASS Shell.

Output

The *Output* tab provides information about the output status of the module. When you click the **[Run]** button, the module switches to the *Output* tab and you see information about the analysis process. If all works well, you will finally see a `Successfully finished` message.

Manual

The *Manual* tab shows the HTML help page of the GRASS module. You can use it to check further module parameters and flags or to get a deeper knowledge about the purpose of the module. At the end of each module manual page you see further links to the `Main Help index`, the `Thematic index` and the `Full index`. These links provide the same information as if you use the module `g.manual`.

참고: Display results immediately

If you want to display your calculation results immediately in your map canvas, you can use the 'View Output' button at the bottom of the module tab.

15.9.2 GRASS module examples

The following examples will demonstrate the power of some of the GRASS modules.

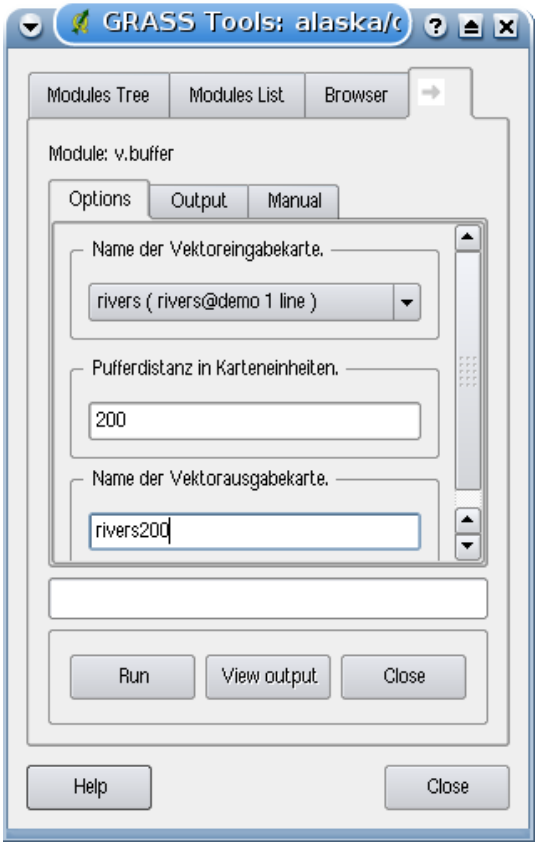


Figure 15.9: GRASS Toolbox Module Options

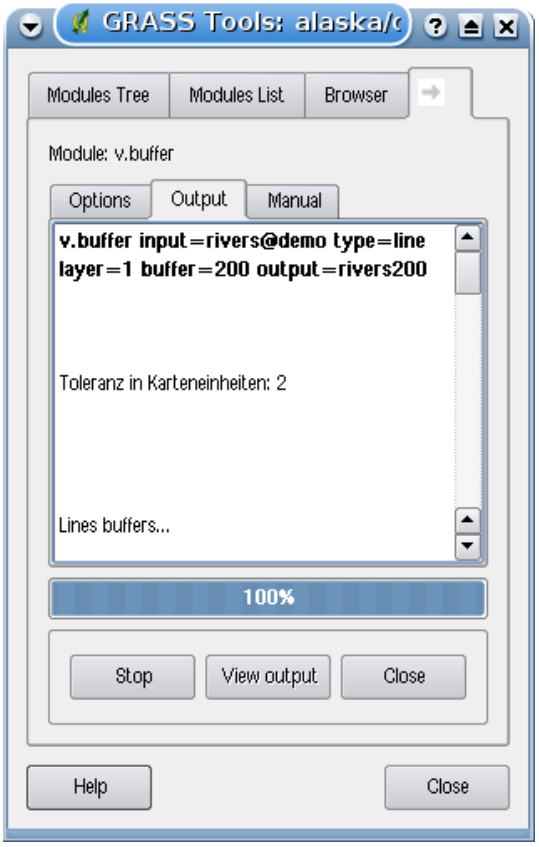


Figure 15.10: GRASS Toolbox Module Output

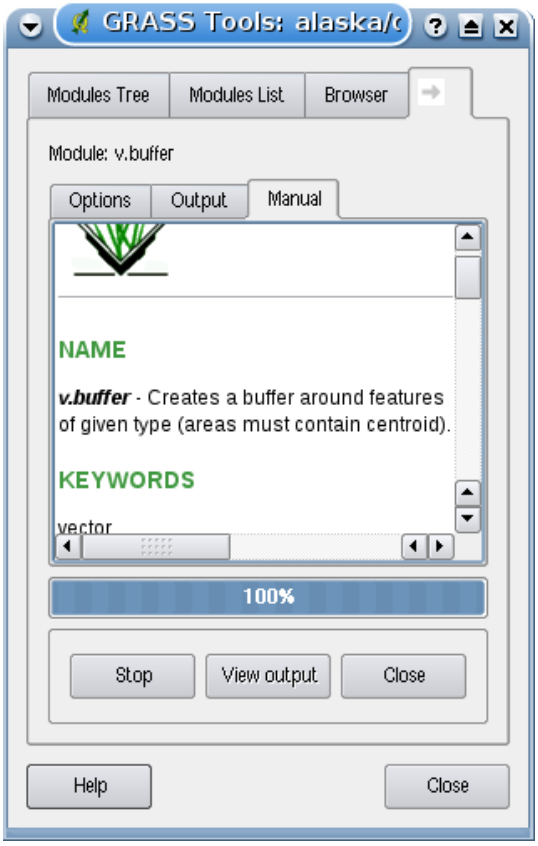


Figure 15.11: GRASS Toolbox Module Manual

Creating contour lines

The first example creates a vector contour map from an elevation raster (DEM). Assuming you have the Alaska `LOCATION` set up as explained in Section *Importing data into a GRASS LOCATION*.

- First open the location by clicking the Open mapset button and choosing the Alaska location.
- Now load the `gtopo30` elevation raster by clicking Add GRASS raster layer and selecting the `gtopo30` raster from the demo location.
- Now open the Toolbox with the Open GRASS tools button.
- In the list of tool categories double click *Raster* → *Surface Management* → *Generate vector contour lines*.
- Now a single click on the tool **r.contour** will open the tool dialog as explained above *Working with GRASS modules*. The `gtopo30` raster should appear as the *Name of input raster*.
- Type into the *Increment between Contour levels* 1,00 the value 100. (This will create contour lines at intervals of 100 meters.)
- Type into the *Name for output vector map* the name `ctour_100`.
- Click **[Run]** to start the process. Wait for several moments until the message `Successfully finished` appears in the output window. Then click **[View Output]** and **[Close]**.

Since this is a large region, it will take a while to display. After it finishes rendering, you can open the layer properties window to change the line color so that the contours appear clearly over the elevation raster, as in *벡터 속성 대화상자*.

Next zoom in to a small mountainous area in the center of Alaska. Zooming in close you will notice that the contours have sharp corners. GRASS offers the **v.generalize** tool to slightly alter vector maps while keeping their

overall shape. The tool uses several different algorithms with different purposes. Some of the algorithms (i.e. Douglas Peuker and Vertex reduction) simplify the line by removing some of the vertices. The resulting vector will load faster. This process will be used when you have a highly detailed vector, but you are creating a very small scale map, so the detail is unnecessary.

참고: The simplify tool

Note that the QGIS fTools plugin has a *Simplify geometries* → tool that works just like the GRASS **v.generalize** Douglas-Peuker algorithm.

However, the purpose of this example is different. The contour lines created by `r.contour` have sharp angles that should be smoothed. Among the **v.generalize** algorithms there is Chaikens which does just that (also Hermite splines). Be aware that these algorithms can **add** additional vertices to the vector, causing it to load even more slowly.

- Open the GRASS toolbox and double click the categories *Vector* → *Develop map* → *Generalization*, then click on the **v.generalize** module to open its options window.
- Check that the 'ctour_100' vector appears as the *Name of input vector*.
- From the list of algorithms choose Chaiken's. Leave all other options at their default, and scroll down to the last row to enter in the field *Name for output vector map* 'ctour_100_smooth', and click **[Run]**.
- The process takes several moments. Once `Successfully finished` appears in the output windows, click **[View output]** and then **[close]**.
- You may change the color of the vector to display it clearly on the raster background and to contrast with the original contour lines. You will notice that the new contour lines have smoother corners than the original while staying faithful to the original overall shape.

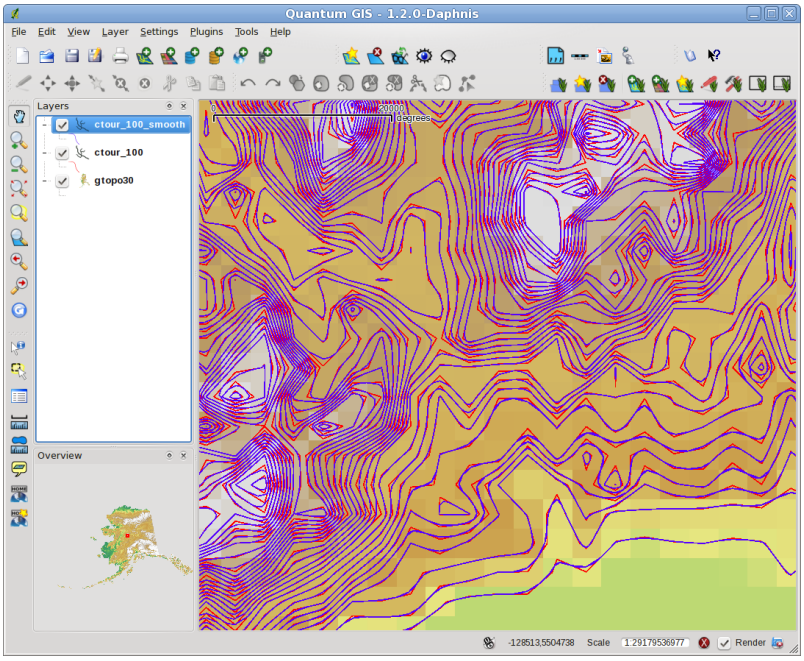


Figure 15.12: GRASS module v.generalize to smooth a vector map

참고: Other uses for r.contour

The procedure described above can be used in other equivalent situations. If you have a raster map of precipitation data, for example, then the same method will be used to create a vector map of isohyetal (constant rainfall) lines.

Creating a Hillshade 3D effect

Several methods are used to display elevation layers and give a 3D effect to maps. The use of contour lines as shown above is one popular method often chosen to produce topographic maps. Another way to display a 3D effect is by hillshading. The hillshade effect is created from a DEM (elevation) raster by first calculating the slope and aspect of each cell, then simulating the sun's position in the sky and giving a reflectance value to each cell. Thus you get sun facing slopes lighted and the slopes facing away from the sun (in shadow) are darkened.

- Begin this example by loading the `gtopo30` elevation raster. Start the GRASS toolbox and under the Raster category double click to open *Spatial analysis* → *Terrain analysis*.
- Then click **r.shaded.relief** to open the module.
- Change the *azimuth angle* 1,00 270 to 315.
- Enter `gtopo30_shade` for the new hillshade raster, and click **[Run]**.
- When the process completes, add the hillshade raster to the map. You should see it displayed in grayscale.
- To view both the hill shading and the colors of the `gtopo30` together shift the hillshade map below the `gtopo30` map in the table of contents, then open the *Properties* window of `gtopo30`, switch to the *transparency* tab and set its transparency level to about 25%.

You should now have the `gtopo30` elevation with its colormap and transparency setting displayed **above** the grayscale hillshade map. In order to see the visual effects of the hillshading, turn off the `gtopo30_shade` map, then turn it back on.

Using the GRASS shell

The GRASS plugin in QGIS is designed for users who are new to GRASS, and not familiar with all the modules and options. As such, some modules in the toolbox do not show all the options available, and some modules do not appear at all. The GRASS shell (or console) gives the user access to those additional GRASS modules that do not appear in the toolbox tree, and also to some additional options to the modules that are in the toolbox with the simplest default parameters. This example demonstrates the use of an additional option in the **r.shaded.relief** module that was shown above.

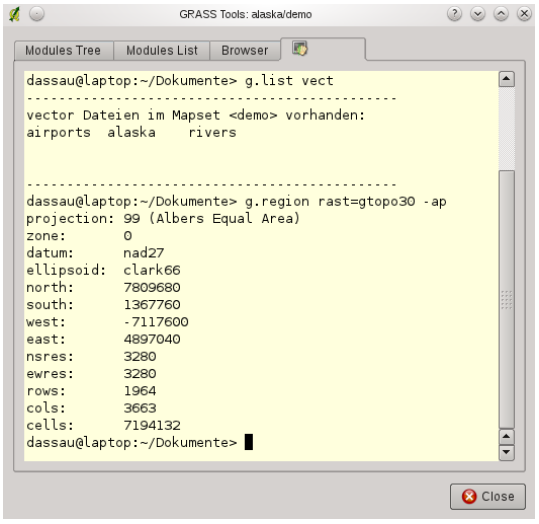


Figure 15.13: The GRASS shell, r.shaded.relief module

The module **r.shaded.relief** can take a parameter `zmult` which multiplies the elevation values relative to the X-Y coordinate units so that the hillshade effect is even more pronounced.

- Load the `gtopo30` elevation raster as above, then start the GRASS toolbox and click on the GRASS shell. In the shell window type the command `r.shaded.relief map=gtopo30 shade=gtopo30_shade2 azimuth=315 zmult=3` and press **[Enter]**.

- After the process finishes shift to the *Browse* tab and double click on the new `gtopo30_shade2` raster to display in QGIS.
- As explained above, shift the shaded relief raster below the gtopo30 raster in the Table of Contents, then check transparency of the colored gtopo30 layer. You should see that the 3D effect stands out more strongly compared to the first shaded relief map.

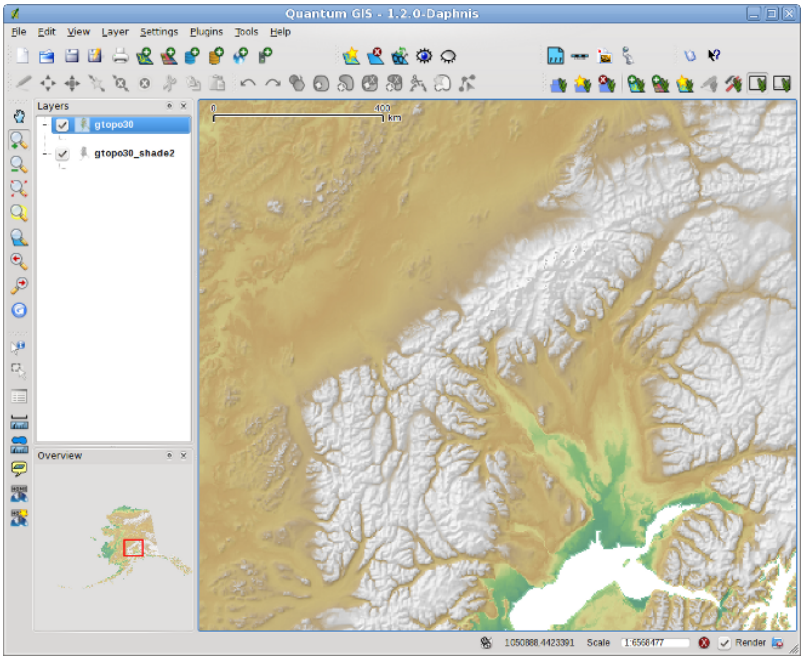


Figure 15.14: Displaying shaded relief created with the GRASS module r.shaded.relief

Raster statistics in a vector map

The next example shows how a GRASS module can aggregate raster data and add columns of statistics for each polygon in a vector map.

- Again using the Alaska data, refer to *Importing data into a GRASS LOCATION* to import the trees shapefile from the `shapefiles` directory into GRASS.
- Now an intermediary step is required: centroids must be added to the imported trees map to make it a complete GRASS area vector (including both boundaries and centroids).
- From the toolbox choose *Vector* → *Manage features*, and open the module **v.centroids**.
- Enter as the *output vector map* 'forest_areas' and run the module.
- Now load the `forest_areas` vector and display the types of forests - deciduous, evergreen, mixed - in different colors: In the layer *Properties* window, *Symbology* tab, choose from *Legend type*  'Unique value' and set the *Classification field* to 'VEGDESC'. (Refer to the explanation of the symbology tab :ref:sec_symbology in the vector section).
- Next reopen the GRASS toolbox and open *Vector* → *Vector update* by other maps.
- Click on the **v.rast.stats** module. Enter `gtopo30`, and `forest_areas`.
- Only one additional parameter is needed: Enter *column prefix* `elev`, and click **[run]**. This is a computationally heavy operation which will run for a long time (probably up to two hours).
- Finally open the `forest_areas` attribute table, and verify that several new columns have been added including `elev_min`, `elev_max`, `elev_mean` etc. for each forest polygon.

15.9.3 Working with the GRASS LOCATION browser

Another useful feature inside the GRASS Toolbox is the GRASS `LOCATION` browser. In figure_grass_module_7 you can see the current working `LOCATION` with its `MAPSETs`.

In the left browser windows you can browse through all `MAPSETs` inside the current `LOCATION`. The right browser window shows some meta information for selected raster or vector layers, e.g. resolution, bounding box, data source, connected attribute table for vector data and a command history.

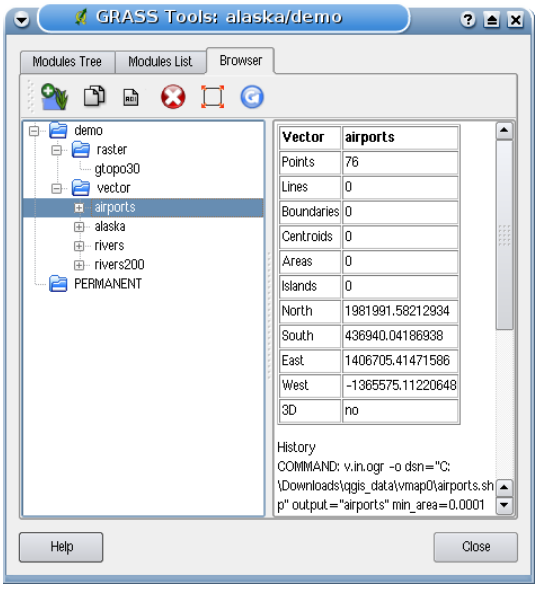


Figure 15.15: GRASS LOCATION browser

The toolbar inside the *Browser* tab offers following tools to manage the selected `LOCATION`:

- *Add selected map to canvas*
- *Copy selected map*
- *Rename selected map*
- *Delete selected map*
- *Set current region to selected map*
- *Refresh browser window*

The *Rename selected map* and *Delete selected map* only work with maps inside your currently selected `MAPSET`. All other tools also work with raster and vector layers in another `MAPSET`.

15.9.4 Customizing the GRASS Toolbox

Nearly all GRASS modules can be added to the GRASS toolbox. A XML interface is provided to parse the pretty simple XML files which configures the modules appearance and parameters inside the toolbox.

A sample XML file for generating the module `v.buffer` (v.buffer.qgm) looks like this:

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE qgisgrassmodule SYSTEM "http://mrcc.com/qgisgrassmodule.dtd">

<qgisgrassmodule label="Vector buffer" module="v.buffer">
```

```
        <option key="input" typeoption="type" layeroption="layer" />
        <option key="buffer"/>
        <option key="output" />
</qgisgrassmodule>
```

The parser reads this definition and creates a new tab inside the toolbox when you select the module. A more detailed description for adding new modules, changing the modules group, etc. can be found on the QGIS wiki at http://hub.qgis.org/projects/quantum-gis/wiki/Adding_New_Tools_to_the_GRASS_Toolbox

CHAPTER 16

OpenStreetMap

In recent years, the OpenStreetMap project has gained popularity because in many countries no free geodata such as digital roadmaps are available. The objective of the OSM project is to create a free editable map of the world from GPS data, aerial photography or from local knowledge. To support this objective, QGIS provides a plugin that enables its users to work with OSM data.

The OpenStreetMap plugin, a core QGIS plugin, provides the basic functionalities for OSM data manipulation; this includes data loading, importing, saving, downloading, editing and uploading data back to the OpenStreetMap server. While implementing the OSM plugin an inspiration was taken from existing OSM data editors. The purpose was to combine their functionalities to get the best possible result.

The following section gives a brief introduction to principles of the OSM project.

Parts of the following paragraphs are copied from the OpenStreetMap web site at http://www.openstreetmap.org.

16.1 The OpenStreetMap project

OpenStreetMap is a project to create a free editable map of the world. The maps are created using data from portable GPS devices, aerial photography, other free sources or simply from local knowledge. The project was started because most maps have legal or technical restrictions on their use, restricting people from using them in creative, productive, or unexpected ways. Both rendered images and the vector dataset of OSM are available for download under a Creative Commons Attribution ShareAlike 2.0 license.

OpenStreetMap was inspired by sites such as Wikipedia - the map display (see Figure_OpenStreetMap_1) features a prominent *Edit* tab and a full revision history is maintained. Registered users can upload GPS track logs and edit the vector data using the given editing tools.

OSM data primitive is an object class that can be stored via the API in the server. The three supported types of data are: **Node**, **Way** and **Relation**.

- **A node** is a latitude/longitude pair of coordinates. It is used as building a block for other features and as a feature itself (Points Of Interest), if they are tagged as required.
- **A way** is a list of at least two nodes that describe a linear feature such as a street, or similar. Nodes can be members of multiple ways.
- **A relation** is a group of zero or more primitives with associated roles. It is used to specify relationships between objects, and may also model an abstract object.

Several different logical features in a common map 'Point Of Interest', 'Street', 'Tram Line', 'Bus Stop' etc.) are defined by these primitives. Map features are well-known in the OSM community and are stored as tags, based on a key and a value. OSM is usually distributed in XML format. XML payload is used for the communication with the OSM server as well.

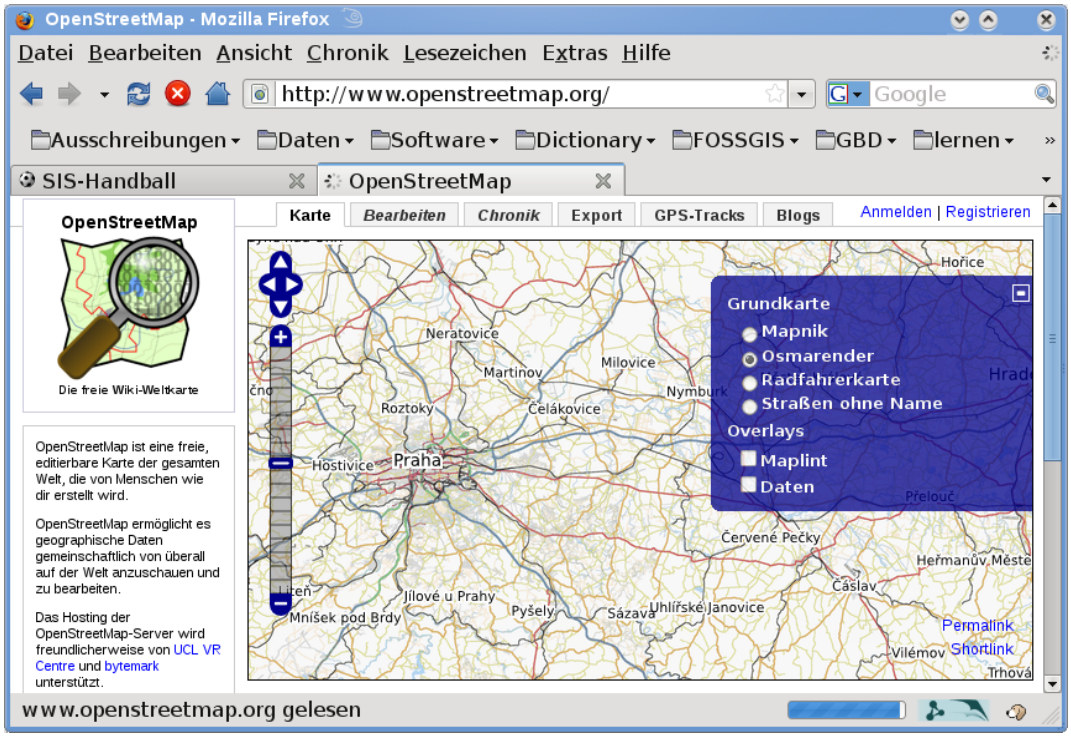


Figure 16.1: OpenStreetMap data in the web

16.2 QGIS - OSM Connection

The first part of this section describes how OSM data primitives are displayed in QGIS vector layers. As previously mentioned, OSM data consists of Nodes, Ways and Relations. In QGIS, they are displayed in three different layer types: Point layer, Line layer and Polygon layer. It is not possible to remove any of these layers and work with the other ones.

- A **Point layer** displays all features of type Node that stands alone. That means that only Nodes that are not included in any Way belongs to the Point layer.
- A **Line layer** displays those OSM features of type Way that are not closed. That means, none of these Ways starts and ends with the same Node.
- A **Polygon layer** displays all Ways that are not included in Line layer.

OpenStreetMap has one more data primitive in addition to the three mentioned above. This is called **Relation**. There is purposely no vector layer to display Relations. A Relation defines a connection between any number of data primitives. After a Point, Line or Polygon is identified on a map, the plugin shows a list of all relations which the identified feature is part of.

It was challenging to design the connection between OSM data and the standard QGIS editing tools. These tools are made to edit a single vector layer at a time, no matter of what feature types it displays. This means that if OSM data are loaded to QGIS through the plugin, you could (theoretically) edit the Point layer, Line layer or Polygon layer with these standard tools separately.

A Line layer consists of two different types of OSM features, Ways and Nodes. In OSM format, a Way is composed of Nodes. If you start editing a Line layer and change the shape of some line, your action affects not only the OSM Way, but also the OSM Nodes that are part of it.

QGIS standard editing tools cannot tell the OSM provider, which members of which line has changed and how. It can tell only what's the new geometry of which line, and that's not enough to propagate changes to the OSM database correctly. The Line layer does also not know the identifiers of the line members. The same problem occurs when you try to edit the Polygon layer.

For this reason, the OSM plugin need its own tools for editing OSM data. While they are used, the OSM layers can be changed correctly. The Plugin editing tools consists of tools for Point, Line, Polygon and Relation creation, deletion and moving.

주석: To create a connection between the OSM plugin and standard editing tools, changes in QuantumGIS core code would be necessary.

16.3 Installation

The OpenStreetMap plugin is a core plugin inside QGIS. The 'OpenStreetMap' plugin can be selected in the Plugin Manager as described in section *Loading a QGIS Core Plugin*.

16.4 Basic user interface

The first time the OSM plugin is started (and after the first data are loaded), several new OSM plugin icons appear in the QGIS toolbar menu together with new dock windows as shown in figure_OpenStreetMap_2.

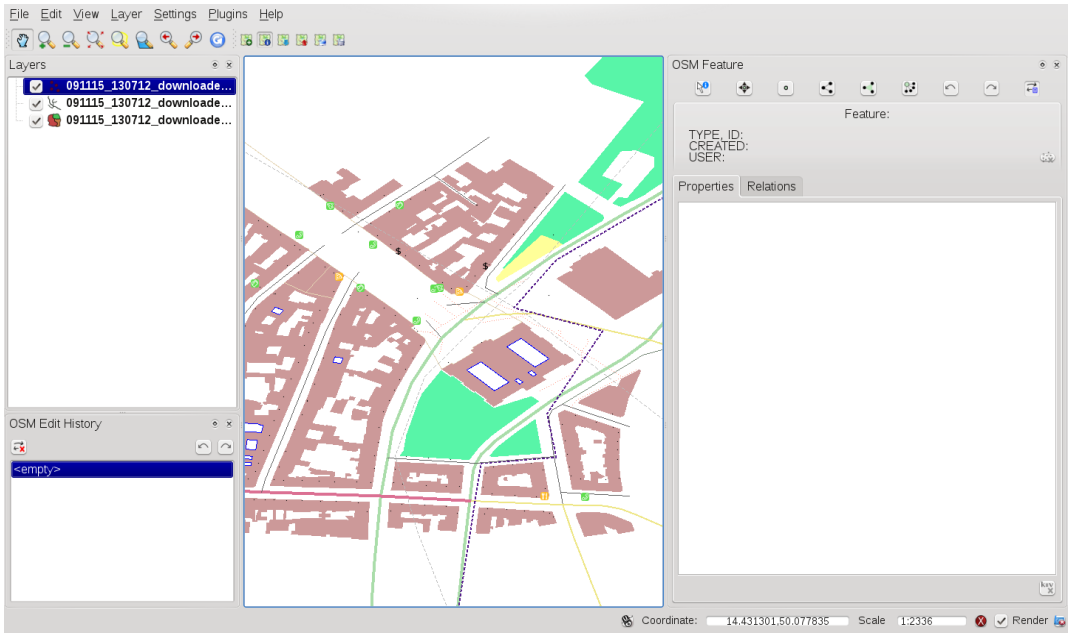


Figure 16.2: OSM plugin user interface

16.4.1 OSM Features widget

The OSM Feature widget helps to identify OSM features. It shows basic information on the feature type and identifier as well as information on who has changed a feature, and when. The OSM Feature widget also provides all editing tools (in the top part of it). More information on those tools can be found in the sections below. The widget is initially disabled. It activates itself after successful loading some OSM data.

16.4.2 OSM Undo/Redo widget

This Undo/Redo widget is used to undo and redo edit actions. It consists not only a classic Undo and Redo button, but also shows a list with a brief description of the edit actions that were done. The OSM Undo/Redo widget is initially closed. You can show it using a button on the OSM Feature widget.

16.4.3 Toolbar menu icons

 Load OSM from file is used to load data from a special OpenStreetMap XML file.

 Show/Hide OSM Feature Manager is used to show or hide the OSM Feature widget. The OSM Feature widget is a panel that helps with OSM feature identification and with OSM data editing.

 Download OSM data is used to download data from the OpenStreetMap server.

 Upload OSM data is used to upload changes (on current data).

 Import data from a layer is used to import data from a vector layer. At least one vector layer must be loaded and current OSM data must be selected.

 Save OSM to file is used to save OSM data back to an XML file.

More detailed information on all the widgets, buttons and dialogs can be found in appropriate sections of this plugin section according to their functionality (editing, identification, etc.).

16.5 Loading OSM data

The first action that should be done after starting the OSM Plugin is opening data from an OSM file. OSM data can be import as shapefile or downloaded directly from the OpenStreetMap server. Here we are focusing on the first mentioned method.

To load data from a file use the  Load OSM from file icon. If there is no such button, maybe someone disabled OpenStreetMap toolbar in your QGIS installation. You can enable it again selecting *Settings* → *Toolbars* → *OpenStreetMap*.

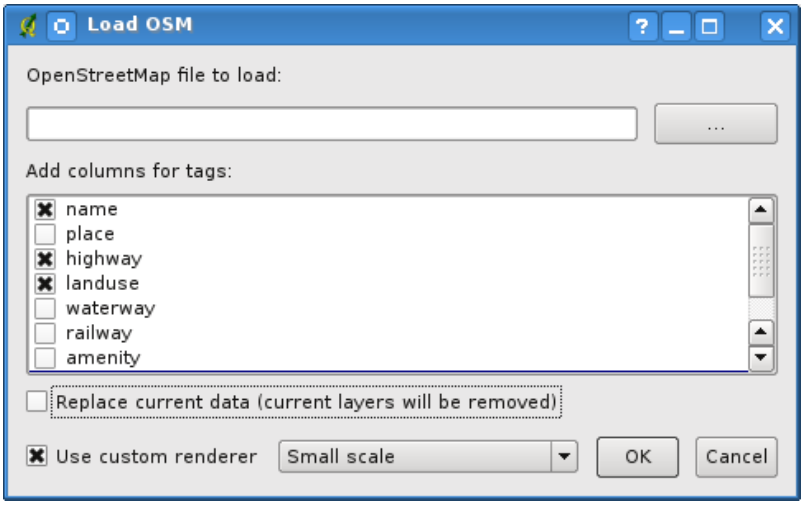


Figure 16.3: Load OSM data dialog

The purpose of its elements is explained below.

OpenStreetMap file to load: Click on the  button to select the file:.*osm* file you want to load data from.

Add columns for tags: This option determines a connection between OSM and QGIS data. Each feature of OSM data has some tags (pairs of key and value), that define the feature properties. Each feature of a QGIS vector layer also has its attributes (key and value). With this option you can define which properties of OSM objects should be visible when displaying detailed information about QGIS features.

Replace current data: Checking this option means that new data should replace current data the user is working with. Layers of current data will be removed and new ones will be loaded. When loading OSM data for the first time, this option is not active, because there is nothing to replace.

Use custom renderer: This option determines how many details of the map will be used. There are three pre-defined OSM styles for map displaying. Use 'Small scale' if you want to view OSM data at low level, to see all details and to edit something. If not you can use 'Medium scale' or 'Large scale'. QGIS 1.8.0 doesn't support changing the renderer style dynamically.

Click **[OK]** to load your data. If this is the first time the OSM file is loaded, the plugin must first parse the database. This may take few seconds or minutes - it depends on the amount of loaded data.

16.6 Viewing OSM data

After the OSM data are loaded, you can identify map features using the appropriate tool. Use the  Identify feature button on the top-left of the OSM Feature widget. Using this tool you can easily explore all map objects. When the mouse cursor is placed over an object, you can see all information on it directly in the OSM Feature widget. There is also a dynamic rubberband displayed on the map so that the user is able to determine which feature is currently identified.

The *Properties* tab of the widget contains of all feature tags. Clicking on the *Relation* tab shows you a list of all relations connected with identified feature.

If you want to hold a feature for a while to be able to read its properties and relations, move the mouse cursor at the same time, try left-clicking while you are over the feature. Identification process will stop until next left-clicking.

Sometimes there is more than one feature at a point where left-clicking was performed. This happens especially when clicking on cross-roads or if you did not zoom enough into the map. In this situation only one of such features is identified (and marked with the rubberband) but the plugin remembers all of them. Then (still in the pause mode) you can cycle through the identified features by right-clicking.

16.7 Editing basic OSM data

'Basic data' in this context means non-relational OSM features - nodes and ways. If you prefer to examine how to perform relational editing, skip this section and move on to the next one.

Basic data editing is a key part of the OSM Plugin. You can change the property, position or shape of any existing basic feature. You can remove features or add new ones. All changes on nodes and ways are remembered by Undo/Redo all changes can be easily uploaded to the OpenStreetMap server.

16.7.1 Changing feature tags

Changing the property/tag of an OSM feature can be done directly in the table of feature tags. The Tags table of basic features can be found on the OSM Feature widget. Don't forget to identify feature first.

If you want to change a tag value, just double-click in the appropriate row of column 'Value' and type, or select a new value. If you want to remove a tag, click in the relevant row, then use the button  Remove selected tags on the right bottom under the table.

To add new tags just type the key and value into the last row of the table where '<next tag value>' is written. Notice that you cannot change the key of an existing tag pair. Conveniently, there are some combo boxes of all existing tag keys and their typical values.

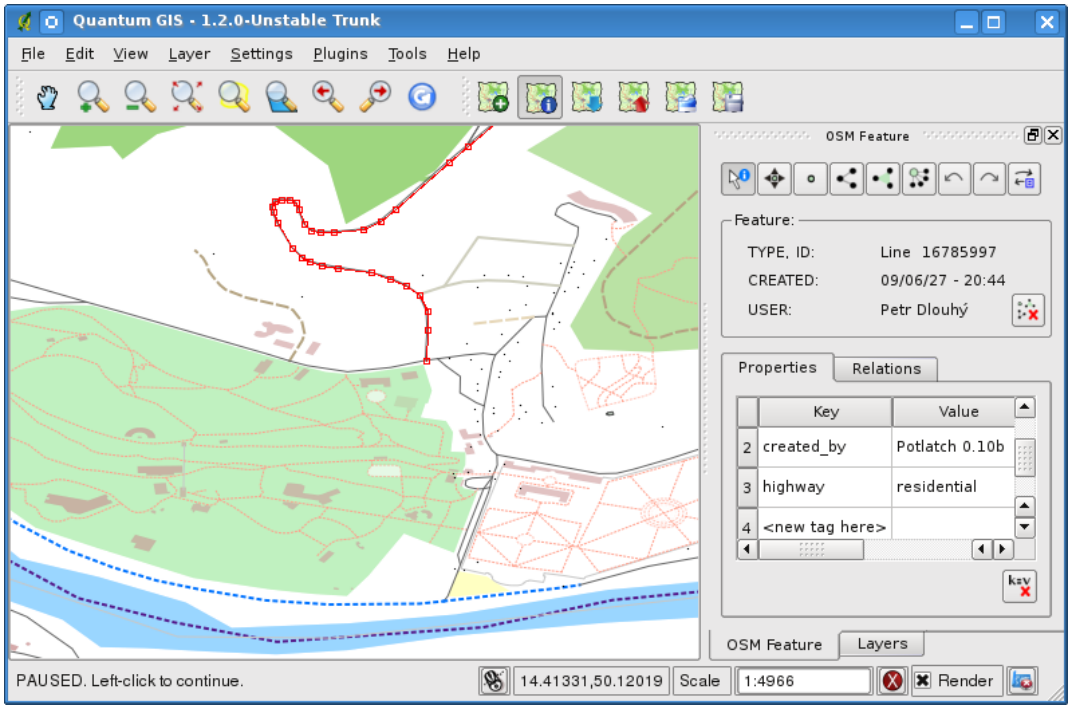


Figure 16.4: Changing an OSM feature tag

16.7.2 Point creation

For point creation there is a Create point button on the OSM Feature widget. To create some points, just click on the button and start clicking on the map. If your cursor is over some map feature, the feature is marked/identified immediately. If you click on the map when a line or polygon is marked, a new point is created directly on such line or polygon as its new member. If your cursor is over an existing point, a new point cannot be created. In such case the OSM plugin will show following message:

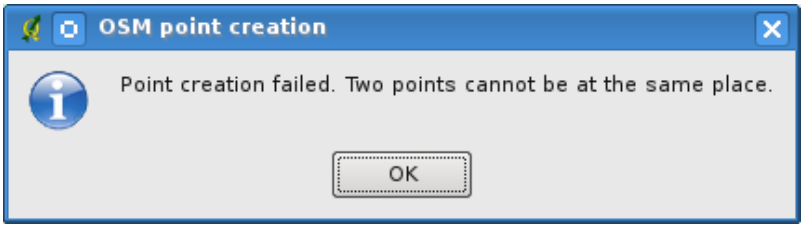


Figure 16.5: OSM point creation message

The mechanism of helping a user to hit the line or polygon is called snapping and is enabled by default. If you want to create a point very close to some line (but not on it) you must disable snapping by holding the `Ctrl` key first.

16.7.3 Line creation

For line creation, there is a Create Line button on the OSM Feature widget. To create a line just click the button and start left-clicking on the map. Each of your left-clicks is remembered as a vertex of the new line. Line creation ends when the first right-click is performed. The new line will immediately appear on the map.

주석: A Line with less than two members cannot be created. In such case the operation is ignored.

Snapping is performed to all map vertices - points from the Point vector layer and all Line and Polygon members. Snapping can be disabled by holding the `Ctrl` key.

16.7.4 Polygon creation

For polygon creation there is a  Create polygon button on the OSM Feature widget. To create a polygon just click the button and start left-clicking on the map. Each of your left-clicks is remembered as a member vertex of the new polygon. The Polygon creation ends when first right-click is performed. The new polygon will immediately appear on the map. Polygon with less than three members cannot be created. In such case operation is ignored. Snapping is performed to all map vertexes - points (from Point vector layer) and all Line and Polygon members. Snapping can be disabled by holding the `Ctrl` key.

16.7.5 Map feature moving

If you want to move a feature (no matter what type) please use the  Move feature button from the OSM Feature widget menu. Then you can browse the map (features are identified dynamically when you go over them) and click on the feature you want to move. If a wrong feature is selected after your click, don't move it from the place. Repeat right-clicking until the correct feature is identified. When selection is done and you move the cursor, you are no more able to change your decision what to move. To confirm the move, click on the left mouse button. To cancel a move, click another mouse button.

If you are moving a feature that is connected to another features, these connections won't be damaged. Other features will just adapt themselves to a new position of a moved feature.

Snapping is also supported in this operation, this means:

- When moving a standalone (not part of any line/polygon) point,snapping to all map segments and vertices is performed.
- When moving a point that is a member of some lines/polygons,snapping to all map segments and vertices is performed, except for vertices of point parents.
- When moving a line/polygon, snapping to all map vertices is performed. Note that the OSM Plugin tries to snap only to the 3 closest-to-cursor vertices of a moved line/polygon, otherwise the operation would by very slow. Snapping can be disabled by holding `Ctrl` key during the operation.

16.7.6 Map feature removing

If you want to remove a feature, you must identify it first. To remove an identified feature, use the  Remove this feature button on the OSM Feature widget. When removing a line/polygon, the line/polygon itself is deleted, so are all its member points that doesn't belong to any other line/polygon.

When removing a point that is member of some lines/polygons, the point is deleted and the geometries of parent lines/polygons are changed. The new parent geometry has less vertices than the old one.

If the parent feature was a polygon with three vertexes, its new geometry has only two vertexes. And because there cannot exist polygon with only two vertices, as described above, the feature type is automatically changed to Line.

If the parent feature was a line with two vertexes, its new geometry has only one vertex. And because there cannot exist a line with only one vertex, the feature type is automatically changed to Point.

16.8 Editing relations

Thanks to existence of OSM relations we can join OSM features into groups and give them common properties - in such way we can model any possible map object: borders of a region (as group of ways and points), routes of a

bus, etc. Each member of a relation has its specific role. There is a pretty good support for OSM Relations in our plugin. Let's see how to examine, create, update or remove them.

16.8.1 Examining relation

If you want to see relation properties, first identify one of its members. After that open the *Relations* tab on the OSM Feature widget. At the top of the tab you can see a list of all relations the identified feature is part of. Please choose the one you want to examine and look at its information below. In the first table called 'Relation tags' you find the properties of the selected relation. In the table called 'Relation members' you see brief information on the relation members. If you click on a member, the plugin will make a rubberband on it in the map.

16.8.2 Relation creation

There are 2 ways to create a relation:

1. You can use the  Create relation button on OSM Feature widget.
2. You can create it from the *Relation* tab of OSM Feature widget using the  Add relation button.

In both cases a dialog will appear. For the second case, the feature that is currently identified is automatically considered to be the first relation member, so the dialog is prefilled a little. When creating a relation, please select its type first. You can select one of predefined relation types or write your own type. After that fill the relation tags and choose its members.

If you have already selected a relation type, try using the  Generate tags button. It will generate typical tags to your relation type. Then you are expected to enter values to the keys. Choosing relation members can be done either by writing member identifiers, types and roles or using the  Identify tool and clicking on map.

Finally when type, tags and members are chosen, the dialog can be submitted. In such case the plugin creates a new relation for you.

16.8.3 Changing relation

If you want to change an existing relation, identify it first (follow steps written above in Section *Examining relation*). After that click on the  Edit relation button. You will find it on the OSM Feature widget. A new dialog appears, nearly the same as for the 'create relation' action. The dialog is pre-filled with information on given relations. You can change relation tags, members or even its type. After submitting the dialog your changes will be committed.

16.9 Downloading OSM data

To download data from OpenStreetMap server click on the  Download OSM data button. If there is no such button, the OSM toolbar may be disabled in your QGIS instalation. You can enable it again at *Settings* → *Toolbars* → *OpenStreetMap* . After clicking the button a dialog appears and provides following functionalities:

Extent: Specifies an area to download data from intervals of latitude and longitude degrees. Because there is some restriction of OpenStreetMap server on how much data can be downloaded, the intervals must not be too wide. More detailed info on extent specification can is shown after clicking the  Help button on the right.

Download to: Here you are expected to write a path to the file where data will be stored. If you can't remember the structure of your disk, don't panic. The browse button  will help you.

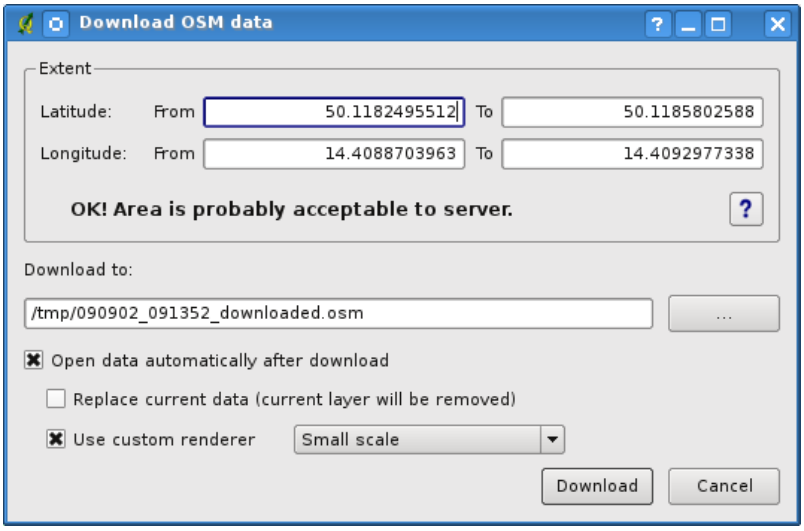


Figure 16.6: OSM download dialog

Open data automatically after download: Determines, if the download process should be followed by loading the data process or not. If you prefer not to load data now, you can do it later by using the Load OSM from file button.

Replace current data: This option is active only if *Open data automatically after download* is checked. Checking this option means that downloaded data should replace current data we are working with now. Layers of the current data will be removed and new ones will be loaded. When starting QGIS and downloading OSM data for the first time, this option is initially inactive, because there is nothing to replace.

Use custom renderer: This option is active only if the *Open data automatically after download* checkbox is checked. It determines how many details will be in the map. There are three predefined OSM styles for map displaying. Use 'Small scale' if you want to view OSM data at low level, to see all details and to edit something. If not you can use 'Medium scale' or 'Large scale'. QGIS 1.8.0 does not support changing the renderer style dynamically.

Click the **[Download]** button to start the download process.

A progress dialog will continuously inform you about how much of data is already downloaded. When an error occurs during the download process, a dialog tells you why. When action finishes successfully both the progress dialog and download dialog will close themselves.

16.10 Uploading OSM data

Note that the upload is always done on current OSM data. Before opening the OSM Upload dialog, please be sure that you really have the right active layer with OSM data.

To upload current data to the OSM server click on the Upload OSM data button. If there is no such button, OSM toolbar in your QGIS installation is disabled. You can enable it again in *Settings* → *Toolbars* → *OpenStreetMap* . After clicking the **[Upload]** button a new dialog will appear.

At the top of the dialog you can check, if you are uploading the correct data. There is a short name of a current database. In the table you find information on how many changes will be uploaded. Statistics are displayed separately for each feature type.

In the *Comment on your changes* box you can write brief information on meaning of your upload operation. Just write in brief what data changes you've done or let the box empty. Fill 'OSM account' arrays so that the server could authenticate you. If you don't have an account on the OSM server, it's the best time to create one at http://www.openstreetmap.org. Finally use **[Upload]** to start an upload operation.

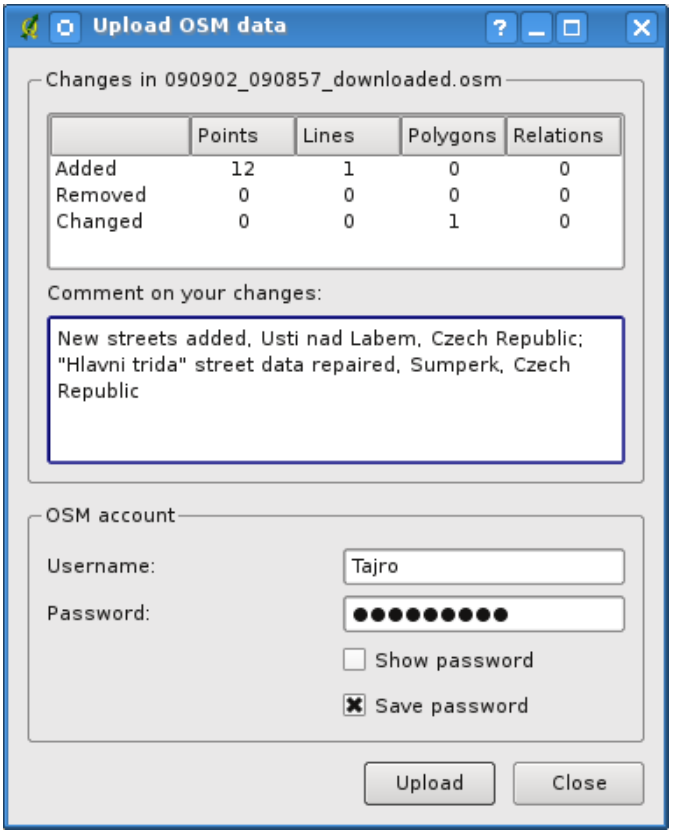


Figure 16.7: OSM upload dialog

16.11 Saving OSM data

To save data from a current map extent to an XML file click on the  Save OSM to file button. If there is no such button, the OSM toolbar in your QuantumGIS installation is probably disabled. You can enable it again in *Settings* → *Toolbars* → *OpenStreetMap*. After clicking on the button a new dialog appears.

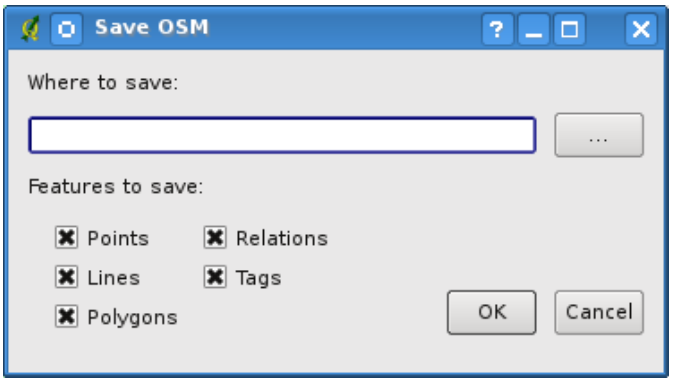


Figure 16.8: OSM saving dialog

Select features you want to save into XML file and the file itself. Use the **[OK]** button to start the operation. The process will create an XML file, in which OSM data from your current map extent are represented. The OSM version of the output file is 0.6. Elements of OSM data (<node>, <way>, <relation>) do not contain information on their changesets and uids. This information are not compulsory yet, see DTD for OSM XML version 0.6. In the output file OSM elements are not ordered.

Notice that not only data from the current extent are saved. Into the output file the whole polygons and lines are

saved even if only a small part of them is visible in the current extent. For each saved line/polygon all its member nodes are saved too.

16.12 Import OSM data

To import OSM data from an opened non-OSM vector layer follow this instructions. Choose current OSM data by clicking on one of their layers. Click on the  Import data from a layer button. If there is no such button, someone has probably disabled the OpenStreetMap toolbar in your QGIS installation. You can enable it again in *Settings* → *Toolbars* → *OpenStreetMap*.

After clicking on the button following message may show up:

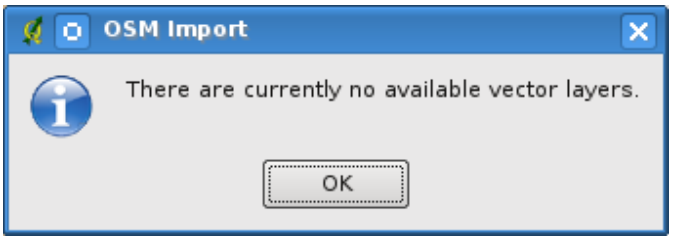


Figure 16.9: OSM import message dialog

In such case there is no vector layer currently loaded. The import must be done from a loaded layer - please load a vector layer from which you want to import data. After a layer is opened, your second try should give you a better result (don't forget to mark the current OSM layer again):

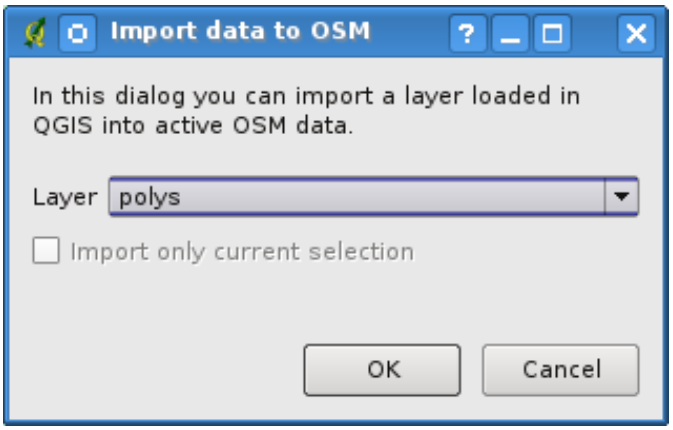


Figure 16.10: Import data to OSM dialog

Use the submit dialog to start the process of OSM data importing. Reject it if you are not sure you want to import something.

CHAPTER 17

SEXTANTE

17.1 Introduction

This chapter introduces SEXTANTE, the powerful geospatial analysis framework of QGIS. SEXTANTE is a geoprocessing environment that can be used to call native and third party algorithms from QGIS, making your spatial analysis tasks more productive and easy to accomplish.

In the following sections we will review how to use the graphical elements of SEXTANTE and take the most out of each one of them.

17.1.1 Basic elements of the SEXTANTE GUI

There are four basic elements in the SEXTANTE GUI, which are used to run SEXTANTE algorithms for different purposes. Choosing one tool or another will depend on the kind of analysis that is to be performed and the particular characteristics of each user and project. All of them (except for the batch processing interface, which is called from the toolbox, as we will see) can be accessed from the *Analysis* menu item (you will see more than four entries. The remaining ones are not used to execute algorithms and will be explained later in this chapter).

- The SEXTANTE toolbox. The main element of the SEXTANTE GUI, it is used to execute a single algorithm or run a batch process based on that algorithm.

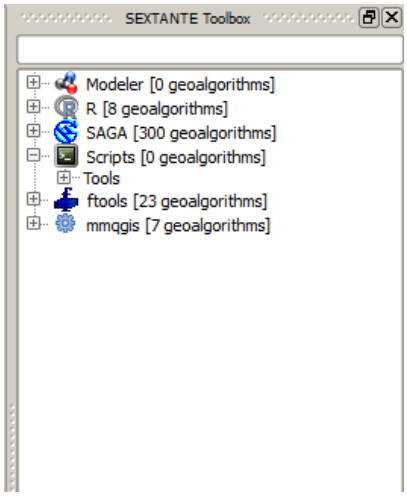


Figure 17.1: SEXTANTE Toolbox

- The SEXTANTE graphical modeler. Several algorithms can be combined graphically using the modeler to define a workflow, creating a single process that involves several sub-processes
- The SEXTANTE history manager. All actions performed using any of the aforementioned elements are stored in a history file and can be later easily reproduced using the history manager

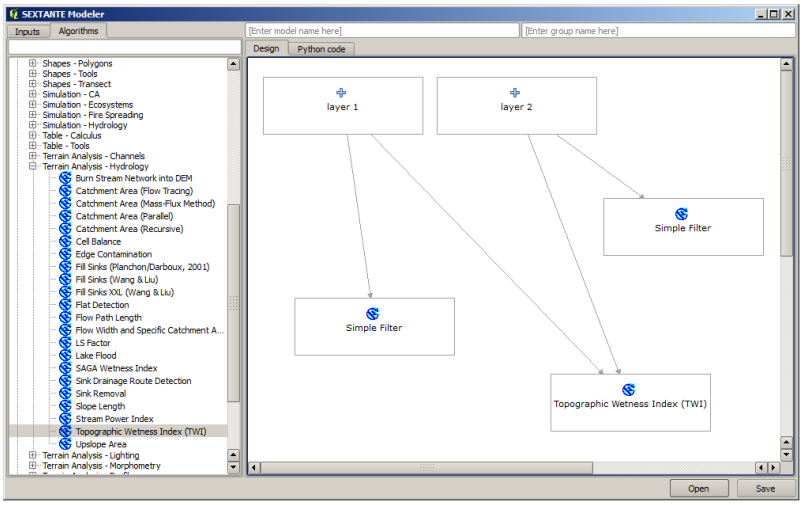


Figure 17.2: SEXTANTE Models

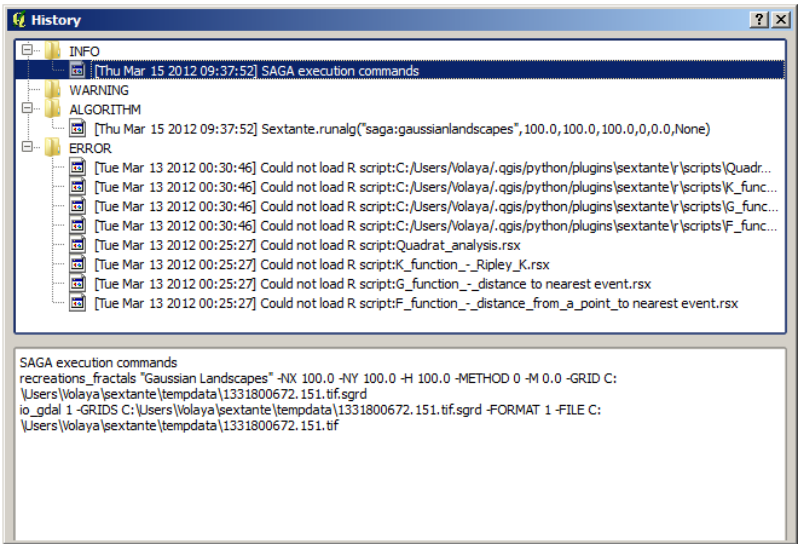


Figure 17.3: SEXTANTE History

- The SEXTANTE batch processing interface manager. This interface allows you to execute batch processes and automate the execution of a single algorithm on multiple datasets.

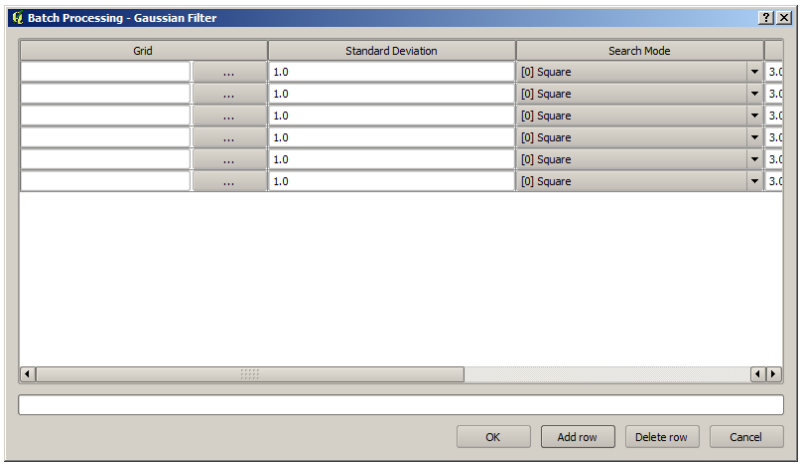


Figure 17.4: SEXTANTE Batch Processing

Along the following sections we will review each one of this elements in detail.

17.2 The SEXTANTE toolbox

17.2.1 Introduction

The *Toolbox* is the main element of the SEXTANTE GUI, and the one that you are more likely to use in your daily work. It shows the list of all available algorithms grouped in different blocks, and is the access point to run them whether as a single process or as a batch process involving several executions of a same algorithm on different sets of inputs.

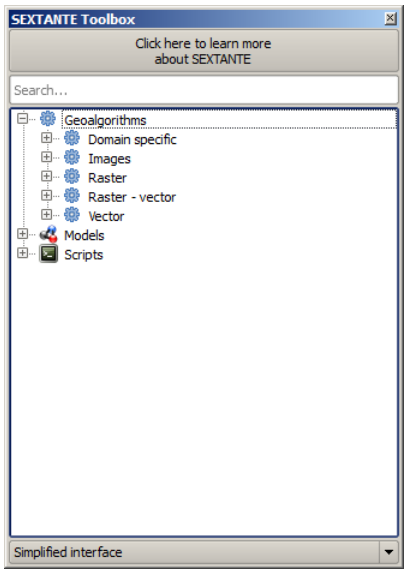


Figure 17.5: SEXTANTE Toolbox

The toolbox contains all the available algorithms, divided into predefined groups. All these groups are found under a single tree entry named *Geoalgorithms*.

Additionally, two more entries are found, namely 'Models' and 'Scripts'. These include user-created algorithms, and allow you to define your own workflows and processing tasks. We will devote a full section to them a bit later.

In the upper part of the toolbox you can find a text box. To reduce the number of algorithms shown in the toolbox and make it easier to find the one you need, you can enter any word or phrase on the text box. Notice that, as you type, the number of algorithms in the toolbox is reduced to just those which contain the text you have entered in their names.

In the lower part you will find a box that allows you to switch between the simplified algorithm list (the one explained above), and the advanced list. If you change to the advanced mode, the toolbox will look like this:

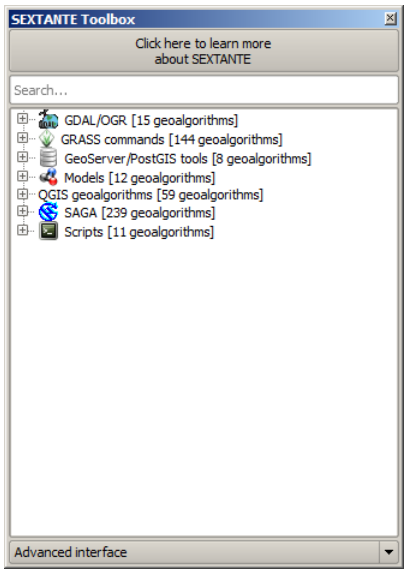


Figure 17.6: SEXTANTE Toolbox (advanced mode)

In the advanced view, each group represents a so-called 'algorithm provider', which is a set of algorithms coming from the same source, for instance, from a third-party application with geoprocessing capabilities. Some of this groups represent algorithms from one of such third-party applications (like SAGA, GRASS or R), while other contain algorithms directly coded along with SEXTANTE elements, not relying on any additional software.

This view is recommended to those users that have a certain knowledge of the applications that are backing those algorithms, since they will be shown with their original names and groups.

Also, some additional algorithms are available in the advanced view, such as LiDAR tools or scripts based on the R statistical computing software, among others. Independent QGIS plugins that add new algorithms to SEXTANTE will only be shown in the advanced view.

In particular, the simplified view contains algorithms from the following providers,

- GRASS
- SAGA
- OTB
- Native QGIS algorithms

  In the particular case of runnign QGIS under Windows. These algorithms are fully-functional in a fresh installation of QGIS and they can be run without requiring any additional installation. Also running them requires no prior knowledge of the external applications they use, making them more accesible for first-time users.

  If you want to use an algorithm not provided by the any of above providers, switch to the advanced mode by selecting the corresponding option at the bottom of the toolbox.

To execute an algorithm, just double-click on its name in the toolbox.

17.2.2 The algorithm dialog

Once you double-click on the name of the algorithm that you want to execute, a dialog similar to the next one is shown (in this case, the dialog corresponds to the SAGA 'Convergence index' algorithm).

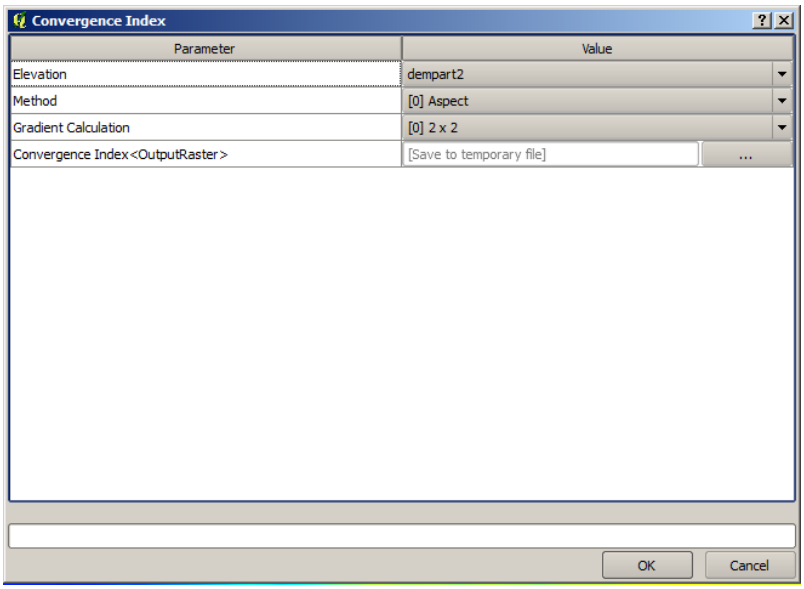


Figure 17.7: Parameters Dialog

This dialog is used to set the input values that the algorithm needs to be executed. It shows a table where input values and configuration parameters are to be set. It, of course, has a different content depending on the requirements of the algorithm to be executed, and is created automatically based on those requirements. On the left side, the name of the parameter is shown. On the right side the value of the parameter can be set.

Although the number and type of parameters depend on the characteristics of the algorithm, the structure is similar for all of them. The parameters found on the table can be of one of the following types.

- A raster layer, to select from a list of all the ones available (currently opened) in QGIS. The selector contains as well a button on its right-hand side, to let you select filenames that represent layers currently not loaded in QGIS.
- A vector layer, to select from a list of all the ones available in the QGIS. Layers not loaded in QGIS can be selected as well, as in the case of raster layers, but only if the algorithm does not require a table field selected from the attributes table of the layer. In that case, only opened layers can be selected, since they need to be open so as to retrieve the list of field names available.

  You will see a button by each vector layer selector. If the algorithm contains several of them, you will be able to toggle just one of them. If the button corresponding to a vector input is toggled, the algorithm will be executed iteratively on each one of its features. We will see more about this kind of execution at the end of this section.
- A table, to select from a list of all the ones available in QGIS. Non-spatial tables are loaded into QGIS like vector layers, and in fact they are treated as such by the program. Currently, the list of available tables that you will see when executing a SEXTANTE algorithm that needs one of them is restricted to tables coming from files in DBase (`.dbf`) or Comma-Separated Values (`.csv`) formats.
- An option, to choose from a selection list of possible options.
- A numerical value, to be introduced in a text box. You will find a button by its side. Clicking on it you will see a dialog that allows you to enter a mathematical expression, so you can use it as a handy calculator. Some useful variables related to data loaded into QGIS can be added to your expression, so you can select a value derived from any of this variables such as the cellsize of a layer or the northern most coordinate of another one.
- A range, with min and max values to be introduced in two text boxes.

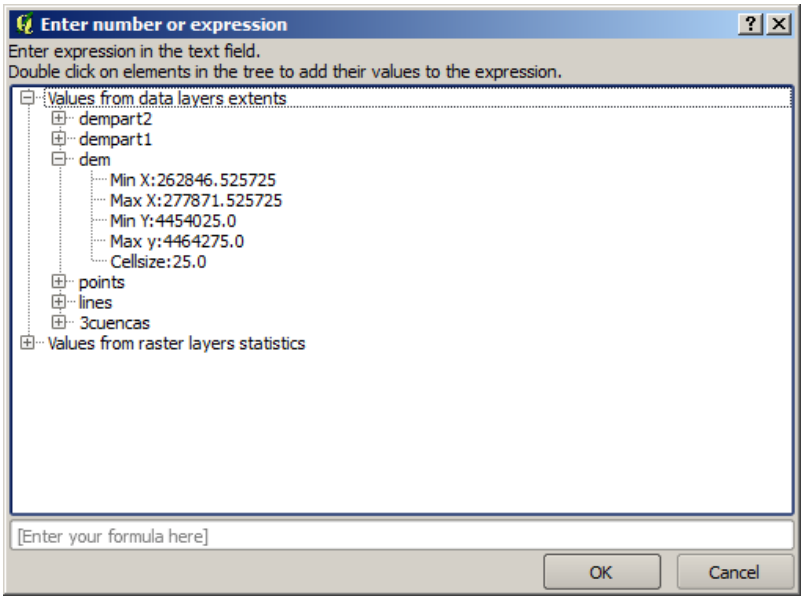


Figure 17.8: Number Selector

- A text string, to be introduced in a text box.
- A field, to choose from the attributes table of a vector layer or a single table selected in another parameter.
- A Coordinate Reference System. You can type the EPSG code directly in the text box, or select it from the CRS selection dialog that appear when you click on the button on the right-hand size
- A extent, to be entered by four number representing its `xmin`, `xmax`, `ymin`, `ymax` limits. Clicking on the button on the right-hand side of the value selector, a pop-up menu will appear, giving you two option: to select the value from a layer or the current canvas extent, or to define it by dragging directly onto the map canvas.

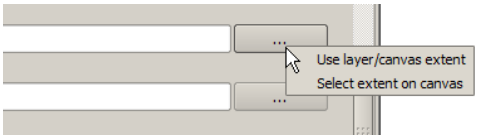


Figure 17.9: SEXTANTE Extent

If you select the first option, you will see a window like the next one.

If you select the second one, the parameters window will hide itself, so you can click and drag onto the canvas. Once you have defined the selected rectangle, the dialog will reappear, containing the values in the extent text box.

- A list of elements (whether raster layers, vector ones or tables), to select from the list of the ones available in QGIS. To make the selection, click on the small button on the left side of the corresponding row to see a dialog like the following one.
- A small table to be edited by the user. These are used to define parameters like lookup tables or convolution kernels, among others.

  Click on the button on the right side to see the table and edit its values.

  Depending on the algorithm, the number of rows can be modified or not, using the buttons on the right side of the window.

You will find a **[Help]** button in the lower part of the parameters dialog. If a help file is available, it will be shown, giving you more information about the algorithms and detailed descriptions of what each parameter does.

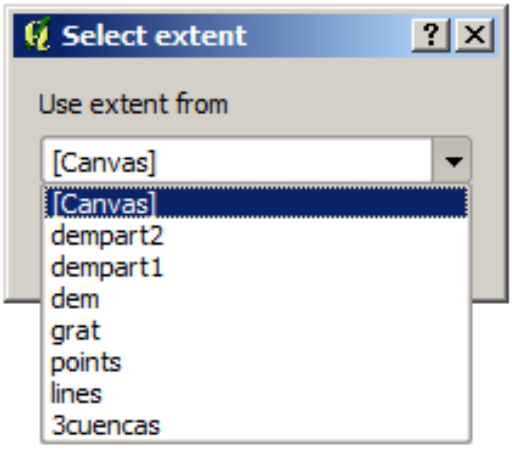


Figure 17.10: SEXTANTE Extent List

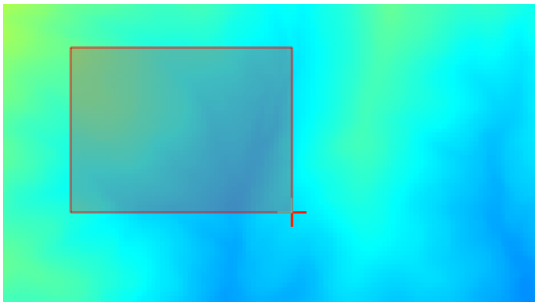


Figure 17.11: Extent Drag

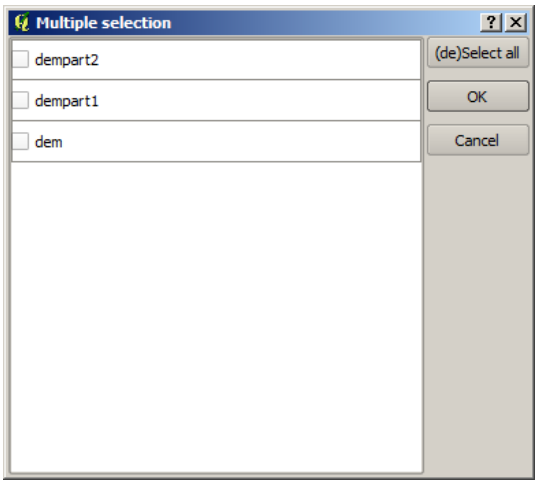


Figure 17.12: Multiple Selection

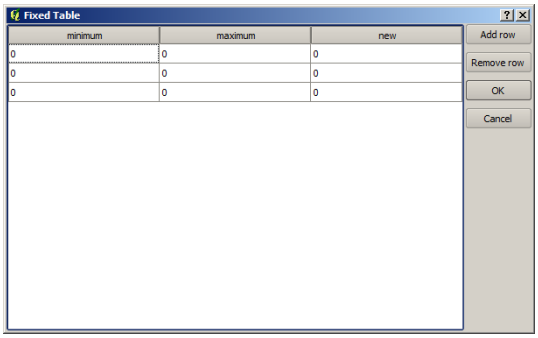


Figure 17.13: Fixed Table

Unfortunately, most algorithms lack good documentation, but if you feel like contributing to the project, this would be a good place to start.

A note on projections

SEXTANTE — and also most of the external applications whose algorithms are available from SEXTANTE — does not perform any reprojection on input layers and assumes that all of them are already in a common coordinate system and ready to be analized. Whenever you use more than one layer as input to an algorithm, whether vector or raster, it is up to you to make sure that they are all in the same coordinate system.

Note that, due to QGIS's on-the-fly reprojecting capabilities, although two layers might seem to overlap and match, that might not be true if their original coordinates are used without reprojecting them onto a common coordinate system. That reprojection should be done manually and then use the resulting files as input to SEXTANTE. Also note that the reprojection process can be performed with SEXTANTE, which incorporates tools to do so.

By default, SEXTANTE will show a description of the CRS of each layer along with its name, making it easy to select layers that share the same CRS to be used as input layers. If you do not want to see this additional information, you can disable this functionality in the SEXTANTE config dialog, unchecking the *Show CRS* option.

If you try to execute an algorithm using as input two or more layers with unmatching CRS's, SEXTANTE will show you the following window:

You still can execute the algorithm, but be aware that in most cases that will produce wrong results, such as empty layers due to input layers not overlapping.

17.2.3 Data objects generated by SEXTANTE algorithms

Data objects generated by SEXTANTE can be of any of the following types:

- A raster layer
- A vector layer
- A table
- An HTML file (used for text and graphical outputs)

They are all saved to disk (there are no in-memory results), and the parameters table will contain a text box corresponding to each one of these outputs, where you can type the output channel to use for saving it. An output channel contains the information needed to save the resulting object somewhere. In the most usual case, you will save it to a file, but the architecture of SEXTANTE allows for any other way of storing it. For instance, a vector layer can be stored in a database or even uploaded to a remote server using a WFS-T service. Although solutions like these are not yet implemented, SEXTANTE is prepared to handle them, and we expect to add new kinds of output channels in a near feature.

To select an output channel, just click on the button on the right side of the text box. That will open a save file dialog, where you can select the desired filepath. Supported file extensions are shown in the file format selector of the dialog, depending on the kind of output and the algorithm.

The format of the output is defined by the filename extension. The supported formats depend on the ones supported by the algorithm itself. To select a format, just select the corresponding file extension (or add it if you are directly typing the filepath instead). If the extension of the filepath you entered does not match any of the supported ones, a default extension (usually `.dbf` for tables, `.tif` for raster layers and `.shp` for vector ones) will be appended to the filepath and the file format corresponding to that extension will be used to save the layer or table.

If you do not enter any filename, the result will be saved as a temporary file and in the corresponding default file format, and will be deleted once you exit QGIS (take care with that in case you save your project and it contains temporary layers).

You can set a default folder for output data objects. Go to the configuration dialog (you can open it from the *Analysis* menu), and in the *General* group you will find a parameter named *Output folder*. This output folder is used as the default path in case you type just a filename with no path (i.e. `myfile.shp`) when executing an algorithm.

Apart from raster layers and tables, SEXTANTE also generates graphics and texts as HTML files. These results are shown at the end of the algorithm execution in a new dialog. This dialog will keep the results produced by SEXTANTE during the current session, and can be shown at any time by selecting the *Analysis* → *SEXTANTE results viewer* from QGIS main menu.

Some external applications might have files (with no particular extension restrictions) as output, but they do not belong to any of the categories above. Those outut files will not be processed by QGIS (opened or included into the current QGIS project), since most of the times correspond to file formats or elements not supported by QGIS. This is, for instance, the case with LAS files used for LiDAR data. The files get created, but you won't see anything new in your QGIS working session.

For all the other types of outputs, you will find a check box that you can use to tell SEXTANTE not whether to load the file once it is generated by the algorithm or not. By default, all files are opened.

SEXTANTE does not support optional outputs, so all outputs are created, but you can uncheck the corresponding check box if you are not interested in a given output, which virtually makes it behave like an optional output (although the layer is created anyway, but if you leave the text box empty, it will be saved to a temporary file and deleted once you exit QGIS)

17.2.4 Configuring SEXTANTE

As it has been mentioned, the configuration menu gives access to a new dialog where you can configure how SEXTANTE works. Configuration parameters are structured in separate blocks that you can select on the left-hand side of the dialog.

Along with the aforementioned *Output folder* entry, the *General* block contains parameters for setting the default rendering style for SEXTANTE layers (that is, layers generated by using algorithms from any of the SEXTANTE components). Just create the style you want using QGIS, save it to a file, and then enter the path to that file in the settings so SEXTANTE can use it. Whenever a layer is loaded by SEXTANTE and added to the QGIS canvas, it will be rendered with that style.

Rendering styles can be configured individually for each algorithm and each one of its outputs. Just right-click on the name of the algorithm in the toolbox and select *Edit rendering styles*. You will see a dialog like the one shown next.

Select the style file (`.qml`) that you want for each output and press **[OK]**.

Other configuration parameters in the *General* group are the following ones:

- *Use filename as layer name*. The name of each resulting layer created by SEXTANTE is defined by the algorithm generating it. In some cases, a fixed name might be used, that meaning that the same name will be used, no matter which input layer is used. In other cases, the name might depend on the name of the input layer or some of the parameters used to run the algorithm. If this checkbox is checked, the name will be taken from the output filename instead. Notice, that, if the output is saved to a temporary file, the filename of this temporary file is usually long and meaningless one intended to avoid collision with other already existing filenames.

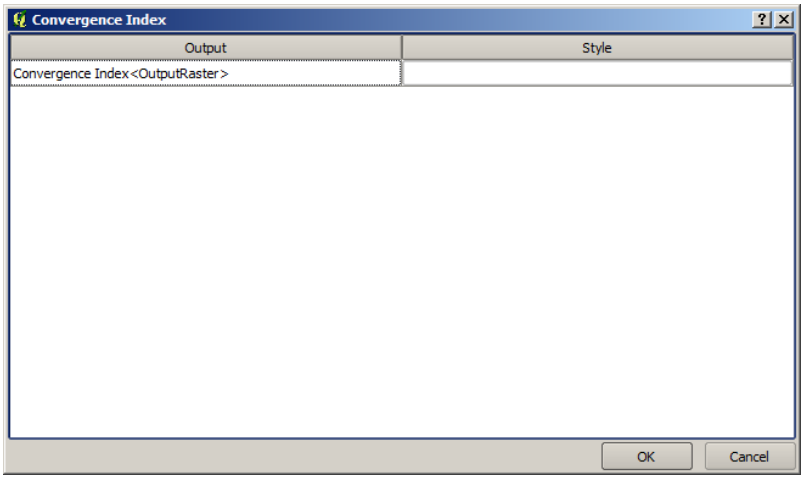


Figure 17.14: Rendering Styles

- *Use only selected features*. If this option is selected, whenever a vector layer is used as input for an algorithm, only its selected features will be used. If the layer has no selected features, all of them will be used.
- *Pre-execution script file* and *Post-execution script file*. This parameters refer to scripts written using the SEXTANTE scripting functionality, and are explained in the section covering scripting and the SEXTANTE console.

Apart from the *General* block in the settings dialog, you will also find one for each algorithm provider. They contain an *Activate* item that you can use to make algorithms appear or not in the toolbox. Also, some algorithm providers have their own configuration items, that we will explain later when covering particular algorithm providers.

17.3 The SEXTANTE graphical modeler

17.3.1 Introduction

The *graphical modeler* allows to create complex models using a simple and easy-to-use interface. When working with a GIS, most analysis operations are not isolated, but part of a chain of operations instead. Using the graphical modeler, that chain of processes can be wrapped into a single process, so it is easier and more convenient to execute than a single process later on a different set on inputs. No matter how many steps and different algorithms it involves, a model is executed as a single algorithm, thus saving time and effort, specially for larger models.

The modeler can be opened from the SEXTANTE menu, but also from the toolbox. In the *Modeler* branch of the algorithms tree you will find a group named *Tools*, which contains an entry called *Create new model*.

The modeler has a working canvas where the structure of the model and the workflow it represents are shown. On the left part of the window, a panel with two tabs can be used to add new elements to the model.

Creating a model involves two steps:

1. *Definition of necessary inputs*. These inputs will be added to the parameters window, so the user can set their values when executing the model. The model itself is a SEXTANTE algorithm, so the parameters window is generated automatically as it happens with all the algorithms included in SEXTANTE.
2. *Definition of the workflow*. Using the input data of the model, the workflow is defined adding algorithms and selecting how they use those inputs or the outputs generated by other algorithms already in the model

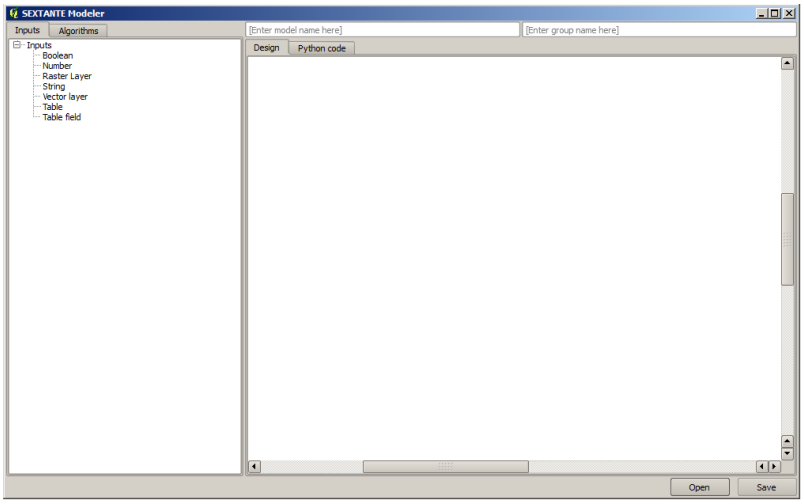


Figure 17.15: Modeler

17.3.2 Definition of inputs

The first step to create a model is to define the inputs it needs. The following elements are found in the *Inputs* tab on the left side of the modeler window:

- Raster layer
- Vector layer
- String
- Table field
- Table
- Numerical value
- Boolean value

Double-clicking on any of them, a dialog is shown to define its characteristics. Depending on the parameter itself, the dialog will contain just one basic element (the description, which is what the user will see when executing the model) or more of them. For instance, when adding a numerical value, as it can be seen in the next figure, apart from the description of the parameter you have to set a default value and a range of valid values.

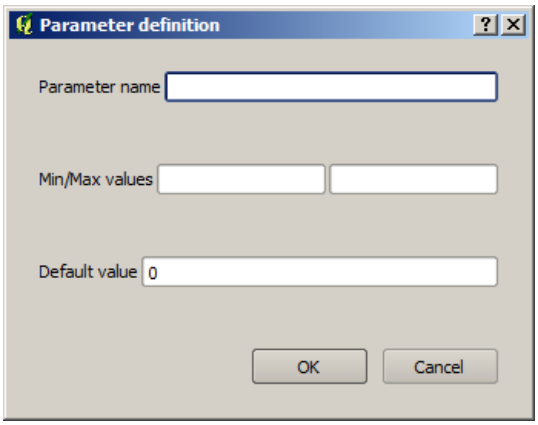


Figure 17.16: Model Parameters

For each added input, a new element is added to the modeler canvas.



Figure 17.17: Model Parameters

17.3.3 Definition of the workflow

Once the inputs have been defined, it is time to define the algorithms to apply on them. Algorithms can be found in the *Algorithms* tab, grouped much in the same way as they are in the toolbox.

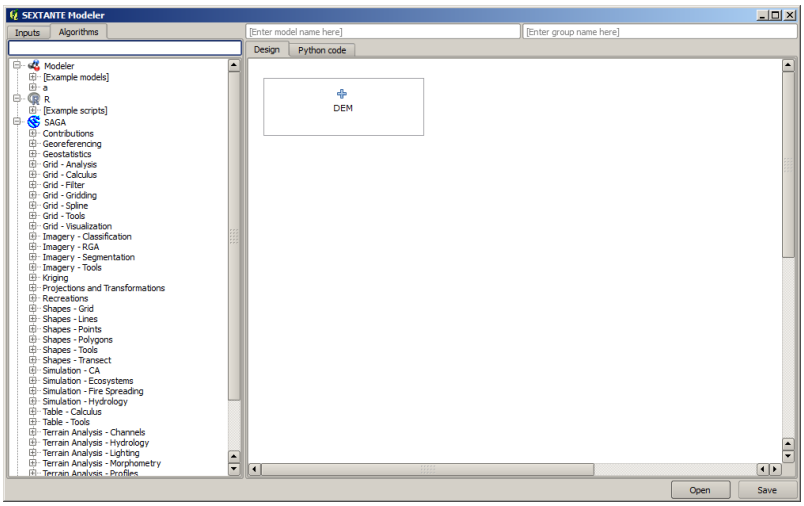


Figure 17.18: Model Parameters

The appearance of the toolbox has two modes here as well: simplified and advanced. However, there is no element to switch between views in the modeler, and you have to do it in the toolbox. The mode that is selected in the SEXTANTE toolbox is the one that will be used for the list of algorithms in the modeler.

To add an algorithm to a model, double-click on its name. An execution dialog will appear, with a content similar to the one found in the execution panel that SEXTANTE shows when executing the algorithm from the toolbox. The one shown next correspond to the SAGA 'Convergence index' algorithm, the same one we saw in the section dedicated to the SEXTANTE toolbox.

As you can see, some differences exist. Instead of the file output box that was used to set the filepath for output layers and tables, a simple text box is. If the layer generated by the algorithm is just a temporary result that will be used as the input of another algorithm and should not be kept as a final result, just do not edit that text box. Typing anything on it means that the result is a final one, and the text that you supply will be the description for the output, which will be the one the user will see when executing the model.

Selecting the value of each parameter is also a bit different, since there are important differences between the context of the modeler and the toolbox one. Let's see how to introduce the values for each type of parameter.

- Layers (raster and vector) and tables. They are selected from a list, but in this case the possible values are not the layers or tables currently loaded in QGIS, but the list of model inputs of the corresponding type, or other layers or tables generated by algorithms already added to the model.
- Numerical values. Literal values can be introduced directly on the text box. But this text box is also a list that can be used to select any of the numerical value inputs of the model. In this case, the parameter will take the value introduced by the user when executing the model.
- String. Like in the case of numerical values, literal strings can be typed, or an input string can be selected.

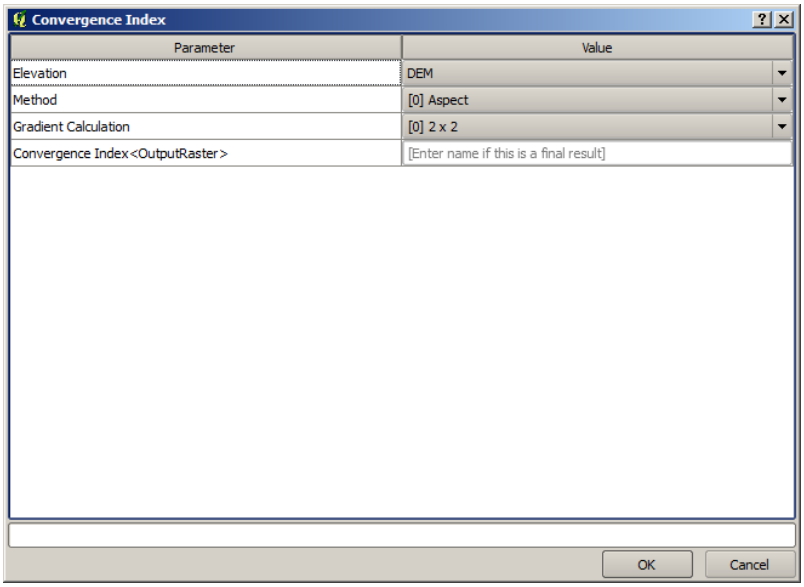


Figure 17.19: Model Parameters

- Table field. The fields of the parent table or layer cannot be known at design-time, since they depend of the selection of the user each time the model is executed. To set the value for this parameter, type the name of a field directly in the text box, or use the list to select a table field input already added to the model. The validity of the selected field will be checked by SEXTANTE at run-time.

In all cases, you will find an additional parameter named *Parent algorithms* that is not available when calling the algorithm from the toolbox. This parameter allows you to define the order in which algorithms are executed, by explicitly defining one algorithm as a parent of the current one, which will force it to be executed before it.

When you use the output of a previous algorithm as the input of your algorithm, that implicitly sets the former as parent of the current one (and places the corresponding arrow in the modeler canvas). However, in some cases an algorithm might depend on another one even if it does not use any output object from it (for instance, and algorithm that executes an SQL sentence on a PostGIS database and another on which import a layer into that same database) In that case, just select it in the *Parent algorithms* parameter and SEXTANTE will take care of executing them in the correct order.

Once all the parameter have been assigned valid values, click on **[OK]** and the algorithm will be added to the canvas. It will be linked to all the other elements in the canvas, whether algorithms or inputs, which provide objects that are used as inputs for that algorithm.

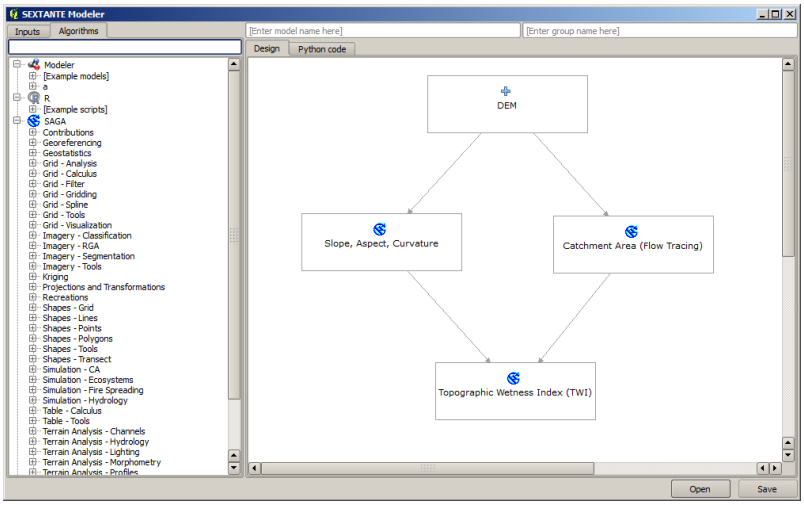


Figure 17.20: Model Parameters

Elements can be dragged to a different position within the canvas, to change the way the module structure is displayed and make it more clear and intuitive. Links between elements are update automatically.

You can run your algorithm anytime clicking on the **[Run]** button. However, in order to use it from the toolbox, it has to be saved and the modeler dialog closed, to allow the toolbox to refresh its contents.

17.3.4 Saving and loading models

Use the **[Save]** button to save the current model and the **[Open]** one to open any model previously saved. Model are saved with the `.model` extension. If the model has been previously saved from the modeler window, you will not be prompted for a filename, since there is already a file associated with that model, and it will be used.

Before saving a model, you have to enter a name and a group for it, using the text boxes in the upper part of the window.

Models saved on the `models` folder (the default folder when you are prompted for a filename to save the model) will appear in the toolbox in the corresponding branch. When the toolbox is invoked, SEXTANTE searches the `models` folder for files with `.model` extension and loads the models they contain. Since a model is itself a SEXTANTE algorithm, it can be added to the toolbox just like any other algorithm.

The models folder can be set from the SEXTANTE configuration dialog, under the *Modeler* group.

Models loaded from the `models` folder appear not only in the toolbox, but also in the algorithms tree in the *Algorithms* tab of the modeler window. That means that you can incorporate a model as a part of a bigger model, just as you add any other algorithm.

In some cases, SEXTANTE might not be able to load a model because it cannot find all the algorithms included in its workflow. If you have used a given algorithm as part of your model, it should be available (that is, it should appear on the toolbox) in order to load that model. Deactivating an algorithm provider in the SEXTANTE configuration window renders all the algorithms in that provider unusable by the modeler, which might cause problems when loading models. Keep that in mind when you have trouble loading or executing models.

17.3.5 Editing a model

You can edit the model you are currently creating, redefining the workflow and the relationships between the algorithms and inputs that define the model itself.

If you right-click on an algorithm in the canvas representing the model, you will see a context menu like the one shown next:

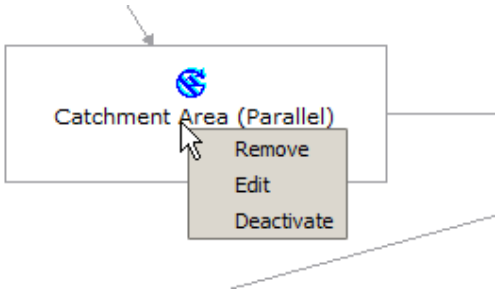


Figure 17.21: Modeler Right Click

Selecting the *Remove* option will cause the selected algorithm to be removed. An algorithm can be removed only if there are no other algorithms depending on it. That is, if no output from the algorithm is used in a different one as input. If you try to remove an algorithm that has others depending on it, SEXTANTE will show you a warning message like the one you can see below:

Selecting the *Edit* option or simply double-clicking on the algorithm icon will show the parameters dialog of the algorithm, so you can change the inputs and parameter values. Not all input elements available in the model will

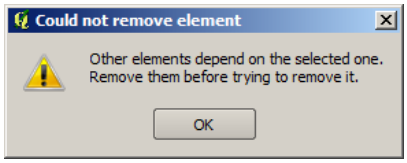


Figure 17.22: Cannot Delete ALG

appear in this case as available inputs. Layers or values generated at a more advanced step in the workflow defined by the model will not be available if they cause circular dependencies.

Select the new values and then click on the **[OK]** button as usual. The connections between the model elements will change accordingly in the modeler canvas.

17.3.6 Activating and deactivating algorithms

Algorithms can be deactivated in the modeler, so they will not be executed once the model is run. This can be used to test just a given part of the model, or when you do not need all the outputs it generates.

To deactivate an algorithm, right-click on its icon in the model canvas and select the *Deactivate* option. You will see that the algorithm is represented now with a red label under its name indicating that is not active.

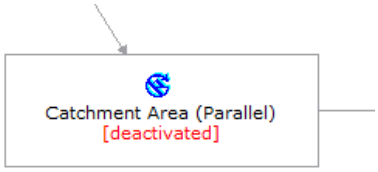


Figure 17.23: Deactivate

All algorithms depending (directly or undirectly) on that algorithm will also appear as inactive, since they cannot be executed now.

To activate an algorithm, just right–click on its icon and select the *Activate* option.

17.3.7 Editing model help files and meta-information

You can document your models from SEXTANTE. Just click on the **[Edit model help]** button and a dialog like the one shown next will appear.

On the right-hand side you will see a simple HTML page, created using the description of the input parameters and outputs of the algorithm, along with some additional items like a general description of the model or its author. The first time you open the help editor all those descriptions are empty, but you can edit them using the elements on the left-hand side of the dialog. Select an element on the upper part and the write its description in the text box below.

Model help is saved in a file in the same folder as the model itself. You do not have to worry about saving it, since it is done automatically.

17.3.8 About available algorithms

You might notice that some algorithms that can be be executed from the toolbox do not appear in the list of available ones when you are designing a model. To be included in a model, and algorithm must have a correct semantic, so as to be properly linked to other in the workflow. If an algorithm does not have such well-defined semantic (for instance, if the number of output layers cannot be know in advance), then it is not possible to use it within a model, and thus does not appear in the list of them that you can find in the modeler dialog.

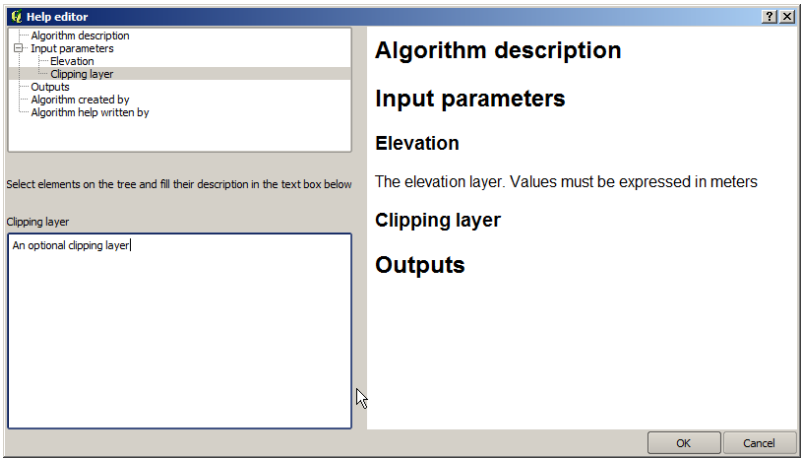


Figure 17.24: Help Edition

Additionally, you will see some algorithms in the modeler that are not found in the toolbox. This algorithms are meant to be used exclusively as part of a model, and they are of no interest in a different context. The 'Calculator' algorithm is an example of that. It is just a simple arithmetic calculator that you can use to modify numerical values (entered by the user or generated by some other algorithm). This tools is really useful within a model, but outside of that context, it doesn't make too much sense.

17.3.9 SEXTANTE models as Python code

Given a model, SEXTANTE can automatically create Python code that performs the same task as the model itself. This code is used to create a console script (we will explain them later in this manual) and you can modify that script to incorporate actions and methods not available in the graphical modeler, such as loops or conditional sentences.

This feature is also a very practical way of learning how to use SEXTANTE from the console and how to create SEXTANTE algorithms using Python code, so you can use it as a learning tool when you start creating your own SEXTANTE scripts.

Save your model in the models folder and go to the toolbox, where it should appear now, ready to be run. Right click on the model name and select *Save as Python script* in the context menu that will pop-up. A dialog will prompt you to introduce the file where you want to save the script.

17.4 The SEXTANTE batch processing interface

17.4.1 Introduction

SEXTANTE algorithms (including models) can be executed as a batch process. That is, they can be executed using not a single set of inputs, but several of them, executing the algorithm as many times as needed. This is useful when processing large amounts of data, since it is not necessary to launch the algorithm many times from the toolbox.

To execute an algorithm as a batch process, right-click on its name in the toolbox and select the *Execute as batch process* option in the pop-up menu that will appear.

17.4.2 The parameters table

Executing a batch process is similar to performing a single execution of an algorithm. Parameter values have to be defined, but in this case we need not just a single value for each parameter, but a set of them instead, one for each time the algorithm has to be executed. Values are introduced using a table like the one shown next.

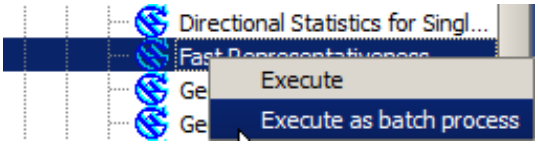


Figure 17.25: Batch Processing Right Click

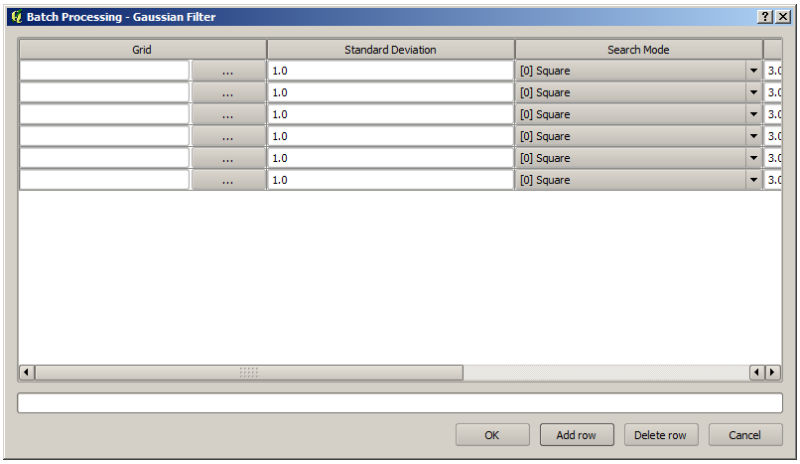


Figure 17.26: Batch Processing

Each line of this table represents a single execution of the algorithm, and each cell contains the value of one of the parameters. It is similar to the parameters dialog that you see when executing an algorithm from the toolbox, but with a different arrangement.

By default, the table contains just two rows. You can add or remove rows using the buttons on the lower part of the window.

Once the size of the table has been set, it has to be filled with the desired values.

17.4.3 Filling the parameters table

For most parameters, setting its value is trivial. Just type the value or select it from the list of available options, depending on the parameter type.

The main differences are found for parameters representing layers or tables, and for output filepaths. Regarding input layers and tables, when an algorithm is executed as part of a batch process those input data objects are taken directly from files, and not from the set of them already opened in QGIS. For this reason, any algorithm can be executed as a batch process even if no data objects at all are opened and the algorithm cannot be run from the toolbox.

Filenames for input data objects are introduced directly typing or, more conveniently, clicking on the ... button on the right hand of the cell, which shows a typical file chooser dialog. Multiple files can be selected at once. If the input parameter represents a single data object and several files are selected, each one of them will be put in a separate row, adding new ones if needed. If it represents a multiple input, all the selected files will be added to a single cell, separated by semicolons (`;`).

Output data objects are always saved to a file and, unlike when executing an algorithm from the toolbox, saving to a temporary one is not permitted. You can type the name directly or use the file chooser dialog that appears when clicking on the accompanying button.

Once you select the file, a new dialog is shown to allow for autocompletion of other cells in the same column (same parameter).

If the default value ('Do not autocomplete') is selected, SEXTANTE will just put the selected filename in the selected cell from the parameters table. If any of the other options is selected, all the cells below the selected one

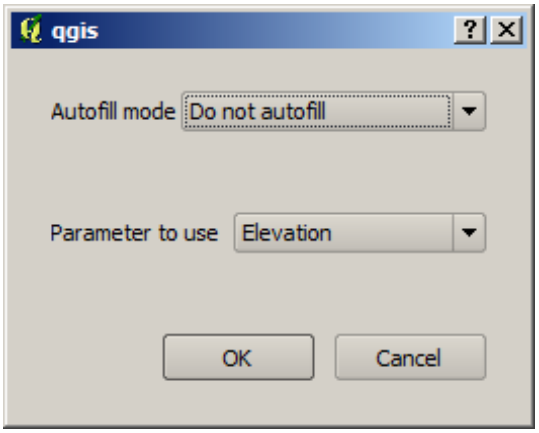


Figure 17.27: Batch Processing Save

will be automatically filled based on a defined criteria. This way, it is much easier to fill the table, and the batch process can be defined with less effort.

Automatic filling can be done simply adding correlative numbers to the selected filepath, or appending the value of another field at the same row. This is particularly useful for naming output data object according to input ones.

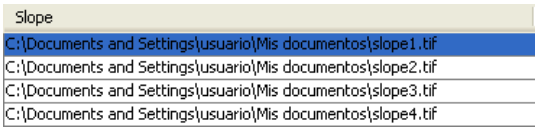


Figure 17.28: Batch Processing File Path

17.4.4 Executing the batch process

To execute the batch process once you have introduced all the necessary values, just click on **[OK]**. SEXTANTE will show the progress of the global batch process in the progress bar in the lower part of the dialog.

17.5 Using SEXTANTE from the console

17.5.1 Introduction

The console allows advanced users to increase their productivity and perform complex operations that cannot be performed using any of the other elements of the SEXTANTE GUI. Models involving several algorithms can be defined using the command-line interface, and additional operations such as loops and conditional sentences can be added to create more flexible and powerful workflows.

There is not a SEXTANTE console in QGIS, but all SEXTANTE commands are available instead from the QGIS built-in Python console. That means that you can incorporate those command to your console work and connect SEXTANTE algorithms to all the other features (including methods from the QGIS API) available from there.

The code that you can execute from the Python console, even if it does not call any SEXTANTE method, can be converted into a new SEXTANTE algorithm that you can later call from the toolbox, the graphical modeler or any other SEXTANTE component, just like you do with any other SEXTANTE algorithm. In fact, some algorithms that you can find in the toolbox are simple scripts.

In this chapter we will see how to use SEXTANTE from the QGIS Python console, and also how to write your own algorithms using Python.

17.5.2 Calling SEXTANTE from the Python console

The first thing you have to do is to import the SEXTANTE functions with the following line:

```
>>> import sextante
```

Now, there is basically just one (interesting) thing you can do with SEXTANTE from the console: to execute an algorithm. That is done using the `runalg()` method, which takes the name of the algorithm to execute as its first parameter, and then a variable number of additional parameter depending on the requirements of the algorithm. So the first thing you need to know is the name of the algorithm to execute. That is not the name you see in the toolbox, but rather a unique command?line name. To find the right name for your algorithm, you can use the `algslist()` method. Type the following line in you console:

```
>>> sextante.alglist()
```

You will see something like this.

```
Accumulated Cost (Anisotropic)----------------->saga:accumulatedcost(anisotropic)
Accumulated Cost (Isotropic)------------------->saga:accumulatedcost(isotropic)
Add Coordinates to points---------------------->saga:addcoordinatestopoints
Add Grid Values to Points---------------------->saga:addgridvaluestopoints
Add Grid Values to Shapes---------------------->saga:addgridvaluestoshapes
Add Polygon Attributes to Points--------------->saga:addpolygonattributestopoints
Aggregate-------------------------------------->saga:aggregate
Aggregate Point Observations------------------->saga:aggregatepointobservations
Aggregation Index------------------------------>saga:aggregationindex
Analytical Hierarchy Process------------------->saga:analyticalhierarchyprocess
Analytical Hillshading------------------------->saga:analyticalhillshading
Average With Mask 1---------------------------->saga:averagewithmask1
Average With Mask 2---------------------------->saga:averagewithmask2
Average With Thereshold 1---------------------->saga:averagewiththereshold1
Average With Thereshold 2---------------------->saga:averagewiththereshold2
Average With Thereshold 3---------------------->saga:averagewiththereshold3
B-Spline Approximation------------------------->saga:b-splineapproximation
...
```

That's a list of all the available algorithms, alphabetically ordered, along with their corresponding command-line names.

You can use a string as a parameter for this method. Instead of returning the full list of algorithm, it will only display those that include that string. If, for instance, you are looking for an algorithm to calculate slope from a DEM, type `alglist("slope")` to get the following result:

```
DTM Filter (slope-based)----------------------->saga:dtmfilter(slope-based)
Downslope Distance Gradient-------------------->saga:downslopedistancegradient
Relative Heights and Slope Positions----------->saga:relativeheightsandslopepositions
Slope Length----------------------------------->saga:slopelength
Slope, Aspect, Curvature----------------------->saga:slopeaspectcurvature
Upslope Area----------------------------------->saga:upslopearea
Vegetation Index[slope based]------------------>saga:vegetationindex[slopebased]
```

This result might change depending on the algorithms you have available.

It is easier now to find the algorithm you are looking for and its command-line name, in this case `saga:slopeaspectcurvature`.

Once you know the command-line name of the algorithm, the next thing to do is to know the right syntax to execute it. That means knowing which parameters are needed and the order in which they have to be passed when calling the `runalg()` method. SEXTANTE has a method to describe an algorithm in detail, which can be used to get a list of the parameters that an algorithms require and the outputs that it will generate. To do it, you can use the `alghelp(name_of_the_algorithm)` method. Use the command-line name of the algorithm, not the full descriptive name.

Calling the method with `saga:slopeaspectcurvature` as parameter, you get the following description.

```
>>> sextante.alghelp("saga:slopeaspectcurvature")
ALGORITHM: Slope, Aspect, Curvature
   ELEVATION <ParameterRaster>
   METHOD <ParameterSelection>
   SLOPE <OutputRaster>
   ASPECT <OutputRaster>
   CURV <OutputRaster>
   HCURV <OutputRaster>
   VCURV <OutputRaster>
```

Now you have everything you need to run any algorithm. As we have already mentioned, there is only one single command to execute algorithms: `runalg()`. Its syntax is as follows:

```
>>> sextante.runalg(name_of_the_algorithm, param1, param2, ..., paramN,
        Output1, Output2, ..., OutputN)
```

The list of parameters and outputs to add depends on the algorithm you want to run, and is exactly the list that the `alghelp()` method gives you, in the same order as shown.

Depending on the type of parameter, values are introduced differently. The next one is a quick review of how to introduce values for each type of input parameter:

- Raster Layer, Vector Layer or Table. Simply use a string with the name that identifies the data object to use (the name it has in the QGIS Table of Contents) or a filename (if the corresponding layer is not opened, it will be opened, but not added to the map canvas). If you have an instance of a QGIS object representing the layer, you can also pass it as parameter. If the input is optional and you do not want to use any data object, use `None`.
- Selection. If an algorithm has a selection parameter, the value of that parameter should be entered using an integer value. To know the available options, you can use the `algoptions()` command, as shown in the following example:

  ```
  >>> sextante.algoptions("saga:slopeaspectcurvature")
  METHOD(Method)
      0 - [0] Maximum Slope (Travis et al. 1975)
      1 - [1] Maximum Triangle Slope (Tarboton 1997)
      2 - [2] Least Squares Fitted Plane (Horn 1981, Costa-Cabral & Burgess 1996)
      3 - [3] Fit 2.Degree Polynom (Bauer, Rohdenburg, Bork 1985)
      4 - [4] Fit 2.Degree Polynom (Heerdegen & Beran 1982)
      5 - [5] Fit 2.Degree Polynom (Zevenbergen & Thorne 1987)
      6 - [6] Fit 3.Degree Polynom (Haralick 1983)
  ```

  In this case, the algorithm has one of such such parameters, with 7 options. Notice that ordering is zero-based.
- Multiple input. The value is a string with input descriptors separated by semicolons (`;`). As in the case of single layers or tables, each input descriptor can be the data object name, or its filepath.
- Table Field from XXX. Use a string with the name of the field to use. This parameter is case-sensitive.
- Fixed Table. Type the list of all table values separated by commas (`,`) and enclosed between quotes (`"`). Values start on the upper row and go from left to right. You can also use a 2D array of values representing the table.
- CRS. Enter the EPSG code number of the desired CRS.
- Extent. You must use a string with `xmin`, `xmax`, `ymin` and `ymax` values separated by commas (`,`).

Boolean, file, string and numerical parameters do not need any additional explanations.

Input parameters such as strings booleans or numerical values have default values. To use them, use `None` in the corresponding parameter entry.

For output data objects, type the filepath to be used to save it, just as it is done from the toolbox. If you want to save the result to a temporary file, use `None`. The extension of the file determines the file format. If you enter a

file extension not included in the ones supported by the algorithm, the default file format for that output type will be used, and its corresponding extension appended to the given filepath.

Unlike when an algorithm is executed from the toolbox, outputs are not added to the map canvas if you execute that same algorithm from the Python Console. If you want to add an output to it, you have to do it yourself after running the algorithm. To do so, you can use QGIS API commands, or, even easier, use one of the handy methods provided by SEXTANTE for such task.

The `runalg` method returns a dictionary with the output names (the ones shown in the algorithm description) as keys and the filepaths of those outputs as values. You can load those layers by passing its filepath to the `load()` method.

17.5.3 Additional functions for handling data

Apart from the functions used to call SEXTANTE algorithms, importing the `sextante` package will also import some additional functions that make it easier to work with data, particularly vector data. They are just convenience functions that wrap some functionality from the QGIS API, usually with a less complex syntax. These functions should be used when developing new algorithms, as they make it easier to operate with input data.

Below is a list of some of this commands. More information can be found in the classes under the `sextante/tools` package, and aso in the example scripts provided with SEXTANTE.

- `getobject(obj)`: Returns a QGIS object (a layer or table) from the passed object, which can be a filename or the name of the object in the QGIS Table of Contents.
- `values(layer, fields)`: Returns the values in the attributes table of a vector layer, for the passed fields. Fields can be passed as field names or as zero-based field indices. Returns a dict of lists, with the passed field identifiers as keys. It considers the existing selection
- `getfeatures(layer)`: Returns an iterator over the features of a vector layer, considering the existing selection.
- `uniquelabels(layer, field)`: Returns a list of unique values for a given attribute. Attribute can be passed as a field name or a zero-based field index. It considers the existing selection

17.5.4 Creating scripts and running them from the toolbox

You can create your own algorithms by writing the corresponding Python code and adding a few extra lines to supply additional information needed by SEXTANTE. You can find a *Create new script* menu under the *Tools* group in the *Script* algorithms block of the toolbox. Double-click on it to open the script edition dialog. That's where you should type your code. Saving the script from there in the `scripts` folder (the default one when you open the save file dialog), with `.py` extension, will automatically create the corresponding algorithm.

The name of the algorithm (the one you will see in the toolbox) is created from the filename, removing its extension and replacing low hyphens with blank spaces.

Let's have the following code, which calculates the Topographic Wetness Index (TWI) directly from a DEM

```
##dem=raster
##twi=output
ret_slope = sextante.runalg("saga:slopeaspectcurvature", dem, 0, None,
                None, None, None, None)
ret_area = sextante.runalg("saga:catchmentarea(mass-fluxmethod)", dem,
                0, False, False, False, False, None, None, None, None, None)
sextante.runalg("saga:topographicwetnessindex(twi), ret_slope['SLOPE'],
                ret_area['AREA'], None, 1, 0, twi)
```

As you can see, it involves 3 algorithms, all of them coming from SAGA. The last one of them calculates the TWI, but it needs a slope layer and a flow accumulation layer. We do not have these ones, but since we have the DEM, we can calculate them calling the corresponding SAGA algorithms.

The part of the code where this processing takes place is not difficult to understand if you have read the previous sections in this chapter. The first lines, however, need some additional explanation. They provide SEXTANTE

the information it needs to turn your code into an algorithm that can be run from any of its components, like the toolbox or the graphical modeler.

These lines start with a double Python comment symbol (##) and have the following structure

```
[parameter_name]=[parameter_type] [optional_values]
```

Here is a list of all the parameter types that SEXTANTE supports in its scripts, their syntax and some examples.

- `raster`. A raster layer
- `vector`. A vector layer
- `table`. A table
- `number`. A numerical value. A default value must be provided. For instance, `depth=number 2.4`
- `string`. A text string. As in the case of numerical values, a default value must be added. For instance, `name=string Victor`
- `boolean`. A boolean value. Add `True` or `False` after it to set the default value. For example, `verbose=boolean True`
- `multiple raster`. A set of input raster layers.
- `multiple vector`. A set of input vector layers.
- `field`. A field in the attributes table of a vector layer. The name of the layer has to be added after the `field` tag. For instance, if you have declared a vector input with `mylayer=vector`, you could use `myfield=field mylayer` to add a field from that layer as parameter.
- `folder`. A folder
- `file`. A filename

The parameter name is the name that will be shown to the user when executing the algorithm, and also the variable name to use in the script code. The value entered by the user for that parameter will be assigned to a variable with that name.

When showing the name of the parameter to the user, SEXTANTE will edit it to improve its appearance, replacing low hyphens with spaces. So, for instance, if you want the user to see a parameter named `A numerical value`, you can use the variable name `A_numerical_value`.

Layers and tables values are strings containing the filepath of the corresponding object. To turn them into a QGIS object, you can use the `sextante.getObjectFromUri()` function. Multiple inputs also have a string value, which contains the filepaths to all selected object, separated by semicolons (`;`).

Outputs are defined in a similar manner, using the following tags:

- `output raster`
- `output vector`
- `output table`
- `output html`
- `output file`
- `output number`
- `output string`

The value assigned to the output variables is always a string with a filepath. It will correspond to a temporary filepath in case the user has not entered any output filename.

When you declare an output, SEXTANTE will try to add it to QGIS once the algorithm is finished. That is the reason why, although the `runalg()` method does not load the layers it produces, the final TWI layer will be loaded, since it is saved to the file entered by the user, which is the value of the corresponding output.

Do not use the `load()` method in your script algorithms, but just when working with the console line. If a layer is created as output of an algorithm, it should be declared as such. Otherwise, you will not be able to properly

use the algorithm in the modeler, since its syntax (as defined by the tags explained above) will not match what the algorithm really creates.

Hidden outputs (numbers and strings) do not have a value. Instead, it is you who has to assign a value to them. To do so, just set the value of a variable with the name you used to declare that output. For instance, if you have used this declaration,

```
##average=output number
```

the following line will set the value of the output to 5:

```
average = 5
```

In addition to the tags for parameters and outputs, you can also define the group under which the algorithm will be shown, using the `group` tag.

If you algorithm takes a long time to process, it is a good idea to inform the user. You have a global named `progress` available, with two available methods: `setText(text)` and `setPercentage(percent)` to modify the progress text and the progress bar.

Several examples are provided with SEXTANTE. Please, check them to see real examples of how to create algorithms using this feature of SEXTANTE. You can right-click on any script algorithm and select *Edit script* to edit its code or just to see it.

17.5.5 Documenting your scripts

As in the case of models, you can create additional documentation for your script, to explain what they do and how to use them. In the script editing dialog you will find a **[Edit script help]** button. Click on it and it will take you to the help editing dialog. Check the chapter about the graphical modeler to know more about this dialog and how to use it.

Help files are saved in the same folder as the script itself, adding the `.help` extension to the filename. Notice that you can edit your script's help before saving it for the first time. If you later close the script editing dialog without saving the script (i.e. you discard it), the help content you wrote will be lost. If your script was already saved and is associated to a filename, saving is done automatically.

17.5.6 Pre- and post-execution script hooks

Script can also be used to set pre- and post-execution hooks that are run before and after an algorithm is run. This can be used to automate tasks that should be performed whenever a SEXTANTE algorithm is executed.

The syntax is identical to the syntax explained above, but an additional global variable named `alg` is available, representing the algorithm that has just been (or is about to be) executed.

In the *General* group of the SEXTANTE config dialog you will find two entries named *Pre-execution script file* and *Post-execution script file* where the filename of the scripts to be run in each case can be entered.

17.6 The SEXTANTE history manager

17.6.1 The SEXTANTE history

Every time you execute a SEXTANTE algorithm, information about the process is stored in the SEXTANTE history manager. Along with the parameters used, the date and time of the execution are also saved.

This way, it is easy to track the and control all the work that has been developed using SEXTANTE, and easily reproduce it.

The SEXTANTE history manager is a set of registry entries grouped according to their date of execution, making it easier to find information about an algorithm executed at any particular moment.

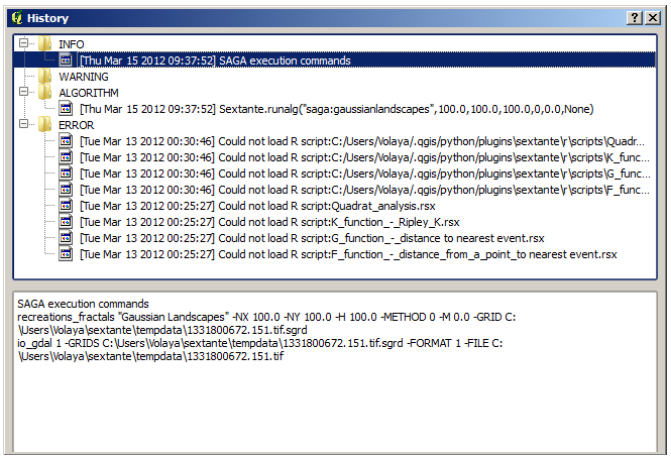


Figure 17.29: History

Process information is kept as a command-line expression, even if the algorithm was launched from the toolbox. This makes it also useful for those learning how to use the command-line interface, since they can call an algorithm using the toolbox and then check the history manager to see how that same algorithm could be called from the command line.

Apart from browsing the entries in the registry, processes can be re-executed, simply double-clicking on the corresponding entry.

Along with algorithm executions, SEXTANTE communicates with the user using the other groups of the registry, namely *Errors*, *Warnings* and *Information*. In case something is not working properly, having a look at the *Errors* might help you to see what is happening. If you get in contact with a SEXTANTE developer to report a bug or error, the information in that group will be very useful for him to find out what is going wrong.

When executing third party algorithms, this is usually done calling their command-line interfaces, which communicate with the user using the console. Although that console is not shown, a full dump of it is stored in the *Information* group each time you run one of those algorithms. If, for instance, you are having problems executing a SAGA algorithm, look for an entry name 'SAGA execution console output' to check all the messages generated by SAGA and try to find out where the problem is.

Some algorithms, even if they can produce a result with the given input data, might add comments or additional information to *Warning* block in case they detect potential problems from that data, in order to warn you about them. Make sure you check those messages in case you are having unexpected results.

17.7 Configuring external applications

17.7.1 Introduction

SEXTANTE can be extended using additional applications, calling them from within SEXTANTE. Currently, SAGA, GRASS, OTB(Orfeo Toolbox) and R are supported, along with some other command-line applications that provide spatial data analysis functionalities. Algorithms relying on an external application are managed by their own algorithm provider.

This chapter will show you how to configure SEXTANTE to include these additional applications, and will explain some particular features of the algorithm based on them. Once you have correctly configured the system, you will be able to execute external algorithms from any SEXTANTE component like the toolbox or the graphical modeler, just like you do with any other SEXTANTE geoalgorithm.

By default, all algorithms that rely on an external appplication not shipped with QGIS are not enabled. You can enable them in the SEXTANTE configuration dialog. Make sure that the corresponding application is already installed in your system. Enabling an algorithm provider without installing the application it needs will cause the algorithms to appear in the toolbox, but an error will be thrown when you try to execute them.

This is because the algorithm descriptions (needed to create the parameters dialog and give SEXTANTE the information it needs about the algorithm) are not included with each appllication, but with SEXTANTE instead. That is,they are part of SEXTANTE, so you have them in your installation even if you have not installed any other software. Running the algorithm, however, needs the application binaries to be installed in your system.

A note for Windows users

If you are not an advanced user and you are running QGIS on windows, you might not be interested in reading the rest of this chapter. Make sure you install QGIS in your system using the OSGeo4W application. That will automatically install SAGA, GRASS and OTB in your system, and configure them so they can be run from SEXTANTE. All the algorithms in the simplified view of the toolbox will be ready to be run, without needing any further configuration.

If you want to know more about how these providers work, or want to use some algorithms not included in the simplified toolbox (such as R scripts), keep on reading.

A note on file formats

When using an external software, opening a file in QGIS does not mean that it can be opened and processed as well on that other software. In most cases, it can read what you have opened in QGIS, but in some cases, that might not be the case. When using databases or uncommon file formats, whether for raster of vector layers, problems might arise. If that happens, try to use well known file formats that you are sure that are understood by both programs, and check to console output (in the history and log dialog) for knowing more about what is going wrong.

Using GRASS raster layers is, for instance, one case in which you might have trouble and not be able to complete your work if you call an external algorithm using such a layer as input. For this reason, these layers will not appear as available to SEXTANTE algorithms.

You should, however, find no problems at all with vector layers, since SEXTANTE automatically converts from the original file format to one accepted by the external application before passing the layer to it. This adds an extra processing time, which might be significant if the layer has a large size, so do not be surprised if it takes more to process a layer from a DB connection that one of a similar size stored in a shapefile.

Providers not using external applications can process any layer that you can open in QGIS, since they open it for analysis through QGIS.

Regarding output formats, all formats supported by QGIS as output can be used, both for raster and vector layers. Some provider do not support certain formats, but all can export to common formats raster layers that can be later transformed by QGIS automatically. As in the case of input layers, if this conversion is needed, that might increase the processing time.

If the extension of the filename specified when calling an algorithm does not match the extension of any of the formats supported by QGIS, then a suffix will be added to set a default format. In the case of raster layers, the `tif` extension is used, while `shp` is used for vector layer.

A note on vector layer selections

External applications are also aware of the selection that exist in vector layers within QGIS. However, that requires rewritting all input vector layers, just as if they were originally in a format not supported by the external application. Only when no selection exist, or the *Use only selected features* option is not enabled in the SEXTANTE general configuration, a layer can be directly passed to an external application.

In other cases, exporting only selected features is needed, which causes execution times to be longer.

17.7.2 SAGA

SAGA algorithms can be run from SEXTANTE if you have SAGA installed in your system and you configure SEXTANTE properly so it can find SAGA executables. In particular, the SAGA command-line executable is

needed to run SAGA algorithms. SAGA binaries are not included with SEXTANTE, so you have to download and install the software yourself. Please check the SAGA website at for more information. SAGA 2.0.8 is needed.

In case of running Once SAGA is installed, and if you are running Windows, open the SEXTANTE configuration dialog. In the *SAGA* block you will find a setting named *SAGA Folder*. Enter the path to the folder where SAGA is installed. Close the configuration dialog and now you are ready to run SAGA algorithms from SEXTANTE.

In case you are using Linux, there is no need to configure that, and you will not see those folders. Instead, you must make sure that SAGA is properly installed and its folder is added to the PATH environment variable. Just open a console and type `saga_cmd` to check that the system can found where SAGA binaries are located.

About SAGA grid system limitations

Most of SAGA algorithms that require several input raster layers, require them to have the same grid system. That is, to cover the same geographic area and have the same cellsize, so their corresponding grids match. When calling SAGA algorithms from SEXTANTE, you can use any layer, regardless of its cellsize and extent. When multiple raster layers are used as input for a SAGA algorithm, SEXTANTE resamples them to a common grid system and then passes them to SAGA (unless the SAGA algorithm can operate with layers from different grid systems).

The definition of that common grid system is controlled by the user, and you will find several parameters in the SAGA group of the setting window to do so. There are two ways of setting the target grid system:

- Setting it manually. You define the extent setting the values of the following parameters:
  - *Resampling min X*
  - *Resampling max X*
  - *Resampling min Y*
  - *Resampling max Y*
  - *Resampling cellsize*

  Notice that SEXTANTE will resample input layers to that extent, even if they do not overlap with it.
- Setting it automatically from input layers. To select this option, just check the *Use min covering grid system for resampling* option. All the other settings will be ignored and the minimum extent that covers all the input layers will be used. The cellsize of the target layer is the maximum of all cellsizes of the input layers.

For algorithms that do not use multiple raster layers, or for those that do not need a unique input grid system, no resampling is performed before calling SAGA, and those parameters are not used.

Limitations for multi-band layers

Unlike QGIS, SAGA has no support for multi-band layers. If you want to use a multiband layer (such as an RGB or multispectral image), you first have to split it into single-banded images. To do so, you can use the 'SAGA/Grid - Tools/Split RGB image' algorithm (which creates 3 images from an RGB image) or the 'SAGA/Grid - Tools/Extract band' algorithm (to extract a single band).

Limitations in cellsize

SAGA assumes that raster layers have the same cellsize in the X and Y axis. If you are working with a layer with different values for its horizontal and vertical cellsizes, you might get unexcepted results. In this case, a warning will be added to the SEXTANTE log, indicating that an input layer might not be suitable to be processed by SAGA.

Logging

When SEXTANTE calls SAGA, it does it using its command-line interface, thus passing a set of commands to perform all the required operation. SAGA show its progress by writing information to the console, which includes

the percentage of processing already done, along with additional content. This output is filtered by SEXTANTE and used to update the progress bar while the algorithm is running.

Both the commands sent by SEXTANTE and the additional information printed by SAGA can be logged along with other SEXTANTE log messages, and you might find them useful to track in detailed what is going on when SEXTANTE runs a SAGA algorithm. you will find two settings, namely *Log console output* and *Log execution commands* to activate that logging mechanism.

Most other providers that use an external application and call it through the command-line have similar options, so you will find them as well in other places in the SEXTANTE settings list.

17.7.3 R. Creating R scripts

R integration in SEXTANTE is different from that of SAGA in that there is not a predefined set of algorithms you can run (except for a few examples). Instead, you should write your scripts and call R commands, much like you would do from R, and in a very similar manner to what we saw in the chapter dedicated to SEXTANTE scripts. This chapter shows you the syntax to use to call those R commands from SEXTANTE and how to use SEXTANTE objects (layers, tables) in them.

The first thing you have to do, as we saw in the case of SAGA, is to tell SEXTANTE where you R binaries are located. You can do so using the *R folder* entry in the SEXTANTE configuration dialog. Once you have set that parameter, you can start creating your own R scripts and executing them.

Once again, this is different in Linux, and you just have to make sure that the R folder is included in the PATH environment variable. If you can start R just typing `R` in a console, then you are ready to go.

To add a new algorithm that calls an R function (or a more complex R script that you have developed and you would like to have available from SEXTANTE), you have to create a script file that tells SEXTANTE how to perform that operation and the corresponding R commands to do so.

Script files have the extension `.rsx` and creating them is pretty easy if you just have a basic knowledge of R syntax and R scripting. They should be stored in the R-scripts folder. You can set this folder in the *R* settings group (available from the SEXTANTE settings dialog), just like you do with the folder for regular SEXTANTE scripts.

Let's have a look at a very simple file script file, which calls the R method `spsample` to create a random grid within the boundary of the polygons in a given polygon layer. This method belong to the `maptools` package. Since almost all the algorithms that you might like to incorporate into SEXTANTE will use or generate spatial data, knowledge of spatial packages like `maptools` and, specially, `sp`, is mandatory.

```
##polyg=vector
##numpoints=number 10
##output=output vector
##sp=group
pts=spsample(polyg,numpoints,type="random")
output=SpatialPointsDataFrame(pts, as.data.frame(pts))
```

The first lines, which start with a double Python comment sign (`##`), tell SEXTANTE the inputs of the algorithm described in the file and the outputs that it will generate. They work exactly with the same syntax as the SEXTANTE scripts that we have already seen, so they will not be described here again. Check the *sextante_scripts* section for more information.

When you declare an input parameter, SEXTANTE uses that information for two things: creating the user interface to ask the user for the value of that parameter and creating a corresponding R variable that can be later used as input for R commands.

In the above example, we are declaring an input of type `vector` named `polyg`. When executing the algorithm, SEXTANTE will open in R the layer selected by the user and store it in a variable also named `polyg`. So the name of a parameter is also the name of the variable that we can use in R for accesing the value of that parameter (thus, you should avoid using reserved R words as parameter names).

Spatial elements such as vector and raster layers are read using the `readOGR()` and `brick()` commands (you do not have to worry about adding those commands to your description file, SEXTANTE will do it) and stored as

Spatial*DataFrame objects. Table fields are stored as strings containing the name of the selected field.

Tables are opened using the read.csv() command. If a table entered by the user is not in CSV format, it will be converted prior to importing it in R.

Additionally, raster files can be read using the readGDAL() command instead of brick(), by using the ##usereadgdal.

If you are an advanced user and do not want SEXTANTE to create the object representing the layer, you can use the ##passfilename tag to tell SEXTANTE that you prefer a string with the filename instead. In this case, it is up to you to open the file before performing any operation on the data it contains.

With the above information, we can now understand the first line of our first example script (the first line not starting with a Python comment).

```
pts=spsample(polyg,numpoints,type="random")
```

The variable polygon already contains a SpatialPolygonsDataFrame object, so it can be used to call the spsample method, just like the numpoints one, which indicates the number of points to add to the created sample grid.

Since we have declared an output of type vector named out, we have to create a variable named out and store a Spatial*DataFrame object in it (in this case, a SpatialPointsDataFrame). You can use any name for your intermediate variables. Just make sure that the variable storing your final result has the same name that you used to declare it, and contains a suitable value.

In this case, the result obtained from the spsample method has to be converted explicitly into a SpatialPointsDataFrame object, since it is itself an object of class ppp, which is not a suitable class to be returned to SEXTANTE.

If your algorithm generates raster layers, the way they are saved will depend on whether you have used or not the #dontuserasterpackage option. In you have used it, layers are saved using the writeGDAL() method. If not, the writeRaster() method from the raster package will be used.

If you have used the #passfilename option, outputs are generated using the raster package (with writeRaster()), even though it is not used for the inputs.

If you algorithm does not generate any layer, but a text result in the console instead, you have to tell SEXTANTE that you want the console to be shown once the execution is finished. To do so, just start the command lines that produce the results you want to print with the > ('greater') sign. The output of all other lines will not be shown. For instance, here is the description file of an algorithm that performs a normality test on a given field (column) of the attributes of a vector layer:

```
##layer=vector
##field=field layer
##nortest=group
library(nortest)
>lillie.test(layer[[field]])
```

The output ot the last line is printed, but the output of the first is not (and neither are the outputs from other command lines added automatically by SEXTANTE).

If your algorithm creates any kind of graphics (using the plot() method), add the following line:

```
##showplots
```

This will cause SEXTANTE to redirect all R graphical outputs to a temporary file, which will be later opened once R execution has finished.

Both graphics and console results will be shown in the SEXTANTE results manager.

For more information, please check the script files provided with SEXTANTE. Most of them are rather simple and will greatly help you understand how to create your own ones.

주석: rgdal and maptools libraries are loaded by default so you do not have to add the corresponding

`library()` commands (you have to make sure, however, that those two packages are installed in your R distribution). However, other additional libraries that you might need have to be explicitly loaded. Just add the necessary commands at the beginning of your script. You also have to make sure that the corresponding packages are installed in the R distribution used by SEXTANTE.

17.7.4 GRASS

Configuring GRASS is not much different from configuring SAGA. First, the path to the GRASS folder has to be defined, but only if you are running Windows. Additionaly, a shell interpreter (usually `msys.exe`, which can be found in most GRASS for Windows distributions) has to be defined and its path set up as well.

By default, SEXTANTE tries to configure its GRASS connector to use the GRASS distribution that ships along with QGIS. This should work without problems in most systems, but if you experience problems, you might have to do it manually. Also, if you want to use a different GRASS version, you can change that setting and point to the folder where that other version is kept. GRASS 6.4 is needed for algorithms to work correctly.

If you are running Linux, you just have to make sure that GRASS is correctly installed, and that it can be run without problem from a console.

GRASS algorithms use a region for calculations. This region can be defined manually using values similar to the ones found in the SAGA configuration, or automatically, taking the minimum extent that covers all the input layers used to execute the algorithm each time. If this is the behaviour you prefer, just check the *Use min covering region* option in the GRASS configuration parameters.

GRASS includes help files describing each algorithm. If you set the *GRASS help folder* parameter, SEXTANTE will open them when you use the **[Show help]** button from the parameters window of the algorithm.

The last parameter that has to be configured is related to the mapset. A mapset is needed to run GRASS, and SEXTANTE creates a temporary one for each execution. You have to tell SEXTANTE if the data you are working with uses geographical (lat/lon) coordinates or projected ones.

17.7.5 GDAL

No additional configuration is needed to run GDAL algorithms, since it is already incorporated to QGIS and SEXTANTE can infere its configuration from it.

17.7.6 Orfeo ToolBox

Orfeo ToolBox (OTB) algorithms can be run from SEXTANTE if you have OTB installed in your system and configured SEXTANTE properly so it can find all necessary files (command-line tools and libraries). Please note that OTB binaries are not included in SEXTANTE, so you have to download and install the software yourself. Please check the OTB website for more information.

Once OTB is installed, start QGIS, open the SEXTANTE configuration dialog and configure OTB algorithm provider. In the *Orfeo Toolbox (image analysis)* block you will find all settings related to OTB. First ensure that algorithms are enabled.

Then configure path to the folder where OTB command-line tools and libraries are installed:

- usually *OTB applications folder* point to `/usr/lib/otb/applications` and *OTB command line tools folder* is `/usr/bin`
- if you use OSGeo4W installer, than install `otb-bin` package and enter `C:\OSGeo4W\apps\orfeotoolbox\applications` as *OTB applications folder* and `C:\OSGeo4W\bin` as *OTB command line tools folder*

17.7.7 TauDEM

To use this provider you need to install TauDEM command line tools.

Windows

Please visit TauDEM homepage for installation instructions and precompiled binaries for 32bit and 64bit systems. **IMPORTANT**: you need TauDEM 5.0.6 executables, version 5.2 currently not supported.

Linux

There are no packages for most Linux distribution, so you should compile TauDEM by yourself. As TauDEM uses MPICH2, first install it using your favorite package manager. Also TauDEM works fine with OpenMPI, so you can use it instead of MPICH2.

Download TauDEM 5.0.6 source code and extract files in some folder.

Open `linearpart.h` file and add after line

```
#include "mpi.h"
```

add new line with

```
#include <stdint.h>
```

so you'll get

```
#include "mpi.h"
#include <stdlib.h>
```

Save changes and close file. Now open `tiffIO.h`, find line `#include "stdint.h"` and replace quotes (`""`) with <>, so you'll get

```
#include <stdint.h>
```

Save changes and close file. Create build directory and cd into it

```
mkdir build
cd build
```

Configure your build with command

```
CXX=mpicxx cmake -DCMAKE_INSTALL_PREFIX=/usr/local ..
```

and then compile

```
make
```

Finaly, to install TauDEM into `/usr/local/bin`, run

```
sudo make install
```

17.8 The SEXTANTE Commander

17.8.1 Introduction

SEXTANTE includes a practical tool that allows you to run algorithms without having to use the toolbox, but just typing the name of the algorithm you want to run.

This tools is known as the *SEXTANTE commander*, and it is just a simple text box with autocompletion where you type the command you want to run.

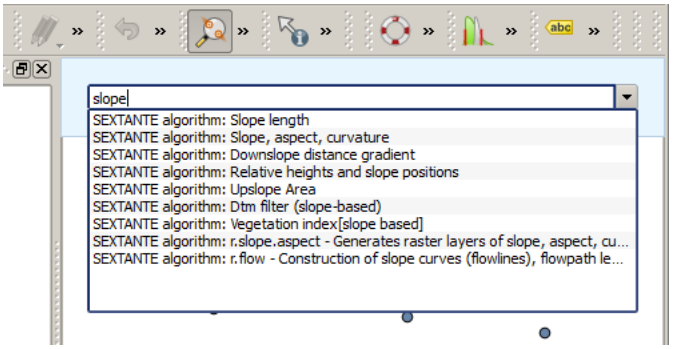


Figure 17.30: The SEXTANTE Commander

The commander is started from the *Analysis* menu or, more practically, pressing `Shift + Ctrl + M` (you can change that default keyboard shortcut in the QGIS configuration if you prefer a different one). Apart from executing SEXTANTE algorithms, the commander gives you access to most of the functionality in QGIS, which means that it gives you a practical and efficient way of running QGIS tasks, and allows you to control QGIS reducing the usage of buttons and menus.

Moreover, the commander is configurable and you can add your custom commands and have them just a few keystrokes away, making it a powerful tool to become more productive in your daily work with QGIS

17.8.2 Available commands

The commands available in the commander fall in the following categories:

- SEXTANTE algorithms. They are shown as `SEXTANTE algorithm: <name of the algorithm>`.
- Menu items. They are shown as `Menu item: <menu entry text>`. All menus items available from the QGIS interface are available, even if they are included in a submenu.
- Python functions. You can create short Python functions that will be then included in the list of available commands. They are shown as `Function: <function name>`

To run any of the above, just start typing and then select the corresponding element from the list of available ones that appears after filtering the whole list of commands with the text you have entered.

In the case of calling a Python function, you can select the entry in the list, which is prefixed by `Function:` (for instance, `Function: removeall`), or just directly type the function name (``removeall` in the previous example). There is no need to add brackets after the function name.

17.8.3 Creating custom functions

Custom functions are added by entering the corresponding Python code in the `commands.py` file that is found in the `.qgis/sextante/commander directory` in your user folder. It is just a simple Python file where you can add the functions that you need.

The file is created the first time you open the commander, with a few example functions. If you haven't launched the commander yet, you can create the file yourself. To edit the commands file, use your favorite text editor. You can also use a built-in editor by calling the `edit` command from the commander. It will open the editor with the commands file, and you can edit it directly and then save your changes.

For instance, you can add the following function, which removes all layers:

```
from qgis.gui import *

def removeall():
```

```
        mapreg = QgsMapLayerRegistry.instance()
        mapreg.removeAllMapLayers()
```

Once you have added the function, it will be available in the commander, and you can invoke it by typing `removeall`. There is no need to do anything apart from writing the function itself.

Functions can receive parameters. Add `*args` to your function definition, to receive argument. When calling the function from the commander, parameters have to be passed separated by spaces.

Here is an example of a function that loads a layer and takes a parameter with the filename of the layer to load.

```
import sextante

def load(*args):
        sextante.load(args[0])
```

If you want to load a layer in `/home/myuser/points.shp`, type `load /home/myuser/points.shp` in the commander text box.

CHAPTER 18

인쇄 조판기

인쇄 조판기는 레이아웃과 인쇄 기능을 제공합니다. 이것을 이용하면 QGIS 지도 캔버스, 범례, 축척 막대, 이미지, 기본도형, 화살표, 텍스트 라벨과 같은 요소를 도면에 추가 할 수 있습니다. 각각의 요소에 대해 크기, 그룹, 정렬, 위치 등을 지정할 수 있으며 레이아웃 작성을 위해 속성을 변경할 수 있습니다. 작성된 레이아웃은 직접 인쇄할 수도 있고, 이미지, 포스트 스크립트, PDF, SVG등의 파일로 내보낼 수도 있습니다. 그리고 레이아웃을 템플릿으로 저장하였다가 다른 섹션에서 이용할 수 있습니다. table_composer_1은 도구 목록을 나열한 것입니다:

아이콘	목적	아이콘	목적
	템플릿 로드		템플릿으로 저장
	이미지로 내보내기		PDF로 내보내기
	SVG로 내보내기		인쇄 또는 포스트스크립트로 내보내기
	전체보기		확대
	축소		뷰 새로 고침
	마지막 변경 되돌리기		마지막 변경 복원
	QGIS 지도 캔버스에 새로운 지도 추가		인쇄 구성에 이미지 추가
	인쇄 구성에 라벨 추가		인쇄 구성에 새로운 범례 추가
	인쇄 구성에 새로운 축적막대 추가		인쇄 구성에 기본도형 추가
	인쇄 구성에 화살표 추가		인쇄 구성에 속성표 추가
	인쇄 구성에서 아이템 선택 및 이동		아이템 내 콘텐츠 이동
	인쇄 구성의 아이템 그룹 지정		인쇄 구성의 아이템 그룹 해제
	선택 아이템을 앞으로 가져오기		선택 아이템을 뒤로 보내기
	선택 아이템을 맨 앞으로 가져오기		선택 아이템을 맨 뒤로 보내기
	선택 아이템을 좌측 정렬		선택 아이템을 우측 정렬
	선택 아이템을 가운데 정렬		선택 아이템을 중간 정렬
	선택 아이템을 위쪽 정렬		선택 아이템을 아래쪽 정렬

Table Composer 1: 인쇄 조판기 도구 목록

모든 인쇄 조판기 도구는 메뉴에 들어 있으며, 도구막대에도 아이콘이 들어 있습니다. 도구 막대위에서 마우스 오른쪽 버튼을 클릭하면 도구막대를 켜거나 끌 수 있습니다.

18.1 새로운 인쇄 조판기 템플릿 열기

인쇄 조판기로 작업을 시작하기 전에 미리 QGIS 지도 캔버스에 인쇄하려는 래스터와 벡터 레이어를 로드하고, 본인의 편의에 적합하게 속성을 설정해 두는 것이 좋습니다. 모든 것이 표현 된 후, 도구모음에서 New Print Composer 아이콘을 누르거나 *File* → *New Print Composer* [새 프린터 구성] 메뉴를 선택합니다.

18.2 인쇄 조판기 이용

인쇄 조판기를 열면 빈 캔버스가 열리는데, 여기에 현재 QGIS 지도 캔버스, 범례, 축척 막대, 이미지, 기본 도형, 화살표, 텍스트 등을 추가할 수 있습니다. Figure_composer_1_은 요소가 추가되기 전의 인쇄 조판기 초기화면으로써 *Snap to grid* [그리드에 스냅] 모드를 활성화시킨 것입니다. 인쇄 조판기에는 다음과 같은 3가지 탭이 있습니다.

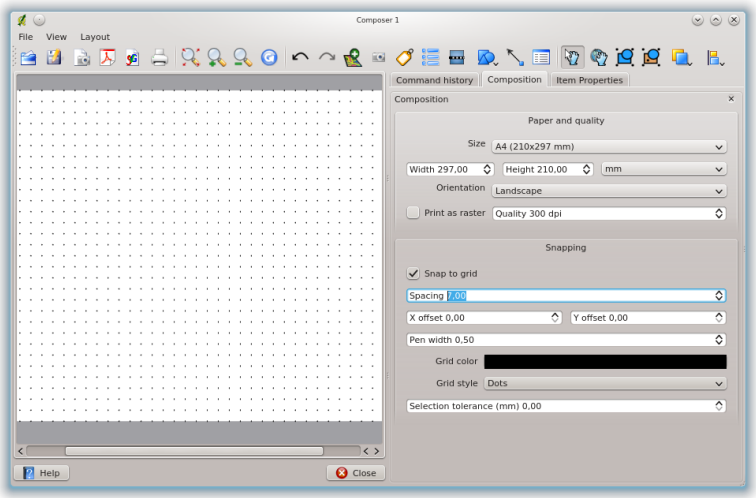


Figure 18.1: 인쇄 조판기

인쇄 조판기에는 다음과 같은 4가지 탭이 있습니다:

- *Composition* [일반정보] 탭에서는 용지 크기, 방향, 인쇄 품질(dpi) 등을 설정하고, 원하는 해상도로 격자에 스냅핑되도록 설정 할 수 있습니다. *Snap to grid* [그리드에 스냅]기능은 격자 해상도가 0 이상일때만 작업이 가능합니다. 아울러 *Print as raster* [래스터로 인쇄] 체크박스를 활성화시키면, 모든 요소를 래스터화시킨 후 인쇄 또는 포스트스크립트로 저장하게 됩니다.
- The *Item Properties* tab displays the properties for the selected map element. Click the Select/Move item icon to select an element (e.g. legend, scalebar or label) on the canvas. Then click the *Item Properties* tab and customize the settings for the selected element.
- The *Command history* tab displays a history of all changes applied to the print composer layout. With a mouse click it is possible to undo and redo layout steps back and forth to a certain status.
- The *Atlas generation* tab allows to enable the generation of an atlas for the current composer and gives access to its parameters.

구성기에 복수의 요소를 추가할 수 있으며, 인쇄 조판기 캔버스에는 지도 뷰, 범례, 축척 막대 등을 여러 개 넣을 수 있습니다. 각 요소는 독립된 속성을 가지며, 지도의 경우에는 영역도 독립적입니다. 구성기 캔버스의 요소를 삭제하고 싶으시다면, `Delete` 또는 the `Backspace` 키를 누르면 됩니다.

18.3 인쇄 조판기에 현재 QGIS 지도 캔버스 추가

인쇄 조판기 도구에 있는 Add new map [새 지도 추가] 아이콘을 클릭하고, 구성기 캔버스 위에 마우스 왼쪽 버튼을 클릭 한 후 사각형으로 드래그하시면 QGIS 지도 캔버스를 추가할 수 있습니다. 현재 지도를 표현하기 위해 지도 *Item Properties* [아이템 속성] 탭에서 3개의 다른 모드를 선택할 수 있습니다:

- **Rectangle** [사각형]은 기본 설정으로, 빈 박스에 '여기에 지도를 인쇄 예정'이라는 메시지가 표시됩니다.
- **Cache** [캐시]는 현재 화면 해상도로 지도를 표현합니다. 구성기 창을 확대하거나 축소할 경우 지도를 다시 그리지 않고, 단순 확대/축소됩니다.
- **Render** [렌더]는 구성기 윈도우를 축소하거나 확대할 경우에 지도를 다시 그리는 것을 의미합니다. 그러나 공간 때문에 최대 해상도까지만 표현됩니다.

인쇄 조판기에 지도가 새로 추가되었을 때의 기본 미리 보기 모드는 **Cache**입니다.

You can resize the map element by clicking on the Select/Move item button, selecting the element, and dragging one of the blue handles in the corner of the map. With the map selected, you can now adapt more properties in the map *Item Properties* tab.

지도 요소 내에서 레이어의 위치를 이동시키려면 먼저 지도 요소를 선택하고 Move item content [아이템 콘텐츠 이동] 아이콘을 클릭한 후에 마우스 왼쪽 버튼으로 지도 요소 프레임 안쪽에서 이동하시면 됩니다. 요소를 원하는 곳에 위치시킨 후에는 요소의 위치를 고정시킬 수 있습니다. 지도 요소를 선택하고 마우스 오른쪽 버튼을 클릭하면 지도요소에 Lock 표시가 나타나고 요소의 위치가 잠금 또는 해제됩니다. *Item Properties* [아이템 속성] 탭의 *Map* (지도) 대화상자에 있는 *Lock layers for map item* [지도 항목 레이어 고정] 체크박스를 활성화하여도 지도 요소를 잠글 수 있습니다.

주석: QGIS에서는 지도 구성기에서 새 라벨링 플러그인(new labeling plugin)으로 라벨을 표현할 수 있지만, 아직 축척이 바르게 작동하지 않습니다. 따라서 경우에 따라서는 표준 라벨로 변경해야 할 수 있습니다.

18.3.1 지도 아이템 탭 — 지도와 지도영역 대화상자

지도 대화상자 (**Map dialog**)

The *Map* dialog of the map *Item Properies* tab provides following functionalities (see figure_composer_2):

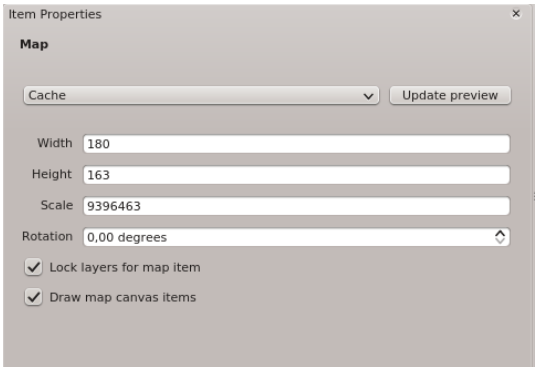


Figure 18.2: 지도 대화상자

- **Preview** [미리보기] 영역에서는 위에서 설명한 것과 같이 미리보기 모드-Rectangle[사각형], Cache[캐시], Render[렌더]-를 지정할 수 있습니다. 지도 화면에 변경된 사항을 적용하고 싶을 때는 **[Update preview]** [미리보기 업데이트] 를 클릭하시면 됩니다.

- **Map** [지도] 영역은 폭과 높이 또는 축척으로 지도 요소의 크기를 조정할 수 있습니다. *Rotation* 1,00 영역에 각도를 지정하면 지도 요소를 시계방향으로 회전시킬 수 있습니다. 주) 좌표 기준선은 회전이 0일 때만 추가할 수 있습니다. 그리고 *Lock layers for map items* [지도 항목 레이어 고정] 과 *Draw map canvas items* [지도 캔버스 아이템 그리기] 체크박스를 활성화 할 수 있습니다.

확대, 축소, 변경, 이동, 벡터 또는 래스터 속성을 변경하여 QGIS 지도 캔버스의 뷰를 변경한 뒤 이를 인쇄 조판기에서 확인하고 싶다면 **[Update preview]** [미리보기 업데이트] 버튼을 클릭하면 됩니다.

지도영역 대화상자 (**Extents dialog**)

지도 아이템 탭의 *Extents* (지도 영역) 대화상자는 다음의 사항을 설정할 수 있습니다 (figure_composer_3 참조):

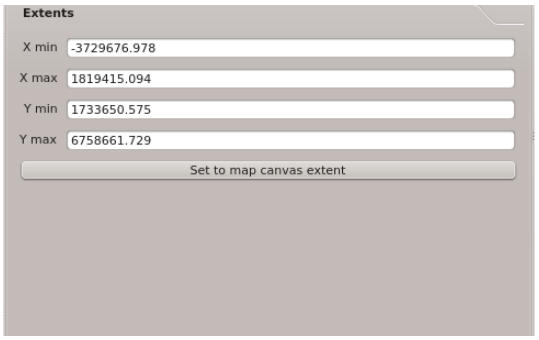


Figure 18.3: 지도영역 대화상자

- **Map extent** 영역에서는 X와 Y의 최소·최대값을 이용하거나 **[Set to map canvas extent]** [지도 캔버스 영역 지정] 버튼을 클릭해서 인쇄 편집기의 지도영역을 변경할 수 있습니다.

QGIS지도캔버스의 확대, 축소, 변경, 이동하거나 벡터 또는 래스터 속성을 변경시켰을 때, 이를 인쇄 조판기에서도 반영하고 싶다면 *Item Properties* [아이템 속성]탭 **[Update preview]** [미리보기 업데이트] 버튼을 클릭하면 됩니다(figure_composer_2 a 참조).

18.3.2 지도 아이템 탭 — 격자와 일반정보 옵션 대화상자

그리드 대화상자 (**Grid dialog**)

The *Grid* dialog of the map *Item Properties* tab provides following functionalities (see Figure_composer_4):

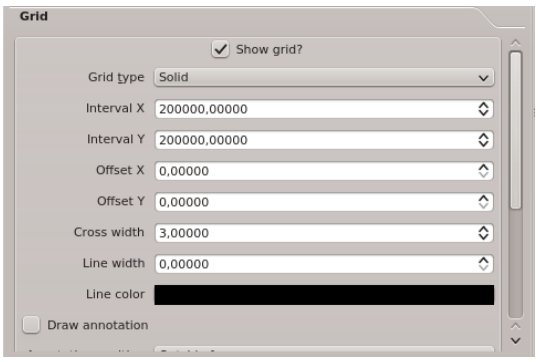


Figure 18.4: 그리드 대화상자

- *Show grid* [그리드를 표시합니까?] 체크박스를 클릭하면 지도요소에 격자를 중첩표시 할 수 있습니다. 격자유형은 실선 또는 교차선이 있습니다. 그리고 X와 Y의 방향 간격, X와Y오프셋, 실선 또는 교차선 격자 유형의 폭을 정의 할 수 있습니다.

- The *Draw annotation* checkbox allows to add coordinates to the map frame. The annotation can be drawn inside or outside the map frame. The annotation direction can be defined as horizontal, vertical, horizontal and vertical, or boundary direction. And finally you can define the grid color, the annotation font, the annotation distance from the map frame and the precision of the drawn coordinates.

일반정보 대화상자 (**General options dialog**)

지도 *Item Properties* 탭의 *General options* (일반 옵션) 대화상자에서는 다음의 사항을 설정할 수 있습니다(Figure_composer_5 참조):

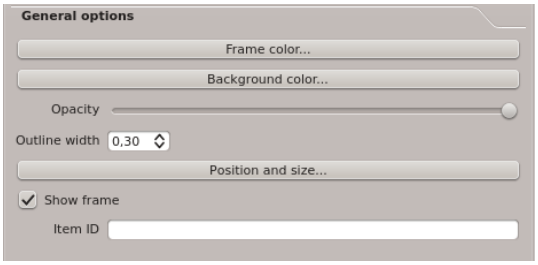


Figure 18.5: 일반 옵션 대화상자

- 요소 프레임의 색상과 외곽선 폭, 라벨의 배경색과 투명도를 설정할 수 있습니다. **[Position and size]** [위치와 크기] 버튼을 누르면 *Set item position* [항목 위치 설정] 대화상자가 열리는데, 참조점 또는 좌표를 이용하여 지도캔버스 위치를 설정할 수 있습니다. 그리고 *Show frame* [프레임 표시] 체크박스를 클릭하면 라벨에 프레임 여부를 지정할 수 있습니다.

18.4 인쇄 조판기에 다른 요소 추가

Besides adding a current QGIS map canvas to the Print Composer, it is also possible to add, position, move and customize legend, scalebar, images and label elements.

18.4.1 이미지 아이템 탭 - 그림 옵션과 일반 옵션 대화상자

To add a label, click the Add label icon, place the element with the left mouse button on the print composer canvas and position and customize their appearance in the label *Item Properties* tab.

라벨 대화상자 (**Label dialog**)

The *Label* dialog of the label item tab provides following functionalities:

- The *Label* dialog offers to add text labels to the composer canvas. You can define the horizontal and vertical alignment, select font and font color for the text and it is possible to define a text margin in mm.

일반정보 대화상자 (**General options dialog**)

The *General options* dialog of the label *Item Properties* tab provides following functionalities:

- 이미지 요소에 대한 프레임 색상과 외곽선 폭, 지도 캔버스의 배경색과 투명도를 설정할 수 있습니다. *Position* [위치와 크기] 버튼을 누르면 *Set items position* [항목 위치 설정] 대화상자가 열리는데, 참조점 또는 좌표를 이용하여 지도캔버스 위치를 설정할 수 있습니다. 그리고 *Show frame* [프레임 표시] 체크박스를 클릭하면 라벨에 프레임 표시 여부를 지정할 수 있습니다. Use the *Item ID* to create a relationship to other print composer items.

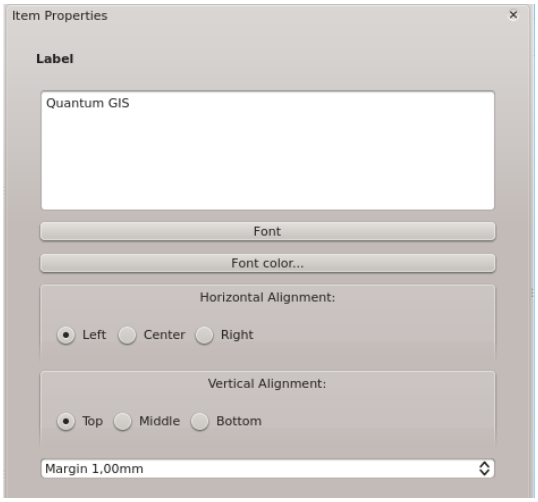


Figure 18.6: 라벨 옵션 대화상자

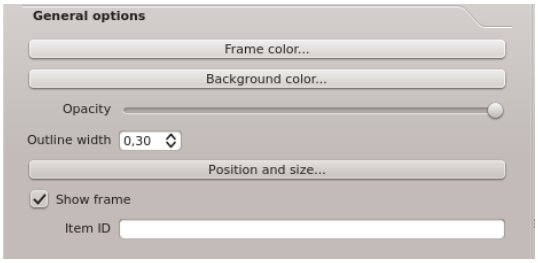


Figure 18.7: 일반 옵션 대화상자

18.4.2 인쇄 조판기 이미지 아이템 탭 ? 그림 옵션과 일반 옵션 대화상자

Add image [이미지 추가] 아이콘을 클릭하고 인쇄 조판기 캔버스 위에서 마우스 왼쪽 버튼을 이용하면 이미지를 추가시킬 수 있습니다. *Item Properties* 탭에서는 이미지의 속성을 설정하거나 변경할 수 있습니다.

그림 옵션 대화상자 (**Picture options dialog**)

The *Picture options* dialog of the image *Item Properties* tab provides following functionalities (see figure_composer_5 a):

- The **Preloaded Images** 영역에는 선택된 디렉토리 내 저장된 모든 그림이 표시됩니다.
- **Options** [옵션] 영역은 현재 선택된 그림의 정보를 표시하고 폭, 높이, 시계방향 회전 값 등을 설정할 수 있습니다. 또한 사용자가 특정 SVG 경로를 추가할 수 있습니다. *Sync with map* [지도에서 동기화] 체크박스를 활성화하면, QGIS 지도 캔버스에 포함된 이미지의 회전(예 북쪽 화살표가 회전)이 동기화됩니다.
- **Search directories** [검색 디렉토리] 영역에서는 SVG 포맷의 이미지가 있는 디렉토리를 그림 데이터베이스에 추가하거나 삭제할 수 있습니다.

일반정보 대화상자 (**General options dialog**)

The *General options* dialog of the image *Item Properties* tab provides following functionalities:

- 이미지 요소에 대한 프레임 색상과 외곽선 폭, 지도 캔버스의 배경색과 투명도를 설정할 수 있습니다. **[Position and size]** [위치와 크기] 버튼을 누르면 *Set items position* [항목 위치 설정] 대화상자가 열리는데, 참조점 또는 좌표를 이용하여 지도캔버스 위치를 설정할 수 있습니다. 그리고 *Show frame* [프레임 표시] 체크박스를 클릭하면 라벨에 프레임 표시 여부를 지정할 수 있습니다. With the *Item ID* to create a relationship to other print composer items

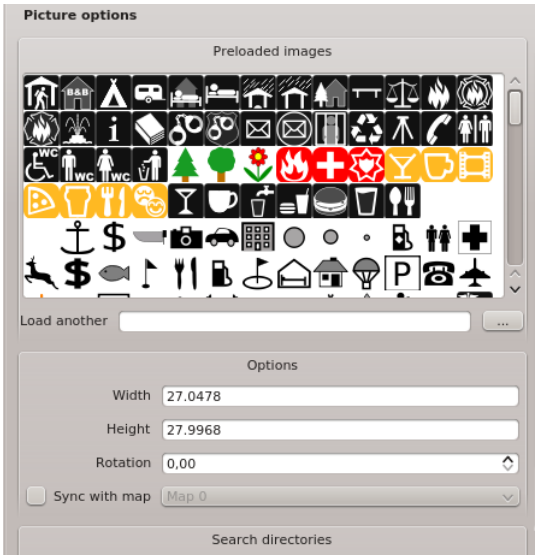


Figure 18.8: 그림 옵션 대화상자

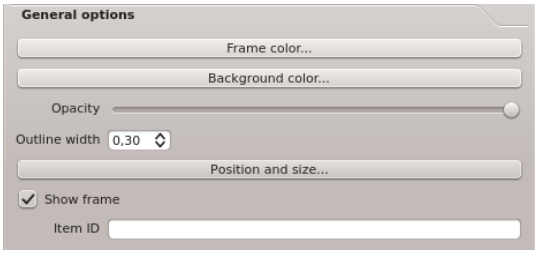


Figure 18.9: 일반 옵션 대화상자

18.4.3 범례 아이템 탭 - 일반, 범례 아이템과 아이템 옵션 대화상자

 Add new legend [새 벡터 범례 추가] 아이콘을 클릭하면 지도 범례를 추가할 수 있습니다. 인쇄 조판기 캔버스 위에서 마우스 왼쪽 버튼을 이용하여 위치시키면 됩니다. *Item Properties* (범례 아이템) 탭에서는 지도 범례의 속성을 정의/변경할 수 있습니다.

일반 대화상자 (**General dialog**)

범례 아이템 탭의 *General* (일반) 대화상자에서는 다음의 사항을 설정할 수 있습니다 (figure_composer_10):

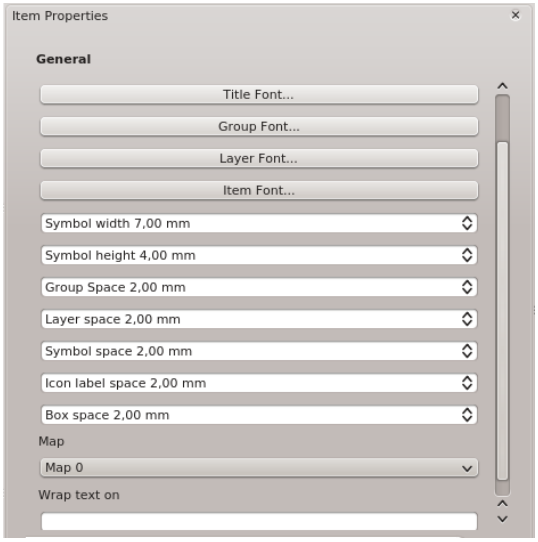


Figure 18.10: 일반 대화상자

- 일반 대화상자에서는 범례 제목을 지정할 수 있습니다. 범례 제목, 레이어와 아이템 이름의 글꼴 등을 변경할 수 있으며, 범례 심볼의 폭과 높이를 변경할 수 있습니다. 그리고 레이어, 심볼, 아이콘 라벨과 박스의 여백을 추가할 수 있습니다. Since QGIS 1.8, you can wrap the text of the legend title to a given character.

범례 항목 대화상자 (**범례 항목 대화상자**)

Item Properties (범례 아이템) 탭의 *Legend items* (범례 아이템) 대화상자에서는 다음의 사항을 설정할 수 있습니다 (figure_composer_11):

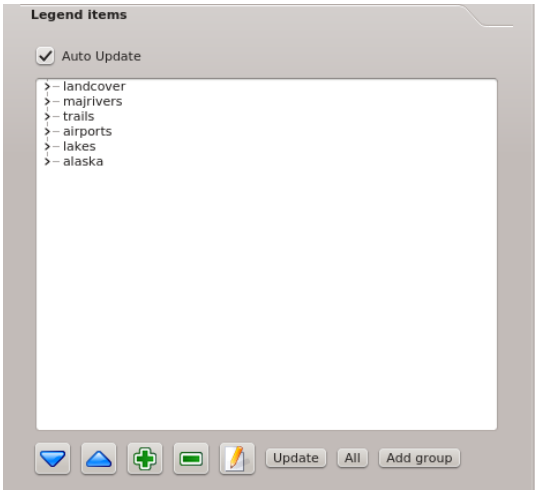


Figure 18.11: 범례 항목 대화상자

- 범례 아이템 창에서는 모든 범례 아이템의 목록이 표시됩니다. 아이템 순서 변경, 레이어 이름 편집, 목록의 아이템 삭제 및 복구 등을 수행할 수 있습니다. QGIS 메인 창에서 심볼 표기법을 변경한 후 인쇄 조판기의 범례 요소에 변경된 사항을 적용시키려면 **[Update]** 를 클릭하면 됩니다. 아이템 순서는 **[Up]** 과 **[Down]** 버튼 또는 드래그 앤 드롭 기능을 이용하여 변경할 수 있습니다.

일반정보 대화상자 (**General options dialog**)

Item Properties 탭의 *General options* (일반 옵션) 대화상자에서는 다음의 사항을 설정할 수 있습니다 (figure_composer_12):

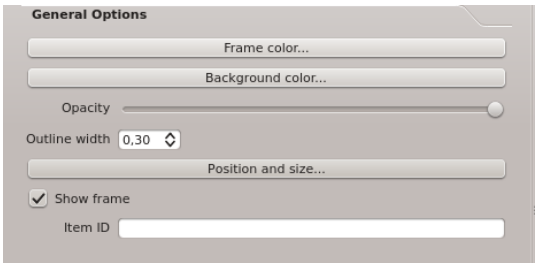


Figure 18.12: 일반 옵션 대화상자

- 범례에 대한 프레임 색상과 외곽선 폭, 지도 캔버스의 배경색과 투명도를 설정할 수 있습니다. **[Position and size]** [위치와 크기] 버튼을 누르면 *Set items position* [항목 위치 설정] 대화상자가 열리는데, 참조점 또는 좌표를 이용하여 지도캔버스 위치를 설정할 수 있습니다. 그리고 *Show frame* [프레임 표시] 체크박스를 클릭하면 라벨에 프레임 표시 여부를 지정할 수 있습니다. Use the *Item ID* to create a relationship to other print composer items.

18.4.4 축척 막대 아이템 탭 - 축척 막대와 일반 옵션 대화상자

To add a scalebar, click the Add new scalebar icon, place the element with the left mouse button on the print composer canvas and position and customize their appearance in the scalebar *Item Properties* tab.

축척 막대 대화상자 (**Scalebar dialog**)

축척 막대 *Item Properties* 탭의 *Scalebar* (축척 막대) 대화상자에서는 다음의 사항을 설정할 수 있습니다 (figure_composer_13):

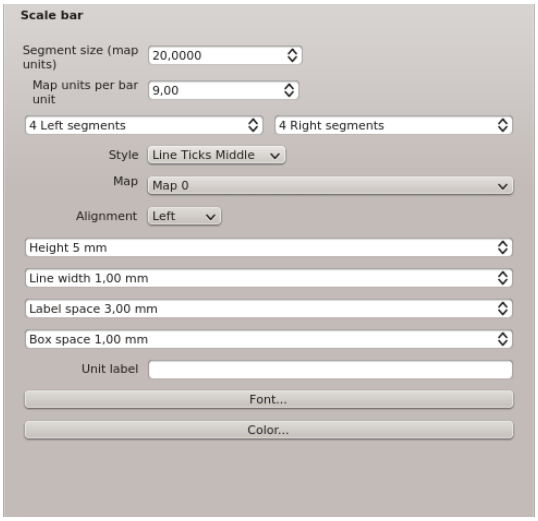


Figure 18.13: 축적 막대 옵션 대화상자

- *Scalebar* (축적 막대) 대화상자는 축척 막대의 한 눈금의 크기와, 한 개의 눈금 당 지도 단위를 정의할 수 있습니다. 또한 0에서 좌측과 우측에 몇 개의 눈금을 그릴지 결정할 수 있습니다.

- 축척 막대의 스타일을 정의할 수 있습니다. 스타일에서는 단일상자 이중상자 형식, 선의중앙/위/아래에 눈금을 그리는 형식, 기타 수를 쓰는 방식 등의 스타일을 선택할 수 있습니다.
- 아울러 축척 막대의 높이, 선 폭, 라벨, 박스 공간 등을 설정할 수 있습니다. 단위 라벨 을 추가하고, 축척 막대의 글꼴과 색상을 정의할 수 있습니다.

일반정보 대화상자 (**General options dialog**)

The *General options* dialog of the scalebar *Item Properties* tab provides following features (see figure_composer_7 b):

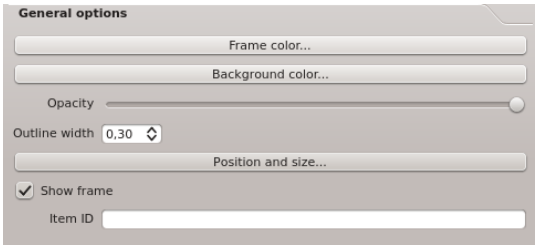


Figure 18.14: 일반 옵션 대화상자

- 축적 막대에 대한 프레임 색상과 외곽선 폭, 지도 캔버스의 배경색과 투명도를 설정할 수 있습니다. **[Position and size]** [위치와 크기] 버튼을 누르면 *Set items position* [항목 위치 설정] 대화상자가 열리는데, 참조점 또는 좌표를 이용하여 지도캔버스 위치를 설정할 수 있습니다. 그리고 *Show frame* [프레임 표시] 체크박스를 클릭하면 라벨에 프레임 표시 여부를 지정할 수 있습니다. Use the *Item ID* to create a relationship to other print composer items.

18.5 Atlas generation

The print composer includes generation functions that allow to create map books in an automated way. The concept is to use a coverage layer, which contains geometries and fields. For each geometry in the coverage layer, a new output will be generated where the content of some canvas maps will be moved to highlight the current geometry. Fields associated to this geometry can be used within text labels.

There can only be one atlas map by print composer. To enable the generation of an atlas and access generation parameters, refer to the *Atlas generation* tab. This tab contains the following widgets (see Figure_composer_15):

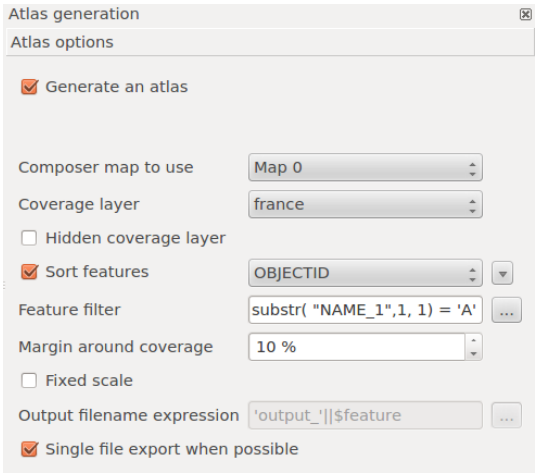


Figure 18.15: Atlas generation tab

- A combobox *Composer map to use*  that allows to choose which map item will be used as the atlas map, i.e. on which map geometries from the coverage layer will be iterated over and displayed.

- A combobox *Coverage layer*  that allows to choose the (vector) layer containing the geometries on which to iterate over.
- An optional *Hidden coverage layer*, that if checked, will hide the coverage layer (but not the other ones) during the generation.
- An optional *Sort features* that, if checked, allows to sort features of the coverage layer. The associated combobox allows to choose which column will be used as the sorting key. Sort order (either ascending or descending) is set by a two-state button that displays an up or a down arrow.
- An optional *Feature filter* text area that allows to specify an expression for filtering features from the coverage layer. If the expression is not empty, only features that evaluate to `True` will be selected. The button on the right allows to display the expression builder.
- An input box *Margin around coverage* that allows to select the amount of space added around each geometry within the allocated map. Its value is meaningful only when using the autoscaling mode.
- A *Fixed scale* that allows to toggle between auto-scale and fixed-scale mode. In fixed scale mode, the map will only be translated for each geometry to be centered. In auto-scale mode, the map's extents are computed in such a way that each geometry will appear in its whole.
- An *Output filename expression* textbox that is used to generate a filename for each geometry if needed. It is based on expressions. This field is meaningful only for rendering to multiple files.
- A *Single file export when possible* that allows to force the generation of a single file if this is possible by the chosen output format (PDF for instance). If this field is checked, the value of the *Output filename expression* field is meaningless.

18.5.1 Generation

The atlas generation is done when the user asks for a print or an export. The behaviour of these functions will be slightly changed if an atlas map has been selected.

For instance, when the user asks for an export to PDF, if an atlas map is defined, the user will be asked for a directory where to save all the generated PDF files (except if the *Single file export when possible* has been selected).

18.6 내비게이션 도구

인쇄 조판기는 다음과 같은 4 가지 내비게이션 도구가 있습니다:

- Zoom in
- Zoom out
- Zoom to full extend
- Refresh the view (if you find the view in an inconsistent state)

18.7 되돌리기와 복원 도구

레이아웃을 설정하는 동안 마지막 변경 사항을 되돌리거나 복원할 수 있습니다. 다음과 같은 도구를 누르시면 됩니다:

- Revert last changes [마지막 변경 취소]

- Restore last changes [마지막 변경 다시 불러오기]

또는 *Command history* [명령 히스토리] 탭에서 원하는 위치를 클릭하면 해당 단계로 복원됩니다 (figure_composer_9 참조).

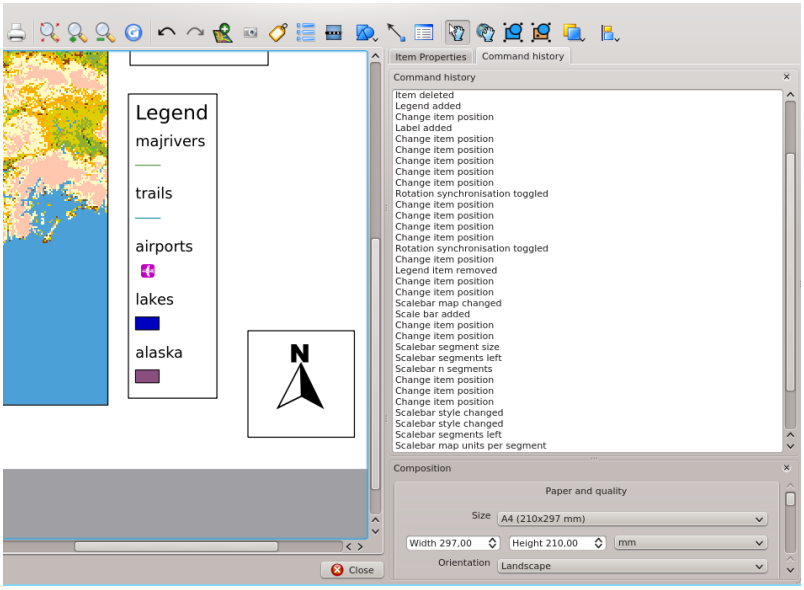


Figure 18.16: 인쇄 조판기의 명령어 히스토리

18.8 기본 도형과 화살표 추가

인쇄 조판기 캔버스에 기본 도형(타원, 사각형, 삼각형)과 화살표를 추가할 수 있습니다.

Shape 대화상자를 이용하면 인쇄 조판기 캔버스에 타원, 사각형, 삼각형을 그릴 수 있습니다. 또한 외곽선이나, 색상, 외곽선 폭, 회전 등을 설정할 수 있습니다.

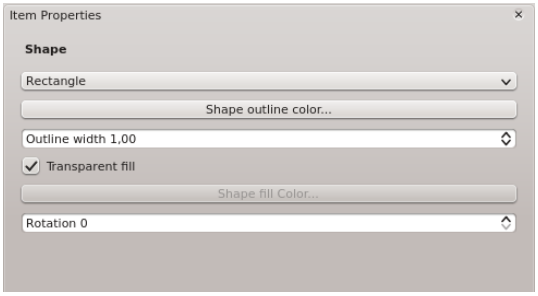


Figure 18.17: Shape 대화상자

Arrow 대화상자는 인쇄 조판기 캔버스 에 화살표를 추가할 수 있습니다. 화살표의 색상, 외곽선 폭, 화살표의 촉의 폭 등을 정의할 수 있으며, 화살표 마커는 기본 마커, 마커 없음, SVG 마커를 이용할 수 있습니다. SVG 마커를 이용할 때에는 추가적으로 컴퓨터에 저장된 SVG 시작과 끝 마커를 지정할 수 있습니다.

18.9 속성표 추가

벡터 속성표를 인쇄 조판기 캔버스에 추가할 수 있습니다.

테이블 대화상자 (**Table dialog**)



Figure 18.18: Arrow 대화상자

속성표 아이템 탭의 *Table* 대화상자에서는 다음의 사항을 설정할 수 있습니다(그림 10.10a 참조):

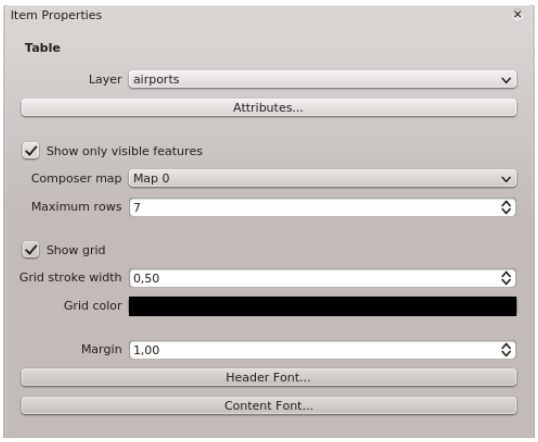


Figure 18.19: 표 대화상자

- The *Table* dialog allows to select the vector layer and columns of the attribute table. Attribute columns can be sorted and you can define to show its values ascending or descending.
- 표시되는 행의 최대수를 정의할 수 있습니다. 또한 현재 구성기 캔버스에 표시되고 있는 객체의 속성만을 표시하도록 설정할 수 있습니다.
- 부가적으로 격자 특성과 헤더와 콘텐츠 폰트를 정의 할 수 있습니다.

일반정보 대화상자 (**General options dialog**)

속성표 아이템 탭의 *General options* (일반 옵션) 대화상자는 다음의 사항을 설정할 수 있습니다(figure_composer_21 참조):

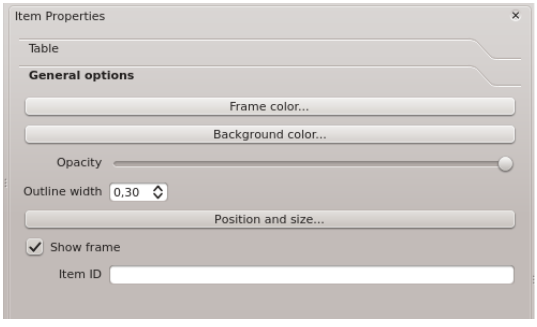


Figure 18.20: 일반 옵션 대화상자

- 표에 대한 프레임 색상과 외곽선 폭, 지도 캔버스의 배경색과 투명도를 설정할 수 있습니다. **[Position and size]** [위치와 크기] 버튼을 누르면 *Set items position* [항목 위치 설정] 대화상자가 열리는데, 참조점 또는 좌표를 이용하여 지도캔버스 위치를 설정할 수 있습니다. 그리고 *Show frame* [프레임 표시] 체크박스를 클릭하면 라벨에 프레임 표시 여부를 지정할 수 있습니다. Use the *Item ID* to create a relationship to other print composer items.

18.10 요소 앞으로 가져오기, 뒤로 보내기, 정렬

요소 앞으로 가져오기 또는 뒤로 보내기 기능은 Raise selected items [선택한 항목을 위로] 풀다운 메뉴에 들어 있습니다. 인쇄 조판기 캔버스에서 원하는 요소를 선택하고 선택된 요소와 다른 요소의 위치관계를 비교해서 앞으로 가져오기 또는 뒤로 보내기 기능을 선택하시면 됩니다. (table_composer_1 참조).

There are several alignment functionalities available within the Align selected items pulldown menu (see table_composer_1). To use an alignment functionality , you first select some elements and then click on the matching alignment icon. All selected will then be aligned within to their common bounding box.

18.11 출력 작성

Figure_composer_22 은 이상에서 설명한 여러가지 요소를 사용한 예시 지도를 인쇄 조판기에 표시한 것입니다.

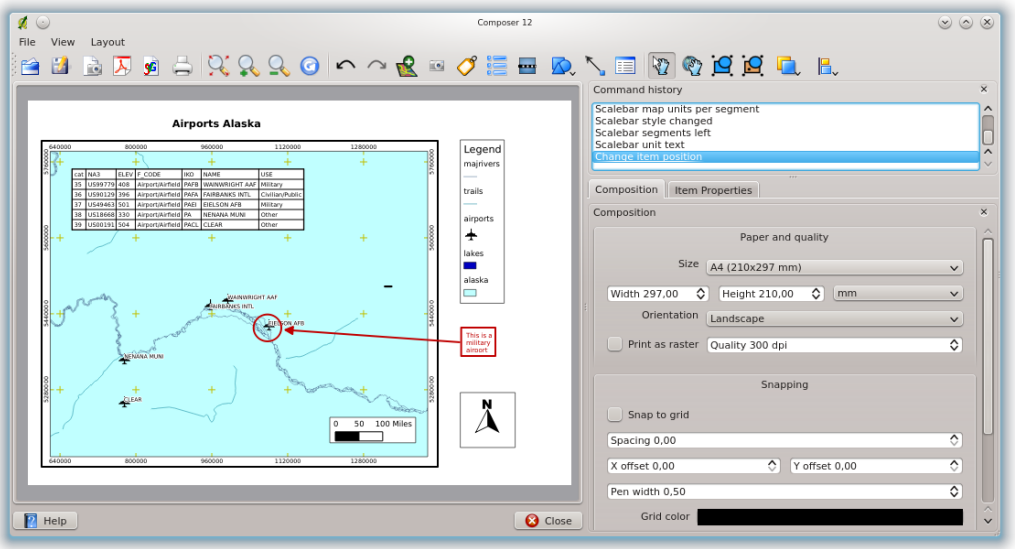


Figure 18.21: 지도 화면, 범례, 축척 막대, 좌표와 텍스트가 추가된 인쇄 조판기

인쇄 조판기에서는 다양한 종류의 출력형식을 생성할 수 있으며, 해상도(인쇄품질)와 용지 크기 등을 지정할 수 있습니다:

- Print 아이콘을 누르면 조판기 레이아웃의 내용을 연결된 프린터 또는 포스트스크립트파일로 출력할 수 있습니다.
- Export as image [이미지로 내보내기] 아이콘을 누르면 PNG, BPM, TIF, JPG 등 다양한 이미지로 내보낼 수 있습니다...
- Export as PDF [PDF로 내보내기]] 아이콘을 이용하면 현재 정의되어 있는 인쇄 조판기 canvas를 PDF 파일로 직접 내보냅니다.

- Export as SVG [SVG로 내보내기] 아이콘은 인쇄 조판기 캔버스의 내용을 SVG(Scalable Vector Graphic) 파일로 저장합니다.

주석: 현재 SVG 출력은 매우 기본적인 기능만 구현되어 있습니다. 이것은 QGIS의 문제가 아니라, QT 라이브러리가 지원하지 않는 문제입니다. 이후 버전에서는 이 문제가 해결되기를 기대합니다.

18.12 인쇄 조판기 레이아웃 저장과 읽기

Save as template [템플릿으로 저장] 아이콘을 선택하면 인쇄 조판기 세션의 현재 상태를 `.qpt` 템플릿으로 저장할 수 있으며, Load from template [템플릿에서 읽기] 아이콘를 이용하여 다른 세션에서 다시 읽어 들일 수 있습니다.

도구 모음에 있는 Composer Manager [구성 관리자] 버튼을 누르거나 *File* → *Composer Manager* [구성 관리자] 메뉴를 이용하면 새로운 구성기 템플릿을 추가하거나 기존 저장된 템플릿의 이름변경이나 삭제 등을 할 수 있습니다.

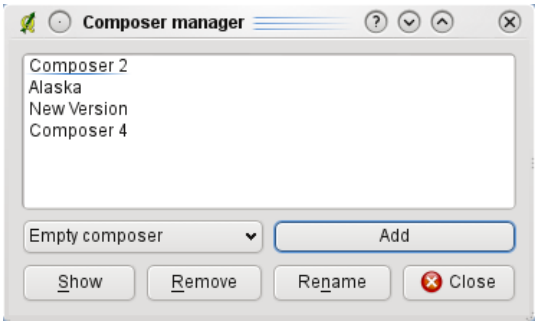


Figure 18.22: 구성기 관리자

CHAPTER 19

Plugins

19.1 QGIS Plugins

QGIS has been designed with a plugin architecture. This allows many new features/functions to be easily added to the application. Many of the features in QGIS are actually implemented as either **core** or **external plugins**.

- **Core Plugins** are maintained by the QGIS Development Team and are automatically part of every QGIS distribution. They are written in one of two languages: C++ or Python. More information about core plugins are provided in Section *Using QGIS Core Plugins*
- **External Plugins** are currently all written in Python. They are stored in external repositories and maintained by the individual authors. They can be added to QGIS using the *Python Plugin Installer*. More information about external plugins is provided in Section *Loading an external QGIS Plugin*.

19.1.1 Managing Plugins

Managing plugins in general means loading or unloading them using the *Plugin Manager*. External plugins can be installed and directly activated or uninstalled using the *Python Plugin Installer*. To deactivate and reactivate external plugins, the *Plugin Manager* is used again.

Loading a QGIS Core Plugin

Loading a QGIS Core Plugin is done from the main menu *Plugins →' |mActionShowPluginManager| :menuselection:'Manage Plugins*.

The *Plugin Manager* lists all the available plugins and their status (loaded or unloaded), including all core plugins and all external plugins that have been installed and automatically activated using the *Python Plugin Installer* (see Section *Loading an external QGIS Plugin*). Those plugins that are already loaded have a check mark to the left of their name. Figure_plugins_1 shows the *Plugin Manager* dialog.

To enable a particular plugin, click on the checkbox to the left of the plugin name, and click **[OK]**. When you exit the application, a list of loaded plugins is retained, and the next time you run QGIS these plugins are automatically loaded.

참고: Crashing Plugins

If you find that QGIS crashes on startup, a plugin may be at fault. You can stop all plugins from loading by editing your stored settings file (see *Options* for location). Locate the plugins settings and change all the plugin values to `false` to prevent them from loading.

For example, to prevent the 'Delimited text' plugin from loading, the entry in `$HOME/.config/QuantumGIS/qgis.conf` on Linux should look like this: `Add Delimited Text Layer=false`.

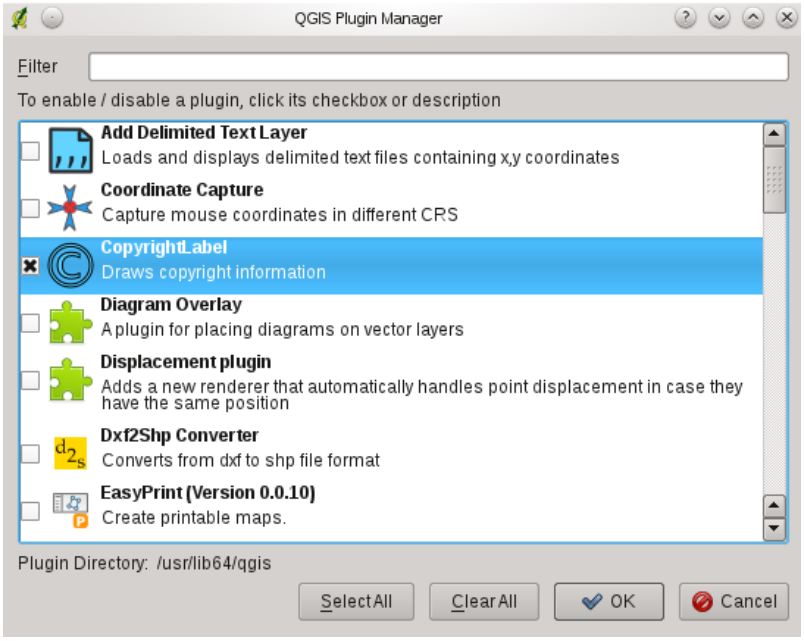


Figure 19.1: Plugin Manager

Do this for each plugin in the [Plugins] section. You can then start QGIS and add the plugins one at a time from the *Plugin Manager* to determine which plugin is causing the problem.

Loading an external QGIS Plugin

External QGIS plugins are written in Python. They are by default stored in either the 'Official' QGIS Repository, or in various other external repositories maintained by individual authors. The 'Official' QGIS Repository is added by default in *Python Plugin installer*.

Detailed documentation about the usage, minimum QGIS version, homepage, authors, and other important information are provided for the 'Official' QGIS Repository at http://plugins.qgis.org/plugins/. For other external repositories, they might be available with the external plugins themselves. In general it is not included in this manual.

주석: Updates of core python plugins may be available in this repository as external overlays.

주석: fTools, Mapserver Export, and the Plugin Installer are Python plugins, but they are also part of the QGIS sources, and are automatically loaded and enabled inside the QGIS Plugin Manager (see Section *Loading an external QGIS Plugin*).

Currently there are over 150 external plugins available from the 'Official' QGIS Repository. Some of these plugins offer functionality that will be required by many users (for example: providing the user with the ability to view and edit OpenStreetMap data, or to add GoogleMap layers) while others offer very specialized functions (for example: Calculate economic pipe diameters for water supply networks).

It is, however, quite straightforward to search through all the available external plugins by providing keywords, choosing a named repository and/or filtering on the status of plugins (currently installed or uninstalled in your system). Searching and filtering is done from the QGIS Python Plugin Installer

---

참고: Add more repositories

To add external author repositories, open the Plugin Installer (*Plugins* → *Fetch Python Plugins*), go to the tab *Repositories*, and click **[Add]**. If you do not want one or more of the added repositories, they can be disabled via the **[Edit...]** button, or completely removed with the **[Delete]** button.

There is only one step required to integrate external plugins into QGIS:

- Download an external plugin from a repository using the *Python Plugin Installer* (see *Using the QGIS Python Plugin Installer*). The new external plugin will be added to the list of available plugins in the *Plugin Manager* and is automatically loaded.

Using the QGIS Python Plugin Installer

In order to download and install an external Python plugin, go to *Plugins* → *Fetch Python Plugins*. The *Plugin Installer* window will appear (figure_plugins_2) with the tab *Plugins*, containing a list of all locally installed Python plugins, as well as plugin available in remote repositories.

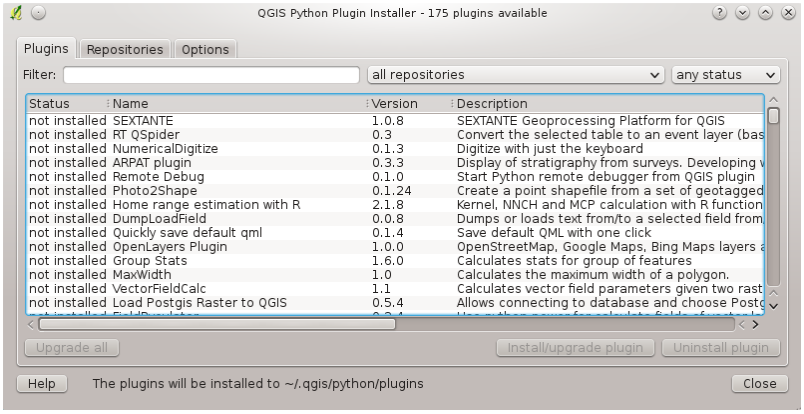


Figure 19.2: Installing external python plugins

Each plugin can be either:

- **not installed** - this means the plugin is available in the repository, but is not installed yet. In order to install it, select the plugin from the list and click the button **[Install plugin]**.
- **new** - this means that the plugin is newly available in the repository.
- **installed** - this indicates that the plugin is already installed. If it is also available in any repository the **[Reinstall plugin]** button will be enabled. If the available version is older than the installed version, the **[Downgrade plugin]** button will appear instead.
- **upgradeable** - this means that the plugin is installed, but there is an updated version available. In this case, the **[Upgrade plugin]** and **[Upgrade all]** buttons will be enabled.
- **invalid** - this means that the plugin is installed, but is unavailable or broken. The reason will be explained in the plugin description field.

Plugins tab

To install a plugin, select it from the list and click the **[Install plugin]** button. The plugin is then activated and installed in its own directory.

- Linux and other unices
  - `/share/qgis/python/plugins`
  - `$HOME/.qgis/python/plugins`
- Mac OS X
  - `/Contents/MacOS/share/qgis/python/plugins`
  - `$HOME/.qgis/python/plugins`

- Windows
  - `C:\Program Files\QGIS\python\plugins`
  - `C:\Documents and Settings\$USERNAME\.qgis\python\plugins`

If the installation is successful, a confirmation message will appear.

If the installation fails, the reason for the failure will be displayed in a warning dialog. Most often, errors are the result of connection problems and/or missing Python modules. In the former case you will likely need to wait before trying the install again, in the latter case, you should install the missing modules relevant to your operating system prior to using the plugin. For Linux, most required modules should be available via a package manager. For install instructions in Windows visit the module home page.

If you are using a proxy, you may need to configure it under *Edit* → *Options* (Gnome, OSX) or *Settings* → *Options* (KDE, Windows)' on the *Proxy* tab.

The **[Uninstall plugin]** button is enabled only if the selected plugin is installed and is not a core plugin. Note that if you have installed an update to a core plugin, you can uninstall this update with the **[Uninstall plugin]** and revert to the version shipped with Quantum GIS. This default version however, cannot be uninstalled.

Repositories tab

The second tab *Repositories*, contains a list of plugin repositories available for the *Plugin Installer*. By default, only the QGIS Official Repository is enabled. You can add several user-contributed repositories, including the central QGIS Contributed Repository and other external repositories by clicking the **[Add ...]** button. The added repositories contain a large number of useful plugins which are not maintained by the QGIS Development Team. As such, we cannot take any responsibility for them. You can also manage the repository list manually, that is add, remove, and edit the entries. Temporarily disabling a particular repository is possible by clicking the **[Edit ...]** button.

Options tab

The *Options* tab is where you can configure the settings of the *Plugin Installer*. The *Check for updates on startup* checkbox tells QGIS to automatically look for plugin updates and news. By default, if this feature is enabled all repositories listed and enabled in the *Repositories* tab are checked for updates each time the program is started. The frequency of update checking can be adjusted using the dropdown menu, and may be adjusted from once a day right up to once a month. If a new plugin or update is available for one of the installed plugins, a notification will appear in the Status Bar. If the checkbox is disabled, looking for updates and news is performed only when the *Plugin Installer* is manually launched from the menu.

Although the plugin installer update can handle ports different from 80, some internet connections will cause problems when attempting to automatically check for updates. In these cases, a *Looking for new plugins...* indicator will remain visible in the Status Bar during your entire QGIS session, and may cause a program crash when exiting. In this case please disable the checkbox.

In addition, you may specify the type of plugins that are displayed by the *Python Plugin Installer*. Under *Allowed plugins*, you can specify whether you would like to:

- *Only show plugins from the official repository*
- *Show all plugins except those marked as experimental*
- *Show all plugins, even those marked as experimental*

참고: Using experimental plugins

Experimental plugins are generally unsuitable for production use. These plugins are in the early stages of development, and should be considered 'incomplete' or 'proof of concept' tools. The QGIS development team does not recommend installing these plugins unless you intend to use them for testing purposes.

19.1.2 Data Providers

Data Providers are 'special' plugins that provides access to a data store. By default, QGIS supports PostGIS layers and disk-based data stores supported by the GDAL/OGR library. A Data Provider plugin extends the ability of QGIS to use other data sources.

Data Provider plugins are registered automatically by QGIS at startup. They are not managed by the Plugin Manager but used behind the scenes when a data type is added as a layer in QGIS.

19.2 Using QGIS Core Plugins

Icon	Plugin	Description	Manual Reference
	Add Delimited Text Layer	Loads text files containing x,y coordinates	*Delimited Text Plugin*
	Coordinate Capture	Capture mouse coordinate in different CRS	*Coordinate Capture Plugin*
	DB Manager	Manage your databases within QGIS	*DB Manager Plugin*
	Diagram Overlay	Add Diagrams for vectors	*Diagram Overlay Plugin*
	DXF2Shape Converter	Converts from DXF to SHP file format	*Dxf2Shp Converter Plugin*
	eVis	Event Visualization Tool	*eVis Plugin*
	fTools	A suite of vector tools	*fTools Plugin*
	GPS Tools	Tools for loading and importing GPS data	*GPS Plugin*
	GRASS	GRASS functionality	*GRASS GIS Integration*
	GDAL Tools	GDAL raster functionality	*GDAL Tools Plugin*
	Georeferencer GDAL	Georeference Raster with GDAL	*Georeferencer Plugin*
	Heatmap	Create heatmap raster from input vector points	*Heatmap Plugin*
	Interpolation plugin	Interpolation on base of vertices of a vector layer	*Interpolation Plugin*
	MapServer Export Plugin	Export a QGIS project file to a MapServer map file	*MapServer Export Plugin*
	Offline Editing	Offline editing and synchronizing with database	*Offline Editing Plugin*
	OpenStreetMap	Access OpenStreetMap	*OpenStreetMap*
	Oracle Spatial Georaster	Access Oracle Spatial GeoRasters	*Oracle GeoRaster Plugin*
	Plugin Installer	Download and install python plugins	*Using the QGIS Python Plugin Installer*
	Raster Terrain Analysis	Compute geomorphological features from DEMs	*Raster Terrain Analysis Plugin*
	Road Graph Plugin	Shortest path analysis	*Road Graph Plugin*
	SPIT	Shapefile to PostGIS Import Tool	*1.1.1. PostgreSQL에 데이터 넣기*
	SQL Anywhere plugin	Access SQL anywhere DB	*SQL Anywhere Plugin*
	Spatial Query	Spatial queries on vectors	*Spatial Query Plugin*
	Zonal Statistics	Calculate raster statistics for vector polygons	*Zonal Statistics Plugin*

19.3 Coordinate Capture Plugin

The coordinate capture plugin is easy to use and provides the ability to display coordinates on the map canvas for two selected Coordinate Reference Systems (CRS).

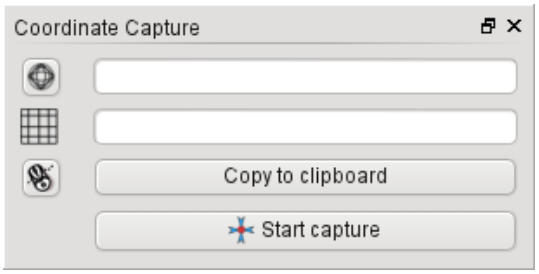


Figure 19.3: Coordinate Capture Plugin

1. Start QGIS, select *Project Properties* from the *Settings* (KDE, Windows) or *File* (Gnome, OSX) menu and click on the *Projection* tab. As an alternative you can also click on the CRS status icon in the lower right-hand corner of the statusbar.
2. Click on the *Enable on the fly projection* checkbox and select a projected coordinate system of your choice (see also *Working with Projections*).
3. Load the coordinate capture plugin in the Plugin Manager (see *Loading a QGIS Core Plugin*) and ensure that the dialog is visible by going to *View* → *Panels* and ensuring that *Coordinate Capture* is enabled. The coordinate capture dialog appears as shown in Figure figure_coordinate_capture_1.
4. Click on the Click to the select the CRS to use for coordinate display icon and select a different CRS from the one you selected above.
5. To start capturing coordinates, click on **[Start capture]**. You can now click anywhere on the map canvas and the plugin will show the coordinates for both of your selected CRS.
6. To enable mouse coordinate tracking click the mouse tracking icon.
7. You can also copy selected coordinates to the clipboard.

19.4 DB Manager Plugin

The DB Manager Plugin is officially part of QGIS core and intends to replace the SPIT Plugin and the PostGIS Manager Plugin and additionally to integrate all other database formats supported by QGIS in one user interface. The DB Manager Plugin provides several features. You can drag layers from the QGIS Browser into the DB Manager and it will import your layer into your spatial database. You can drag and drop tables between spatial databases and they will get imported. You can also use the DB Manager to execute SQL queries against your spatial database and then view the spatial output for queries by adding the results to QGIS as a query layer.

The *Database* menu allows to connect to an existing database, to start the SQL-window and to exit the DB Manager Plugin. The *Schema* menu includes tools to create and delete (empty) schemas and, if topology is available (e.g. PostGIS 2) to start a *TopoViewer*. The menu *Table* allows to create and edit tables and to delete tables and views. It is also possible to empty tables and to move tables from one to another schema. Finally you can also run Vacuum Analyze and add Versioning Support to a table.

The *Tree* window lists all existing databases supported by QGIS. With a double-click you can connect to the database. With the right-mouse button you can rename and delete existing schemas and tables. Tables can also be added to the QGIS canvas with the context menu.

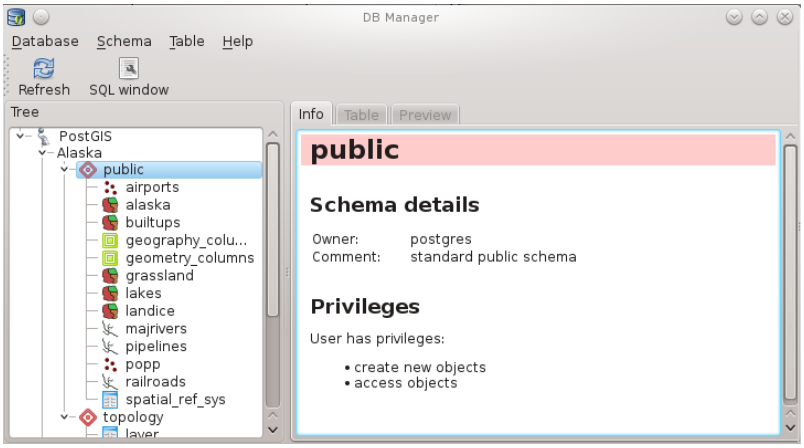


Figure 19.4: DB Manager dialog (KDE)

If connected to a database, the **main** window of the DB Manager offers three tabs. The *Info* tab provides information about the table and its geometry as well as about existing Fields, Constraints and Indexes. It also allows to run Vacuum Analyze and to create a spatial index on a selected table, if not already done. The *Table* tab shows all attributes and the *Preview* tab renders the geometries as preview.

19.5 Delimited Text Plugin

The Delimited Text plugin allows you to load a delimited text file as a layer in QGIS.

19.5.1 Requirements

To view a delimited text file as layer, the text file must contain:

1. A delimited header row of field names. This must be the first line in the text file.
2. The header row must contain an X and Y field. These fields can have any name.
3. The x and y coordinates must be specified as a number. The coordinate system is not important.

As an example of a valid text file we import the elevation point data file `elevp.csv` coming with the QGIS sample dataset (See Section *Sample Data*):

```
X;Y;ELEV
-300120;7689960;13
-654360;7562040;52
1640;7512840;3
[...]
```

Some items of note about the text file are:

1. The example text file uses `;` (semicolon) as delimiter. Any character can be used to delimit the fields.
2. The first row is the header row. It contains the fields `X`, `Y` and `ELEV`.
3. No quotes (`"`) are used to delimit text fields.
4. The x coordinates are contained in the `X` field.
5. The y coordinates are contained in the `Y` field.

19.5.2 Using the Plugin

To use the plugin you must first enable it as described in Section *Managing Plugins*.

Click the new toolbar icon Add Delimited Text Layer to open the *Delimited Text* dialog as shown in figure_delimited_text_1.

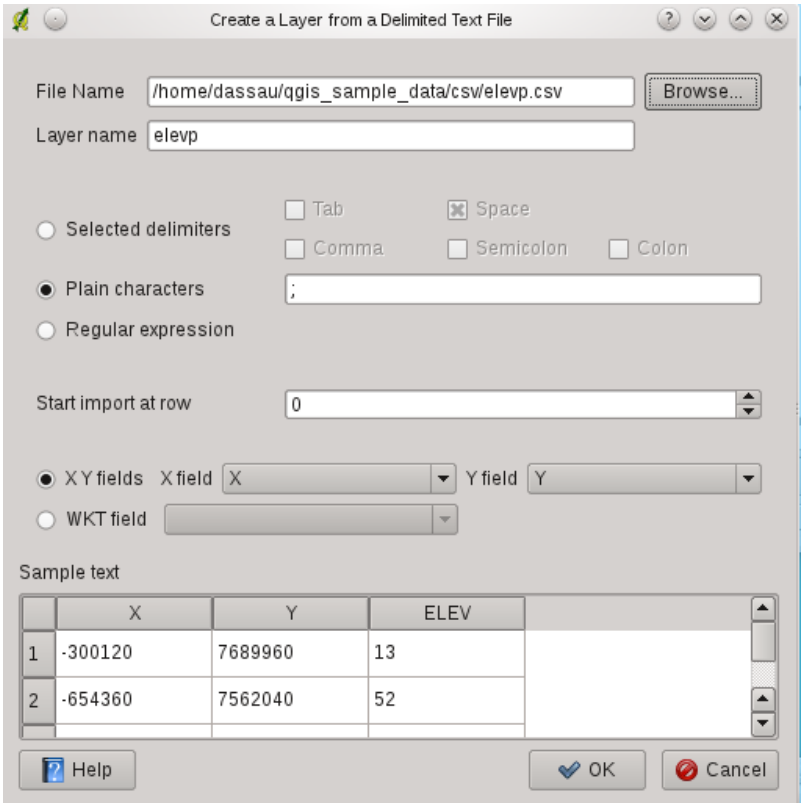


Figure 19.5: Delimited Text Dialog

First select the file (e.g., `qgis_sample_data/csv/elevp.csv`) to import by clicking on the **[Browse]** button. Once the file is selected, the plugin attempts to parse the file using the last used delimiter, in this case a semicolon (`;`). To properly parse the file, it is important to select the correct delimiter. To change the delimiter to tab use `\t` (this is a regular expression for the tab character).

Once the file is parsed, choose the `X` and `Y` fields from the dropdown lists and if available also the WKT field for the CRS information. Finally enter a Layer name (e.g., `elevp`) as shown in figure_delimited_text_1 . To add the layer to the map, click **[OK]**. The delimited text file now behaves as any other map layer in QGIS.

19.6 Diagram Overlay Plugin

 The *Diagram Overlay* Plugin allows you to add a graphic overlay to a vector layer (see figure_overlay_1). It provides additional and not yet implemented features to the *Diagrams* tab, described in secion *다이어그램 탭(Diagram Tab)*.

Before starting, the *Diagram Overlay* Plugin needs to be activated using the Plugin Manager (see Section *Loading a QGIS Core Plugin*). It will then appear as *Overlay* tab in the *Layer Properties* dialog next to the *Diagrams* tab.

The *Overlay* tab provides support for Pie charts, Bar charts and proportional SVG symbols.

Similar to the *Diagrams* tab, we will demonstrate an example and overlay the alaska boundary layer a piechart diagram showing some temperature data from a climate vector layer. Both vector layers are part of the QGIS sample dataset (see Section *Sample Data*).

1. First click on the  Load Vector icon, browse to the QGIS sample dataset folder and load the two vector shape layers `alaska.shp` and `climate.shp`.

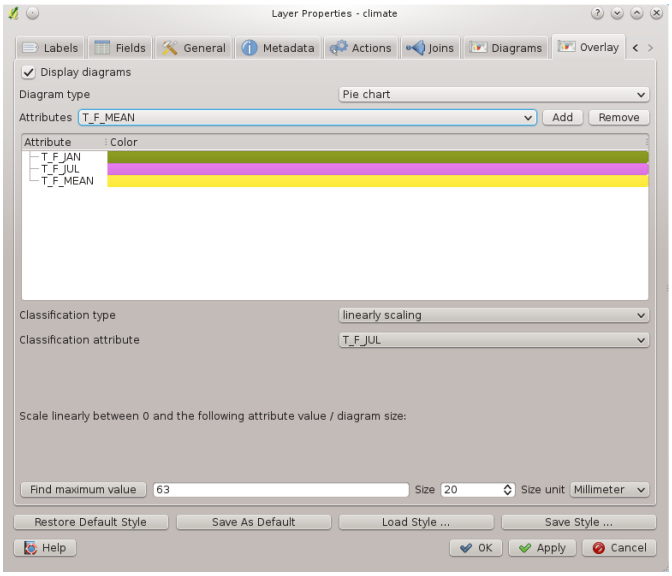


Figure 19.6: Vector properties dialog with overlay tab

2. Double click the `climate` layer in the map legend to open the *Layer Properties* dialog.
3. Click on the *Overlay* tab, activate *Display diagrams* and select 'Pie Chart' from *Diagram type* combobox.
4. We want to display the values of the three columns `T_F_JAN`, `T_F_JUL` and `T_F_MEAN`. First select `T_F_JAN` as Attributes and click the **[Add]** button, then `T_F_JUL` and finally `T_F_MEAN`.
5. Diagram size is based on linear scaling of a classification attribute. We choose `T_F_JUL`, click on **[Find maximum value]** and set size to `20` and *Size unit* to 'Millimeter'.
6. Now click **[Apply]** to display the diagram in the QGIS main window.
7. You can now adapt the chart size, or change the attribute colors double clicking on the color values in the attribute field. Figure_overlay_2 gives an impression.
8. Finally click **[Ok]**.

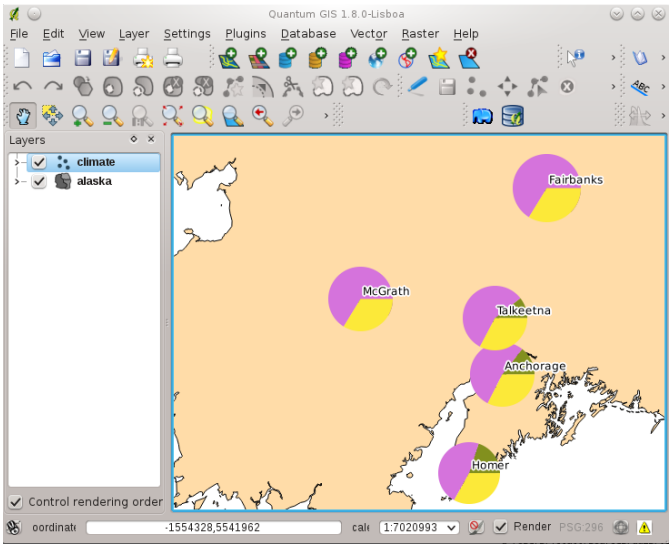


Figure 19.7: Pie chart diagram from temperature data overlayed on a map

Additionally in the *Settings* → *Options* dialog, there is a `Overlay` tab where it is possible to select the placement

algorithm of the diagrams. The 'Central point' method is a generic one, the others use algorithms of the PAL library. They also consider diagram objects and labels in different layers.

19.7 Dxf2Shp Converter Plugin

The dxf2shape converter plugin can be used to convert vector data from DXF to Shapefile format. It requires the following parameters to be specified before running:

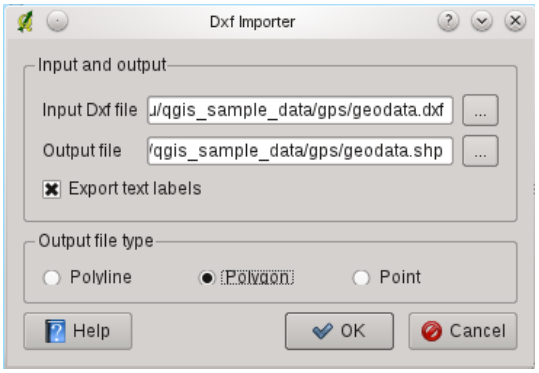


Figure 19.8: Dxf2Shape Converter Plugin

- **Input DXF file**: Enter path to the DXF file to be converted
- **Output Shp file**: Enter desired name of the Shapefile to be created
- **Output file type**: Specify the geometry type of the output Shapefile. Currently supported types are polyline, polygon, and point.
- **Export text labels**: When this checkbox is enabled, an additional Shapefile point layer will be created, and the associated dbf table will contain information about the "TEXT" fields found in the dxf file, and the text strings themselves.

19.7.1 Using the Plugin

1. Start QGIS, load the Dxf2Shape plugin in the Plugin Manager (see Section *Loading a QGIS Core Plugin*) and click on the Dxf2Shape Converter icon which appears in the QGIS toolbar menu. The Dxf2Shape plugin dialog appears as shown in Figure_dxf2shape_1.
2. Enter input DXF file, a name for the output Shapefile and the Shapefile type.
3. Enable the *Export text labels* checkbox if you want to create an extra point layer with labels.
4. Click **[OK]**.

19.8 eVis Plugin

The Biodiversity Informatics Facility at the American Museum of Natural History's (AMNH) Center for Biodiversity and Conservation (CBC) (this section is derived from Horning, N., K. Koy, P. Ersts. 2009. eVis (v1.1.0) User's Guide. American Museum of Natural History, Center for Biodiversity and Conservation. Available from http://biodiversityinformatics.amnh.org/, and released under the GNU FDL.) has developed the Event Visualization Tool (eVis), another software tool to add to the suite of conservation monitoring and decision support tools for guiding protected area and landscape planning. This plugin enables users to easily link geocoded (i.e., referenced with latitude and longitude or X and Y coordinates) photographs, and other supporting documents, to vector data in QGIS.

eVis is now automatically installed and enabled in new versions of QGIS, and as with all plugins, it can be disabled and enabled using the Plugin Manager (See *Managing Plugins*).

The eVis plugin is made up of three modules: the 'Database Connection tool', 'Event ID tool', and the 'Event Browser'. These work together to allow viewing of geocoded photographs and other documents that are linked to features stored in vector files, databases, or spreadsheets.

19.8.1 Event Browser

The Event Browser module provides the functionality to display geocoded photographs that are linked to vector features displayed in the QGIS map window. Point data, for example, can be from a vector file that can be input using QGIS or it can be from the result of a database query. The vector feature must have attribute information associated with it to describe the location and name of the file containing the photograph and, optionally, the compass direction the camera was pointed when the image was acquired. Your vector layer must be loaded into QGIS before running the Event Browser.

Launch the Event Browser module

To launch the Event browser module either click on the  eVis Event Browser icon or click on *Database* → *eVis* → *eVis Event Browser*. This will open the *Generic Event Browser* window.

The *Generic Event Browser* window has three tabs displayed at the top of the window. The *Display* tab is used to view the photograph and its associated attribute data. The *Options* tab provides a number of settings that can be adjusted to control the behavior of the eVis plugin. Lastly, the *Configure External Applications* tab is used to maintain a table of file extensions and their associated application to allow eVis to display documents other than images.

Understanding the Display window

To see the *Display* window click on the *Display* tab in the *Generic Event Browser* window. The *Display* window is used to view geocoded photographs and their associated attribute data.

1. **Display window**: A window where the photograph will appear.
2. **Zoom in button**: Zoom in to see more detail. If the entire image cannot be displayed in the display window, scroll bars will appear on the left and bottom sides of the window to allow you to pan around the image.
3. **Zoom out button**: Zoom out to see more area.
4. **Zoom to full extent** button: Displays the full extent of the photograph.
5. **Attribute information window**: All of the attribute information for the point associated with the photograph being viewed is displayed here. If the file type being referenced in the displayed record is not an image but is of a file type defined in the *Configure External Applications* tab then when you double-click on the value of the field containing the path to the file the application to open the file will be launched to view or hear the contents of the file. If the file extension is recognized the attribute data will be displayed in green.
6. **Navigation buttons**: Use the Previous and Next buttons to load the previous or next feature when more than one feature is selected.
7. **Feature indicator**: This heading indicates which feature is being displayed and how many features are available for display.

Understanding the Options window

1. **File path**: A dropdown list to specify the attribute field that contains the directory path or URL for the photographs or other documents being displayed. If the location is a relative path then the checkbox must be clicked. The base path for a relative path can be entered in the *Base Path* text box below. Information

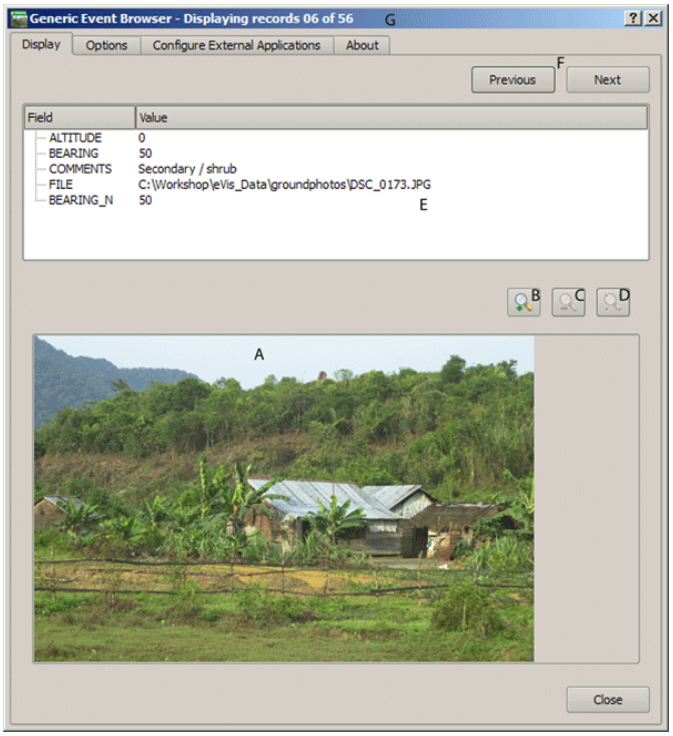


Figure 19.9: The *eVis* display window

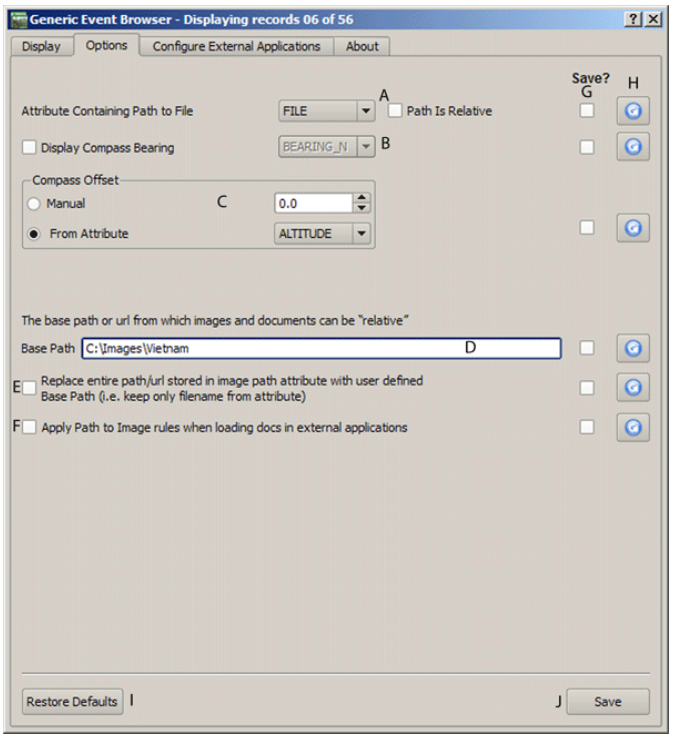


Figure 19.10: The *eVis* Options window

about the different options for specifying the file location are noted in the section *Specifying the location and name of a photograph* below.

2. **Compass bearing display field**: A dropdown list to specify the attribute field that contains the compass bearing associated with the photograph being displayed. If compass bearing information is available it is necessary to click the checkbox to the left of the dropdown menu title.
3. **Compass offset setting**: Compass offsets can be used to compensate for declination (adjust bearings collected using magnetic bearings to true north bearings). Click the *Manual* radiobutton to enter the offset in the text box or click the *From Attribute* radiobutton to select the attribute field containing the offsets. For both of these options east declinations should be entered using positive values and west declinations should use negative values.
4. **Directory base path**: The base path onto which the relative path defined in Figure_eVis_2 (A) will be appended.
5. **Replace path**: If this checkbox is checked, only the file name from the A will be appended to the Base Path.
6. **Apply rule to all documents**: If checked, the same path rules that are defined for photographs will be used for non-image documents such as movies, text documents, and sound files. If not checked the path rules will only apply to photographs and other documents will ignore the Base Path parameter.
7. **Save settings**: If the checkbox is checked the values for the associated parameters will be saved for the next session when the window is closed or when the **[Save]** button below is pressed.
8. **Reset values**: Resets the values on this line to the default setting.
9. **Restore faults**: This will reset all of the fields to their default settings. It has the same effect as clicking all of the **[Reset]** buttons.
10. **Save**: This will save the settings without closing the *Options* pane.

Understanding the Configure External Applications window

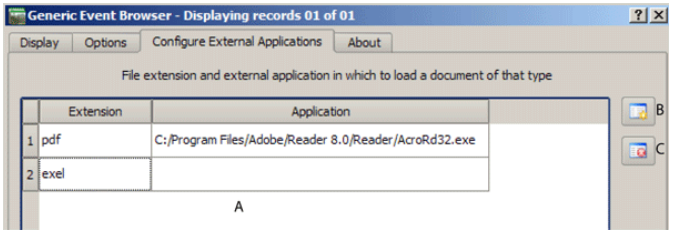


Figure 19.11: The *eVis* External Applications window

1. **File reference table**: A table containing file types that can be opened using eVis. Each file type needs a file extension and the path to an application that can open that type of file. This provides the capability of opening a broad range of files such as movies, sound recordings, and text documents instead of only images.
2. **Add new file type**: Add a new file type with a unique extension and the path for the application that can open the file.
3. **Delete current row**: Delete the file type highlighted in the table and defined by a file extension and a path to an associated application.

19.8.2 Specifying the location and name of a photograph

The location and name of the photograph can be stored using an absolute or relative path or a URL if the photograph is available on a web server. Examples of the different approaches are listed in Table evis_examples.

```
X        Y        FILE                                                                BEARING
780596   1784017  C:\Workshop\eVis_Data\groundphotos\DSC_0168.JPG                      275
780596   1784017  /groundphotos/DSC_0169.JPG                                           80
780819   1784015  http://biodiversityinformatics.amnh.org/evis_testdata/DSC_0170.JPG   10
780596   1784017  pdf:http://www.testsite.com/attachments.php?attachment_id=12         76
```

19.8.3 Specifying the location and name of other supporting documents

Supporting documents such as text documents, videos, and sound clips can also be displayed or played by eVis. To do this it is necessary to add an entry in the file reference table that can be accessed from the *Configure External Applications* window in the *Generic Event Browser* that matches the file extension to an application that can be used to open the file. It is also necessary to have the path or URL to the file in the attribute table for the vector layer. One additional rule that can be used for URLs that don't contain a file extension for the document you want to open is to specify the file extension before the URL. The format is — `file extension:URL`. The URL is preceded by the file extension and a colon, and is particularly useful for accessing documents from Wikis and other web sites that use a database to manage the web pages (see Table evis_examples).

19.8.4 Using the Generic Event Browser

When the *Event Browser* window opens a photograph will appear in the display window if the document referenced in the vector file attribute table is an image and if the file location information in the *Options* window is properly set. If a photograph is expected and it does not appear it will be necessary to adjust the parameters in the *Options* window.

If a supporting document (or an image that does not have a file extension recognized by eVis) is referenced in the attribute table the field containing the file path will be highlighted in green in the attribute information window if that file extension is defined in the file reference table located in the *Configure External Applications* window. To open the document double-click on the green-highlighted line in the attribute information window. If a supporting document is referenced in the attribute information window and the file path is not highlighted in green then it will be necessary to add an entry for the file's filename extension in the *Configure External Applications* window. If the file path is highlighted in green but does not open when double-clicked it will be necessary to adjust the parameters in the *Options* window so the file can be located by eVis.

If no compass bearing is provided in the *Options* window a red asterisk will be displayed on top of the vector feature that is associated with the photograph being displayed. If a compass bearing is provided then an arrow will appear pointing in the direction indicated by the value in the compass bearing display field in the *Generic Event Browser* window. The arrow will be centered over the point that is associated with the photograph or other document.

To close the *Generic Event Browser* window click on the **[Close]** button from the *Display* window.

19.8.5 Event ID Tool

The 'Event ID' module allows you to display a photograph by clicking on a feature displayed in the QGIS map window. The vector feature must have attribute information associated with it to describe the location and name of the file containing the photograph and optionally the compass direction the camera was pointed when the image was acquired. This layer must be loaded into QGIS before running the 'Event ID' tool.

Launch the Event ID module

To launch the 'Event ID' module either click on the  Event ID icon or click on *Plugins* → *eVis* → *Event ID Tool*. This will cause the cursor to change to an arrow with an 'i' on top of it signifying that the ID tool is active.

To view the photographs linked to vector features in the active vector layer displayed in the QGIS map window, move the Event ID cursor over the feature and then click the mouse. After clicking on the feature, the *Generic Event Browser* window is opened and the photographs on or near the clicked locality are available for display

in the browser. If more than one photograph is available, you can cycle through the different features using the **[Previous]** and **[Next]** buttons. The other controls are described in the ref:*evis_browser* section of this guide.

19.8.6 Database connection

The 'Database Connection' module provides tools to connect to and query a database or other ODBC resource, such as a spreadsheet.

eVis can directly connect to four types of databases: Microsoft Access, PostgreSQL, MySQL, SQLite, and can also read from ODBC connections. When reading from an ODBC database (such as an Excel spreadsheet) it is necessary to configure your ODBC driver for the operating system you are using.

Launch the Database Connection module

To launch the 'Database Connection' module either click on the appropriate icon  eVis Database Connection or click on *Plugins* → *eVis* → *Database Connection*. This will launch the *Database Connection* window. The window has three tabs: *Predefined Queries*, *Database Connection*, and *SQL Query*. The *Output Console* window at the bottom of the window displays the status of actions initiated by the different sections of this module.

Connect to a database

Click on the *Database Connection* tab to open the database connection interface. Next, use the *Database Type*  combobox to select the type of database that you want to connect to. If a password or username is required, that information can be entered in the *Username* and *Password* textboxes.

Enter the database host in the *Database Host* textbox. This option is not available if you selected 'MS Access' as the database type. If the database resides on your desktop you should enter "localhost".

Enter the name of the database in the *Database Name* textbox. If you selected 'ODBC' as the database type, you need to enter the data source name.

When all of the parameters are filled in, click on the **[Connect]** button. If the connection is successful, a message will be written in the *Output Console* window stating that the connection was established. If a connection was not established you will need to check that the correct parameters were entered above.

1. **Database Type**: A dropdown list to specify the type of database that will be used.
2. **Database Host**: The name of the database host.
3. **Port** The port number if a MySQL or PostgreSQL database type is selected.
4. **Database Name** The name of the database.
5. **Connect** A button to connect to the database using the parameters defined above.
6. **Output Console** The console window where messages related to processing are displayed.
7. **Username**: Username for use when a database is password protected.
8. **Password**: Password for use when a database is password protected.
9. **Predefined Queries**: Tab to open the "Predefined Queries" window.
10. **Database Connection**: Tab to open the "Database Connection" window.
11. **SQL Query**: Tab to open the "SQL Query" window.
12. **Help**: Displays the on line help.
13. **OK**: Close the main "Database Connection" window.

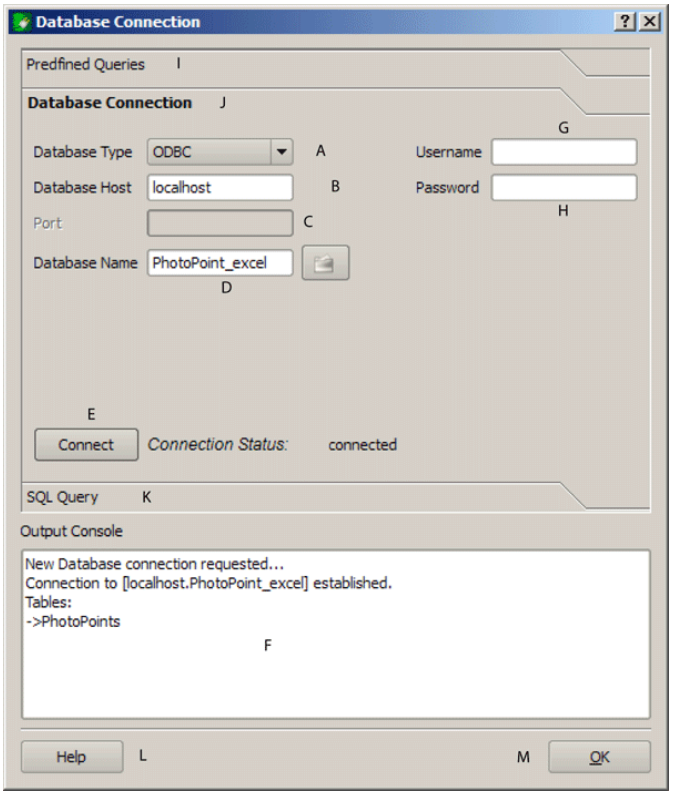


Figure 19.12: The *eVis* Database connection window

Running SQL queries

SQL queries are used to extract information from a database or ODBC resource. In eVis the output from these queries is a vector layer added to the QGIS map window. Click on the *SQL Query* tab to display the SQL query interface. SQL commands can be entered in this text window. A helpful tutorial on SQL commands is available at http://www.w3schools.com/sql. For example, to extract all of the data from a worksheet in an Excel file, `select * from [sheet1$]` where `sheet1` is the name of the worksheet.

Click on the **[Run Query]** button to execute the command. If the query is successful a *Database File Selection* window will be displayed. If the query is not successful an error message will appear in the *Output Console* window.

In the *Database File Selection* window, enter the name of the layer that will be created from the results of the query in the *Name of New Layer* textbox.

1. **SQL Query Text Window**: A screen to type SQL queries.
2. **Run Query**: Button to execute the query entered in the *SQL Query Window*.
3. **Console Window**: The console window where messages related to processing are displayed.
4. **Help**: Displays the on line help.
5. **OK**: Closes the main *Database Connection* window.

Use the *X Coordinate*  and *Y Coordinate*  comboboxes to select the field from the database that store the `X` (or longitude) and `Y` (or latitude) coordinates. Clicking on the **[OK]** button causes the vector layer created from the SQL query to be displayed in the QGIS map window.

To save this vector file for future use, you can use the QGIS 'Save as...' command that is accessed by right clicking on the layer name in the QGIS map legend and then selecting 'Save as...'

참고: Creating a vector layer from a Microsoft Excel Worksheet

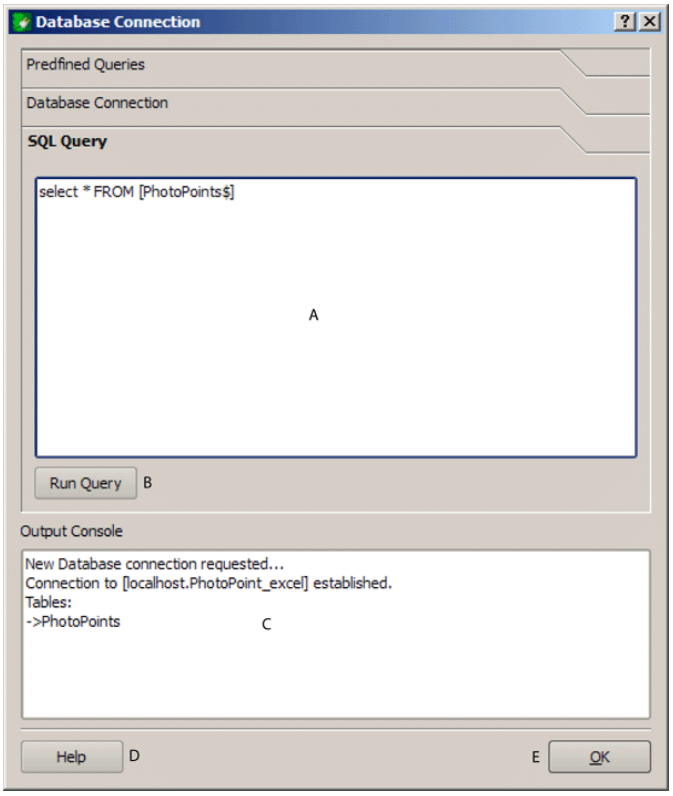


Figure 19.13: The eVis SQL query tab

When creating a vector layer from a Microsoft Excel Worksheet you might see that unwanted zeros ("0") have been inserted in the attribute table rows beneath valid data.This can be caused by deleting the values for these cells in Excel using the `Backspace` key. To correct this problem you need to open the Excel file (you'll need to close QGIS if there if you are connected to the file to allow you to edit the file) and then use *Edit* → *Delete* to remove the blank rows from the file. To avoid this problem you can simply delete several rows in the Excel Worksheet using *Edit* → *Delete* before saving the file.

Running predefined queries

With predefined queries you can select previously written queries stored in XML format in a file. This is particularly helpful if you are not familiar with SQL commands. Click on the *Predefined Queries* tab to display the predefined query interface.

To load a set of predefined queries click on the Open File icon. This opens the *Open File* window which is used to locate the file containing the SQL queries. When the queries are loaded their titles, as defined in the XML file, will appear in the dropdown menu located just below the Open File icon, the full description of the query is displayed in the text window under the dropdown menu.

Select the query you want to run from the dropdown menu and then click on the *SQL Query* tab to see that the query has been loaded into the query window. If it is the first time you are running a predefined query or are switching databases, you need to be sure to connect to the database.

Click on the **[Run Query]** button in the *SQL Query* tab to execute the command. If the query is successful a *Database File Selection* window will be displayed. If the query is not successful an error message will appear in the *Output Console* window.

1. **Open Query File**: Launches the "Open File" file browser to search for the XML file holding the predefined queries.
2. **Predefined Queries**: A dropdown list with all of the queries defined by the predefined queries XML file.

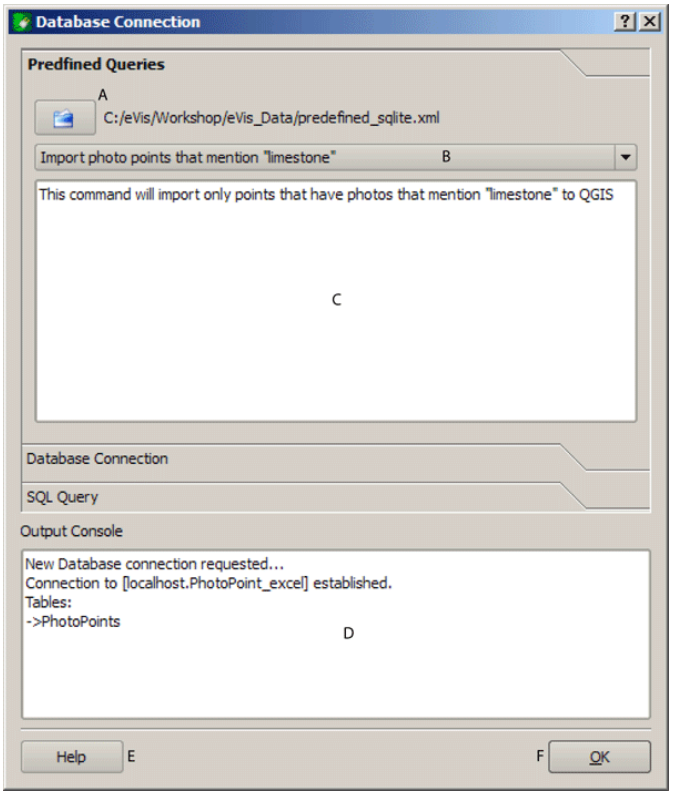


Figure 19.14: The *eVis* Predefined queries tab

3. **Query description**: A short description of the query. This description is from the predefined queries XML file.
4. **Console Window**: The console window where messages related to processing are displayed.
5. **Help**: Displays the on line help.
6. **OK**: Closes the main "Database Connection" window.

XML format for eVis predefined queries

The XML tags read by eVis

Tag	Description
query	Defines the beginning and end of a query statement.
shortdescription	A short description of the query that appears in the eVis dropdown menu.
description	A more detailed description of the query displayed in the Predefined Query text window.
databasetype	The database type as defined in the Database Type dropdown menu in the Database Connection tab.
databaseport	The port as defined in the Port textbox in the Database Connection tab.
databasename	The database name as defined in the Database Name textbox in the Database Connection tab.
databaseusername	The database username as defined in the Username textbox in the Database Connection tab.
databasepassword	The database password as defined in the Password textbox in the Database Connection tab.
sqlstatement	The SQL command.
autoconnect	A flag (“true”” or “false”) to specify if the above tags should be used to automatically connect to database without running the database connection routine in the Database Connection tab.

A complete sample XML file with three queries is displayed below:

```
<?xml version="1.0"?>
<doc>
 <query>
   <shortdescription>Import all photograph points</shortdescription>
   <description>This command will import all of the data in the SQLite database to QGIS
      </description>
   <databasetype>SQLITE</databasetype>
   <databasehost />
   <databaseport />
   <databasename>C:\textbackslash Workshop/textbackslash
eVis\_Data\textbackslash PhotoPoints.db</databasename>
   <databaseusername />
   <databasepassword />
   <sqlstatement>SELECT Attributes.*, Points.x, Points.y FROM Attributes LEFT JOIN
      Points ON Points.rec_id=Attributes.point_ID</sqlstatement>
   <autoconnect>false</autoconnect>
 </query>
  <query>
   <shortdescription>Import photograph points "looking across Valley"</shortdescription>
   <description>This command will import only points that have photographs "looking across
      a valley" to QGIS</description>
   <databasetype>SQLITE</databasetype>
   <databasehost />
   <databaseport />
   <databasename>C:\Workshop\eVis_Data\PhotoPoints.db</databasename>
   <databaseusername />
   <databasepassword />
   <sqlstatement>SELECT Attributes.*, Points.x, Points.y FROM Attributes LEFT JOIN
      Points ON Points.rec_id=Attributes.point_ID where COMMENTS='Looking across
      valley'</sqlstatement>
   <autoconnect>false</autoconnect>
 </query>
 <query>
   <shortdescription>Import photograph points that mention "limestone"</shortdescription>
   <description>This command will import only points that have photographs that mention
      "limestone" to QGIS</description>
   <databasetype>SQLITE</databasetype>
   <databasehost />
   <databaseport />
```

```
   <databasename>C:\Workshop\eVis_Data\PhotoPoints.db</databasename>
   <databaseusername />
   <databasepassword />
   <sqlstatement>SELECT Attributes.*, Points.x, Points.y FROM Attributes LEFT JOIN
      Points ON Points.rec_id=Attributes.point_ID where COMMENTS like '%limestone%'
      </sqlstatement>
   <autoconnect>false</autoconnect>
 </query>
</doc>
```

19.9 fTools Plugin

The goal of the fTools python plugin is to provide a one-stop resource for many common vector-based GIS tasks, without the need for additional software, libraries, or complex workarounds. It provides a growing suite of spatial data management and analysis functions that are both fast and functional.

fTools is now automatically installed and enabled in new versions of QGIS, and as with all plugins, it can be disabled and enabled using the Plugin Manager (See Section *Loading a QGIS Core Plugin*). When enabled, the fTools plugin adds a *Vector* menu to QGIS, providing functions ranging from Analysis and Research Tools to Geometry and Geoprocessing Tools, as well as several useful Data Management Tools.

19.9.1 Analysis tools

Icon	Tool	Purpose
	Distance Matrix	Measure distances between two point layers, and output results as a) Square distance matrix, b) Linear distance matrix, or c) Summary of distances. Can limit distances to the k nearest features.
	Sum line length	Calculate the total sum of line lengths for each polygon of a polygon vector layer.
	Points in polygon	Count the number of points that occur in each polygon of an input polygon vector layer.
	List unique values	List all unique values in an input vector layer field
	Basic statistic	Compute basic statistics (mean, std dev, N, sum, CV) on an input field.
	Nearest Neighbor analysis	Compute nearest neighbor statistics assess the level of clustering in a point vector layer
	Mean coordinate(s)	Compute either the normal or weighted mean center of an entire vector layer, or multiple features based on a unique ID field.
	Line intersections	Locate intersections between lines, and output results as a point shapefile. Useful for locating road or stream intersections, ignores line intersections with length > 0.

Table Ftools 1: fTools Analysis tools

19.9.2 Research tools

Icon	Tool	Purpose
	Random selection	Randomly select n number of features, or n percentage of features
	Random selection within subsets	Randomly select features within subsets based on a unique ID field.
	Random points	Generate pseudo-random points over a given input layer.
	Regular points	Generate a regular grid of points over a specified region and export them as a point shapefile.
	Vector grid	Generate a line or polygon grid based on user specified grid spacing.
	Select by location	Select features based on their location relative to another layer to form a new selection, or add or subtract from the current selection.
	Polygon from layer extent	Create a single rectangular polygon layer from the extent of an input raster or vector layer.

Table Ftools 2: fTools Research tools

19.9.3 Geoprocessing tools

Icon	Tool	Purpose
	Convex hull(s)	Create minimum convex hull(s) for an input layer, or based on an ID field.
	Buffer(s)	Create buffer(s) around features based on distance, or distance field.
	Intersect	Overlay layers such that output contains areas where both layers intersect.
	Union	Overlay layers such that output contains intersecting and non-intersecting areas.
	Symmetrical difference	Overlay layers such that output contains those areas of the input and difference layers that do not intersect.
	Clip	Overlay layers such that output contains areas that intersect the clip layer.
	Difference	Overlay layers such that output contains areas not intersecting the clip layer.
	Dissolve	Merge features based on input field. All features with identical input values are combined to form one single feature.

Table Ftools 3: fTools Geoprocessing tools

19.9.4 Geometry tools

Icon	Tool	Purpose
	Check geometry	Check polygons for intersections, closed-holes, and fix node ordering.
	Export/Add geometry columns	Add vector layer geometry info to point (XCOORD, YCOORD), line (LENGTH), or polygon (AREA, PERIMETER) layer.
	Polygon centroids	Calculate the true centroids for each polygon in an input polygon layer.
	Delaunay triangulation	Calculate and output (as polygons) the delaunay triangulation of an input point vector layer.
	Voronoi Polygons	Calculate voronoi polygons of an input point vector layer.
	Simplify geometry	Generalize lines or polygons with a modified Douglas-Peucker algorithm.
	Densify geometry	Densify lines or polygons by adding vertices
	Multipart to singleparts	Convert multipart features to multiple singlepart features. Creates simple polygons and lines.
	Singleparts to multipart	Merge multiple features to a single multipart feature based on a unique ID field.
	Polygons to lines	Convert polygons to lines, multipart polygons to multiple singlepart lines.
	Lines to polygons	Convert lines to polygons, multipart lines to multiple singlepart polygons.
	Extract nodes	Extract nodes from line and polygon layers and output them as points.

Table Ftools 4: fTools Geometry tools

주석: The *Simplify geometry* tool can be used to remove duplicate nodes in line and polygon geometries, just set the *Simplify tolerance* parameter to 0 and this will do the trick.

19.9.5 Data management tools

Icon	Tool	Purpose
	Define projection	Specify the CRS for shapefiles whose CRS has not been defined.
	Join attributes by location	Join additional attributes to vector layer based on spatial relationship. Attributes from one vector layer are appended to the attribute table of another layer and exported as a shapefile.
	Split vector layer	Split input layer into multiple separate layers based on input field.
	Merge shapefiles to one	Merge several shapefiles within a folder into a new shapefile based on the layer type (point, line, area).
	Create spatial index	Create a spatial index for OGR supported formats.

Table Ftools 5: fTools Data management tools

19.10 GDAL Tools Plugin

19.10.1 What is GDALTools?

The GDAL Tools plugin offers a GUI to the collection of tools in the Geospatial Data Abstraction Library, http://gdal.osgeo.org . These are raster management tools to query, re-project, warp and merge a wide variety of raster formats. Also included are tools to create a contour (vector) layer, or a shaded relief from a raster DEM, and to make a vrt (Virtual Raster Tile in XML format) from a collection of one or more raster files. These tools are available when the plugin is installed and activated.

The GDAL Library

The GDAL library consists of a set of command line programs, each with a large list of options. Users comfortable with running commands from a terminal may prefer the command line, with access to the full set of options. The GDALTools plugin offers an easy interface to the tools, exposing only the most popular options.

19.10.2 List of GDAL tools

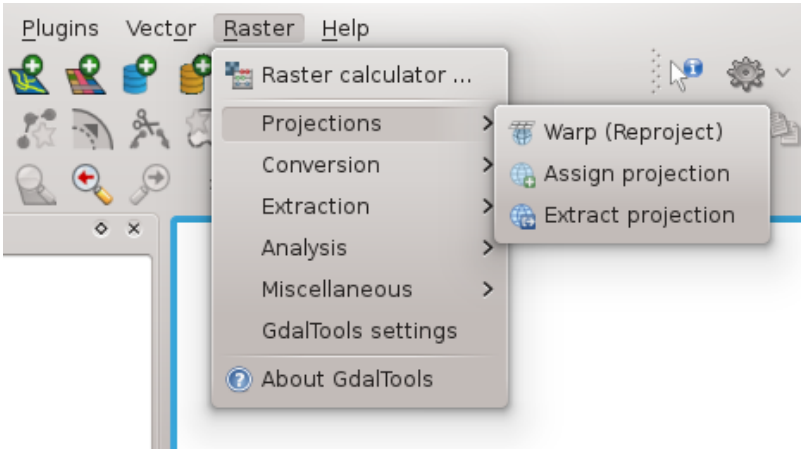


Figure 19.15: The *GDALTools* menu list

Projections

Warp (Reproject)	This utility is an image mosaicing, reprojection and warping utility. The program can reproject to any supported projection, and can also apply GCPs stored with the image if the image is "raw" with control information. For more information you can read on the GDAL website http://www.gdal.org/gdalwarp.html
Assign projection	This tool allows to assign projection to rasters that already georeferenced but miss projection information. Also with it help it is possible to alter existing projection definition. Both single file and batch mode are supported. For more information please visit utility page at GDAL site http://www.gdal.org/gdalwarp.html
Extract projection	This utility helps you to extract projection information from an input file. If you want to extract projection from a whole directory you can use the Batch mode. It creates both `.prj` and `.wld` files.

Conversion

Rasterize	This program burns vector geometries (points, lines and polygons) into the raster band(s) of a raster image. Vectors are read from OGR supported vector formats. Note that the vector data must in the same coordinate system as the raster data; on the fly reprojection is not provided. For more information see http://www.gdal.org/gdal-rasterize.html
Polygonize	This utility creates vector polygons for all connected regions of pixels in the raster sharing a common pixel value. Each polygon is created with an attribute indicating the pixel value of that polygon. The utility will create the output vector datasource if it does not already exist, defaulting to ESRI shapefile format. See also http://www.gdal.org/gdal_polygonize.html
Translate	This utility can be used to convert raster data between different formats, potentially performing some operations like subsettings, resampling, and rescaling pixels in the process. For more information you can read on http://www.gdal.org/gdal_translate.html
RGB to PCT	This utility will compute an optimal pseudo-color table for a given RGB image using a median cut algorithm on a downsampled RGB histogram. Then it converts the image into a pseudo-colored image using the color table. This conversion utilizes Floyd-Steinberg dithering (error diffusion) to maximize output image visual quality. The utility is also desribed at http://www.gdal.org/rgb2pct.html
PCT to RGB	This utility will convert a pseudocolor band on the input file into an output RGB file of the desired format. For more information see http://www.gdal.org/pct2rgb.html

Extraction

Contour	This program generates a vector contour file from the input raster elevation model (DEM). On http://www.gdal.org/gdal_contour.html you can find more information.
Clipper	This utility allows to clip (extract subset) raster using selected extent or based on mask layer bounds. More information can be found at http://www.gdal.org/gdal_translate.html.

Analysis

Sieve	This utility removes raster polygons smaller than a provided threshold size (in pixels) and replaces them with the pixel value of the largest neighbor polygon. The result can be written back to the existing raster band, or copied into a new file. For more information see http://www.gdal.org/gdal_sieve.html .
Near Black	This utility will scan an image and try to set all pixels that are nearly black (or nearly white) around the edge to exactly black (or white). This is often used to "fix up" lossy compressed aerial photos so that color pixels can be treated as transparent when mosaicing. See also http://www.gdal.org/nearblack.html .
Fill nodata	This utility fills selection raster regions (usually nodata areas) by interpolation from valid pixels around the edges of the area. On http://www.gdal.org/gdal_fillnodata.html you can find more information.
Proximity	This utility generates a raster proximity map indicating the distance from the center of each pixel to the center of the nearest pixel identified as a target pixel. Target pixels are those in the source raster for which the raster pixel value is in the set of target pixel values. For more information see http://www.gdal.org/gdal_proximity.html .
Grid	This utility creates regular grid (raster) from the scattered data read from the OGR datasource. Input data will be interpolated to fill grid nodes with values, you can choose from various interpolation methods. The utility is also described on the GDAL website http://www.gdal.org/gdal_grid.html .
DEM terrain models	Tools to analyze and visualize DEMs. It can create a shaded relief, a slope, an aspect, a color relief, a Terrain Ruggedness Index, a Topographic Position Index and a roughness map from any GDAL-supported elevation raster. For more information you can read on http://www.gdal.org/gdaldem.html

Miscellaneous

Build Virtual Raster	This program builds a VRT (Virtual Dataset) that is a mosaic of the list of input gdal datasets. See also http://www.gdal.org/gdalbuildvrt.html .
Merge	This utility will automatically mosaic a set of images. All the images must be in the same coordinate system and have a matching number of bands, but they may be overlapping, and at different resolutions. In areas of overlap, the last image will be copied over earlier ones. The utility is also described on http://www.gdal.org/gdal_merge.html .
Information	This utility lists various information about a GDAL supported raster dataset. On http://www.gdal.org/gdalinfo.html you can find more information.
Build Overviews	The gdaladdo utility can be used to build or rebuild overview images for most supported file formats with one of several downsampling algorithms. For more information see http://www.gdal.org/gdaladdo.html .
Tile Index	This utility builds a shapefile with a record for each input raster file, an attribute containing the filename, and a polygon geometry outlining the raster. See also http://www.gdal.org/gdaltindex.html .

19.11 Georeferencer Plugin

The Georeferencer Plugin is a tool for generating world files for rasters. It allows you to reference rasters to geographic or projected coordinate systems by creating a new GeoTiff or by adding a world file to the existing image. The basic approach to georeferencing a raster is to locate points on the raster for which you can accurately determine their coordinates.

Features

Icon	Purpose	Icon	Purpose
	Open raster		Start georeferencing
	Generate GDAL Script		Load GCP Points
	Save GCP Points As		Transformation settings
	Add Point		Delete Point
	Move GCP Point		Pan
	Zoom In		Zoom Out
	Zoom To Layer		Zoom Last
	Zoom Next		Link Georeferencer to QGIS
	Link QGIS to Georeferencer		

Table Georeferencer 1: Georeferencer Tools

19.11.1 Usual procedure

As X and Y coordinates (DMS (dd mm ss.ss), DD (dd.dd) or projected coordinates (mmmm.mm) which correspond with the selected point on the image, two alternative procedures can be used:

- The raster itself sometimes provides crosses with coordinates "written" on the image. In this case you can enter the coordinates manually.
- Using already georeferenced layers, this can be either vector or raster data that contain the same objects/features that you have on the image that you want to georeference and the projection you want to have your image. In this case you can enter the coordinates by clicking on the reference dataset loaded in QGIS map canvas.

The usual procedure for georeferencing an image involves selecting multiple points on the raster, specifying their coordinates, and choosing a relevant transformation type. Based on the input parameters and data, the plugin will compute the world file parameters. The more coordinates you provide, the better the result will be.

The first step is to start QGIS, load the Georeferencer Plugin (see Section *Loading a QGIS Core Plugin*) and click on the  Georeferencer icon which appears in the QGIS toolbar menu. The Georeferencer Plugin dialog appears as shown in figure_georeferencer_1.

For this example, we are using a topo sheet of South Dakota from SDGS. It can later be visualized together with the data from the GRASS `spearfish60` location. You can download the topo sheet here: http://grass.osgeo.org/sampledata/spearfish_toposheet.tar.gz.

Entering ground control points (GCPs)

1. To start georeferencing an unreferenced raster, we must load it using the  button. The raster will show up in the main working area of the dialog. Once the raster is loaded, we can start to enter reference points.

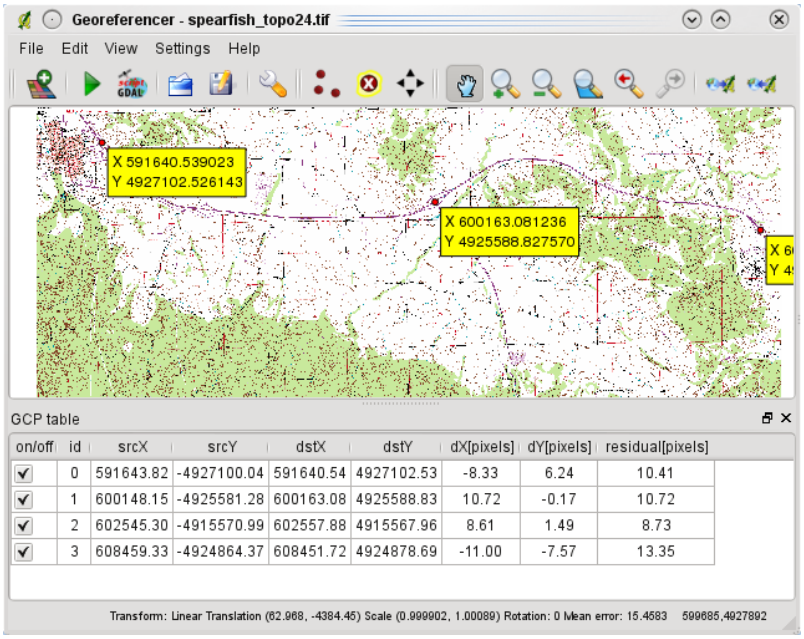


Figure 19.16: Georeferencer Plugin Dialog

2. Using the Add Point button, add points to the main working area and enter their coordinates (see Figure figure_georeferencer_2). For this procedure you have three options:
   - Click on a point in the raster image and enter the X and Y coordinates manually.
   - Click on a point in the raster image and choose the button from map canvas to add the X and Y coordinates with the help of a georeferenced map already loaded in the QGIS map canvas.
   - With the button, you can move the GCPs in both windows, if they are at the wrong place.
3. Continue entering points. You should have at least 4 points, and the more coordinates you can provide, the better the result will be. There are additional tools on the plugin dialog to zoom and pan the working area in order to locate a relevant set of GCP points.

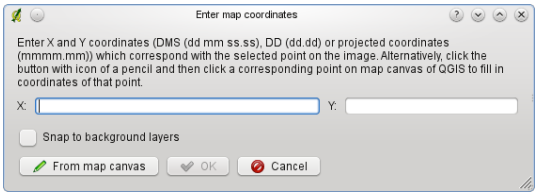


Figure 19.17: Add points to the raster image

The points that are added to the map will be stored in a separate text file (`[filename].points`) usually together with the raster image. This allows us to reopen the Georeferencer plugin at a later date and add new points or delete existing ones to optimize the result. The points file contains values of the form: `mapX, mapY, pixelX, pixelY`. You can use the Load GCP Points and Save GCP Points buttons to manage the files.

Defining the transformation settings

After you have added your GCPs to the raster image, you need to define the transformation settings for the georeferencing process.

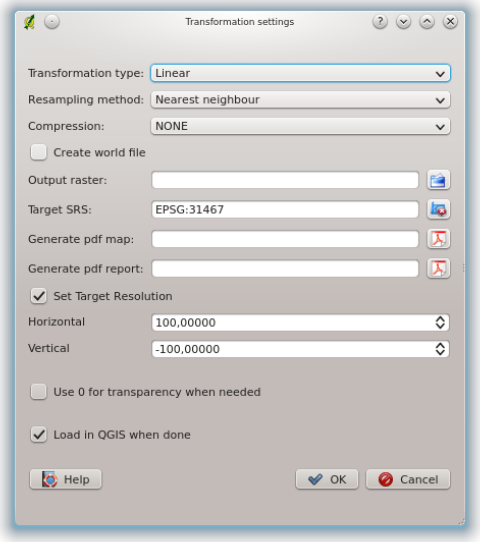


Figure 19.18: Defining the georeferencer transformation settings

Available Transformation algorithms

Depending on how many ground control point you have captured, you may want to use different transformation algorithms. Choice of transformation algorithm is also dependent on the type and quality of input data and the amount of geometric distortion that you are willing to introduce to final result.

Currently, following algorithms are available:

- The **Linear algorithm** is used to create a world-file, and is different from the other algorithms, as it does not actually transform the raster. This algorithm likely won't be sufficient if you are dealing with scanned material.
- The **Helmert transformation** performs simple scaling and rotation transformations.
- The **Polynomial algorithms** 1-3 are among the most widely used algorithms introduced to match source and destination ground control points. The most widely used polynomial algorithm is the second order polynomial transformation, which allows some curvature. First order polynomial transformation (affine) preserves colliniarity and allows scaling, translation and rotation only.
- The **Thin plate spline (TPS) algorithm** is a more modern georeferencing method, which is able to introduce local deformations in the data. This algorithm is useful when very low quality originals are being georeferenced.
- The **Projective transformation** is a linear rotation and translation of coordinates.

Define the Resampling method

The type of resampling you choose will likely depending on your input data and the ultimate objective of the exercise. If you don't want to change statistics of the image, you might want to choose 'Nearest neighbour', whereas a 'Cubic resampling' will likely provide a more smoothed result.

It is prossible to choose between five different resampling methods.

1. Nearest neighbour
2. Linear
3. Cubic
4. Cubic Spline

5. Lanczos

Define the transformation settings

There are several options that need to be defined for the georeferenced output raster.

- The checkbox *Create world file* is only available, if you decide to use the linear transformation type, because this means that the raster image actually won't be transformed. In this case, the field *Output raster* is not activated, because only a new world-file will be created.
- For all other transformation type you have to define an *Output raster*. As default a new file ([filename]_modified) will be created in the same folder together with the original raster image.
- As a next step you have to define the *Target SRS* (Spatial Reference System) for the georeferenced raster (see section *Working with Projections*).
- If you like, you can **generate a pdf map** and also **a pdf report**. The report includes information about the used transformation parameters. An image of the residuals and a list with all GCPs and their RMS errors.
- Furthermore you can activate the *Set Target Resolution* checkbox and define pixel resolution of the output raster. Default horizontal and vertical resolution is 1,
- The *Use 0 for transparency when needed* can be activated, if pixels with the value 0 shall be visualized transparent. In our example toposheet all white areas would be transparent.
- Finally *Load in QGIS when done* loads the output raster automatically into the QGIS map canvas when the transformation is done.

Show and adapt raster properties

Clicking on the *Raster properties* dialog in the *Settings* menu opens the raster properties of the layer that you want to georeference.

Configure the georeferencer

- You can define if you want to show GCP coordiniates and/or IDs.
- As residual units pixels and map units can be chosen.
- For the PDF report a left and right margin can be defined and you can also set the paper size for the PDF map.
- Finally you can activate to *show georeferencer window docked*.

Running the transformation

After all GCPs have been collected and all transformation settings are defined, just press the button Start georeferencing to create the new georeferenced raster.

19.12 Interpolation Plugin

The Interplation plugin can be used to generate a TIN or IDW interpolation of a point vector layer. It is very simple to handle and provides an intuitive graphical user interface for creating interpolated raster layers (See Figure_interpolation_1). The plugin requires the following parameters to be specified before running:

- **Input vector layer**: Specify the input point vector layer(s) from a list of loaded point layers. If several layers are specified, then data from all layers is used for interpolation. Note: It is possible to insert lines or polygons as constraints for the triangulation, by specifying either "points", "structure lines" or "break lines" in the *Type*  combobox.
- **Interpolation attribute**: Select attribute column to be used for interpolation or enable the  *Use Z-Coordinate* checkbox to use the layers stored Z values.
- **Interpolation Method**: Select interpolation method. This can be either 'Triangulated Irregular Network (TIN)' or 'Inverse Distance Weighted (IDW)'.
- **Number of columns/rows**: Specify the number row and colums for the output raster file.
- **Output file**: Specify a name for the output raster file.

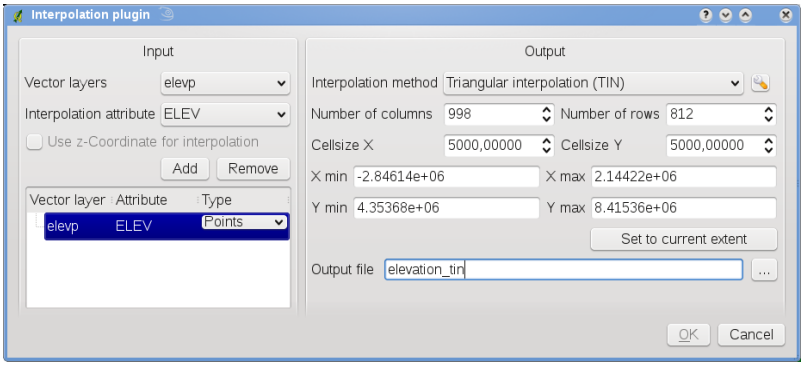


Figure 19.19: Interpolation Plugin 

19.12.1 Using the plugin

1. Start QGIS and load a point vector layer (e.g., `elevp.csv`).
2. Load the Interpolation plugin in the Plugin Manager (see Section *Loading a QGIS Core Plugin*) and click on the  Interpolation icon which appears in the QGIS toolbar menu. The Interpolation plugin dialog appears as shown in Figure_interpolation_1.
3. Select an input layer (e.g., *elevp* ) and column (e.g., `ELEV`) for interpolation.
4. Select an interpolation method (e.g. 'Triangulated Irregular Network (TIN)'), and specify a cellsize of 5000 as well as the raster output filename (e.g., `elevation_tin`).
5. Click **[OK]**.
6. For the current example, double click `elevation_tin` in the layer list to open the raster *Layer Properties* dialog and select 'Pseudocolor' as *Color Map*  in the *Symbology* tab. Or you can define a new color table as described in section *Working with Raster Data*.

19.13 MapServer Export Plugin

You can use QGIS to 'compose' your map by adding and arranging layers, symbolizing them, customizing the colors and then creating a map file for MapServer.

주석: Currently the plugin only works when you use 'Old Symbology' in QGIS. 'New Symbology' is not yet supported.

19.13.1 Creating the Project File

The MapServer Export Plugin operates on a saved QGIS project file and **not** on the current contents of the map canvas and legend. This has been a source of confusion for a number of users. As described below, before you start using the MapServer Export Plugin, you need to arrange the raster and vector layers you want to use in MapServer and save this status in a QGIS project file.

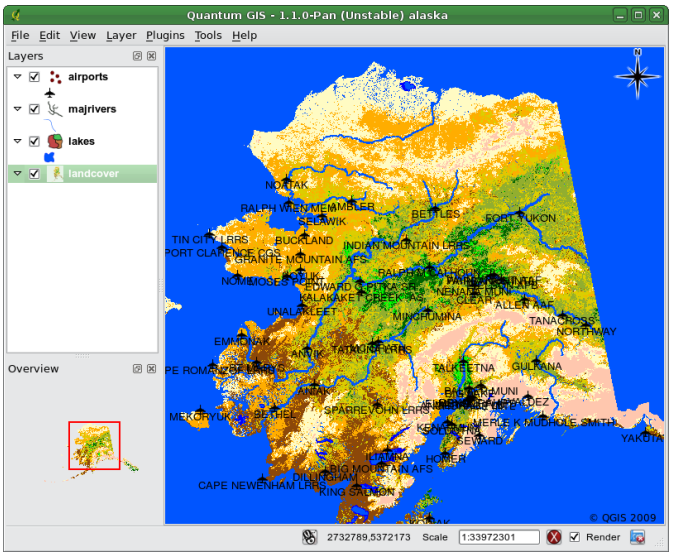


Figure 19.20: Arrange raster and vector layers for QGIS project file

In this example, we demonstrate the four steps required to create a simple project file which can be used to create the MapServer map file. We use raster and vector files from the QGIS sample dataset *Sample Data*.

1. Add the raster layer `landcover.tif` clicking on the Add Raster Layer icon.
2. Add the vector Shapefiles `lakes.shp`, `majrivers.shp` and `airports.shp` from the QGIS sample dataset clicking on the Add Vector Layer icon.
3. Change the colors and symbolize the data as you like (for example see figure_mapserver_export_1)
4. Save a new project named `mapserverproject.qgs` using *File* → *Save Project*.

19.13.2 Creating the Map File

To use the Mapserver Export plugin from within QGIS, you need to enable the MapServer Export Plugin first using the Plugin Manager (see *Loading a QGIS Core Plugin*).

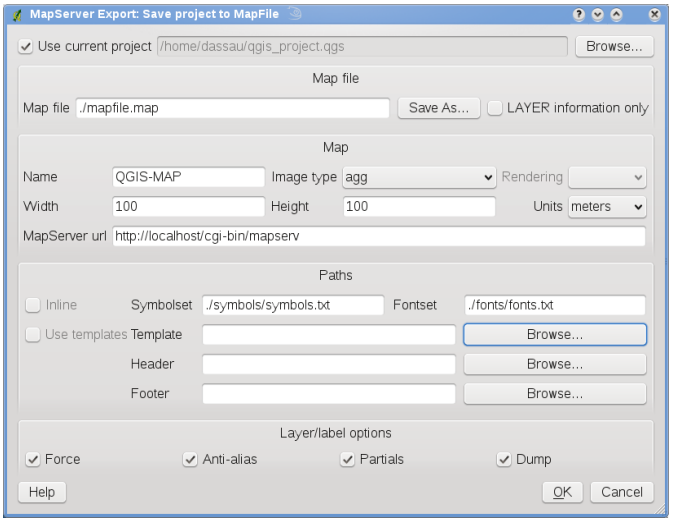


Figure 19.21: Export to MapServer Dialog

Map file	Enter the name for the map file to be created. You can use the button at the right to browse for the directory where you want the map file created.
Qgis project file	Enter the full path to the QGIS project file (.qgs) you want to export. You can use the button at the right to browse for the QGIS project file.
Map Name	A name for the map. This name is prefixed to all images generated by the mapserver.
Map Width	Width of the output image in pixels.
Map Height	Height of the output image in pixels.
Map Units	Units of measure used for output.
Image type	Format for the output image generated by MapServer
Web Template	Full path to the MapServer template file to be used with the map file
Web Header	Full path to the MapServer header file to be used with the map file
Web Footer	Full path to the MapServer footer file to be used with the map file

Only the inputs *Map file* and *QGIS project file* are required to create a map file, however by omitting the other parameters, you may end up creating a non-functional map file, depending on your intended use.

Although QGIS is good at creating a map file from your project file, it may require some tweaking to get the results you want. For this example, we will create a map file using the project file `mapserverproject.qgs` we just created (see Figure_mapserver_export_2):

1. Click the  MapServer Export icon in the toolbar menu to start the MapServer dialog (see Figure_mapserver_export_2).
2. Enter the name (e.g., `qgisproject.map`) for your new map file.
3. Browse and find the QGIS project file (e.g., `mapserverproject.qgs`) you previously saved.
4. Enter a name (e.g., `MyMap`) for the map.
5. Enter the width and height (e.g., `600` for the width and `400` for the height) for your output image.
6. For this example, the layers are in meters, so we change the units to meters.

7. Choose *png* for the image type.
8. Click **[OK]** to generate the new map file `qgisproject.map`. QGIS displays the success of your efforts.

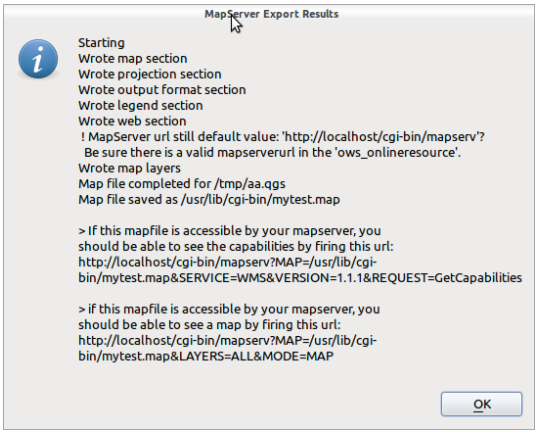


Figure 19.22: Export to MapServer Succesfull Dialog

You can view the map file in any text editor or visualizer. If you take a look, you'll notice that the export tool adds the metadata needed to enable our map file for WMS.

19.13.3 Troubleshooting

If you get error messages from mapserver like:

```
loadSymbolSet(): Unable to access file. (./symbols/symbols.txt)
```

or:

```
msLoadFontset(): Unable to access file. Error opening fontset ./fonts/fonts.txt.
```

This means that that the map file is requesting for fonts or symbol definition files but that Mapserver cannot find those.

Either comment the lines containing those names in the map file (this is possible if you do not have labels or use styles from the `symbols.txt`). Or create those files (see below).

The most simple `fonts.txt` file (because arial is used in the QGIS generated map files) contains the following line

```
# either relative to the map file or a full path:
arial /usr/share/fonts/truetype/msttcorefonts/arial.ttf
```

The most simple `symbols.txt` file contains the definition of a circle symbol (because a circle symbol is used for point layers)

```
SYMBOLSET
    SYMBOL
    NAME "circle"
    TYPE ellipse
    FILLED true
    POINTS
    1 1
    END
    END
END
```

19.13.4 Testing the Map File

We can now test our work.

If the map file is accessible for you mapserver cgi you can use the one of the URL's from the success dialog.

Another option is using the **shp2img** tool to create an image from the map file. The **shp2img** utility is part of MapServer and FWTools. To create an image from our map:

- Open a terminal window
- If you didn't save your map file in your home directory, change to the folder where you saved it.
- Run `shp2img -m qgisproject.map -o mapserver\_test.png` and display the image

This creates a PNG with all the layers included in the QGIS project file. In addition, the extent of the PNG will be the same as when we saved the project. As you can see in figure_mapserver_export_4, all information except the airport symbols are included.

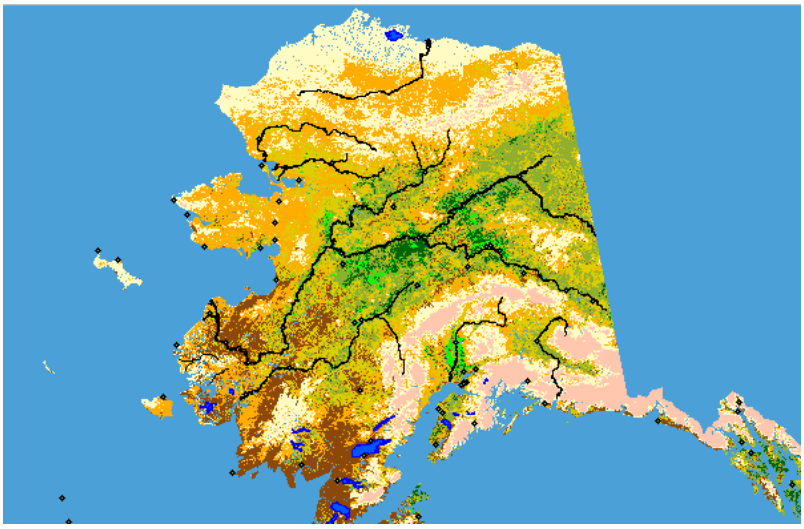


Figure 19.23: Test PNG created by shp2img with all MapServer Export layers

19.13.5 Using Map File

If you plan to use the map file to serve WMS requests, you probably don't have to tweak anything. If you plan to use it with a mapping template or a custom interface, you may have a bit of manual work to do. To see how easy it is to go from QGIS to serving maps on the web, take a look at Christopher Schmidt's 5 minute flash video. He used an older version of QGIS (version 0.8), but the demo applies equally well to newer versions.

19.14 Offline Editing Plugin

For data collection, it is a common situation to work with a laptop or a cell phone offline in the field. Upon returning to the network, the changes need to be synchronized with the master data source, e.g. a PostGIS database. If several persons are working simultaneously on the same datasets, it is difficult to merge the edits by hand, even if people don't change the same features.

The Offline Editing Plugin automates the synchronisation by copying the content of a datasource (usually PostGIS or WFS-T) to a SpatiaLite database and storing the offline edits to dedicated tables. After being connected to the network again, it is possible to apply the offline edits to the master dataset.

19.14.1 Using the plugin

- Open some vector layers, e.g. from a PostGIS or WFS-T datasource
- Save it as a project
- Press the Convert to offline project icon and select the layers to save. The content of the layers is saved to SpatiaLite tables.
- Edit the layers offline.
- After being connected again, upload the changes with the Synchronize button.

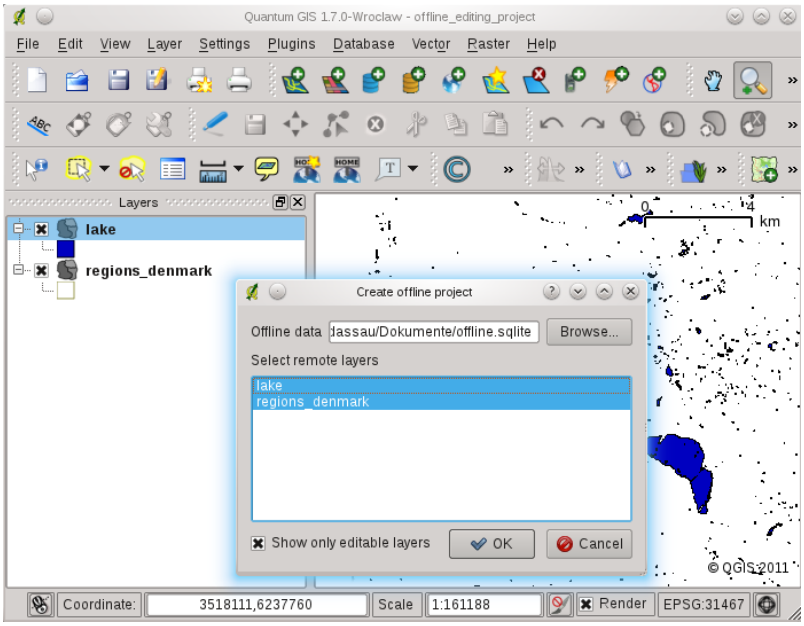


Figure 19.24: Create an offline project from PostGIS or WFS layers

19.15 Oracle GeoRaster Plugin

In Oracle databases, raster data can be stored in SDO_GEORASTER objects available with the Oracle Spatial extension. In QGIS, the OracleGeoRasterPlugin is supported by GDAL, and depends on Oracle's database product being installed and working on your machine. While Oracle is proprietary software, they provide their software free for development and testing purposes. Here is one simple example of how to load raster images to GeoRaster:

```
$ gdal_translate -of georaster input_file.tif geor:scott/tiger@orcl
```

This will load the raster into the default GDAL_IMPORT table, as a column named `RASTER`.

19.15.1 Managing connections

Firstly, the Oracle GeoRaster Plugin must be enabled using the Plugin Manager (see Section *Loading a QGIS Core Plugin*). The first time you load a GeoRaster in QGIS, you must create a connection to the Oracle database that contains the data. To do this, begin by clicking on the Select GeoRaster toolbar button, it will open the *Select Oracle Spatial GeoRaster* dialog window. Click on **[New]** to open the dialog window, and specify the connection parameters (See Figure_oracle_raster_1):

- **Name**: Enter a name for the database connection

- **Database instance**: Enter the name of the database that you will connect to
- **Username**: Specify your own username that you will use to access the database
- **Password**: The password associated with your username that is required to access the database

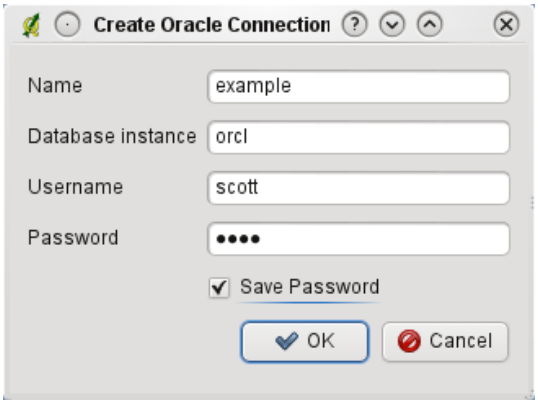


Figure 19.25: Create Oracle connection dialog

Now, back on the main *Oracle Spatial GeoRaster* dialog window (see Figure_oracle_raster_2), use the drop-down list to choose one connection, and use the **[Connect]** button to establish a connection. You may also **[Edit]** the connection by opening the previous dialog and making changes to the connection information, or use the **[Delete]** button to remove the connection from the drop-down list.

19.15.2 Selecting a GeoRaster

Once a connection has been established, the sub-datasets window will show the names of all the tables that contains GeoRaster columns in that database in the format of a GDAL subdataset name.

Click on one of the listed subdatasets and then click on **[Select]** to choose the table name. Now another list of subdatasets will show with the names of GeoRaster columns on that table. This is usually a short list, since most users will not have more than one or two GeoRaster columns on the same table.

Click on one of the listed subdatasets and then click on **[Select]** to choose one of the the table/column combination. The dialog will now show all the rows that contains GeoRaster objects. Note that the subdataset list will now show the Raster Data Table and Raster Id's pairs.

At anytime the Selection entry can be edited in order to go directly to a known GeoRaster or to go back to the beginning and select another table name.

The Selection data entry can also be used to enter a `WHERE` clause at the end of the identification string, e.g. `georscott/tiger@orcl,gdal_import,raster,geoid=`. See http://www.gdal.org/frmt_georaster.html for more information.

19.15.3 Displaying GeoRaster

Finally, by selecting a GeoRaster from the list of Raster Data Table and Raster Id's, the raster image will be loaded into QGIS.

The *Select Oracle Spatial GeoRaster* dialog can be closed now and next time it opens it will keep the same connection, and will show the same previous list of subdataset making it very easy to open up another image from the same context.

주석: GeoRasters that contains pyramids will display much faster but the pyramids need to be generated outside of QGIS using Oracle PL/SQL or gdaladdo.

The following is example using `gdaladdo`:

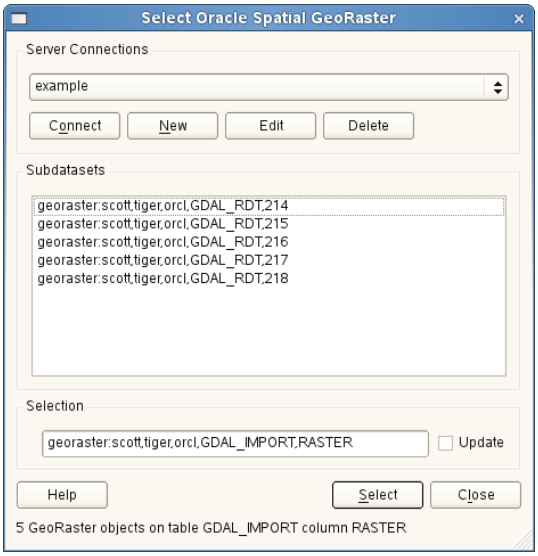


Figure 19.26: Select Oracle GeoRaster dialog

```
gdaladdo georaster:scott/tiger@orcl,georaster\_table,georaster,georid=6 -r nearest 2 4 6 8 16 32
```

This is an example using PL/SQL:

```
$ sqlplus scott/tiger
SQL> DECLARE
 gr sdo_georaster;
BEGIN
    SELECT image INTO gr FROM cities WHERE id = 1 FOR UPDATE;
    sdo_geor.generatePyramid(gr, 'rLevel=5, resampling=NN');
    UPDATE cities SET image = gr WHERE id = 1;
    COMMIT;
END;
```

19.16 Raster Terrain Analysis Plugin

The Raster Terrain Analysis Plugin can be used to calculate the slope, aspect, hillshade, ruggedness index and relief for digital elevation models (DEM). It is very simple to handle and provides an intuitive graphical user interface for creating new raster layers (See Figure_raster_terrain_1).

Description of the analysis:

- **Slope**: Calculates slope angle for each cell in degrees (based on first order derivative estimation).
- **Aspect**: Exposition (starting with 0 for north direction, in degrees counterclockwise).
- **Hillshade**: Create shaded map using light and shadow to provide a more three-dimensional appearance for a shaded relief map.
- **Ruggedness Index**: A quantitative measurement of terrain heterogeneity as described by Riley et al. (1999). It is calculated for every location, by summarizing the change in elevation within the 3x3 pixel grid.
- **Relief**: Creating a shaded relief map from digital elevation data. Implemented is a method to choose the elevation colors analysing the frequency distribution.

19.16.1 Using the plugin

1. Start QGIS and load the `gtopo30` raster layer from the GRASS sample location.

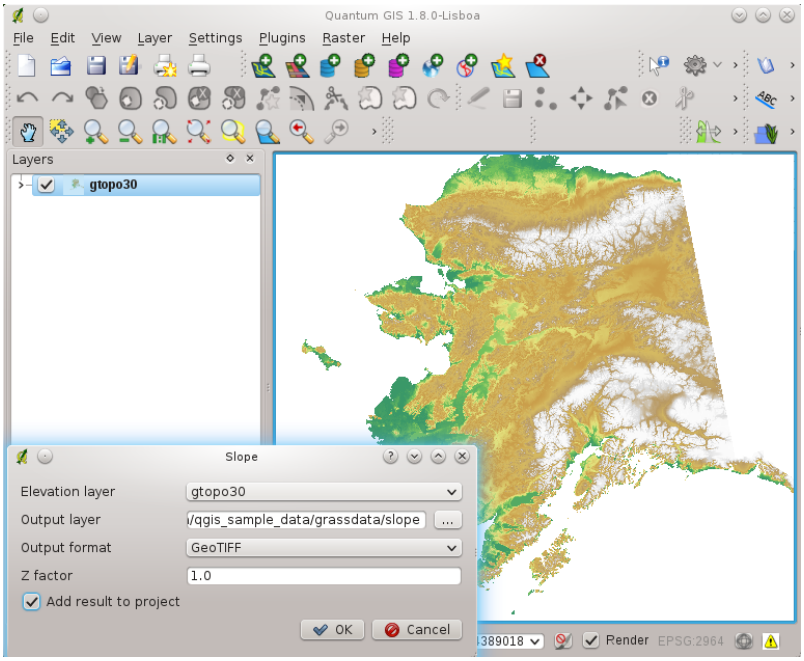


Figure 19.27: Raster Terrain Modelling Plugin (slope calculation)

2. Load the Raster Terrain Analysis plugin in the Plugin Manager (see Section *Loading a QGIS Core Plugin*)
3. Select an analysis method from menu (e.g. *Raster* → *Terrain Analysis* → *Slope*). The *Slope* dialog appears as shown in Figure_raster_terrain_1.
4. Specify an output file path, and an output file type.
5. Click **[OK]**.

19.17 Heatmap Plugin

The Heatmap plugin allows to create a heatmap from a point vector map. A heatmap is a raster map showing the density or magnitude of point related information. From the result "hotspots" can easily be identified.

19.17.1 Activate the Heatmap plugin

First this core plugin needs to be activated using the Plugin Manager (see Section *Loading a QGIS Core Plugin*).

After activation the heatmap icon can be found in the Raster Toolbar.

Select from menu *View* → *Toolbars* → *Raster* to activate the Raster Toolbar when it is not yet activated.

19.17.2 Using the Heatmap plugin

The Heatmap toolbutton starts the dialog of the Heatmap plugin (see figure_heatmap_2).

The dialog has the following options:

- **Output Raster**: Using the ... button you select the folder and the name of the output raster the Heatmap plugin will generate. It is not necessary to give a file extension
- **Output Format**: Selection of the output format. Although all formats supported by GDAL can be choosen, GeoTIFF is most in cases the best format to choose.

- **Radius**: can be used to give the radius in meters or mapunit. When the radius is choosen too small you will not get a nice heatmap. In places where more circles around points overlap eachother the brighter hotspots will occur!
- **Decay Ratio**: is used to show with which ratio the heat dies out from the center.
  - When 0 is given (=minimum) the heat will be concentrated in the centre of given radius and be completely extinguished at the edge.
  - When 10 is given (=maximum) the heat is concentrated at the edge and in the centre it is completely extinguished. Higher given values are allowed but have no effect.
  - When 1 is given the heat is spread evenly over the whole circle.
  - When a negative Decay Ratio is given you will create a Coolmap instead of a Heatmap!

When the  *Advanced* checkbox is checked it will give acces to additional advanced options.

- **Row** and **Column**, can be used to change the pixelsize of the output raster. More rows and colums means a smaller pixelsize and the size of the output file will become bigger and processing a bit slower. When doubling the number of rows this will automatically double the size of Columns and the cell sizes will also be halved. The area of the output raster will remain the same!
- **Cell Size X** and **Cell Size Y** influence the pixelsize of the output raster and will also change the rows and columns.

The point vector layer may have attribute fields that can be used to create a heatmap:

- **Use Radius from Field**: the radius set from an attribute field.
- **Use Weight from field**: the weight factor set from an attribute field.

When the output raster is given, the **[OK]** button can be used to create a heatmap. The first result is a grey surface that still needs to be given additional treatment to make it a heatmap.

경고: Changing the raster size properties
When changing the raster size properties, this also changes the resulting output. The hotspot areas will become bigger.

19.17.3 Creating a Heatmap

For the following example, we will use the `airports` vector point layer from the QGIS sample dataset (see *Sample Data*). Another exellent QGIS tutorial on making heatmaps can be found on http://qgis.spatialthoughts.com.

In Figure_Heatmap_1 the airports of Alaska are shown.

1. Select the  toolbutton. The heatmap dialog starts (see Figure_Heatmap_2).
2. In field *Input Point Vector*  select `airpoirt` from the list of point layers loaded in current project.
3. In field *Output Raster* give the name and location using the  button of the output raster. Give the output raster file the name `heatmap_airports` to give an file extension is not necessary.
4. In field *Output Format* choose `GeoTIFF`.
5. In the field *Radius* give `1000000` meters.
6. The original *Decay Ratio* of `0.1` is fine.
7. Click on **[OK]** to create and load the new raster map (see Figure_Heatmap_3).

The result after loading the heatmap is probably not the result which was expected. It is still a grey even surface not showing any variation. First some additional configuration of this layer needs to be done to make the heatmap present itself.

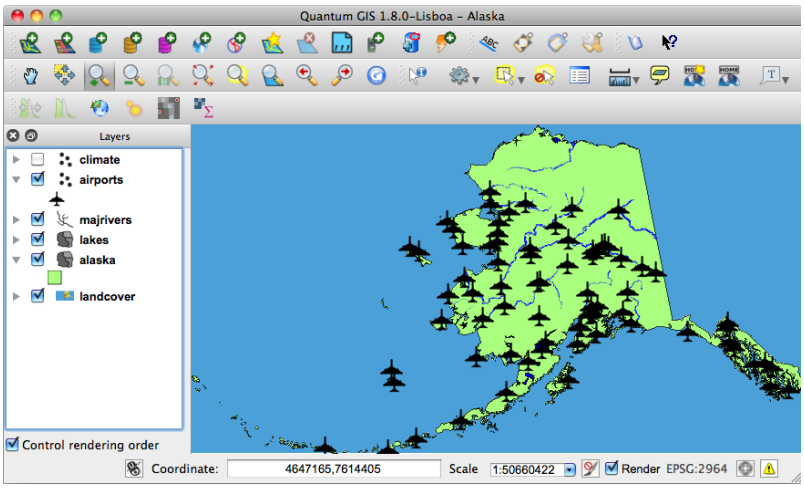


Figure 19.28: Airports of Alaska

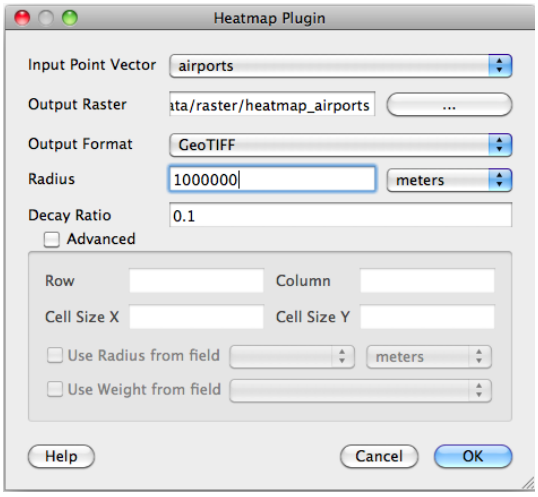


Figure 19.29: The Heatmap Dialog

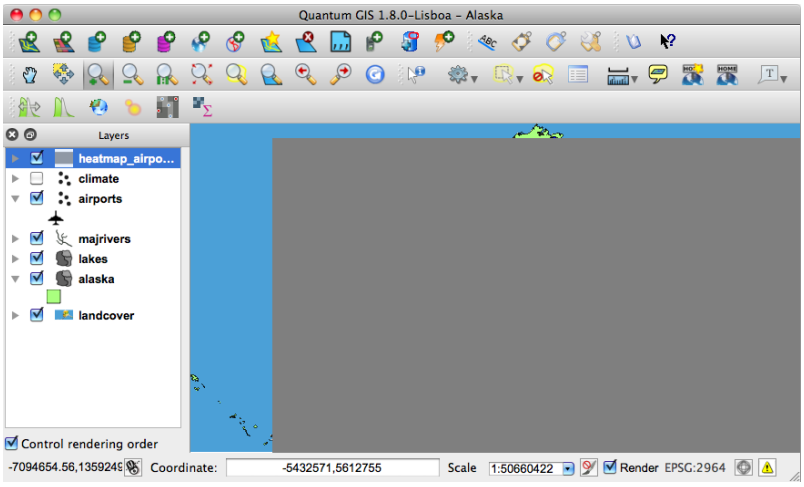


Figure 19.30: The heatmap after loading looks like a grey surface

1. Open the properties dialog of layer `heatmap_airports` (select the layer `heatmap_airports`, open context menu with right mouse button and select *Properties*).
2. Select the *Symbology* tab.
3. Change *Color map*  from 'Grayscale' to 'Pseudocolor'.
4. Press the **[Apply]** button.
5. Switch to the *Transparency* tab and set *Global transparency*  to 40%.
6. Press **[OK]**

The final result is shown in Figure_Heatmap_4.

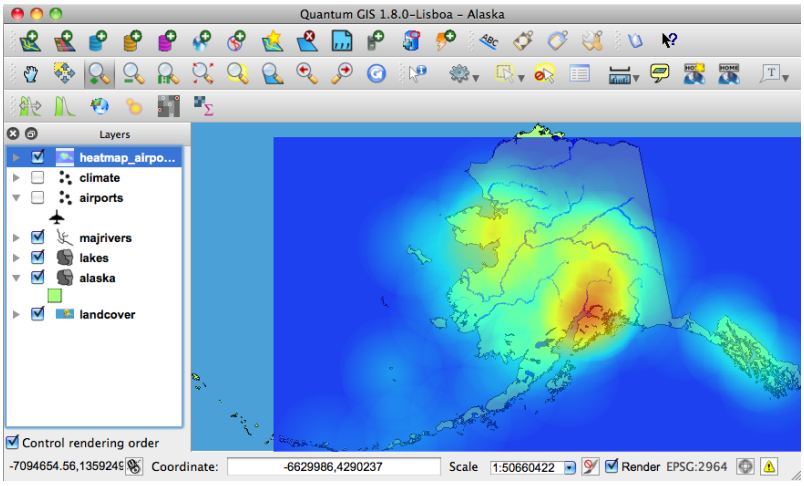


Figure 19.31: Final result of heatmap created of airports of Alaska

19.18 Road Graph Plugin

The Road Graph Plugin is a C++ plugin for QGIS, that calculates the shortest path between two points on any polyline layer and plots this path over the road network.

Main features:

- calculate path, it's length and travel time
- optimize by length or by travel time
- export path to a vector layer
- highlight roads directions (this is slow and used mainly for debug purposes and for the settings testing)

As a roads layer you can use any polyline vector layer in any QGIS supported format. Two lines with a common point are considered connected. Please note, it is required to use layer CRS as project CRS while editing roads layer. This is due to the fact that recalculation of the coordinates between different CRS introduce some errors that can result in discontinuities, even when 'snapping' is used.

In the layer attribute table the following fields can be used:

- speed on road section — numeric field;
- direction — any type, that can be casted to string. Forward and reverse directions are correspond to the one-way road, both directions — two-way road

If some fields don't have any value or do not exist — default values are used. You can change defaults and some plugin settings in plugin settings dialog.

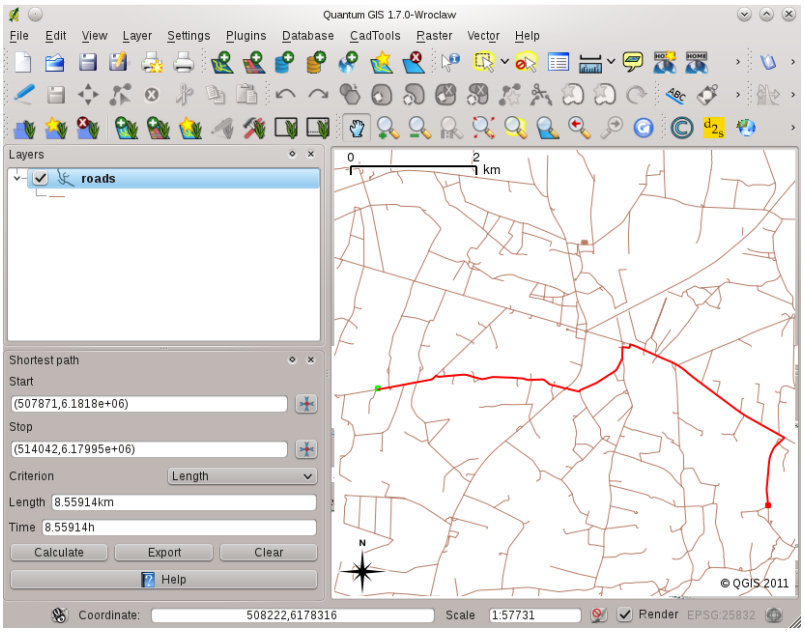


Figure 19.32: Road Graph Plugin

19.18.1 Using the plugin

After plugin activation you will see an additional panel on the left side of the main QGIS window. Now make some definitions to the *Road graph plugin settings* dialog in the menu *Vector* → *Road Graph*.

Select a Start and a Stop point in the road network layer and click on **[Calculate]**.

19.19 Spatial Query Plugin

The Spatial Query plugin allows to make a spatial query (select features) in a target layer with reference to another layer. The functionality is based on the GEOS library and depends on the selected source feature layer.

Possible operators are:

- Contains
- Equals
- Overlap
- Crosses
- Intersects
- Is disjoint
- Touches
- Within

19.19.1 Using the plugin

As an example we want to find regions in the Alaska dataset that contain airports. Following steps are necessary:

1. Start QGIS and load the vector layers `regions.shp` and `airports.shp`.

2. Load the Spatial Query plugin in the Plugin Manager (see Section *Loading a QGIS Core Plugin*) and click on the Spatial Query icon which appears in the QGIS toolbar menu. The plugin dialog appears.
3. Select layer `regions` as source layer and `airports` as reference feature layer.
4. Select 'Contains' as operator and click **[Apply]**.

Now you get a list of feature IDs from the query and you have several options as shown in figure_spatial_query_1.

- Click on the Create layer with list of items
- Select an ID from the list and click on Create layer with selected
- Select the 'Remove from current selection' in the field *And use the result to* .
- Additionally you can *Zoom to item* or display *Log messages*.

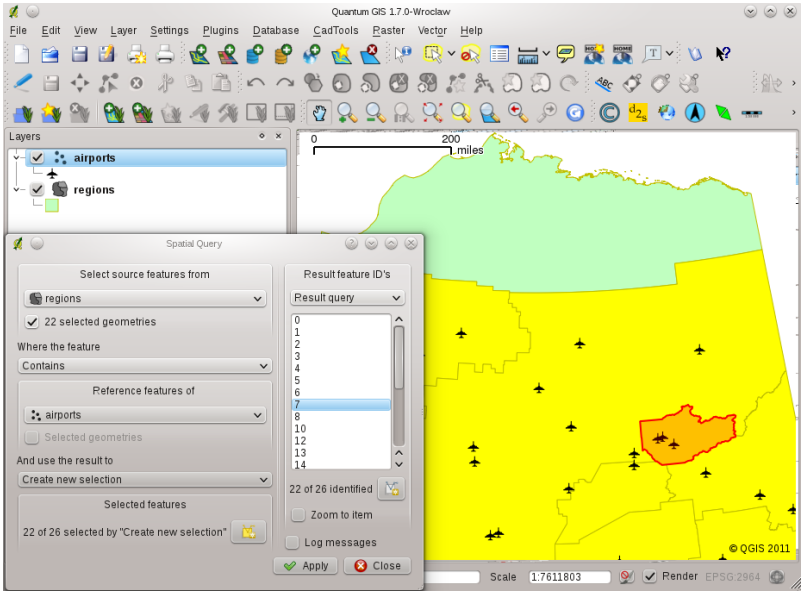


Figure 19.33: Spatial Query analysis - regions contain airports

19.20 SPIT Plugin

QGIS comes with a plugin named SPIT (Shapefile to PostGIS Import Tool). SPIT can be used to load multiple shapefiles at one time and includes support for schemas. To use SPIT, open the Plugin Manager from the *Plugins* menu, check the box next to the *SPIT plugin* and click **[OK]**. The SPIT icon will be added to the plugin toolbar.

To import a shapefile, click on the SPIT tool in the toolbar to open the *SPIT - Shapefile to PostGIS Import Tool* dialog. Select the PostGIS database you want to connect to and click on **[Connect]**. If you want, you can define or change some import options. Now you can add one or more files to the queue by clicking on the **[Add]** button. To process the files, click on the **[OK]** button. The progress of the import as well as any errors/warnings will be displayed as each shapefile is processed.

참고: Importing Shapefiles Containing PostgreSQL Reserved Words

If a shapefile is added to the queue containing fields that are reserved words in the PostgreSQL database a dialog will popup showing the status of each field. You can edit the field names prior to import and change any that are reserved words (or change any other field names as desired). Attempting to import a shapefile with reserved words as field names will likely fail.

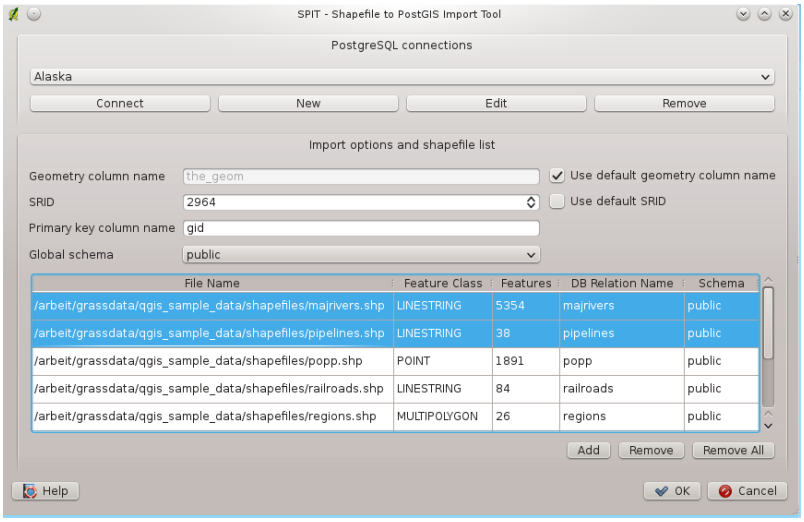


Figure 19.34: Using SPIT Plugin to import Shape files to PostGIS

19.21 SQL Anywhere Plugin

SQL Anywhere is a proprietary relational database management system (RDBMS) from Sybase. SQL Anywhere includes spatial support including OGC, shape files etc. and built in functions to export to KML, GML and SVG formats.

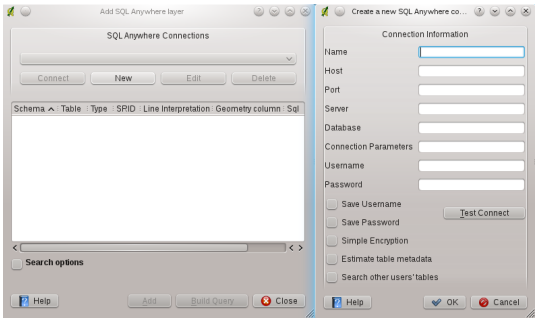


Figure 19.35: SQL Anywhere dialog (KDE)

The SQL Anywhere allows to connect to spatialy enabled SQL Anywhere databases. The *Add SQL Anywhere layer* dialog is similar in functionality to the dialogs for PostGIS and SpatiaLite.

19.22 Zonal Statistics Plugin

With the Zonal Statistics Plugin you can analyze the results of a thematic classification. It allows to calculate several values of the pixels of a raster layer with the help of a polygonal vector layer (see figure_zonal_statistics). You can calculate the sum, the mean value and the total count of the pixels that are within a polygon. The plugin generates output columns in the vector layer with a user-defined prefix.

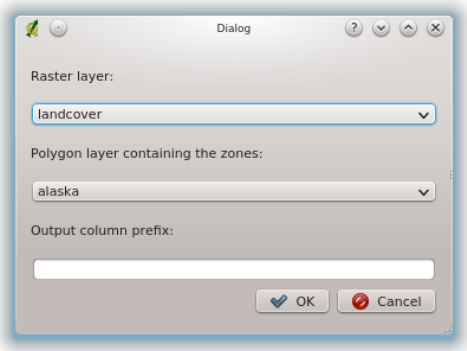


Figure 19.36: Zonal statistics dialog (KDE)

CHAPTER 20

Help and Support

20.1 Mailing lists

QGIS is under active development and as such it won't always work like you expect it to. The preferred way to get help is by joining the qgis-users mailing list. Your questions will reach a broader audience and answers will benefit others.

20.1.1 qgis-users

This mailing list is used for discussion of QGIS in general, as well as specific questions regarding its installation and use. You can subscribe to the qgis-users mailing list by visiting the following URL: http://lists.osgeo.org/mailman/listinfo/qgis-user

20.1.2 fossgis-talk-liste

For the german speaking audience the german FOSSGIS e.V. provides the fossgis-talk-liste mailing list. This mailing list is used for discussion of open source GIS in general including QGIS. You can subscribe to the fossgis-talk-liste mailing list by visiting the following URL: https://lists.fossgis.de/mailman/listinfo/fossgis-talk-liste

20.1.3 qgis-developer

If you are a developer facing problems of a more technical nature, you may want to join the qgis-developer mailing list here: http://lists.osgeo.org/mailman/listinfo/qgis-developer

20.1.4 qgis-commit

Each time a commit is made to the QGIS code repository an email is posted to this list. If you want to be up to date with every change to the current code base, you can subscribe to this list at: http://lists.osgeo.org/mailman/listinfo/qgis-commit

20.1.5 qgis-trac

This list provides email notification related to project management, including bug reports, tasks, and feature requests. You can subscribe to this list at: http://lists.osgeo.org/mailman/listinfo/qgis-trac

20.1.6 qgis-community-team

This list deals with topics like documentation, context help, user-guide, online experience including web sites, blog, mailing lists, forums, and translation efforts. If you like to work on the user-guide as well, this list is a good starting point to ask your questions. You can subscribe to this list at: http://lists.osgeo.org/mailman/listinfo/qgis-community-team

20.1.7 qgis-release-team

This list deals with topics like the release process, packaging binaries for various OS and announcing new releases to the world at large. You can subscribe to this list at: http://lists.osgeo.org/mailman/listinfo/qgis-release-team

20.1.8 qgis-tr

This list deals with the translation efforts. If you like to work on the translation of the manuals or the graphical user interface (GUI), this list is a good starting point to ask your questions. You can subscribe to this list at: http://lists.osgeo.org/mailman/listinfo/qgis-tr

20.1.9 qgis-edu

This list deals with QGIS education efforts. If you like to work on QGIS education materials, this list is a good starting point to ask your questions. You can subscribe to this list at: http://lists.osgeo.org/mailman/listinfo/qgis-edu

20.1.10 qgis-psc

This list is used to discuss Steering Committee issues related to overall management and direction of Quantum GIS. You can subscribe to this list at: http://lists.osgeo.org/mailman/listinfo/qgis-psc

You are welcome to subscribe to any of the lists. Please remember to contribute to the list by answering questions and sharing your experiences. Note that the qgis-commit and qgis-trac are designed for notification only and not meant for user postings.

20.2 IRC

We also maintain a presence on IRC - visit us by joining the #qgis channel on irc.freenode.net. Please wait around for a response to your question as many folks on the channel are doing other things and it may take a while for them to notice your question. Commercial support for QGIS is also available. Check the website http://qgis.org/en/commercial-support.html for more information.

If you missed a discussion on IRC, not a problem! We log all discussion so you can easily catch up. Just go to http://logs.qgis.org and read the IRC-logs.

20.3 BugTracker

While the qgis-users mailing list is useful for general 'how do I do XYZ in QGIS' type questions, you may wish to notify us about bugs in QGIS. You can submit bug reports using the QGIS bug tracker at http://hub.qgis.org/projects/quantum-gis/issues. When creating a new ticket for a bug, please provide an email address where we can request additional information.

Please bear in mind that your bug may not always enjoy the priority you might hope for (depending on its severity). Some bugs may require significant developer effort to remedy and the manpower is not always available for this.

Feature requests can be submitted as well using the same ticket system as for bugs. Please make sure to select the type `Feature`.

If you have found a bug and fixed it yourself you can submit this patch also. Again, the lovely redmine ticketsystem at http://hub.qgis.org/projects/quantum-gis/issues has this type as well. Check `Patch supplied` checkbox and attach your patch before submitting bug. Someone of the developers will review it and apply it to QGIS. Please don't be alarmed if your patch is not applied straight away — developers may be tied up with other committments.

20.4 Blog

The QGIS-community also runs a weblog at http://www.qgis.org/planet which has some interesting articles for users and developers as well provided by other blogs in the community. You are invited to contribute your own QGIS blog!

20.5 Plugins

The website http://plugins.qgis.org provides the official QGIS plugins web portal. Here you find a list of all stable and experimental QGIS plugins available via the 'Official QGIS Plugin Repository'.

20.6 Wiki

Lastly, we maintain a WIKI web site at http://hub.qgis.org/projects/quantum-gis/wiki where you can find a variety of useful information relating to QGIS development, release plans, links to download sites, message translation-hints and so on. Check it out, there are some goodies inside!

CHAPTER 21

Appendix

21.1 GNU General Public License

Version 2, June 1991

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